

T3 UV one-loop RGE term comparison

Each running coupling has its own table. Rows are model configurations and columns are additive structures in the beta function. Every populated cell contains the complete model-specific term, including its numerical coefficient. A dash means that the term is absent for that configuration. All equations use  $16\pi^2\,\mu\,dX/d\mu=\beta_X^{(1)}$ . Gauge-coupling results use the same RGBeta  $g^2\rightarrow g$  conversion as in the previous report.

$\beta_{g_Y}^{(1)}$

Search name: gY

$16\pi^2\,\mu\frac{dg_Y}{d\mu}=\beta_{g_Y}^{(1)}$

| Column Representative term structure |                     |
|--------------------------------------|---------------------|
| T1                                   | $\frac{55g_Y^3}{6}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{55g_Y^3}{6}$  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{43g_Y^3}{6}$  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{59g_Y^3}{6}$  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{103g_Y^3}{6}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{175g_Y^3}{6}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\frac{17g_Y^3}{2}$  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\frac{43g_Y^3}{6}$  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\frac{17g_Y^3}{2}$  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\frac{25g_Y^3}{2}$  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\frac{115g_Y^3}{6}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $\frac{21g_Y^3}{2}$  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $\frac{43g_Y^3}{6}$  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $\frac{21g_Y^3}{2}$  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $\frac{41g_Y^3}{2}$  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------|
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $\frac{223g_Y^3}{6}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $\frac{59g_Y^3}{6}$  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $\frac{43g_Y^3}{6}$  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $\frac{55g_Y^3}{6}$  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $\frac{95g_Y^3}{6}$  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $\frac{163g_Y^3}{6}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{59g_Y^3}{6}$  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{22g_Y^3}{3}$  |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{59g_Y^3}{6}$  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{52g_Y^3}{3}$  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{179g_Y^3}{6}$ |

$\beta_{g_2}^{(1)}$

Search name: g2

$16\pi^2\mu\frac{dg_2}{d\mu}=\beta_{g_2}^{(1)}$

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $-\frac{g_2^3}{2}$            |
| T2     | $-\frac{3g_2^3}{2}$           |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                 | T2                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------|----------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{g_2^3}{2}$ | —                    |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                  | $-\frac{3g_2^3}{2}$  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{g_2^3}{2}$ | —                    |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{g_2^3}{2}$ | —                    |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{g_2^3}{2}$ | —                    |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                  | $-\frac{17g_2^3}{6}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                  | $-\frac{17g_2^3}{6}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                  | $-\frac{17g_2^3}{6}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                  | $-\frac{17g_2^3}{6}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                  | $-\frac{17g_2^3}{6}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                  | $\frac{31g_2^3}{6}$  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                  | $\frac{7g_2^3}{6}$   |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                  | $\frac{31g_2^3}{6}$  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                  | $\frac{31g_2^3}{6}$  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                  | $\frac{31g_2^3}{6}$  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{g_2^3}{2}$ | —                    |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                  | $-\frac{3g_2^3}{2}$  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{g_2^3}{2}$ | —                    |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{g_2^3}{2}$ | —                    |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                 | T2                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------|---------------------|
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{g_2^3}{2}$ | —                   |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{g_2^3}{6}$  | —                   |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                  | $-\frac{5g_2^3}{6}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{g_2^3}{6}$  | —                   |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{g_2^3}{6}$  | —                   |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{g_2^3}{6}$  | —                   |

$\beta_{g_3}^{(1)}$

Search name:  $g_3^3$

$$16\pi^2\,\mu\frac{dg_3}{d\mu}=\beta_{g_3}^{(1)}$$

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $-7g_3^3$                     |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1        |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-7g_3^3$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-7g_3^3$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-7g_3^3$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-7g_3^3$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-7g_3^3$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-7g_3^3$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-7g_3^3$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $-7g_3^3$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-7g_3^3$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-7g_3^3$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-7g_3^3$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-7g_3^3$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-7g_3^3$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-7g_3^3$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-7g_3^3$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-7g_3^3$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-7g_3^3$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-7g_3^3$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-7g_3^3$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-7g_3^3$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-7g_3^3$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-7g_3^3$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-7g_3^3$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-7g_3^3$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-7g_3^3$ |

Search name: yu

## Terms 1 of 3

[illegible]

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1                                                                             | T2                                                                             | T3                                                                              |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| T3-E  | 0        | 3         | 0             | 3         | 1             | 2     | $\frac{1}{2}$ | $3 \operatorname{Tr} \left( Y_d \left( Y_d \right)^\dagger \right) (Y_u)_{ij}$ | $+ \operatorname{Tr} \left( Y_e \left( Y_e \right)^\dagger \right) (Y_u)_{ij}$ | $+3 \operatorname{Tr} \left( Y_u \left( Y_u \right)^\dagger \right) (Y_u)_{ij}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | $3 \operatorname{Tr} \left( Y_d \left( Y_d \right)^\dagger \right) (Y_u)_{ij}$ | $+ \operatorname{Tr} \left( Y_e \left( Y_e \right)^\dagger \right) (Y_u)_{ij}$ | $+3 \operatorname{Tr} \left( Y_u \left( Y_u \right)^\dagger \right) (Y_u)_{ij}$ |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | $3 \operatorname{Tr} \left( Y_d \left( Y_d \right)^\dagger \right) (Y_u)_{ij}$ | $+ \operatorname{Tr} \left( Y_e \left( Y_e \right)^\dagger \right) (Y_u)_{ij}$ | $+3 \operatorname{Tr} \left( Y_u \left( Y_u \right)^\dagger \right) (Y_u)_{ij}$ |

## Terms 2 of 3

| Column | Representative term structure                                               |  |
|--------|-----------------------------------------------------------------------------|--|
| T1     | $-\frac{3}{2}((Y_d (Y_d)^\dagger Y_u)_{ij} - (Y_u (Y_u)^\dagger Y_u)_{ij})$ |  |
| T2     | $-\frac{9}{4}g_2^2 (Y_u)_{ij}$                                              |  |
| T3     | $-8g_3^2 (Y_u)_{ij}$                                                        |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                          | T2                             | T3                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------------------------------------|--------------------------------|----------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{3}{2}((Y_d (Y_d)^\dagger Y_u)_{ij} - (Y_u (Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2 (Y_u)_{ij}$ | $-8g_3^2 (Y_u)_{ij}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{3}{2}((Y_d (Y_d)^\dagger Y_u)_{ij} - (Y_u (Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2 (Y_u)_{ij}$ | $-8g_3^2 (Y_u)_{ij}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{3}{2}((Y_d (Y_d)^\dagger Y_u)_{ij} - (Y_u (Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2 (Y_u)_{ij}$ | $-8g_3^2 (Y_u)_{ij}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{3}{2}((Y_d (Y_d)^\dagger Y_u)_{ij} - (Y_u (Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2 (Y_u)_{ij}$ | $-8g_3^2 (Y_u)_{ij}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{3}{2}((Y_d (Y_d)^\dagger Y_u)_{ij} - (Y_u (Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2 (Y_u)_{ij}$ | $-8g_3^2 (Y_u)_{ij}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-\frac{3}{2}((Y_d (Y_d)^\dagger Y_u)_{ij} - (Y_u (Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2 (Y_u)_{ij}$ | $-8g_3^2 (Y_u)_{ij}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{3}{2}((Y_d (Y_d)^\dagger Y_u)_{ij} - (Y_u (Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2 (Y_u)_{ij}$ | $-8g_3^2 (Y_u)_{ij}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $-\frac{3}{2}((Y_d (Y_d)^\dagger Y_u)_{ij} - (Y_u (Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2 (Y_u)_{ij}$ | $-8g_3^2 (Y_u)_{ij}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-\frac{3}{2}((Y_d (Y_d)^\dagger Y_u)_{ij} - (Y_u (Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2 (Y_u)_{ij}$ | $-8g_3^2 (Y_u)_{ij}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-\frac{3}{2}((Y_d (Y_d)^\dagger Y_u)_{ij} - (Y_u (Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2 (Y_u)_{ij}$ | $-8g_3^2 (Y_u)_{ij}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-\frac{3}{2}((Y_d (Y_d)^\dagger Y_u)_{ij} - (Y_u (Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2 (Y_u)_{ij}$ | $-8g_3^2 (Y_u)_{ij}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-\frac{3}{2}((Y_d (Y_d)^\dagger Y_u)_{ij} - (Y_u (Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2 (Y_u)_{ij}$ | $-8g_3^2 (Y_u)_{ij}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-\frac{3}{2}((Y_d (Y_d)^\dagger Y_u)_{ij} - (Y_u (Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2 (Y_u)_{ij}$ | $-8g_3^2 (Y_u)_{ij}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-\frac{3}{2}((Y_d (Y_d)^\dagger Y_u)_{ij} - (Y_u (Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2 (Y_u)_{ij}$ | $-8g_3^2 (Y_u)_{ij}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-\frac{3}{2}((Y_d (Y_d)^\dagger Y_u)_{ij} - (Y_u (Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2 (Y_u)_{ij}$ | $-8g_3^2 (Y_u)_{ij}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{3}{2}((Y_d (Y_d)^\dagger Y_u)_{ij} - (Y_u (Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2 (Y_u)_{ij}$ | $-8g_3^2 (Y_u)_{ij}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-\frac{3}{2}((Y_d (Y_d)^\dagger Y_u)_{ij} - (Y_u (Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2 (Y_u)_{ij}$ | $-8g_3^2 (Y_u)_{ij}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{3}{2}((Y_d (Y_d)^\dagger Y_u)_{ij} - (Y_u (Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2 (Y_u)_{ij}$ | $-8g_3^2 (Y_u)_{ij}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{3}{2}((Y_d (Y_d)^\dagger Y_u)_{ij} - (Y_u (Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2 (Y_u)_{ij}$ | $-8g_3^2 (Y_u)_{ij}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{3}{2}((Y_d (Y_d)^\dagger Y_u)_{ij} - (Y_u (Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2 (Y_u)_{ij}$ | $-8g_3^2 (Y_u)_{ij}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{3}{2}((Y_d (Y_d)^\dagger Y_u)_{ij} - (Y_u (Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2 (Y_u)_{ij}$ | $-8g_3^2 (Y_u)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1                                                                          | T2                             | T3                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|---------------|-----------------------------------------------------------------------------|--------------------------------|----------------------|
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0             | $-\frac{3}{2}((Y_d (Y_d)^\dagger Y_u)_{ij} - (Y_u (Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2 (Y_u)_{ij}$ | $-8g_3^2 (Y_u)_{ij}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$ | $-\frac{3}{2}((Y_d (Y_d)^\dagger Y_u)_{ij} - (Y_u (Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2 (Y_u)_{ij}$ | $-8g_3^2 (Y_u)_{ij}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1             | $-\frac{3}{2}((Y_d (Y_d)^\dagger Y_u)_{ij} - (Y_u (Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2 (Y_u)_{ij}$ | $-8g_3^2 (Y_u)_{ij}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$ | $-\frac{3}{2}((Y_d (Y_d)^\dagger Y_u)_{ij} - (Y_u (Y_u)^\dagger Y_u)_{ij})$ | $-\frac{9}{4}g_2^2 (Y_u)_{ij}$ | $-8g_3^2 (Y_u)_{ij}$ |

Terms 3 of 3

| Column | Representative term structure   |
|--------|---------------------------------|
| T1     | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                              |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{17}{12}g_Y^2(Y_u)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1                                |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|-----------------------------------|
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | $-\frac{17}{12}g_Y{}^2(Y_u)_{ij}$ |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | $-\frac{17}{12}g_Y{}^2(Y_u)_{ij}$ |

Search name: yd

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[illegible]

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1                                                                         | T2                                                                           | T3                                                                           |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|----------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| T3-E  | 0        | 3         | 0             | 3         | 1             | 2     | $\frac{1}{2}$ | $\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)\left(Y_d\right)_{ij}$ | $+3\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)\left(Y_d\right)_{ij}$ | $+3\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)\left(Y_d\right)_{ij}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | $\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)\left(Y_d\right)_{ij}$ | $+3\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)\left(Y_d\right)_{ij}$ | $+3\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)\left(Y_d\right)_{ij}$ |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | $\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)\left(Y_d\right)_{ij}$ | $+3\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)\left(Y_d\right)_{ij}$ | $+3\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)\left(Y_d\right)_{ij}$ |

Terms 2 of 3

| Column | Representative term structure                                                         |
|--------|---------------------------------------------------------------------------------------|
| T1     | $+ \frac{3}{2} ((Y_d \ (Y_d)^\dagger \ Y_d)_{ij} - (Y_u \ (Y_u)^\dagger \ Y_d)_{ij})$ |
| T2     | $- \frac{9}{4} g_2^2 \ (Y_d)_{ij}$                                                    |
| T3     | $- 8 g_3^2 \ (Y_d)_{ij}$                                                              |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                    | T2                                 | T3                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------------------------------|------------------------------------|--------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+ \frac{3}{2} ((Y_d \ (Y_d)^\dagger \ Y_d)_{ij} - (Y_u \ (Y_u)^\dagger \ Y_d)_{ij})$ | $- \frac{9}{4} g_2^2 \ (Y_d)_{ij}$ | $- 8 g_3^2 \ (Y_d)_{ij}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+ \frac{3}{2} ((Y_d \ (Y_d)^\dagger \ Y_d)_{ij} - (Y_u \ (Y_u)^\dagger \ Y_d)_{ij})$ | $- \frac{9}{4} g_2^2 \ (Y_d)_{ij}$ | $- 8 g_3^2 \ (Y_d)_{ij}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+ \frac{3}{2} ((Y_d \ (Y_d)^\dagger \ Y_d)_{ij} - (Y_u \ (Y_u)^\dagger \ Y_d)_{ij})$ | $- \frac{9}{4} g_2^2 \ (Y_d)_{ij}$ | $- 8 g_3^2 \ (Y_d)_{ij}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+ \frac{3}{2} ((Y_d \ (Y_d)^\dagger \ Y_d)_{ij} - (Y_u \ (Y_u)^\dagger \ Y_d)_{ij})$ | $- \frac{9}{4} g_2^2 \ (Y_d)_{ij}$ | $- 8 g_3^2 \ (Y_d)_{ij}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+ \frac{3}{2} ((Y_d \ (Y_d)^\dagger \ Y_d)_{ij} - (Y_u \ (Y_u)^\dagger \ Y_d)_{ij})$ | $- \frac{9}{4} g_2^2 \ (Y_d)_{ij}$ | $- 8 g_3^2 \ (Y_d)_{ij}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+ \frac{3}{2} ((Y_d \ (Y_d)^\dagger \ Y_d)_{ij} - (Y_u \ (Y_u)^\dagger \ Y_d)_{ij})$ | $- \frac{9}{4} g_2^2 \ (Y_d)_{ij}$ | $- 8 g_3^2 \ (Y_d)_{ij}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+ \frac{3}{2} ((Y_d \ (Y_d)^\dagger \ Y_d)_{ij} - (Y_u \ (Y_u)^\dagger \ Y_d)_{ij})$ | $- \frac{9}{4} g_2^2 \ (Y_d)_{ij}$ | $- 8 g_3^2 \ (Y_d)_{ij}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $+ \frac{3}{2} ((Y_d \ (Y_d)^\dagger \ Y_d)_{ij} - (Y_u \ (Y_u)^\dagger \ Y_d)_{ij})$ | $- \frac{9}{4} g_2^2 \ (Y_d)_{ij}$ | $- 8 g_3^2 \ (Y_d)_{ij}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $+ \frac{3}{2} ((Y_d \ (Y_d)^\dagger \ Y_d)_{ij} - (Y_u \ (Y_u)^\dagger \ Y_d)_{ij})$ | $- \frac{9}{4} g_2^2 \ (Y_d)_{ij}$ | $- 8 g_3^2 \ (Y_d)_{ij}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+ \frac{3}{2} ((Y_d \ (Y_d)^\dagger \ Y_d)_{ij} - (Y_u \ (Y_u)^\dagger \ Y_d)_{ij})$ | $- \frac{9}{4} g_2^2 \ (Y_d)_{ij}$ | $- 8 g_3^2 \ (Y_d)_{ij}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+ \frac{3}{2} ((Y_d \ (Y_d)^\dagger \ Y_d)_{ij} - (Y_u \ (Y_u)^\dagger \ Y_d)_{ij})$ | $- \frac{9}{4} g_2^2 \ (Y_d)_{ij}$ | $- 8 g_3^2 \ (Y_d)_{ij}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+ \frac{3}{2} ((Y_d \ (Y_d)^\dagger \ Y_d)_{ij} - (Y_u \ (Y_u)^\dagger \ Y_d)_{ij})$ | $- \frac{9}{4} g_2^2 \ (Y_d)_{ij}$ | $- 8 g_3^2 \ (Y_d)_{ij}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $+ \frac{3}{2} ((Y_d \ (Y_d)^\dagger \ Y_d)_{ij} - (Y_u \ (Y_u)^\dagger \ Y_d)_{ij})$ | $- \frac{9}{4} g_2^2 \ (Y_d)_{ij}$ | $- 8 g_3^2 \ (Y_d)_{ij}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $+ \frac{3}{2} ((Y_d \ (Y_d)^\dagger \ Y_d)_{ij} - (Y_u \ (Y_u)^\dagger \ Y_d)_{ij})$ | $- \frac{9}{4} g_2^2 \ (Y_d)_{ij}$ | $- 8 g_3^2 \ (Y_d)_{ij}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+ \frac{3}{2} ((Y_d \ (Y_d)^\dagger \ Y_d)_{ij} - (Y_u \ (Y_u)^\dagger \ Y_d)_{ij})$ | $- \frac{9}{4} g_2^2 \ (Y_d)_{ij}$ | $- 8 g_3^2 \ (Y_d)_{ij}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $+ \frac{3}{2} ((Y_d \ (Y_d)^\dagger \ Y_d)_{ij} - (Y_u \ (Y_u)^\dagger \ Y_d)_{ij})$ | $- \frac{9}{4} g_2^2 \ (Y_d)_{ij}$ | $- 8 g_3^2 \ (Y_d)_{ij}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $+ \frac{3}{2} ((Y_d \ (Y_d)^\dagger \ Y_d)_{ij} - (Y_u \ (Y_u)^\dagger \ Y_d)_{ij})$ | $- \frac{9}{4} g_2^2 \ (Y_d)_{ij}$ | $- 8 g_3^2 \ (Y_d)_{ij}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $+ \frac{3}{2} ((Y_d \ (Y_d)^\dagger \ Y_d)_{ij} - (Y_u \ (Y_u)^\dagger \ Y_d)_{ij})$ | $- \frac{9}{4} g_2^2 \ (Y_d)_{ij}$ | $- 8 g_3^2 \ (Y_d)_{ij}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $+ \frac{3}{2} ((Y_d \ (Y_d)^\dagger \ Y_d)_{ij} - (Y_u \ (Y_u)^\dagger \ Y_d)_{ij})$ | $- \frac{9}{4} g_2^2 \ (Y_d)_{ij}$ | $- 8 g_3^2 \ (Y_d)_{ij}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $+ \frac{3}{2} ((Y_d \ (Y_d)^\dagger \ Y_d)_{ij} - (Y_u \ (Y_u)^\dagger \ Y_d)_{ij})$ | $- \frac{9}{4} g_2^2 \ (Y_d)_{ij}$ | $- 8 g_3^2 \ (Y_d)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                        | T2                              | T3                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------------------------------------------|---------------------------------|----------------------|
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+\frac{3}{2} \left( (Y_d (Y_d)^\dagger Y_d)_{ij} - (Y_u (Y_u)^\dagger Y_d)_{ij} \right)$ | $-\frac{9}{4} g_2^2 (Y_d)_{ij}$ | $-8g_3^2 (Y_d)_{ij}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+\frac{3}{2} \left( (Y_d (Y_d)^\dagger Y_d)_{ij} - (Y_u (Y_u)^\dagger Y_d)_{ij} \right)$ | $-\frac{9}{4} g_2^2 (Y_d)_{ij}$ | $-8g_3^2 (Y_d)_{ij}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+\frac{3}{2} \left( (Y_d (Y_d)^\dagger Y_d)_{ij} - (Y_u (Y_u)^\dagger Y_d)_{ij} \right)$ | $-\frac{9}{4} g_2^2 (Y_d)_{ij}$ | $-8g_3^2 (Y_d)_{ij}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+\frac{3}{2} \left( (Y_d (Y_d)^\dagger Y_d)_{ij} - (Y_u (Y_u)^\dagger Y_d)_{ij} \right)$ | $-\frac{9}{4} g_2^2 (Y_d)_{ij}$ | $-8g_3^2 (Y_d)_{ij}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+\frac{3}{2} \left( (Y_d (Y_d)^\dagger Y_d)_{ij} - (Y_u (Y_u)^\dagger Y_d)_{ij} \right)$ | $-\frac{9}{4} g_2^2 (Y_d)_{ij}$ | $-8g_3^2 (Y_d)_{ij}$ |

Terms 3 of 3

| Column | Representative term structure  |
|--------|--------------------------------|
| T1     | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{5}{12}g_Y^2(Y_d)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1                               |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|----------------------------------|
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | $-\frac{5}{12}g_Y{}^2(Y_d)_{ij}$ |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | $-\frac{5}{12}g_Y{}^2(Y_d)_{ij}$ |

Search name: ye

## Terms 1 of 3

[illegible]

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1                                                                             | T2                                                                              | T3                                                                             |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| T3-E  | 0        | 3         | 0             | 3         | 1             | 2     | $\frac{1}{2}$ | $3 \operatorname{Tr} \left( Y_d \left( Y_d \right)^\dagger \right) (Y_e)_{ij}$ | $+3 \operatorname{Tr} \left( Y_u \left( Y_u \right)^\dagger \right) (Y_e)_{ij}$ | $+ \operatorname{Tr} \left( Y_e \left( Y_e \right)^\dagger \right) (Y_e)_{ij}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | $3 \operatorname{Tr} \left( Y_d \left( Y_d \right)^\dagger \right) (Y_e)_{ij}$ | $+3 \operatorname{Tr} \left( Y_u \left( Y_u \right)^\dagger \right) (Y_e)_{ij}$ | $+ \operatorname{Tr} \left( Y_e \left( Y_e \right)^\dagger \right) (Y_e)_{ij}$ |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | $3 \operatorname{Tr} \left( Y_d \left( Y_d \right)^\dagger \right) (Y_e)_{ij}$ | $+3 \operatorname{Tr} \left( Y_u \left( Y_u \right)^\dagger \right) (Y_e)_{ij}$ | $+ \operatorname{Tr} \left( Y_e \left( Y_e \right)^\dagger \right) (Y_e)_{ij}$ |

Terms 2 of 3

| Column | Representative term structure                                                                                                                                                    |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\frac{1}{4}(2\left(y_1\left(y_1\right)^{\dagger} Y_e\right)_{i j}+3\left(y_2\left(y_2\right)^{\dagger} Y_e\right)_{i j}+6\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j})$ |
| T2     | $-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$                                                                                                                                       |
| T3     | $-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$                                                                                                                                      |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                                                | T2                                         | T3                                          |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|---------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+\frac{1}{4}(2\left(y_1\left(y_1\right)^{\dagger} Y_e\right)_{i j}+3\left(y_2\left(y_2\right)^{\dagger} Y_e\right)_{i j}+6\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j})$ | $-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$ | $-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+\frac{1}{4}(2\left(y_1\left(y_1\right)^{\dagger} Y_e\right)_{i j}+3\left(y_2\left(y_2\right)^{\dagger} Y_e\right)_{i j}+6\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j})$ | $-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$ | $-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+\frac{1}{4}(2\left(y_1\left(y_1\right)^{\dagger} Y_e\right)_{i j}+3\left(y_2\left(y_2\right)^{\dagger} Y_e\right)_{i j}+6\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j})$ | $-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$ | $-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+\frac{1}{4}(2\left(y_1\left(y_1\right)^{\dagger} Y_e\right)_{i j}+3\left(y_2\left(y_2\right)^{\dagger} Y_e\right)_{i j}+6\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j})$ | $-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$ | $-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+\frac{1}{4}(2\left(y_1\left(y_1\right)^{\dagger} Y_e\right)_{i j}+3\left(y_2\left(y_2\right)^{\dagger} Y_e\right)_{i j}+6\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j})$ | $-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$ | $-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                                                 | $-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$ | $-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                                                                                                                                 | $-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$ | $-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                                                 | $-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$ | $-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                                                 | $-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$ | $-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                                                 | $-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$ | $-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                                                 | $-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$ | $-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                                                                                                                                 | $-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$ | $-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                                                 | $-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$ | $-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                                                 | $-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$ | $-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                                                 | $-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$ | $-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $+\frac{1}{4}(3\left(y_1\left(y_1\right)^{\dagger} Y_e\right)_{i j}+2\left(y_2\left(y_2\right)^{\dagger} Y_e\right)_{i j}+6\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j})$ | $-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$ | $-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $+\frac{1}{4}(3\left(y_1\left(y_1\right)^{\dagger} Y_e\right)_{i j}+2\left(y_2\left(y_2\right)^{\dagger} Y_e\right)_{i j}+6\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j})$ | $-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$ | $-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $+\frac{1}{4}(3\left(y_1\left(y_1\right)^{\dagger} Y_e\right)_{i j}+2\left(y_2\left(y_2\right)^{\dagger} Y_e\right)_{i j}+6\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j})$ | $-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$ | $-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $+\frac{1}{4}(3\left(y_1\left(y_1\right)^{\dagger} Y_e\right)_{i j}+2\left(y_2\left(y_2\right)^{\dagger} Y_e\right)_{i j}+6\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j})$ | $-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$ | $-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $+\frac{1}{4}(3\left(y_1\left(y_1\right)^{\dagger} Y_e\right)_{i j}+2\left(y_2\left(y_2\right)^{\dagger} Y_e\right)_{i j}+6\left(Y_e\left(Y_e\right)^{\dagger} Y_e\right)_{i j})$ | $-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$ | $-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                 | $-\frac{9}{4} g_2^2\left(Y_e\right)_{i j}$ | $-\frac{15}{4} g_Y^2\left(Y_e\right)_{i j}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1 | T2                            | T3                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|---------------|----|-------------------------------|--------------------------------|
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0             | —  | $-\frac{9}{4}g_2^2(Y_e)_{ij}$ | $-\frac{15}{4}g_Y^2(Y_e)_{ij}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$ | —  | $-\frac{9}{4}g_2^2(Y_e)_{ij}$ | $-\frac{15}{4}g_Y^2(Y_e)_{ij}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1             | —  | $-\frac{9}{4}g_2^2(Y_e)_{ij}$ | $-\frac{15}{4}g_Y^2(Y_e)_{ij}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$ | —  | $-\frac{9}{4}g_2^2(Y_e)_{ij}$ | $-\frac{15}{4}g_Y^2(Y_e)_{ij}$ |

Terms 3 of 3

| Column | Representative term structure                                                                                   |
|--------|-----------------------------------------------------------------------------------------------------------------|
| T1     | $+\frac{1}{2}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+3\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                              |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                               |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                               |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                               |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                               |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                               |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+\frac{1}{2}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+3\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+\frac{1}{2}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+3\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $+\frac{1}{2}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+3\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $+\frac{1}{2}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+3\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+\frac{1}{2}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+3\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+\frac{3}{4}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+2\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+\frac{3}{4}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+2\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $+\frac{3}{4}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+2\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $+\frac{3}{4}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+2\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+\frac{3}{4}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+2\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                               |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                               |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                               |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                               |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                               |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+\frac{3}{4}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+2\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+\frac{3}{4}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+2\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+\frac{3}{4}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+2\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+\frac{3}{4}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+2\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+\frac{3}{4}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+2\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$ |

$\beta_{y_1}^{(1)}$

Search name:  $y_1$

$16\pi^2\,\mu\frac{dy_1}{d\mu}=\beta_{y_1}^{(1)}$

Terms 1 of 5

| Column | Representative term structure                            |
|--------|----------------------------------------------------------|
| T1     | $2\,\mathrm{Tr}\Big(y_1\,(y_1)^\dagger\Big)\,(y_1)_{ij}$ |
| T2     | $+\frac{3}{4}\Big(y_2\,(y_2)^\dagger\,y_1\Big)_{ij}$     |
| T3     | $+\frac{1}{2}\Big(Y_e\,(Y_e)^\dagger\,y_1\Big)_{ij}$     |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                       | T2                                                   | T3                                                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------|------------------------------------------------------|------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $2\,\mathrm{Tr}\Big(y_1\,(y_1)^\dagger\Big)\,(y_1)_{ij}$ | $+\frac{3}{4}\Big(y_2\,(y_2)^\dagger\,y_1\Big)_{ij}$ | $+\frac{1}{2}\Big(Y_e\,(Y_e)^\dagger\,y_1\Big)_{ij}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $2\,\mathrm{Tr}\Big(y_1\,(y_1)^\dagger\Big)\,(y_1)_{ij}$ | —                                                    | —                                                    |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $2\,\mathrm{Tr}\Big(y_1\,(y_1)^\dagger\Big)\,(y_1)_{ij}$ | $+\frac{3}{4}\Big(y_2\,(y_2)^\dagger\,y_1\Big)_{ij}$ | $+\frac{1}{2}\Big(Y_e\,(Y_e)^\dagger\,y_1\Big)_{ij}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $2\,\mathrm{Tr}\Big(y_1\,(y_1)^\dagger\Big)\,(y_1)_{ij}$ | $+\frac{3}{4}\Big(y_2\,(y_2)^\dagger\,y_1\Big)_{ij}$ | $+\frac{1}{2}\Big(Y_e\,(Y_e)^\dagger\,y_1\Big)_{ij}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $2\,\mathrm{Tr}\Big(y_1\,(y_1)^\dagger\Big)\,(y_1)_{ij}$ | $+\frac{3}{4}\Big(y_2\,(y_2)^\dagger\,y_1\Big)_{ij}$ | $+\frac{1}{2}\Big(Y_e\,(Y_e)^\dagger\,y_1\Big)_{ij}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                        | —                                                    | —                                                    |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                        | —                                                    | —                                                    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                        | —                                                    | —                                                    |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                        | —                                                    | —                                                    |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                        | —                                                    | —                                                    |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                        | —                                                    | —                                                    |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                        | —                                                    | —                                                    |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                        | —                                                    | —                                                    |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                        | —                                                    | —                                                    |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                        | —                                                    | —                                                    |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                        | —                                                    | —                                                    |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                        | —                                                    | —                                                    |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                        | —                                                    | —                                                    |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                        | —                                                    | —                                                    |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                        | —                                                    | —                                                    |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                        | —                                                    | —                                                    |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                        | —                                                    | —                                                    |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                        | —                                                    | —                                                    |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                        | —                                                    | —                                                    |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                        | —                                                    | —                                                    |

Terms 2 of 5

| Column | Representative term structure                                |
|--------|--------------------------------------------------------------|
| T1     | $+ \left( y_1 \left( y_1 \right)^{\dagger} y_1 \right)_{ij}$ |
| T2     | $-\frac{9}{2} g_2^2 \left( y_1 \right)_{ij}$                 |
| T3     | $-\frac{3}{2} g_Y^2 \left( y_1 \right)_{ij}$                 |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                           | T2                                            | T3                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------|-----------------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+ \left( y_1 \left( y_1 \right)^{\dagger} y_1 \right)_{ij}$ | $-\frac{9}{2} g_2^2 \left( y_1 \right)_{ij}$  | $-\frac{3}{2} g_Y^2 \left( y_1 \right)_{ij}$  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                            | $-\frac{9}{2} g_2^2 \left( y_1 \right)_{ij}$  | $-\frac{3}{4} g_Y^2 \left( y_1 \right)_{ij}$  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+ \left( y_1 \left( y_1 \right)^{\dagger} y_1 \right)_{ij}$ | $-\frac{9}{2} g_2^2 \left( y_1 \right)_{ij}$  | $-\frac{3}{2} g_Y^2 \left( y_1 \right)_{ij}$  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+ \left( y_1 \left( y_1 \right)^{\dagger} y_1 \right)_{ij}$ | $-\frac{9}{2} g_2^2 \left( y_1 \right)_{ij}$  | $-\frac{15}{4} g_Y^2 \left( y_1 \right)_{ij}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+ \left( y_1 \left( y_1 \right)^{\dagger} y_1 \right)_{ij}$ | $-\frac{9}{2} g_2^2 \left( y_1 \right)_{ij}$  | $-\frac{15}{2} g_Y^2 \left( y_1 \right)_{ij}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                            | $-\frac{9}{4} g_2^2 \left( y_1 \right)_{ij}$  | $-\frac{3}{2} g_Y^2 \left( y_1 \right)_{ij}$  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                            | $-\frac{9}{4} g_2^2 \left( y_1 \right)_{ij}$  | $-\frac{3}{4} g_Y^2 \left( y_1 \right)_{ij}$  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                            | $-\frac{9}{4} g_2^2 \left( y_1 \right)_{ij}$  | $-\frac{3}{2} g_Y^2 \left( y_1 \right)_{ij}$  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                            | $-\frac{9}{4} g_2^2 \left( y_1 \right)_{ij}$  | $-\frac{15}{4} g_Y^2 \left( y_1 \right)_{ij}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                            | $-\frac{9}{4} g_2^2 \left( y_1 \right)_{ij}$  | $-\frac{15}{2} g_Y^2 \left( y_1 \right)_{ij}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                            | $-\frac{33}{4} g_2^2 \left( y_1 \right)_{ij}$ | $-\frac{3}{2} g_Y^2 \left( y_1 \right)_{ij}$  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                            | $-\frac{33}{4} g_2^2 \left( y_1 \right)_{ij}$ | $-\frac{3}{4} g_Y^2 \left( y_1 \right)_{ij}$  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                            | $-\frac{33}{4} g_2^2 \left( y_1 \right)_{ij}$ | $-\frac{3}{2} g_Y^2 \left( y_1 \right)_{ij}$  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                            | $-\frac{33}{4} g_2^2 \left( y_1 \right)_{ij}$ | $-\frac{15}{4} g_Y^2 \left( y_1 \right)_{ij}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                            | $-\frac{33}{4} g_2^2 \left( y_1 \right)_{ij}$ | $-\frac{15}{2} g_Y^2 \left( y_1 \right)_{ij}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                            | $-\frac{9}{2} g_2^2 \left( y_1 \right)_{ij}$  | $-\frac{3}{2} g_Y^2 \left( y_1 \right)_{ij}$  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                            | $-\frac{9}{2} g_2^2 \left( y_1 \right)_{ij}$  | $-\frac{3}{4} g_Y^2 \left( y_1 \right)_{ij}$  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                            | $-\frac{9}{2} g_2^2 \left( y_1 \right)_{ij}$  | $-\frac{3}{2} g_Y^2 \left( y_1 \right)_{ij}$  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                            | $-\frac{9}{2} g_2^2 \left( y_1 \right)_{ij}$  | $-\frac{15}{4} g_Y^2 \left( y_1 \right)_{ij}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                            | $-\frac{9}{2} g_2^2 \left( y_1 \right)_{ij}$  | $-\frac{15}{2} g_Y^2 \left( y_1 \right)_{ij}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                            | $-\frac{9}{2} g_2^2 \left( y_1 \right)_{ij}$  | $-\frac{3}{2} g_Y^2 \left( y_1 \right)_{ij}$  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1 | T2                              | T3                               |
|-------|----------|-----------|----------------|-----------|---------------|-------|---------------|----|---------------------------------|----------------------------------|
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0             | —  | $-\frac{9}{2}g_2{}^2(y_1)_{ij}$ | $-\frac{3}{4}g_Y{}^2(y_1)_{ij}$  |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$ | —  | $-\frac{9}{2}g_2{}^2(y_1)_{ij}$ | $-\frac{3}{2}g_Y{}^2(y_1)_{ij}$  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1             | —  | $-\frac{9}{2}g_2{}^2(y_1)_{ij}$ | $-\frac{15}{4}g_Y{}^2(y_1)_{ij}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$ | —  | $-\frac{9}{2}g_2{}^2(y_1)_{ij}$ | $-\frac{15}{2}g_Y{}^2(y_1)_{ij}$ |

Terms 3 of 5

| Column | Representative term structure                                                                                                                                                                                                   |  |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| T1     | $+\frac{1}{4}(3\left(y_1\left(y_2\right)^{\dagger}y_2\right)_{ij}+3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+4\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$ |  |
| T2     | $\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$                                                                                                                                                      |  |
| T3     | $+\frac{1}{2}((y_2\left(y_2\right)^{\dagger}y_1)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+3\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$                                                                  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                                                                                              | T2                                                                          | T3                                                                                                                                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                               | —                                                                           | —                                                                                                                                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+\frac{1}{4}(3\left(y_1\left(y_2\right)^{\dagger}y_2\right)_{ij}+3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+4\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$ | —                                                                           | —                                                                                                                                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                               | —                                                                           | —                                                                                                                                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                               | —                                                                           | —                                                                                                                                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                               | —                                                                           | —                                                                                                                                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                               | $\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$  | $+\frac{1}{2}((y_2\left(y_2\right)^{\dagger}y_1)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+3\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                                                                                                                                                                               | $+\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$ | —                                                                                                                                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                                                                                               | $\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$  | $+\frac{1}{2}((y_2\left(y_2\right)^{\dagger}y_1)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+3\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                                                                                               | $\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$  | $+\frac{1}{2}((y_2\left(y_2\right)^{\dagger}y_1)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+3\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                                                                                               | $\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$  | $+\frac{1}{2}((y_2\left(y_2\right)^{\dagger}y_1)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+3\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                               | —                                                                           | —                                                                                                                                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+\frac{1}{4}(2\left(y_1\left(y_2\right)^{\dagger}y_2\right)_{ij}+3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+5\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$ | —                                                                           | —                                                                                                                                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                                                                                               | —                                                                           | —                                                                                                                                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                                                                                               | —                                                                           | —                                                                                                                                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                                                                                               | —                                                                           | —                                                                                                                                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                               | $\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$  | $+\frac{1}{2}((y_2\left(y_2\right)^{\dagger}y_1)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+3\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                               | $\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$  | —                                                                                                                                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                               | $\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$  | $+\frac{1}{2}((y_2\left(y_2\right)^{\dagger}y_1)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+3\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                               | $\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$  | $+\frac{1}{2}((y_2\left(y_2\right)^{\dagger}y_1)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+3\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                               | $\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$  | $+\frac{1}{2}((y_2\left(y_2\right)^{\dagger}y_1)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+3\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                               | $\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$  | —                                                                                                                                                              |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+\frac{1}{4}(3\left(y_1\left(y_2\right)^{\dagger}y_2\right)_{ij}+3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+6\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$ | $+\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$ | —                                                                                                                                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                               | $\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$  | —                                                                                                                                                              |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                               | $\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$  | —                                                                                                                                                              |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                               | $\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$  | —                                                                                                                                                              |

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| Column | Representative term structure                                                                                                                                                                                                            |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\text{Tr}\left(y_1\left(y_2\right)^{\dagger}\right)\left(y_2\right)_{ij}$                                                                                                                                                               |
| T2     | $+\frac{1}{2}\left(2\left(y_1\left(y_2\right)^{\dagger}y_2\right)_{ij}+\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+3\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij}\right)$ |
| T3     | $\frac{3}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$                                                                                                                                                    |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                         | T2                                                                                                                                                                                                                                       | T3                                                                                     |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                          | —                                                                                                                                                                                                                                        | —                                                                                      |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                          | —                                                                                                                                                                                                                                        | —                                                                                      |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                          | —                                                                                                                                                                                                                                        | —                                                                                      |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                          | —                                                                                                                                                                                                                                        | —                                                                                      |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                          | —                                                                                                                                                                                                                                        | —                                                                                      |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                          | —                                                                                                                                                                                                                                        | —                                                                                      |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\text{Tr}\left(y_1\left(y_2\right)^{\dagger}\right)\left(y_2\right)_{ij}$ | $+\frac{1}{2}\left(2\left(y_1\left(y_2\right)^{\dagger}y_2\right)_{ij}+\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+3\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij}\right)$ | —                                                                                      |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                          | —                                                                                                                                                                                                                                        | —                                                                                      |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                          | —                                                                                                                                                                                                                                        | —                                                                                      |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                          | —                                                                                                                                                                                                                                        | —                                                                                      |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                          | —                                                                                                                                                                                                                                        | $\frac{3}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                          | —                                                                                                                                                                                                                                        | $+\frac{3}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                          | —                                                                                                                                                                                                                                        | $\frac{3}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                          | —                                                                                                                                                                                                                                        | $\frac{3}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                          | —                                                                                                                                                                                                                                        | $\frac{3}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                          | —                                                                                                                                                                                                                                        | —                                                                                      |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                          | —                                                                                                                                                                                                                                        | —                                                                                      |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                          | —                                                                                                                                                                                                                                        | —                                                                                      |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                          | —                                                                                                                                                                                                                                        | —                                                                                      |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                          | —                                                                                                                                                                                                                                        | —                                                                                      |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                          | —                                                                                                                                                                                                                                        | —                                                                                      |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\text{Tr}\left(y_1\left(y_2\right)^{\dagger}\right)\left(y_2\right)_{ij}$ | —                                                                                                                                                                                                                                        | —                                                                                      |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                          | —                                                                                                                                                                                                                                        | —                                                                                      |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                          | —                                                                                                                                                                                                                                        | —                                                                                      |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                          | —                                                                                                                                                                                                                                        | —                                                                                      |

Terms 5 of 5

| Column | Representative term structure                                                                                                                                                                                     |  |  |  |  |  |  |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|--|
| T1     | $+\frac{1}{4}(3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+5\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$                                       |  |  |  |  |  |  |
| T2     | $-\frac{3}{2}\mathrm{Tr}\left(y_1\left(y_2\right)^{\dagger}\right)\left(y_2\right)_{ij}$                                                                                                                          |  |  |  |  |  |  |
| T3     | $+\frac{1}{2}((y_1\left(y_2\right)^{\dagger}y_2)_{ij}+\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+3\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$ |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$                                                                                                                                                                                   | T1 | T2                                                                                      | T3                                                                                                                                                                                                                |
|-------|----------|-----------|----------------|-----------|---------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$                                                                                                                                                                          | —  | —                                                                                       | —                                                                                                                                                                                                                 |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0                                                                                                                                                                                       | —  | —                                                                                       | —                                                                                                                                                                                                                 |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$                                                                                                                                                                           | —  | —                                                                                       | —                                                                                                                                                                                                                 |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1                                                                                                                                                                                       | —  | —                                                                                       | —                                                                                                                                                                                                                 |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$                                                                                                                                                                           | —  | —                                                                                       | —                                                                                                                                                                                                                 |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$                                                                                                                                                                          | —  | —                                                                                       | —                                                                                                                                                                                                                 |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0                                                                                                                                                                                       | —  | —                                                                                       | —                                                                                                                                                                                                                 |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$                                                                                                                                                                           | —  | —                                                                                       | —                                                                                                                                                                                                                 |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1                                                                                                                                                                                       | —  | —                                                                                       | —                                                                                                                                                                                                                 |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$                                                                                                                                                                           | —  | —                                                                                       | —                                                                                                                                                                                                                 |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}+\frac{1}{4}(3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+5\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$ | —  | —                                                                                       | —                                                                                                                                                                                                                 |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0                                                                                                                                                                                       | —  | $\frac{3}{2}\mathrm{Tr}\left(y_1\left(y_2\right)^{\dagger}\right)\left(y_2\right)_{ij}$ | —                                                                                                                                                                                                                 |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}+\frac{1}{4}(3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+5\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$  | —  | —                                                                                       | —                                                                                                                                                                                                                 |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | $1+\frac{1}{4}(3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+5\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$            | —  | —                                                                                       | —                                                                                                                                                                                                                 |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}+\frac{1}{4}(3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+5\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$  | —  | —                                                                                       | —                                                                                                                                                                                                                 |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$                                                                                                                                                                          | —  | —                                                                                       | —                                                                                                                                                                                                                 |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0                                                                                                                                                                                       | —  | —                                                                                       | $+\frac{1}{2}((y_1\left(y_2\right)^{\dagger}y_2)_{ij}+\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+3\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$                                                                                                                                                                           | —  | —                                                                                       | —                                                                                                                                                                                                                 |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1                                                                                                                                                                                       | —  | —                                                                                       | —                                                                                                                                                                                                                 |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$                                                                                                                                                                           | —  | —                                                                                       | —                                                                                                                                                                                                                 |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}+\frac{1}{4}(3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+6\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$ | —  | —                                                                                       | —                                                                                                                                                                                                                 |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0                                                                                                                                                                                       | —  | —                                                                                       | —                                                                                                                                                                                                                 |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}+\frac{1}{4}(3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+6\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$  | —  | —                                                                                       | —                                                                                                                                                                                                                 |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | $1+\frac{1}{4}(3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+6\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$            | —  | —                                                                                       | —                                                                                                                                                                                                                 |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}+\frac{1}{4}(3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+6\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$  | —  | —                                                                                       | —                                                                                                                                                                                                                 |

$16\pi^2\,\mu\frac{dy_2}{d\mu}=\beta_{y_2}^{(1)}$

Terms 1 of 5

| Column | Representative term structure                                                                                   |
|--------|-----------------------------------------------------------------------------------------------------------------|
| T1     | $\text{Tr}\Big(y_2\ (y_2)^\dagger\Big)\,(y_2)_{ij}$                                                             |
| T2     | $+\frac{1}{2}((y_1\ (y_1)^\dagger\ y_2)_{ij}+(Y_e\ (Y_e)^\dagger\ y_2)_{ij}+3\,(y_2\ (y_2)^\dagger\ y_2)_{ij})$ |
| T3     | $-\frac{9}{2}g_2^2\,(y_2)_{ij}$                                                                                 |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                   | T2                                                                                                              | T3                               |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|----------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\text{Tr}\Big(y_2\ (y_2)^\dagger\Big)\,(y_2)_{ij}$  | $+\frac{1}{2}((y_1\ (y_1)^\dagger\ y_2)_{ij}+(Y_e\ (Y_e)^\dagger\ y_2)_{ij}+3\,(y_2\ (y_2)^\dagger\ y_2)_{ij})$ | $-\frac{9}{2}g_2^2\,(y_2)_{ij}$  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\text{Tr}\Big(y_2\ (y_2)^\dagger\Big)\,(y_2)_{ij}$  | —                                                                                                               | $-\frac{9}{2}g_2^2\,(y_2)_{ij}$  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\text{Tr}\Big(y_2\ (y_2)^\dagger\Big)\,(y_2)_{ij}$  | $+\frac{1}{2}((y_1\ (y_1)^\dagger\ y_2)_{ij}+(Y_e\ (Y_e)^\dagger\ y_2)_{ij}+3\,(y_2\ (y_2)^\dagger\ y_2)_{ij})$ | $-\frac{9}{2}g_2^2\,(y_2)_{ij}$  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\text{Tr}\Big(y_2\ (y_2)^\dagger\Big)\,(y_2)_{ij}$  | $+\frac{1}{2}((y_1\ (y_1)^\dagger\ y_2)_{ij}+(Y_e\ (Y_e)^\dagger\ y_2)_{ij}+3\,(y_2\ (y_2)^\dagger\ y_2)_{ij})$ | $-\frac{9}{2}g_2^2\,(y_2)_{ij}$  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\text{Tr}\Big(y_2\ (y_2)^\dagger\Big)\,(y_2)_{ij}$  | $+\frac{1}{2}((y_1\ (y_1)^\dagger\ y_2)_{ij}+(Y_e\ (Y_e)^\dagger\ y_2)_{ij}+3\,(y_2\ (y_2)^\dagger\ y_2)_{ij})$ | $-\frac{9}{2}g_2^2\,(y_2)_{ij}$  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\text{Tr}\Big(y_2\ (y_2)^\dagger\Big)\,(y_2)_{ij}$  | $+\frac{1}{2}((y_1\ (y_1)^\dagger\ y_2)_{ij}+(Y_e\ (Y_e)^\dagger\ y_2)_{ij}+3\,(y_2\ (y_2)^\dagger\ y_2)_{ij})$ | $-\frac{9}{4}g_2^2\,(y_2)_{ij}$  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+\text{Tr}\Big(y_2\ (y_2)^\dagger\Big)\,(y_2)_{ij}$ | —                                                                                                               | $-\frac{9}{4}g_2^2\,(y_2)_{ij}$  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\text{Tr}\Big(y_2\ (y_2)^\dagger\Big)\,(y_2)_{ij}$  | $+\frac{1}{2}((y_1\ (y_1)^\dagger\ y_2)_{ij}+(Y_e\ (Y_e)^\dagger\ y_2)_{ij}+3\,(y_2\ (y_2)^\dagger\ y_2)_{ij})$ | $-\frac{9}{4}g_2^2\,(y_2)_{ij}$  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\text{Tr}\Big(y_2\ (y_2)^\dagger\Big)\,(y_2)_{ij}$  | $+\frac{1}{2}((y_1\ (y_1)^\dagger\ y_2)_{ij}+(Y_e\ (Y_e)^\dagger\ y_2)_{ij}+3\,(y_2\ (y_2)^\dagger\ y_2)_{ij})$ | $-\frac{9}{4}g_2^2\,(y_2)_{ij}$  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\text{Tr}\Big(y_2\ (y_2)^\dagger\Big)\,(y_2)_{ij}$  | $+\frac{1}{2}((y_1\ (y_1)^\dagger\ y_2)_{ij}+(Y_e\ (Y_e)^\dagger\ y_2)_{ij}+3\,(y_2\ (y_2)^\dagger\ y_2)_{ij})$ | $-\frac{9}{4}g_2^2\,(y_2)_{ij}$  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                    | —                                                                                                               | $-\frac{33}{4}g_2^2\,(y_2)_{ij}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                    | —                                                                                                               | $-\frac{33}{4}g_2^2\,(y_2)_{ij}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                    | —                                                                                                               | $-\frac{33}{4}g_2^2\,(y_2)_{ij}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                    | —                                                                                                               | $-\frac{33}{4}g_2^2\,(y_2)_{ij}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                    | —                                                                                                               | $-\frac{33}{4}g_2^2\,(y_2)_{ij}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                    | —                                                                                                               | $-\frac{9}{2}g_2^2\,(y_2)_{ij}$  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                    | —                                                                                                               | $-\frac{9}{2}g_2^2\,(y_2)_{ij}$  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                    | —                                                                                                               | $-\frac{9}{2}g_2^2\,(y_2)_{ij}$  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                    | —                                                                                                               | $-\frac{9}{2}g_2^2\,(y_2)_{ij}$  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                    | —                                                                                                               | $-\frac{9}{2}g_2^2\,(y_2)_{ij}$  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                    | T2 | T3                              |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------|----|---------------------------------|
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\text{Tr}\left(y_2 (y_2)^\dagger\right) (y_2)_{ij}$  | —  | $-\frac{9}{2} g_2^2 (y_2)_{ij}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+\text{Tr}\left(y_2 (y_2)^\dagger\right) (y_2)_{ij}$ | —  | $-\frac{9}{2} g_2^2 (y_2)_{ij}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\text{Tr}\left(y_2 (y_2)^\dagger\right) (y_2)_{ij}$  | —  | $-\frac{9}{2} g_2^2 (y_2)_{ij}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\text{Tr}\left(y_2 (y_2)^\dagger\right) (y_2)_{ij}$  | —  | $-\frac{9}{2} g_2^2 (y_2)_{ij}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\text{Tr}\left(y_2 (y_2)^\dagger\right) (y_2)_{ij}$  | —  | $-\frac{9}{2} g_2^2 (y_2)_{ij}$ |

Terms 2 of 5

| Column | Representative term structure                                                                                                                                                           |  |  |  |  |  |  |  |  |  |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|--|--|--|--|
| T1     | $-\frac{3}{2}g_Y^2\left(y_2\right)_{ij}$                                                                                                                                                |  |  |  |  |  |  |  |  |  |
| T2     | $+\frac{1}{2}((y_1\left(y_1\right)^\dagger y_2)_{ij}+(y_2\left(y_1\right)^\dagger y_1)_{ij}+(Y_e\left(Y_e\right)^\dagger y_2)_{ij}+3\left(y_2\left(y_2\right)^\dagger y_2\right)_{ij})$ |  |  |  |  |  |  |  |  |  |
| T3     | $\text{Tr}(y_2\left(y_1\right)^\dagger)(y_1)_{ij}$                                                                                                                                      |  |  |  |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                        | T2                                                                                                                                                                                      | T3                                                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{3}{2}g_Y^2\left(y_2\right)_{ij}$  | —                                                                                                                                                                                       | —                                                  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{3}{4}g_Y^2\left(y_2\right)_{ij}$  | $+\frac{1}{2}((y_1\left(y_1\right)^\dagger y_2)_{ij}+(y_2\left(y_1\right)^\dagger y_1)_{ij}+(Y_e\left(Y_e\right)^\dagger y_2)_{ij}+3\left(y_2\left(y_2\right)^\dagger y_2\right)_{ij})$ | —                                                  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{3}{2}g_Y^2\left(y_2\right)_{ij}$  | —                                                                                                                                                                                       | —                                                  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{15}{4}g_Y^2\left(y_2\right)_{ij}$ | —                                                                                                                                                                                       | —                                                  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{15}{2}g_Y^2\left(y_2\right)_{ij}$ | —                                                                                                                                                                                       | —                                                  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-\frac{3}{2}g_Y^2\left(y_2\right)_{ij}$  | —                                                                                                                                                                                       | —                                                  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{3}{4}g_Y^2\left(y_2\right)_{ij}$  | —                                                                                                                                                                                       | $\text{Tr}(y_2\left(y_1\right)^\dagger)(y_1)_{ij}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $-\frac{3}{2}g_Y^2\left(y_2\right)_{ij}$  | —                                                                                                                                                                                       | —                                                  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-\frac{15}{4}g_Y^2\left(y_2\right)_{ij}$ | —                                                                                                                                                                                       | —                                                  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-\frac{15}{2}g_Y^2\left(y_2\right)_{ij}$ | —                                                                                                                                                                                       | —                                                  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-\frac{3}{2}g_Y^2\left(y_2\right)_{ij}$  | —                                                                                                                                                                                       | —                                                  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-\frac{3}{4}g_Y^2\left(y_2\right)_{ij}$  | —                                                                                                                                                                                       | —                                                  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-\frac{3}{2}g_Y^2\left(y_2\right)_{ij}$  | —                                                                                                                                                                                       | —                                                  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-\frac{15}{4}g_Y^2\left(y_2\right)_{ij}$ | —                                                                                                                                                                                       | —                                                  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-\frac{15}{2}g_Y^2\left(y_2\right)_{ij}$ | —                                                                                                                                                                                       | —                                                  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{3}{2}g_Y^2\left(y_2\right)_{ij}$  | —                                                                                                                                                                                       | —                                                  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-\frac{3}{4}g_Y^2\left(y_2\right)_{ij}$  | —                                                                                                                                                                                       | —                                                  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{3}{2}g_Y^2\left(y_2\right)_{ij}$  | —                                                                                                                                                                                       | —                                                  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{15}{4}g_Y^2\left(y_2\right)_{ij}$ | —                                                                                                                                                                                       | —                                                  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{15}{2}g_Y^2\left(y_2\right)_{ij}$ | —                                                                                                                                                                                       | —                                                  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{3}{2}g_Y^2\left(y_2\right)_{ij}$  | —                                                                                                                                                                                       | —                                                  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1                               | T2 | T3                                                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|---------------|----------------------------------|----|----------------------------------------------------|
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0             | $-\frac{3}{4}g_Y{}^2(y_2)_{ij}$  | —  | $\text{Tr}\left(y_2(y_1)^\dagger\right)(y_1)_{ij}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$ | $-\frac{3}{2}g_Y{}^2(y_2)_{ij}$  | —  | —                                                  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1             | $-\frac{15}{4}g_Y{}^2(y_2)_{ij}$ | —  | —                                                  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$ | $-\frac{15}{2}g_Y{}^2(y_2)_{ij}$ | —  | —                                                  |

Terms 3 of 5

| Column | Representative term structure                                                                                                                                       |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $+ \frac{1}{2} ((y_1 \ (y_1)^\dagger \ y_2)_{ij} + 2 \ (y_2 \ (y_1)^\dagger \ y_1)_{ij} + (Y_e \ (Y_e)^\dagger \ y_2)_{ij} + 3 \ (y_2 \ (y_2)^\dagger \ y_2)_{ij})$ |
| T2     | $\frac{3}{2} \text{Tr} (y_2 \ (y_2)^\dagger) \ (y_2)_{ij}$                                                                                                          |
| T3     | $+ \frac{1}{4} (3 \ (y_1 \ (y_1)^\dagger \ y_2)_{ij} + 2 \ (Y_e \ (Y_e)^\dagger \ y_2)_{ij} + 5 \ (y_2 \ (y_2)^\dagger \ y_2)_{ij})$                                |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                                  | T2                                                           | T3                                                                                                                                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                   | —                                                            | —                                                                                                                                    |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                   | —                                                            | —                                                                                                                                    |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                   | —                                                            | —                                                                                                                                    |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                   | —                                                            | —                                                                                                                                    |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                   | —                                                            | —                                                                                                                                    |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                                   | —                                                            | —                                                                                                                                    |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+ \frac{1}{2} ((y_1 \ (y_1)^\dagger \ y_2)_{ij} + 2 \ (y_2 \ (y_1)^\dagger \ y_1)_{ij} + (Y_e \ (Y_e)^\dagger \ y_2)_{ij} + 3 \ (y_2 \ (y_2)^\dagger \ y_2)_{ij})$ | —                                                            | —                                                                                                                                    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                                   | —                                                            | —                                                                                                                                    |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                                   | —                                                            | —                                                                                                                                    |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                                   | —                                                            | —                                                                                                                                    |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                                   | $\frac{3}{2} \text{Tr} (y_2 \ (y_2)^\dagger) \ (y_2)_{ij}$   | $+ \frac{1}{4} (3 \ (y_1 \ (y_1)^\dagger \ y_2)_{ij} + 2 \ (Y_e \ (Y_e)^\dagger \ y_2)_{ij} + 5 \ (y_2 \ (y_2)^\dagger \ y_2)_{ij})$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                                                                                                                   | $+ \frac{3}{2} \text{Tr} (y_2 \ (y_2)^\dagger) \ (y_2)_{ij}$ | —                                                                                                                                    |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                                   | $\frac{3}{2} \text{Tr} (y_2 \ (y_2)^\dagger) \ (y_2)_{ij}$   | $+ \frac{1}{4} (3 \ (y_1 \ (y_1)^\dagger \ y_2)_{ij} + 2 \ (Y_e \ (Y_e)^\dagger \ y_2)_{ij} + 5 \ (y_2 \ (y_2)^\dagger \ y_2)_{ij})$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                                   | $\frac{3}{2} \text{Tr} (y_2 \ (y_2)^\dagger) \ (y_2)_{ij}$   | $+ \frac{1}{4} (3 \ (y_1 \ (y_1)^\dagger \ y_2)_{ij} + 2 \ (Y_e \ (Y_e)^\dagger \ y_2)_{ij} + 5 \ (y_2 \ (y_2)^\dagger \ y_2)_{ij})$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                                   | $\frac{3}{2} \text{Tr} (y_2 \ (y_2)^\dagger) \ (y_2)_{ij}$   | $+ \frac{1}{4} (3 \ (y_1 \ (y_1)^\dagger \ y_2)_{ij} + 2 \ (Y_e \ (Y_e)^\dagger \ y_2)_{ij} + 5 \ (y_2 \ (y_2)^\dagger \ y_2)_{ij})$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                   | —                                                            | —                                                                                                                                    |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                   | —                                                            | —                                                                                                                                    |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                   | —                                                            | —                                                                                                                                    |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                   | —                                                            | —                                                                                                                                    |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                   | —                                                            | —                                                                                                                                    |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                   | —                                                            | $+ \frac{1}{4} (3 \ (y_1 \ (y_1)^\dagger \ y_2)_{ij} + 2 \ (Y_e \ (Y_e)^\dagger \ y_2)_{ij} + 6 \ (y_2 \ (y_2)^\dagger \ y_2)_{ij})$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                   | —                                                            | —                                                                                                                                    |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                   | —                                                            | $+ \frac{1}{4} (3 \ (y_1 \ (y_1)^\dagger \ y_2)_{ij} + 2 \ (Y_e \ (Y_e)^\dagger \ y_2)_{ij} + 6 \ (y_2 \ (y_2)^\dagger \ y_2)_{ij})$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                   | —                                                            | $+ \frac{1}{4} (3 \ (y_1 \ (y_1)^\dagger \ y_2)_{ij} + 2 \ (Y_e \ (Y_e)^\dagger \ y_2)_{ij} + 6 \ (y_2 \ (y_2)^\dagger \ y_2)_{ij})$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                   | —                                                            | $+ \frac{1}{4} (3 \ (y_1 \ (y_1)^\dagger \ y_2)_{ij} + 2 \ (Y_e \ (Y_e)^\dagger \ y_2)_{ij} + 6 \ (y_2 \ (y_2)^\dagger \ y_2)_{ij})$ |

Terms 4 of 5

| Column | Representative term structure                                                                                                                                                                                                              |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$                                                                                                                                                      |
| T2     | $+\frac{1}{4}\left(3\left(y_1\left(y_1\right)^{\dagger}y_2\right)_{ij}+2\left(y_2\left(y_1\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_2\right)_{ij}+5\left(y_2\left(y_2\right)^{\dagger}y_2\right)_{ij}\right)$ |
| T3     | $2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)\left(y_2\right)_{ij}$                                                                                                                                                                |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                    | T2                                                                                                                                                                                                                                         | T3                                                                          |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                     | —                                                                                                                                                                                                                                          | —                                                                           |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                     | —                                                                                                                                                                                                                                          | —                                                                           |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                     | —                                                                                                                                                                                                                                          | —                                                                           |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                     | —                                                                                                                                                                                                                                          | —                                                                           |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                     | —                                                                                                                                                                                                                                          | —                                                                           |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                     | —                                                                                                                                                                                                                                          | —                                                                           |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                                     | —                                                                                                                                                                                                                                          | —                                                                           |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                     | —                                                                                                                                                                                                                                          | —                                                                           |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                     | —                                                                                                                                                                                                                                          | —                                                                           |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                     | —                                                                                                                                                                                                                                          | —                                                                           |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                     | —                                                                                                                                                                                                                                          | —                                                                           |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$ | $+\frac{1}{4}\left(3\left(y_1\left(y_1\right)^{\dagger}y_2\right)_{ij}+2\left(y_2\left(y_1\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_2\right)_{ij}+5\left(y_2\left(y_2\right)^{\dagger}y_2\right)_{ij}\right)$ | —                                                                           |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                     | —                                                                                                                                                                                                                                          | —                                                                           |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                     | —                                                                                                                                                                                                                                          | —                                                                           |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                     | —                                                                                                                                                                                                                                          | —                                                                           |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                     | —                                                                                                                                                                                                                                          | $2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)\left(y_2\right)_{ij}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                     | $+\frac{1}{4}\left(3\left(y_1\left(y_1\right)^{\dagger}y_2\right)_{ij}+3\left(y_2\left(y_1\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_2\right)_{ij}+4\left(y_2\left(y_2\right)^{\dagger}y_2\right)_{ij}\right)$ | $2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)\left(y_2\right)_{ij}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                     | —                                                                                                                                                                                                                                          | $2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)\left(y_2\right)_{ij}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                     | —                                                                                                                                                                                                                                          | $2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)\left(y_2\right)_{ij}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                     | —                                                                                                                                                                                                                                          | $2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)\left(y_2\right)_{ij}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                     | —                                                                                                                                                                                                                                          | —                                                                           |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                     | $+\frac{1}{4}\left(3\left(y_1\left(y_1\right)^{\dagger}y_2\right)_{ij}+3\left(y_2\left(y_1\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_2\right)_{ij}+6\left(y_2\left(y_2\right)^{\dagger}y_2\right)_{ij}\right)$ | —                                                                           |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                     | —                                                                                                                                                                                                                                          | —                                                                           |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                     | —                                                                                                                                                                                                                                          | —                                                                           |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                     | —                                                                                                                                                                                                                                          | —                                                                           |

Terms 5 of 5

| Column | Representative term structure                                            |  |
|--------|--------------------------------------------------------------------------|--|
| T1     | $+ \frac{3}{4} \left( y_1 \left( y_1 \right)^{\dagger} y_2 \right)_{ij}$ |  |
| T2     | $+ \frac{1}{2} \left( Y_e \left( Y_e \right)^{\dagger} y_2 \right)_{ij}$ |  |
| T3     | $+ \left( y_2 \left( y_2 \right)^{\dagger} y_2 \right)_{ij}$             |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                       | T2                                                                       | T3                                                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------------|--------------------------------------------------------------------------|--------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                        | —                                                                        | —                                                            |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                        | —                                                                        | —                                                            |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                        | —                                                                        | —                                                            |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                        | —                                                                        | —                                                            |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                        | —                                                                        | —                                                            |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                        | —                                                                        | —                                                            |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                        | —                                                                        | —                                                            |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                        | —                                                                        | —                                                            |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                        | —                                                                        | —                                                            |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                        | —                                                                        | —                                                            |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                        | —                                                                        | —                                                            |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                        | —                                                                        | —                                                            |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                        | —                                                                        | —                                                            |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                        | —                                                                        | —                                                            |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                        | —                                                                        | —                                                            |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $+ \frac{3}{4} \left( y_1 \left( y_1 \right)^{\dagger} y_2 \right)_{ij}$ | $+ \frac{1}{2} \left( Y_e \left( Y_e \right)^{\dagger} y_2 \right)_{ij}$ | $+ \left( y_2 \left( y_2 \right)^{\dagger} y_2 \right)_{ij}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                        | —                                                                        | —                                                            |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $+ \frac{3}{4} \left( y_1 \left( y_1 \right)^{\dagger} y_2 \right)_{ij}$ | $+ \frac{1}{2} \left( Y_e \left( Y_e \right)^{\dagger} y_2 \right)_{ij}$ | $+ \left( y_2 \left( y_2 \right)^{\dagger} y_2 \right)_{ij}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $+ \frac{3}{4} \left( y_1 \left( y_1 \right)^{\dagger} y_2 \right)_{ij}$ | $+ \frac{1}{2} \left( Y_e \left( Y_e \right)^{\dagger} y_2 \right)_{ij}$ | $+ \left( y_2 \left( y_2 \right)^{\dagger} y_2 \right)_{ij}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $+ \frac{3}{4} \left( y_1 \left( y_1 \right)^{\dagger} y_2 \right)_{ij}$ | $+ \frac{1}{2} \left( Y_e \left( Y_e \right)^{\dagger} y_2 \right)_{ij}$ | $+ \left( y_2 \left( y_2 \right)^{\dagger} y_2 \right)_{ij}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                        | —                                                                        | —                                                            |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                        | —                                                                        | —                                                            |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                        | —                                                                        | —                                                            |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                        | —                                                                        | —                                                            |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                        | —                                                                        | —                                                            |

$\beta_{M_F}^{(1)}$

Search name: MF

$16\pi^2\,\mu\frac{dM_F}{d\mu}=\beta_{M_F}^{(1)}$

Terms 1 of 5

| Column | Representative term structure                                                                                      |
|--------|--------------------------------------------------------------------------------------------------------------------|
| T1     | $\frac{1}{4}(2\left((y_1)^T\left(y_1\right)^*M_F\right)_{ij}+3\left(M_F\left(y_2\right)^{\dagger}y_2\right)_{ij})$ |
| T2     | $-\frac{9}{2}g_2^2\left(M_F\right)_{ij}$                                                                           |
| T3     | $-\frac{3}{2}g_Y^2\left(M_F\right)_{ij}$                                                                           |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                 | T2                                       | T3                                        |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------------------------------------------------------|------------------------------------------|-------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\frac{1}{4}(2\left((y_1)^T\left(y_1\right)^*M_F\right)_{ij}+3\left(M_F\left(y_2\right)^{\dagger}y_2\right)_{ij})$ | $-\frac{9}{2}g_2^2\left(M_F\right)_{ij}$ | $-\frac{3}{2}g_Y^2\left(M_F\right)_{ij}$  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                  | —                                        | —                                         |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\frac{1}{4}(2\left((y_1)^T\left(y_1\right)^*M_F\right)_{ij}+3\left(M_F\left(y_2\right)^{\dagger}y_2\right)_{ij})$ | $-\frac{9}{2}g_2^2\left(M_F\right)_{ij}$ | $-\frac{3}{2}g_Y^2\left(M_F\right)_{ij}$  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\frac{1}{4}(2\left((y_1)^T\left(y_1\right)^*M_F\right)_{ij}+3\left(M_F\left(y_2\right)^{\dagger}y_2\right)_{ij})$ | $-\frac{9}{2}g_2^2\left(M_F\right)_{ij}$ | —                                         |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\frac{1}{4}(2\left((y_1)^T\left(y_1\right)^*M_F\right)_{ij}+3\left(M_F\left(y_2\right)^{\dagger}y_2\right)_{ij})$ | $-\frac{9}{2}g_2^2\left(M_F\right)_{ij}$ | $-\frac{27}{2}g_Y^2\left(M_F\right)_{ij}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                  | —                                        | $-\frac{3}{2}g_Y^2\left(M_F\right)_{ij}$  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                                                                  | —                                        | —                                         |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                  | —                                        | $-\frac{3}{2}g_Y^2\left(M_F\right)_{ij}$  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                  | —                                        | —                                         |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                  | —                                        | $-\frac{27}{2}g_Y^2\left(M_F\right)_{ij}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                  | —                                        | $-\frac{3}{2}g_Y^2\left(M_F\right)_{ij}$  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                                                                  | —                                        | —                                         |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                  | —                                        | $-\frac{3}{2}g_Y^2\left(M_F\right)_{ij}$  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                  | —                                        | —                                         |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                  | —                                        | $-\frac{27}{2}g_Y^2\left(M_F\right)_{ij}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $\frac{1}{4}(3\left((y_1)^T\left(y_1\right)^*M_F\right)_{ij}+2\left(M_F\left(y_2\right)^{\dagger}y_2\right)_{ij})$ | $-\frac{9}{2}g_2^2\left(M_F\right)_{ij}$ | $-\frac{3}{2}g_Y^2\left(M_F\right)_{ij}$  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                  | —                                        | —                                         |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $\frac{1}{4}(3\left((y_1)^T\left(y_1\right)^*M_F\right)_{ij}+2\left(M_F\left(y_2\right)^{\dagger}y_2\right)_{ij})$ | $-\frac{9}{2}g_2^2\left(M_F\right)_{ij}$ | $-\frac{3}{2}g_Y^2\left(M_F\right)_{ij}$  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $\frac{1}{4}(3\left((y_1)^T\left(y_1\right)^*M_F\right)_{ij}+2\left(M_F\left(y_2\right)^{\dagger}y_2\right)_{ij})$ | $-\frac{9}{2}g_2^2\left(M_F\right)_{ij}$ | —                                         |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $\frac{1}{4}(3\left((y_1)^T\left(y_1\right)^*M_F\right)_{ij}+2\left(M_F\left(y_2\right)^{\dagger}y_2\right)_{ij})$ | $-\frac{9}{2}g_2^2\left(M_F\right)_{ij}$ | $-\frac{27}{2}g_Y^2\left(M_F\right)_{ij}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                  | $-\frac{9}{2}g_2^2\left(M_F\right)_{ij}$ | $-\frac{3}{2}g_Y^2\left(M_F\right)_{ij}$  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                  | —                                        | —                                         |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1 | T2                            | T3                             |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|----|-------------------------------|--------------------------------|
| T3-E  | 0        | 3         | 0             | 3         | 1             | 2     | $\frac{1}{2}$ | —  | $-\frac{9}{2}g_2^2(M_F)_{ij}$ | $-\frac{3}{2}g_Y^2(M_F)_{ij}$  |
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | —  | $-\frac{9}{2}g_2^2(M_F)_{ij}$ | —                              |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | —  | $-\frac{9}{2}g_2^2(M_F)_{ij}$ | $-\frac{27}{2}g_Y^2(M_F)_{ij}$ |

Terms 2 of 5

| Column | Representative term structure                                                                                                                                                                  |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\frac{1}{4}(2\left((M_F)^T\ (y_1)^\dagger\ y_1\right)_{ij}+2\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}+3\left((M_F)^T\ (y_2)^\dagger\ y_2\right)_{ij}+3\left((y_2)^T\ (y_2)^*\ M_F\right)_{ij})$ |
| T2     | $-\frac{9}{4}g_2^2((M_F)^T)_{ij}+(M_F)_{ij})$                                                                                                                                                  |
| T3     | $-6g_Y^2\ (M_F)_{ij}$                                                                                                                                                                          |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                                                             | T2                                            | T3                    |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|-----------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                              | —                                             | —                     |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\frac{1}{4}(2\left((M_F)^T\ (y_1)^\dagger\ y_1\right)_{ij}+2\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}+3\left((M_F)^T\ (y_2)^\dagger\ y_2\right)_{ij}+3\left((y_2)^T\ (y_2)^*\ M_F\right)_{ij})$ | $-\frac{9}{4}g_2^2((M_F)^T)_{ij}+(M_F)_{ij})$ | —                     |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                              | —                                             | —                     |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                              | —                                             | $-6g_Y^2\ (M_F)_{ij}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                              | —                                             | —                     |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                                                              | —                                             | —                     |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                                                                                                                                              | —                                             | —                     |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                                                              | —                                             | —                     |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                                                              | —                                             | $-6g_Y^2\ (M_F)_{ij}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                                                              | —                                             | —                     |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                                                              | —                                             | —                     |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                                                                                                                                              | —                                             | —                     |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                                                              | —                                             | —                     |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                                                              | —                                             | $-6g_Y^2\ (M_F)_{ij}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                                                              | —                                             | —                     |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                              | —                                             | —                     |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $\frac{1}{4}(3\left((M_F)^T\ (y_1)^\dagger\ y_1\right)_{ij}+3\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}+2\left((M_F)^T\ (y_2)^\dagger\ y_2\right)_{ij}+2\left((y_2)^T\ (y_2)^*\ M_F\right)_{ij})$ | $-\frac{9}{4}g_2^2((M_F)^T)_{ij}+(M_F)_{ij})$ | —                     |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                              | —                                             | —                     |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                              | —                                             | $-6g_Y^2\ (M_F)_{ij}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                              | —                                             | —                     |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                              | —                                             | —                     |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                              | $-\frac{9}{4}g_2^2((M_F)^T)_{ij}+(M_F)_{ij})$ | —                     |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                              | —                                             | —                     |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                              | —                                             | $-6g_Y^2\ (M_F)_{ij}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                              | —                                             | —                     |

Terms 3 of 5

| Column | Representative term structure                       |
|--------|-----------------------------------------------------|
| T1     | $\left( (y_1)^T \ (y_1)^* \ M_F \right)_{ij}$       |
| T2     | $+ \left( M_F \ (y_2)^\dagger \ y_2 \right)_{ij}$   |
| T3     | $\left( (M_F)^T \ (y_1)^\dagger \ y_1 \right)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                              | T2                                                | T3                                                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------|---------------------------------------------------|-----------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                               | —                                                 | —                                                   |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                               | —                                                 | —                                                   |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                               | —                                                 | —                                                   |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                               | —                                                 | —                                                   |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                               | —                                                 | —                                                   |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\left( (y_1)^T \ (y_1)^* \ M_F \right)_{ij}$   | $+ \left( M_F \ (y_2)^\dagger \ y_2 \right)_{ij}$ | —                                                   |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+ \left( (y_1)^T \ (y_1)^* \ M_F \right)_{ij}$ | —                                                 | $\left( (M_F)^T \ (y_1)^\dagger \ y_1 \right)_{ij}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\left( (y_1)^T \ (y_1)^* \ M_F \right)_{ij}$   | $+ \left( M_F \ (y_2)^\dagger \ y_2 \right)_{ij}$ | —                                                   |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\left( (y_1)^T \ (y_1)^* \ M_F \right)_{ij}$   | $+ \left( M_F \ (y_2)^\dagger \ y_2 \right)_{ij}$ | —                                                   |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\left( (y_1)^T \ (y_1)^* \ M_F \right)_{ij}$   | $+ \left( M_F \ (y_2)^\dagger \ y_2 \right)_{ij}$ | —                                                   |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                               | —                                                 | —                                                   |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                               | —                                                 | —                                                   |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                               | —                                                 | —                                                   |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                               | —                                                 | —                                                   |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                               | —                                                 | —                                                   |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                               | —                                                 | —                                                   |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                               | —                                                 | —                                                   |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                               | —                                                 | —                                                   |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                               | —                                                 | —                                                   |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                               | —                                                 | —                                                   |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                               | —                                                 | —                                                   |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                               | —                                                 | —                                                   |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                               | —                                                 | —                                                   |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                               | —                                                 | —                                                   |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                               | —                                                 | —                                                   |

Terms 4 of 5

| Column | Representative term structure                                                                                                            |
|--------|------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $+ \left( (M_F)^T \left( y_2 \right)^\dagger y_2 \right)_{ij}$                                                                           |
| T2     | $+ \left( (y_2)^T \left( y_2 \right)^* M_F \right)_{ij}$                                                                                 |
| T3     | $\frac{1}{2} \left( \left( y_1 \right)^T \left( y_1 \right)^* M_F \right)_{ij} + \left( M_F \left( y_2 \right)^\dagger y_2 \right)_{ij}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                             | T2                                                       | T3                                                                                                                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                        | —                                                                                                                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                        | —                                                                                                                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                        | —                                                                                                                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                        | —                                                                                                                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                        | —                                                                                                                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                              | —                                                        | —                                                                                                                                        |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+ \left( (M_F)^T \left( y_2 \right)^\dagger y_2 \right)_{ij}$ | $+ \left( (y_2)^T \left( y_2 \right)^* M_F \right)_{ij}$ | —                                                                                                                                        |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                              | —                                                        | —                                                                                                                                        |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                              | —                                                        | —                                                                                                                                        |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                              | —                                                        | —                                                                                                                                        |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                              | —                                                        | $\frac{1}{2} \left( \left( y_1 \right)^T \left( y_1 \right)^* M_F \right)_{ij} + \left( M_F \left( y_2 \right)^\dagger y_2 \right)_{ij}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                              | —                                                        | —                                                                                                                                        |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                              | —                                                        | $\frac{1}{2} \left( \left( y_1 \right)^T \left( y_1 \right)^* M_F \right)_{ij} + \left( M_F \left( y_2 \right)^\dagger y_2 \right)_{ij}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                              | —                                                        | $\frac{1}{2} \left( \left( y_1 \right)^T \left( y_1 \right)^* M_F \right)_{ij} + \left( M_F \left( y_2 \right)^\dagger y_2 \right)_{ij}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                              | —                                                        | $\frac{1}{2} \left( \left( y_1 \right)^T \left( y_1 \right)^* M_F \right)_{ij} + \left( M_F \left( y_2 \right)^\dagger y_2 \right)_{ij}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                        | —                                                                                                                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                        | —                                                                                                                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                        | —                                                                                                                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                        | —                                                                                                                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                        | —                                                                                                                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                              | —                                                        | $\frac{3}{4} \left( \left( y_1 \right)^T \left( y_1 \right)^* M_F \right)_{ij} + \left( M_F \left( y_2 \right)^\dagger y_2 \right)_{ij}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                              | —                                                        | —                                                                                                                                        |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                              | —                                                        | $\frac{3}{4} \left( \left( y_1 \right)^T \left( y_1 \right)^* M_F \right)_{ij} + \left( M_F \left( y_2 \right)^\dagger y_2 \right)_{ij}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                              | —                                                        | $\frac{3}{4} \left( \left( y_1 \right)^T \left( y_1 \right)^* M_F \right)_{ij} + \left( M_F \left( y_2 \right)^\dagger y_2 \right)_{ij}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                              | —                                                        | $\frac{3}{4} \left( \left( y_1 \right)^T \left( y_1 \right)^* M_F \right)_{ij} + \left( M_F \left( y_2 \right)^\dagger y_2 \right)_{ij}$ |

Terms 5 of 5

| Column | Representative term structure                                                                                                                                                                               |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $-12g_2^2\left(M_F\right)_{ij}$                                                                                                                                                                             |
| T2     | $\frac{1}{2}((M_F)^T\left(y_1\right)^\dagger y_1)_{ij}+\left((y_1)^T\left(y_1\right)^*M_F\right)_{ij}+\left((M_F)^T\left(y_2\right)^\dagger y_2\right)_{ij}+\left((y_2)^T\left(y_2\right)^*M_F\right)_{ij}$ |
| T3     | $-6g_2^2((M_F)^T)_{ij}+(M_F)_{ij}$                                                                                                                                                                          |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                              | T2                                                                                                                                                                                                          | T3                                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                               | —                                                                                                                                                                                                           | —                                  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                               | —                                                                                                                                                                                                           | —                                  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                               | —                                                                                                                                                                                                           | —                                  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                               | —                                                                                                                                                                                                           | —                                  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                               | —                                                                                                                                                                                                           | —                                  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                               | —                                                                                                                                                                                                           | —                                  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                               | —                                                                                                                                                                                                           | —                                  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                               | —                                                                                                                                                                                                           | —                                  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                               | —                                                                                                                                                                                                           | —                                  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                               | —                                                                                                                                                                                                           | —                                  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-12g_2^2\left(M_F\right)_{ij}$ | —                                                                                                                                                                                                           | —                                  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                               | $\frac{1}{2}((M_F)^T\left(y_1\right)^\dagger y_1)_{ij}+\left((y_1)^T\left(y_1\right)^*M_F\right)_{ij}+\left((M_F)^T\left(y_2\right)^\dagger y_2\right)_{ij}+\left((y_2)^T\left(y_2\right)^*M_F\right)_{ij}$ | $-6g_2^2((M_F)^T)_{ij}+(M_F)_{ij}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-12g_2^2\left(M_F\right)_{ij}$ | —                                                                                                                                                                                                           | —                                  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-12g_2^2\left(M_F\right)_{ij}$ | —                                                                                                                                                                                                           | —                                  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-12g_2^2\left(M_F\right)_{ij}$ | —                                                                                                                                                                                                           | —                                  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                               | —                                                                                                                                                                                                           | —                                  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                               | —                                                                                                                                                                                                           | —                                  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                               | —                                                                                                                                                                                                           | —                                  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                               | —                                                                                                                                                                                                           | —                                  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                               | —                                                                                                                                                                                                           | —                                  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                               | —                                                                                                                                                                                                           | —                                  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                               | $\frac{3}{4}((M_F)^T\left(y_1\right)^\dagger y_1)_{ij}+\left((y_1)^T\left(y_1\right)^*M_F\right)_{ij}+\left((M_F)^T\left(y_2\right)^\dagger y_2\right)_{ij}+\left((y_2)^T\left(y_2\right)^*M_F\right)_{ij}$ | —                                  |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                               | —                                                                                                                                                                                                           | —                                  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                               | —                                                                                                                                                                                                           | —                                  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                               | —                                                                                                                                                                                                           | —                                  |

$\beta_{m_{S_1}^2}^{(1)}$

Search name: mS1Sq

$$16\pi^2\mu\frac{dm_{S_1}^2}{d\mu}=\beta_{m_{S_1}^2}^{(1)}$$

Terms 1 of 4

| Column | Representative term structure                                      |
|--------|--------------------------------------------------------------------|
| T1     | $m_{S_1}^2(4\text{Tr}\Big(y_1\ (y_1)^\dagger\Big)+4\lambda_{S_1})$ |
| T2     | $-8\text{Tr}\Big(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\Big)$        |
| T3     | $-6g_Y{}^2m_{S_1}^2$                                               |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                 | T2                                                          | T3                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------|-------------------------------------------------------------|----------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $m_{S_1}^2(4\text{Tr}\Big(y_1\ (y_1)^\dagger\Big)+4\lambda_{S_1})$ | $-8\text{Tr}\Big(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\Big)$ | $-6g_Y{}^2m_{S_1}^2$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $m_{S_1}^2(4\text{Tr}\Big(y_1\ (y_1)^\dagger\Big)+4\lambda_{S_1})$ | —                                                           | —                    |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $m_{S_1}^2(4\text{Tr}\Big(y_1\ (y_1)^\dagger\Big)+4\lambda_{S_1})$ | $-8\text{Tr}\Big(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\Big)$ | —                    |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $m_{S_1}^2(4\text{Tr}\Big(y_1\ (y_1)^\dagger\Big)+4\lambda_{S_1})$ | $-8\text{Tr}\Big(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\Big)$ | —                    |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $m_{S_1}^2(4\text{Tr}\Big(y_1\ (y_1)^\dagger\Big)+4\lambda_{S_1})$ | $-8\text{Tr}\Big(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\Big)$ | $-6g_Y{}^2m_{S_1}^2$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $m_{S_1}^2(2\text{Tr}\Big(y_1\ (y_1)^\dagger\Big)+6\lambda_{S_1})$ | $-4\text{Tr}\Big(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\Big)$ | $-6g_Y{}^2m_{S_1}^2$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $m_{S_1}^2(2\text{Tr}\Big(y_1\ (y_1)^\dagger\Big)+6\lambda_{S_1})$ | —                                                           | —                    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $m_{S_1}^2(2\text{Tr}\Big(y_1\ (y_1)^\dagger\Big)+6\lambda_{S_1})$ | $-4\text{Tr}\Big(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\Big)$ | —                    |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $m_{S_1}^2(2\text{Tr}\Big(y_1\ (y_1)^\dagger\Big)+6\lambda_{S_1})$ | $-4\text{Tr}\Big(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\Big)$ | —                    |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $m_{S_1}^2(2\text{Tr}\Big(y_1\ (y_1)^\dagger\Big)+6\lambda_{S_1})$ | $-4\text{Tr}\Big(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\Big)$ | $-6g_Y{}^2m_{S_1}^2$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $m_{S_1}^2(3\text{Tr}\Big(y_1\ (y_1)^\dagger\Big)+6\lambda_{S_1})$ | $-6\text{Tr}\Big(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\Big)$ | $-6g_Y{}^2m_{S_1}^2$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $m_{S_1}^2(3\text{Tr}\Big(y_1\ (y_1)^\dagger\Big)+6\lambda_{S_1})$ | —                                                           | —                    |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $m_{S_1}^2(3\text{Tr}\Big(y_1\ (y_1)^\dagger\Big)+6\lambda_{S_1})$ | $-6\text{Tr}\Big(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\Big)$ | —                    |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $m_{S_1}^2(3\text{Tr}\Big(y_1\ (y_1)^\dagger\Big)+6\lambda_{S_1})$ | $-6\text{Tr}\Big(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\Big)$ | —                    |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $m_{S_1}^2(3\text{Tr}\Big(y_1\ (y_1)^\dagger\Big)+6\lambda_{S_1})$ | $-6\text{Tr}\Big(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\Big)$ | $-6g_Y{}^2m_{S_1}^2$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                  | $-4\text{Tr}\Big(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\Big)$ | $-6g_Y{}^2m_{S_1}^2$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                  | —                                                           | —                    |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                  | $-4\text{Tr}\Big(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\Big)$ | —                    |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                  | $-4\text{Tr}\Big(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\Big)$ | —                    |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                  | $-4\text{Tr}\Big(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\Big)$ | $-6g_Y{}^2m_{S_1}^2$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                  | $-4\text{Tr}\Big(M_F\ (M_F)^\dagger\ (y_1)^T\ (y_1)^*\Big)$ | $-6g_Y{}^2m_{S_1}^2$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                  | —                                                           | —                    |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1 | T2                                                                      | T3                   |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|----|-------------------------------------------------------------------------|----------------------|
| T3-E  | 0        | 3         | 0             | 3         | 1             | 2     | $\frac{1}{2}$ | —  | $-4 \operatorname{Tr} \left( M_F (M_F)^\dagger (y_1)^T (y_1)^* \right)$ | —                    |
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | —  | $-4 \operatorname{Tr} \left( M_F (M_F)^\dagger (y_1)^T (y_1)^* \right)$ | —                    |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | —  | $-4 \operatorname{Tr} \left( M_F (M_F)^\dagger (y_1)^T (y_1)^* \right)$ | $-6 g_Y^2 m_{S_1}^2$ |

Terms 2 of 4

| Column | Representative term structure                                                        |
|--------|--------------------------------------------------------------------------------------|
| T1     | $+6\lambda_{12}m_{S_2}^2$                                                            |
| T2     | $-16\text{Tr}\left(M_F\left(M_F\right)^*\left(y_1\right)^T\left(y_1\right)^*\right)$ |
| T3     | $-16\text{Tr}\left(M_F\left(y_1\right)^\dagger y_1\left(M_F\right)^*\right)$         |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                        | T2                                                                                   | T3                                                                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+6\lambda_{12}m_{S_2}^2$ | —                                                                                    | —                                                                            |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+6\lambda_{12}m_{S_2}^2$ | $-16\text{Tr}\left(M_F\left(M_F\right)^*\left(y_1\right)^T\left(y_1\right)^*\right)$ | $-16\text{Tr}\left(M_F\left(y_1\right)^\dagger y_1\left(M_F\right)^*\right)$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+6\lambda_{12}m_{S_2}^2$ | —                                                                                    | —                                                                            |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+6\lambda_{12}m_{S_2}^2$ | —                                                                                    | —                                                                            |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+6\lambda_{12}m_{S_2}^2$ | —                                                                                    | —                                                                            |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+4\lambda_{12}m_{S_2}^2$ | —                                                                                    | —                                                                            |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+4\lambda_{12}m_{S_2}^2$ | $-8\text{Tr}\left(M_F\left(M_F\right)^*\left(y_1\right)^T\left(y_1\right)^*\right)$  | $-8\text{Tr}\left(M_F\left(y_1\right)^\dagger y_1\left(M_F\right)^*\right)$  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $+4\lambda_{12}m_{S_2}^2$ | —                                                                                    | —                                                                            |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $+4\lambda_{12}m_{S_2}^2$ | —                                                                                    | —                                                                            |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+4\lambda_{12}m_{S_2}^2$ | —                                                                                    | —                                                                            |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+4\lambda_{12}m_{S_2}^2$ | —                                                                                    | —                                                                            |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+4\lambda_{12}m_{S_2}^2$ | $-12\text{Tr}\left(M_F\left(M_F\right)^*\left(y_1\right)^T\left(y_1\right)^*\right)$ | $-12\text{Tr}\left(M_F\left(y_1\right)^\dagger y_1\left(M_F\right)^*\right)$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $+4\lambda_{12}m_{S_2}^2$ | —                                                                                    | —                                                                            |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $+4\lambda_{12}m_{S_2}^2$ | —                                                                                    | —                                                                            |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+4\lambda_{12}m_{S_2}^2$ | —                                                                                    | —                                                                            |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $+2\lambda_{12}m_{S_2}^2$ | —                                                                                    | —                                                                            |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $+2\lambda_{12}m_{S_2}^2$ | $-8\text{Tr}\left(M_F\left(M_F\right)^*\left(y_1\right)^T\left(y_1\right)^*\right)$  | $-8\text{Tr}\left(M_F\left(y_1\right)^\dagger y_1\left(M_F\right)^*\right)$  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $+2\lambda_{12}m_{S_2}^2$ | —                                                                                    | —                                                                            |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $+2\lambda_{12}m_{S_2}^2$ | —                                                                                    | —                                                                            |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $+2\lambda_{12}m_{S_2}^2$ | —                                                                                    | —                                                                            |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                         | —                                                                                    | —                                                                            |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                         | $-8\text{Tr}\left(M_F\left(M_F\right)^*\left(y_1\right)^T\left(y_1\right)^*\right)$  | $-8\text{Tr}\left(M_F\left(y_1\right)^\dagger y_1\left(M_F\right)^*\right)$  |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                         | —                                                                                    | —                                                                            |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                         | —                                                                                    | —                                                                            |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                         | —                                                                                    | —                                                                            |

Terms 3 of 4

| Column | Representative term structure                                                                  |
|--------|------------------------------------------------------------------------------------------------|
| T1     | $-\frac{3g_Y^2m_{S_1}^2}{2}$                                                                   |
| T2     | $-\frac{9g_2^2m_{S_1}^2}{2}$                                                                   |
| T3     | $m_{S_1}^2(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+8\lambda_{S_1}+4\lambda_{S_1}^{\text{Adj}})$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                           | T2                           | T3                                                                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------|------------------------------|------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                            | —                            | —                                                                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{3g_Y^2m_{S_1}^2}{2}$ | —                            | —                                                                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                            | —                            | —                                                                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{3g_Y^2m_{S_1}^2}{2}$ | —                            | —                                                                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                            | —                            | —                                                                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                            | $-\frac{9g_2^2m_{S_1}^2}{2}$ | —                                                                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{3g_Y^2m_{S_1}^2}{2}$ | $-\frac{9g_2^2m_{S_1}^2}{2}$ | —                                                                                              |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                            | $-\frac{9g_2^2m_{S_1}^2}{2}$ | —                                                                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-\frac{3g_Y^2m_{S_1}^2}{2}$ | $-\frac{9g_2^2m_{S_1}^2}{2}$ | —                                                                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                            | $-\frac{9g_2^2m_{S_1}^2}{2}$ | —                                                                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                            | $-\frac{9g_2^2m_{S_1}^2}{2}$ | —                                                                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-\frac{3g_Y^2m_{S_1}^2}{2}$ | $-\frac{9g_2^2m_{S_1}^2}{2}$ | —                                                                                              |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                            | $-\frac{9g_2^2m_{S_1}^2}{2}$ | —                                                                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-\frac{3g_Y^2m_{S_1}^2}{2}$ | $-\frac{9g_2^2m_{S_1}^2}{2}$ | —                                                                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                            | $-\frac{9g_2^2m_{S_1}^2}{2}$ | —                                                                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                            | —                            | $m_{S_1}^2(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+8\lambda_{S_1}+4\lambda_{S_1}^{\text{Adj}})$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-\frac{3g_Y^2m_{S_1}^2}{2}$ | —                            | $m_{S_1}^2(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+8\lambda_{S_1}+4\lambda_{S_1}^{\text{Adj}})$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                            | —                            | $m_{S_1}^2(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+8\lambda_{S_1}+4\lambda_{S_1}^{\text{Adj}})$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{3g_Y^2m_{S_1}^2}{2}$ | —                            | $m_{S_1}^2(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+8\lambda_{S_1}+4\lambda_{S_1}^{\text{Adj}})$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                            | —                            | $m_{S_1}^2(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+8\lambda_{S_1}+4\lambda_{S_1}^{\text{Adj}})$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                            | T2 | T3                                                                                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------|----|----------------------------------------------------------------------------------------------|
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                             | —  | $m_{S_1}^2 (2 \text{Tr} (y_1 (y_1)^\dagger) + 8\lambda_{S_1} + 4\lambda_{S_1}^{\text{Adj}})$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{3g_Y^2 m_{S_1}^2}{2}$ | —  | $m_{S_1}^2 (2 \text{Tr} (y_1 (y_1)^\dagger) + 8\lambda_{S_1} + 4\lambda_{S_1}^{\text{Adj}})$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                             | —  | $m_{S_1}^2 (2 \text{Tr} (y_1 (y_1)^\dagger) + 8\lambda_{S_1} + 4\lambda_{S_1}^{\text{Adj}})$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{3g_Y^2 m_{S_1}^2}{2}$ | —  | $m_{S_1}^2 (2 \text{Tr} (y_1 (y_1)^\dagger) + 8\lambda_{S_1} + 4\lambda_{S_1}^{\text{Adj}})$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                             | —  | $m_{S_1}^2 (2 \text{Tr} (y_1 (y_1)^\dagger) + 8\lambda_{S_1} + 4\lambda_{S_1}^{\text{Adj}})$ |

Terms 4 of 4

| Column | Representative term structure                            |
|--------|----------------------------------------------------------|
| T1     | $-12g_2^2m_{S_1}^2$                                      |
| T2     | $+m_{S_2}^2(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                  | T2                                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------|----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                   | —                                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                   | —                                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                   | —                                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                   | —                                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                   | —                                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                   | —                                                        |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                   | —                                                        |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                   | —                                                        |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                   | —                                                        |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                   | —                                                        |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                   | —                                                        |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                   | —                                                        |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                   | —                                                        |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                   | —                                                        |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                   | —                                                        |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-12g_2^2m_{S_1}^2$ | —                                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-12g_2^2m_{S_1}^2$ | —                                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-12g_2^2m_{S_1}^2$ | —                                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-12g_2^2m_{S_1}^2$ | —                                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-12g_2^2m_{S_1}^2$ | —                                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-12g_2^2m_{S_1}^2$ | $+m_{S_2}^2(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-12g_2^2m_{S_1}^2$ | $+m_{S_2}^2(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-12g_2^2m_{S_1}^2$ | $+m_{S_2}^2(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-12g_2^2m_{S_1}^2$ | $+m_{S_2}^2(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-12g_2^2m_{S_1}^2$ | $+m_{S_2}^2(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$ |

$\beta_{m_{S_2}^2}^{(1)}$

Search name: mS2Sq

$$16\pi^2\mu\frac{dm_{S_2}^2}{d\mu}=\beta_{m_{S_2}^2}^{(1)}$$

Terms 1 of 4

| Column | Representative term structure                                                                              |
|--------|------------------------------------------------------------------------------------------------------------|
| T1     | $m_{S_2}^2(2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$ |
| T2     | $-4\text{Tr}\left(M_F\left(y_2\right)^\dagger y_2\left(M_F\right)^\dagger\right)$                          |
| T3     | $-12g_2^2m_{S_2}^2$                                                                                        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                         | T2                                                                                | T3                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $m_{S_2}^2(2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$ | $-4\text{Tr}\left(M_F\left(y_2\right)^\dagger y_2\left(M_F\right)^\dagger\right)$ | $-12g_2^2m_{S_2}^2$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $m_{S_2}^2(2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$ | —                                                                                 | $-12g_2^2m_{S_2}^2$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $m_{S_2}^2(2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$ | $-4\text{Tr}\left(M_F\left(y_2\right)^\dagger y_2\left(M_F\right)^\dagger\right)$ | $-12g_2^2m_{S_2}^2$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $m_{S_2}^2(2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$ | $-4\text{Tr}\left(M_F\left(y_2\right)^\dagger y_2\left(M_F\right)^\dagger\right)$ | $-12g_2^2m_{S_2}^2$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $m_{S_2}^2(2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$ | $-4\text{Tr}\left(M_F\left(y_2\right)^\dagger y_2\left(M_F\right)^\dagger\right)$ | $-12g_2^2m_{S_2}^2$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                          | $-4\text{Tr}\left(M_F\left(y_2\right)^\dagger y_2\left(M_F\right)^\dagger\right)$ | —                   |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                                                          | —                                                                                 | —                   |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                          | $-4\text{Tr}\left(M_F\left(y_2\right)^\dagger y_2\left(M_F\right)^\dagger\right)$ | —                   |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                          | $-4\text{Tr}\left(M_F\left(y_2\right)^\dagger y_2\left(M_F\right)^\dagger\right)$ | —                   |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                          | $-4\text{Tr}\left(M_F\left(y_2\right)^\dagger y_2\left(M_F\right)^\dagger\right)$ | —                   |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                          | $-6\text{Tr}\left(M_F\left(y_2\right)^\dagger y_2\left(M_F\right)^\dagger\right)$ | —                   |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                                                          | —                                                                                 | —                   |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                          | $-6\text{Tr}\left(M_F\left(y_2\right)^\dagger y_2\left(M_F\right)^\dagger\right)$ | —                   |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                          | $-6\text{Tr}\left(M_F\left(y_2\right)^\dagger y_2\left(M_F\right)^\dagger\right)$ | —                   |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                          | $-6\text{Tr}\left(M_F\left(y_2\right)^\dagger y_2\left(M_F\right)^\dagger\right)$ | —                   |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                          | $-8\text{Tr}\left(M_F\left(y_2\right)^\dagger y_2\left(M_F\right)^\dagger\right)$ | —                   |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                          | —                                                                                 | —                   |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                          | $-8\text{Tr}\left(M_F\left(y_2\right)^\dagger y_2\left(M_F\right)^\dagger\right)$ | —                   |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                          | $-8\text{Tr}\left(M_F\left(y_2\right)^\dagger y_2\left(M_F\right)^\dagger\right)$ | —                   |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                          | $-8\text{Tr}\left(M_F\left(y_2\right)^\dagger y_2\left(M_F\right)^\dagger\right)$ | —                   |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $m_{S_2}^2(2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$ | $-4\text{Tr}\left(M_F\left(y_2\right)^\dagger y_2\left(M_F\right)^\dagger\right)$ | $-12g_2^2m_{S_2}^2$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $m_{S_2}^2(2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$ | —                                                                                 | $-12g_2^2m_{S_2}^2$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $m_{S_2}^2(2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+8\lambda_{S_2}+4\lambda_{S_2}^{\text{Adj}})$ | $-4\text{Tr}\left(M_F\left(y_2\right)^\dagger y_2\left(M_F\right)^\dagger\right)$ | $-12g_2^2m_{S_2}^2$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1                                                                                                        | T2                                                                | T3                   |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|----------------------|
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | $m_{S_2}^2 (2 \text{Tr} \left( y_2 (y_2)^\dagger \right) + 8\lambda_{S_2} + 4\lambda_{S_2}^{\text{Adj}})$ | $-4 \text{Tr} \left( M_F (y_2)^\dagger y_2 (M_F)^\dagger \right)$ | $-12g_2^2 m_{S_2}^2$ |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | $m_{S_2}^2 (2 \text{Tr} \left( y_2 (y_2)^\dagger \right) + 8\lambda_{S_2} + 4\lambda_{S_2}^{\text{Adj}})$ | $-4 \text{Tr} \left( M_F (y_2)^\dagger y_2 (M_F)^\dagger \right)$ | $-12g_2^2 m_{S_2}^2$ |

Terms 2 of 4

| Column | Representative term structure                                                       |
|--------|-------------------------------------------------------------------------------------|
| T1     | $+2\lambda_{12}m_{S_1}^2$                                                           |
| T2     | $-8\text{Tr}\left(M_F\left(M_F\right)^*\left(y_2\right)^T\left(y_2\right)^*\right)$ |
| T3     | $-8\text{Tr}\left(M_F\left(y_2\right)^\dagger y_2\left(M_F\right)^*\right)$         |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                        | T2                                                                                   | T3                                                                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+2\lambda_{12}m_{S_1}^2$ | —                                                                                    | —                                                                            |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+2\lambda_{12}m_{S_1}^2$ | $-8\text{Tr}\left(M_F\left(M_F\right)^*\left(y_2\right)^T\left(y_2\right)^*\right)$  | $-8\text{Tr}\left(M_F\left(y_2\right)^\dagger y_2\left(M_F\right)^*\right)$  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+2\lambda_{12}m_{S_1}^2$ | —                                                                                    | —                                                                            |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+2\lambda_{12}m_{S_1}^2$ | —                                                                                    | —                                                                            |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+2\lambda_{12}m_{S_1}^2$ | —                                                                                    | —                                                                            |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+4\lambda_{12}m_{S_1}^2$ | —                                                                                    | —                                                                            |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+4\lambda_{12}m_{S_1}^2$ | $-8\text{Tr}\left(M_F\left(M_F\right)^*\left(y_2\right)^T\left(y_2\right)^*\right)$  | $-8\text{Tr}\left(M_F\left(y_2\right)^\dagger y_2\left(M_F\right)^*\right)$  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $+4\lambda_{12}m_{S_1}^2$ | —                                                                                    | —                                                                            |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $+4\lambda_{12}m_{S_1}^2$ | —                                                                                    | —                                                                            |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+4\lambda_{12}m_{S_1}^2$ | —                                                                                    | —                                                                            |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+4\lambda_{12}m_{S_1}^2$ | —                                                                                    | —                                                                            |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+4\lambda_{12}m_{S_1}^2$ | $-12\text{Tr}\left(M_F\left(M_F\right)^*\left(y_2\right)^T\left(y_2\right)^*\right)$ | $-12\text{Tr}\left(M_F\left(y_2\right)^\dagger y_2\left(M_F\right)^*\right)$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $+4\lambda_{12}m_{S_1}^2$ | —                                                                                    | —                                                                            |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $+4\lambda_{12}m_{S_1}^2$ | —                                                                                    | —                                                                            |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+4\lambda_{12}m_{S_1}^2$ | —                                                                                    | —                                                                            |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $+6\lambda_{12}m_{S_1}^2$ | —                                                                                    | —                                                                            |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $+6\lambda_{12}m_{S_1}^2$ | $-16\text{Tr}\left(M_F\left(M_F\right)^*\left(y_2\right)^T\left(y_2\right)^*\right)$ | $-16\text{Tr}\left(M_F\left(y_2\right)^\dagger y_2\left(M_F\right)^*\right)$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $+6\lambda_{12}m_{S_1}^2$ | —                                                                                    | —                                                                            |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $+6\lambda_{12}m_{S_1}^2$ | —                                                                                    | —                                                                            |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $+6\lambda_{12}m_{S_1}^2$ | —                                                                                    | —                                                                            |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                         | —                                                                                    | —                                                                            |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                         | $-8\text{Tr}\left(M_F\left(M_F\right)^*\left(y_2\right)^T\left(y_2\right)^*\right)$  | $-8\text{Tr}\left(M_F\left(y_2\right)^\dagger y_2\left(M_F\right)^*\right)$  |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                         | —                                                                                    | —                                                                            |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                         | —                                                                                    | —                                                                            |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                         | —                                                                                    | —                                                                            |

Terms 3 of 4

| Column | Representative term structure                                                  |
|--------|--------------------------------------------------------------------------------|
| T1     | $-\frac{3g_Y^2m_{S_2}^2}{2}$                                                   |
| T2     | $-6g_Y^2m_{S_2}^2$                                                             |
| T3     | $m_{S_2}^2(2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\lambda_{S_2})$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                            | T2                  | T3                                                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------|---------------------|--------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                             | —                   | —                                                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{3g_Y^2m_{S_2}^2}{2}$  | —                   | —                                                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                             | $-6g_Y^2m_{S_2}^2$  | —                                                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{27g_Y^2m_{S_2}^2}{2}$ | —                   | —                                                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                             | $-24g_Y^2m_{S_2}^2$ | —                                                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                             | —                   | $m_{S_2}^2(2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\lambda_{S_2})$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{3g_Y^2m_{S_2}^2}{2}$  | —                   | $m_{S_2}^2(2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\lambda_{S_2})$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                             | $-6g_Y^2m_{S_2}^2$  | $m_{S_2}^2(2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\lambda_{S_2})$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-\frac{27g_Y^2m_{S_2}^2}{2}$ | —                   | $m_{S_2}^2(2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\lambda_{S_2})$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                             | $-24g_Y^2m_{S_2}^2$ | $m_{S_2}^2(2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\lambda_{S_2})$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                             | —                   | $m_{S_2}^2(3\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\lambda_{S_2})$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-\frac{3g_Y^2m_{S_2}^2}{2}$  | —                   | $m_{S_2}^2(3\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\lambda_{S_2})$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                             | $-6g_Y^2m_{S_2}^2$  | $m_{S_2}^2(3\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\lambda_{S_2})$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-\frac{27g_Y^2m_{S_2}^2}{2}$ | —                   | $m_{S_2}^2(3\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\lambda_{S_2})$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                             | $-24g_Y^2m_{S_2}^2$ | $m_{S_2}^2(3\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\lambda_{S_2})$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                             | —                   | $m_{S_2}^2(4\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+4\lambda_{S_2})$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-\frac{3g_Y^2m_{S_2}^2}{2}$  | —                   | $m_{S_2}^2(4\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+4\lambda_{S_2})$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                             | $-6g_Y^2m_{S_2}^2$  | $m_{S_2}^2(4\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+4\lambda_{S_2})$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{27g_Y^2m_{S_2}^2}{2}$ | —                   | $m_{S_2}^2(4\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+4\lambda_{S_2})$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                             | $-24g_Y^2m_{S_2}^2$ | $m_{S_2}^2(4\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+4\lambda_{S_2})$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                             | —                   | —                                                                              |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{3g_Y^2m_{S_2}^2}{2}$  | —                   | —                                                                              |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1                              | T2                    | T3 |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|---------------------------------|-----------------------|----|
| T3-E  | 0        | 3         | 0             | 3         | 1             | 2     | $\frac{1}{2}$ | —                               | $-6g_Y{}^2m_{S_2}^2$  | —  |
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | $-\frac{27g_Y{}^2m_{S_2}^2}{2}$ | —                     | —  |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | —                               | $-24g_Y{}^2m_{S_2}^2$ | —  |

Terms 4 of 4

| Column | Representative term structure                            |
|--------|----------------------------------------------------------|
| T1     | $-\frac{9g_2^2m_{S_2}^2}{2}$                             |
| T2     | $+m_{S_1}^2(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                           | T2                                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------|----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                            | —                                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                            | —                                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                            | —                                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                            | —                                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                            | —                                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-\frac{9g_2^2m_{S_2}^2}{2}$ | —                                                        |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{9g_2^2m_{S_2}^2}{2}$ | —                                                        |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $-\frac{9g_2^2m_{S_2}^2}{2}$ | —                                                        |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-\frac{9g_2^2m_{S_2}^2}{2}$ | —                                                        |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-\frac{9g_2^2m_{S_2}^2}{2}$ | —                                                        |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-\frac{9g_2^2m_{S_2}^2}{2}$ | —                                                        |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-\frac{9g_2^2m_{S_2}^2}{2}$ | —                                                        |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-\frac{9g_2^2m_{S_2}^2}{2}$ | —                                                        |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-\frac{9g_2^2m_{S_2}^2}{2}$ | —                                                        |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-\frac{9g_2^2m_{S_2}^2}{2}$ | —                                                        |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                            | —                                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                            | —                                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                            | —                                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                            | —                                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                            | —                                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                            | $+m_{S_1}^2(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                            | $+m_{S_1}^2(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                            | $+m_{S_1}^2(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                            | $+m_{S_1}^2(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                            | $+m_{S_1}^2(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$ |

Search name: lambdaH

$$16\pi^2 \mu \frac{d\lambda_H}{d\mu} = \beta_{\lambda_H}^{(1)}$$

## Terms 1 of 7

| Column | Representative term structure                                                                                                               |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_H (12 \operatorname{Tr} (Y_d (Y_d)^\dagger) + 4 \operatorname{Tr} (Y_e (Y_e)^\dagger) + 12 \operatorname{Tr} (Y_u (Y_u)^\dagger))$ |
| T2     | $-12 \operatorname{Tr} (Y_d (Y_d)^\dagger Y_d (Y_d)^\dagger)$                                                                               |
| T3     | $-4 \operatorname{Tr} (Y_e (Y_e)^\dagger Y_e (Y_e)^\dagger)$                                                                                |

[illegible]

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1                                                                                                              | T2                                                   | T3                                                  |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------|-----------------------------------------------------|
| T3-E  | 0        | 3         | 0             | 3         | 1             | 2     | $\frac{1}{2}$ | $\lambda_H(12 \text{Tr}(Y_d (Y_d)^\dagger) + 4 \text{Tr}(Y_e (Y_e)^\dagger) + 12 \text{Tr}(Y_u (Y_u)^\dagger))$ | $-12 \text{Tr}(Y_d (Y_d)^\dagger Y_d (Y_d)^\dagger)$ | $-4 \text{Tr}(Y_e (Y_e)^\dagger Y_e (Y_e)^\dagger)$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | $\lambda_H(12 \text{Tr}(Y_d (Y_d)^\dagger) + 4 \text{Tr}(Y_e (Y_e)^\dagger) + 12 \text{Tr}(Y_u (Y_u)^\dagger))$ | $-12 \text{Tr}(Y_d (Y_d)^\dagger Y_d (Y_d)^\dagger)$ | $-4 \text{Tr}(Y_e (Y_e)^\dagger Y_e (Y_e)^\dagger)$ |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | $\lambda_H(12 \text{Tr}(Y_d (Y_d)^\dagger) + 4 \text{Tr}(Y_e (Y_e)^\dagger) + 12 \text{Tr}(Y_u (Y_u)^\dagger))$ | $-12 \text{Tr}(Y_d (Y_d)^\dagger Y_d (Y_d)^\dagger)$ | $-4 \text{Tr}(Y_e (Y_e)^\dagger Y_e (Y_e)^\dagger)$ |

Terms 2 of 7

| Column | Representative term structure                                  |
|--------|----------------------------------------------------------------|
| T1     | $-12\text{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big)$ |
| T2     | $+4\lambda_{T3}\ (\lambda_{T3})^*$                             |
| T3     | $+\frac{9g_2^4}{4}$                                            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                             | T2                                 | T3                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------|------------------------------------|---------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-12\text{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big)$ | $+4\lambda_{T3}\ (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-12\text{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big)$ | $+4\lambda_{T3}\ (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-12\text{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big)$ | $+4\lambda_{T3}\ (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-12\text{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big)$ | $+4\lambda_{T3}\ (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-12\text{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big)$ | $+4\lambda_{T3}\ (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-12\text{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big)$ | $+4\lambda_{T3}\ (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-12\text{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big)$ | $+4\lambda_{T3}\ (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $-12\text{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big)$ | $+4\lambda_{T3}\ (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-12\text{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big)$ | $+4\lambda_{T3}\ (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-12\text{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big)$ | $+4\lambda_{T3}\ (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-12\text{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big)$ | $+4\lambda_{T3}\ (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-12\text{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big)$ | $+4\lambda_{T3}\ (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-12\text{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big)$ | $+4\lambda_{T3}\ (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-12\text{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big)$ | $+4\lambda_{T3}\ (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-12\text{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big)$ | $+4\lambda_{T3}\ (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-12\text{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big)$ | $+4\lambda_{T3}\ (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-12\text{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big)$ | $+4\lambda_{T3}\ (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-12\text{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big)$ | $+4\lambda_{T3}\ (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-12\text{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big)$ | $+4\lambda_{T3}\ (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-12\text{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big)$ | $+4\lambda_{T3}\ (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                 | T2                                | T3                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------|-----------------------------------|---------------------|
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-12 \text{Tr} \left( Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$ | $+4\lambda_{T3} (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-12 \text{Tr} \left( Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$ | $+4\lambda_{T3} (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-12 \text{Tr} \left( Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$ | $+4\lambda_{T3} (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-12 \text{Tr} \left( Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$ | $+4\lambda_{T3} (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-12 \text{Tr} \left( Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$ | $+4\lambda_{T3} (\lambda_{T3})^*$ | $+\frac{9g_2^4}{4}$ |

Terms 3 of 7

| Column | Representative term structure                    |
|--------|--------------------------------------------------|
| T1     | $+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$ |
| T2     | $-9g_2^2\lambda_H$                               |
| T3     | $+\frac{3g_Y^4}{4}$                              |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                               | T2                 | T3                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------|--------------------|---------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$ | $-9g_2^2\lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$ | $-9g_2^2\lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$ | $-9g_2^2\lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$ | $-9g_2^2\lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$ | $-9g_2^2\lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$ | $-9g_2^2\lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$ | $-9g_2^2\lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$ | $-9g_2^2\lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$ | $-9g_2^2\lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$ | $-9g_2^2\lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$ | $-9g_2^2\lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$ | $-9g_2^2\lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$ | $-9g_2^2\lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$ | $-9g_2^2\lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$ | $-9g_2^2\lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $+g_Y^2\left(\frac{3g_2^2}{2}-3\lambda_H\right)$ | $-9g_2^2\lambda_H$ | $+\frac{3g_Y^4}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                    | T2                  | T3                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------|---------------------|---------------------|
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_H \right)$ | $-9g_2^2 \lambda_H$ | $+\frac{3g_Y^4}{4}$ |

Terms 4 of 7

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $+12\lambda_H^2$              |
| T2     | $+2\lambda_{H1}^2$            |
| T3     | $+6\lambda_{H2}^2$            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1               | T2                 | T3                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------|--------------------|--------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+12\lambda_H^2$ | $+2\lambda_{H1}^2$ | $+6\lambda_{H2}^2$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+12\lambda_H^2$ | $+2\lambda_{H1}^2$ | $+6\lambda_{H2}^2$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+12\lambda_H^2$ | $+2\lambda_{H1}^2$ | $+6\lambda_{H2}^2$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+12\lambda_H^2$ | $+2\lambda_{H1}^2$ | $+6\lambda_{H2}^2$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+12\lambda_H^2$ | $+2\lambda_{H1}^2$ | $+6\lambda_{H2}^2$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+12\lambda_H^2$ | $+4\lambda_{H1}^2$ | $+4\lambda_{H2}^2$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+12\lambda_H^2$ | $+4\lambda_{H1}^2$ | $+4\lambda_{H2}^2$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $+12\lambda_H^2$ | $+4\lambda_{H1}^2$ | $+4\lambda_{H2}^2$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $+12\lambda_H^2$ | $+4\lambda_{H1}^2$ | $+4\lambda_{H2}^2$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+12\lambda_H^2$ | $+4\lambda_{H1}^2$ | $+4\lambda_{H2}^2$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+12\lambda_H^2$ | $+4\lambda_{H1}^2$ | $+4\lambda_{H2}^2$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+12\lambda_H^2$ | $+4\lambda_{H1}^2$ | $+4\lambda_{H2}^2$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $+12\lambda_H^2$ | $+4\lambda_{H1}^2$ | $+4\lambda_{H2}^2$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $+12\lambda_H^2$ | $+4\lambda_{H1}^2$ | $+4\lambda_{H2}^2$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+12\lambda_H^2$ | $+4\lambda_{H1}^2$ | $+4\lambda_{H2}^2$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $+12\lambda_H^2$ | $+6\lambda_{H1}^2$ | $+2\lambda_{H2}^2$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $+12\lambda_H^2$ | $+6\lambda_{H1}^2$ | $+2\lambda_{H2}^2$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $+12\lambda_H^2$ | $+6\lambda_{H1}^2$ | $+2\lambda_{H2}^2$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $+12\lambda_H^2$ | $+6\lambda_{H1}^2$ | $+2\lambda_{H2}^2$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $+12\lambda_H^2$ | $+6\lambda_{H1}^2$ | $+2\lambda_{H2}^2$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+12\lambda_H^2$ | $+6\lambda_{H1}^2$ | $+6\lambda_{H2}^2$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+12\lambda_H^2$ | $+6\lambda_{H1}^2$ | $+6\lambda_{H2}^2$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+12\lambda_H^2$ | $+6\lambda_{H1}^2$ | $+6\lambda_{H2}^2$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+12\lambda_H^2$ | $+6\lambda_{H1}^2$ | $+6\lambda_{H2}^2$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+12\lambda_H^2$ | $+6\lambda_{H1}^2$ | $+6\lambda_{H2}^2$ |

Terms 5 of 7

| Column | Representative term structure            |
|--------|------------------------------------------|
| T1     | $+3\lambda_{H2}^{\text{Adj}^2}$          |
| T2     | $+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$ |
| T3     | $+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                              | T2                                       | T3                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------|------------------------------------------|------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+3\lambda_{H2}^{\text{Adj}^2}$ | —                                        | —                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+3\lambda_{H2}^{\text{Adj}^2}$ | —                                        | —                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+3\lambda_{H2}^{\text{Adj}^2}$ | —                                        | —                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+3\lambda_{H2}^{\text{Adj}^2}$ | —                                        | —                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+3\lambda_{H2}^{\text{Adj}^2}$ | —                                        | —                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                               | $+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$ | $+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                               | $+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$ | $+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                               | $+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$ | $+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                               | $+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$ | $+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                               | $+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$ | $+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                               | $+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$ | $+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                               | $+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$ | $+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                               | $+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$ | $+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                               | $+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$ | $+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                               | $+\frac{\lambda_{H1}^{\text{Adj}^2}}{4}$ | $+\frac{\lambda_{H2}^{\text{Adj}^2}}{4}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                               | —                                        | —                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                               | —                                        | —                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                               | —                                        | —                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                               | —                                        | —                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                               | —                                        | —                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+3\lambda_{H2}^{\text{Adj}^2}$ | —                                        | —                                        |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+3\lambda_{H2}^{\text{Adj}^2}$ | —                                        | —                                        |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+3\lambda_{H2}^{\text{Adj}^2}$ | —                                        | —                                        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1                              | T2 | T3 |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|---------------------------------|----|----|
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | $+3\lambda_{H2}^{\text{Adj}^2}$ | —  | —  |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | $+3\lambda_{H2}^{\text{Adj}^2}$ | —  | —  |

Terms 6 of 7

| Column | Representative term structure                                                                                                                |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $+8\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left( \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$             |
| T2     | $+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ |
| T3     | $+8\lambda_{H^\dagger H^\dagger S_2 S_2} \left( \lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$                                             |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                               | T2                                                                                                                                           | T3                                                                                               |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+8\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left( \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$ | $+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ | $+8\lambda_{H^\dagger H^\dagger S_2 S_2} \left( \lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+8\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left( \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$ | $+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ | $+8\lambda_{H^\dagger H^\dagger S_2 S_2} \left( \lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                | —                                                                                                                                            | —                                                                                                |

Terms 7 of 7

| Column | Representative term structure   |
|--------|---------------------------------|
| T1     | $+3\lambda_{H1}^{\text{Adj}^2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                              |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                               |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                               |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                               |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                               |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                               |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                               |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                               |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                               |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                               |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                               |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                               |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                               |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                               |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                               |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                               |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $+3\lambda_{H1}^{\text{Adj}^2}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $+3\lambda_{H1}^{\text{Adj}^2}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $+3\lambda_{H1}^{\text{Adj}^2}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $+3\lambda_{H1}^{\text{Adj}^2}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $+3\lambda_{H1}^{\text{Adj}^2}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+3\lambda_{H1}^{\text{Adj}^2}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+3\lambda_{H1}^{\text{Adj}^2}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+3\lambda_{H1}^{\text{Adj}^2}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+3\lambda_{H1}^{\text{Adj}^2}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+3\lambda_{H1}^{\text{Adj}^2}$ |

$$16\pi^2\mu\frac{d\lambda_{S_1}}{d\mu}=\beta_{\lambda_{S_1}}^{(1)}$$

Terms 1 of 10

| Column | Representative term structure                                 |
|--------|---------------------------------------------------------------|
| T1     | $8\lambda_{S_1}\text{Tr}\big(y_1\,(y_1)^\dagger\big)$         |
| T2     | $-8\text{Tr}\big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\big)$ |
| T3     | $+12g_Y{}^4$                                                  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                    | T2                                                            | T3           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------|---------------------------------------------------------------|--------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $8\lambda_{S_1}\text{Tr}\big(y_1\,(y_1)^\dagger\big)$ | $-8\text{Tr}\big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\big)$ | $+12g_Y{}^4$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $8\lambda_{S_1}\text{Tr}\big(y_1\,(y_1)^\dagger\big)$ | $-8\text{Tr}\big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\big)$ | —            |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $8\lambda_{S_1}\text{Tr}\big(y_1\,(y_1)^\dagger\big)$ | $-8\text{Tr}\big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\big)$ | —            |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $8\lambda_{S_1}\text{Tr}\big(y_1\,(y_1)^\dagger\big)$ | $-8\text{Tr}\big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\big)$ | —            |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $8\lambda_{S_1}\text{Tr}\big(y_1\,(y_1)^\dagger\big)$ | $-8\text{Tr}\big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\big)$ | $+12g_Y{}^4$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $4\lambda_{S_1}\text{Tr}\big(y_1\,(y_1)^\dagger\big)$ | $-4\text{Tr}\big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\big)$ | $+12g_Y{}^4$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $4\lambda_{S_1}\text{Tr}\big(y_1\,(y_1)^\dagger\big)$ | $-4\text{Tr}\big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\big)$ | —            |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $4\lambda_{S_1}\text{Tr}\big(y_1\,(y_1)^\dagger\big)$ | $-4\text{Tr}\big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\big)$ | —            |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $4\lambda_{S_1}\text{Tr}\big(y_1\,(y_1)^\dagger\big)$ | $-4\text{Tr}\big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\big)$ | —            |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $4\lambda_{S_1}\text{Tr}\big(y_1\,(y_1)^\dagger\big)$ | $-4\text{Tr}\big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\big)$ | $+12g_Y{}^4$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $6\lambda_{S_1}\text{Tr}\big(y_1\,(y_1)^\dagger\big)$ | $-5\text{Tr}\big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\big)$ | $+12g_Y{}^4$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $6\lambda_{S_1}\text{Tr}\big(y_1\,(y_1)^\dagger\big)$ | $-5\text{Tr}\big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\big)$ | —            |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $6\lambda_{S_1}\text{Tr}\big(y_1\,(y_1)^\dagger\big)$ | $-5\text{Tr}\big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\big)$ | —            |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $6\lambda_{S_1}\text{Tr}\big(y_1\,(y_1)^\dagger\big)$ | $-5\text{Tr}\big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\big)$ | —            |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $6\lambda_{S_1}\text{Tr}\big(y_1\,(y_1)^\dagger\big)$ | $-5\text{Tr}\big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\big)$ | $+12g_Y{}^4$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                     | $-3\text{Tr}\big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\big)$ | $+12g_Y{}^4$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                     | $-3\text{Tr}\big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\big)$ | —            |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                     | $-3\text{Tr}\big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\big)$ | —            |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                     | $-3\text{Tr}\big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\big)$ | —            |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                     | $-3\text{Tr}\big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\big)$ | $+12g_Y{}^4$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                     | $-3\text{Tr}\big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\big)$ | $+12g_Y{}^4$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                     | $-3\text{Tr}\big(y_1\,(y_1)^\dagger\,y_1\,(y_1)^\dagger\big)$ | —            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1 | T2                                                                                                  | T3          |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|----|-----------------------------------------------------------------------------------------------------|-------------|
| T3-E  | 0        | 3         | 0             | 3         | 1             | 2     | $\frac{1}{2}$ | —  | $-3 \operatorname{Tr} \left( y_1 \left( y_1 \right)^\dagger y_1 \left( y_1 \right)^\dagger \right)$ | —           |
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | —  | $-3 \operatorname{Tr} \left( y_1 \left( y_1 \right)^\dagger y_1 \left( y_1 \right)^\dagger \right)$ | —           |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | —  | $-3 \operatorname{Tr} \left( y_1 \left( y_1 \right)^\dagger y_1 \left( y_1 \right)^\dagger \right)$ | $+12 g_Y^4$ |

Terms 2 of 10

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $-12g_Y^2\lambda_{S_1}$       |
| T2     | $+6\lambda_{12}^2$            |
| T3     | $+4\lambda_{H1}^2$            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                      | T2                 | T3                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------|--------------------|--------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-12g_Y^2\lambda_{S_1}$ | $+6\lambda_{12}^2$ | $+4\lambda_{H1}^2$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-3g_Y^2\lambda_{S_1}$  | $+6\lambda_{12}^2$ | $+4\lambda_{H1}^2$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                       | $+6\lambda_{12}^2$ | $+4\lambda_{H1}^2$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-3g_Y^2\lambda_{S_1}$  | $+6\lambda_{12}^2$ | $+4\lambda_{H1}^2$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-12g_Y^2\lambda_{S_1}$ | $+6\lambda_{12}^2$ | $+4\lambda_{H1}^2$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                       | $+4\lambda_{12}^2$ | $+4\lambda_{H1}^2$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                       | $+4\lambda_{12}^2$ | $+4\lambda_{H1}^2$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                       | $+4\lambda_{12}^2$ | $+4\lambda_{H1}^2$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                       | $+4\lambda_{12}^2$ | $+4\lambda_{H1}^2$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                       | $+4\lambda_{12}^2$ | $+4\lambda_{H1}^2$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                       | $+4\lambda_{12}^2$ | $+4\lambda_{H1}^2$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                       | $+4\lambda_{12}^2$ | $+4\lambda_{H1}^2$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                       | $+4\lambda_{12}^2$ | $+4\lambda_{H1}^2$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                       | $+4\lambda_{12}^2$ | $+4\lambda_{H1}^2$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                       | $+4\lambda_{12}^2$ | $+4\lambda_{H1}^2$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                       | $+2\lambda_{12}^2$ | $+4\lambda_{H1}^2$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                       | $+2\lambda_{12}^2$ | $+4\lambda_{H1}^2$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                       | $+2\lambda_{12}^2$ | $+4\lambda_{H1}^2$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                       | $+2\lambda_{12}^2$ | $+4\lambda_{H1}^2$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                       | $+2\lambda_{12}^2$ | $+4\lambda_{H1}^2$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                       | $+6\lambda_{12}^2$ | $+4\lambda_{H1}^2$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                       | $+6\lambda_{12}^2$ | $+4\lambda_{H1}^2$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                       | $+6\lambda_{12}^2$ | $+4\lambda_{H1}^2$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                       | $+6\lambda_{12}^2$ | $+4\lambda_{H1}^2$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                       | $+6\lambda_{12}^2$ | $+4\lambda_{H1}^2$ |

Terms 3 of 10

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $+10\lambda_{S_1}^2$          |
| T2     | $+\frac{3g_Y^4}{4}$           |
| T3     | $+\frac{9g_2^4}{4}$           |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                   | T2                  | T3                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------|---------------------|---------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+10\lambda_{S_1}^2$ | —                   | —                   |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+10\lambda_{S_1}^2$ | $+\frac{3g_Y^4}{4}$ | —                   |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+10\lambda_{S_1}^2$ | —                   | —                   |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+10\lambda_{S_1}^2$ | $+\frac{3g_Y^4}{4}$ | —                   |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+10\lambda_{S_1}^2$ | —                   | —                   |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+12\lambda_{S_1}^2$ | —                   | $+\frac{9g_2^4}{4}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+12\lambda_{S_1}^2$ | $+\frac{3g_Y^4}{4}$ | $+\frac{9g_2^4}{4}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $+12\lambda_{S_1}^2$ | —                   | $+\frac{9g_2^4}{4}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $+12\lambda_{S_1}^2$ | $+\frac{3g_Y^4}{4}$ | $+\frac{9g_2^4}{4}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+12\lambda_{S_1}^2$ | —                   | $+\frac{9g_2^4}{4}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+12\lambda_{S_1}^2$ | —                   | $+\frac{9g_2^4}{4}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+12\lambda_{S_1}^2$ | $+\frac{3g_Y^4}{4}$ | $+\frac{9g_2^4}{4}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $+12\lambda_{S_1}^2$ | —                   | $+\frac{9g_2^4}{4}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $+12\lambda_{S_1}^2$ | $+\frac{3g_Y^4}{4}$ | $+\frac{9g_2^4}{4}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+12\lambda_{S_1}^2$ | —                   | $+\frac{9g_2^4}{4}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $+14\lambda_{S_1}^2$ | —                   | —                   |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $+14\lambda_{S_1}^2$ | $+\frac{3g_Y^4}{4}$ | —                   |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $+14\lambda_{S_1}^2$ | —                   | —                   |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $+14\lambda_{S_1}^2$ | $+\frac{3g_Y^4}{4}$ | —                   |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $+14\lambda_{S_1}^2$ | —                   | —                   |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+14\lambda_{S_1}^2$ | —                   | —                   |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+14\lambda_{S_1}^2$ | $+\frac{3g_Y^4}{4}$ | —                   |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1                   | T2                  | T3 |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|----------------------|---------------------|----|
| T3-E  | 0        | 3         | 0             | 3         | 1             | 2     | $\frac{1}{2}$ | $+14\lambda_{S_1}^2$ | —                   | —  |
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | $+14\lambda_{S_1}^2$ | $+\frac{3g_Y^4}{4}$ | —  |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | $+14\lambda_{S_1}^2$ | —                   | —  |

Terms 4 of 10

| Column | Representative term structure               |
|--------|---------------------------------------------|
| T1     | $+g_Y^2\left(6g_2^2-12\lambda_{S_1}\right)$ |
| T2     | $-9g_2^2\lambda_{S_1}$                      |
| T3     | $+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$    |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                          | T2                     | T3                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------|------------------------|------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                      | —                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                      | —                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                      | —                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                      | —                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                      | —                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+g_Y^2\left(6g_2^2-12\lambda_{S_1}\right)$ | $-9g_2^2\lambda_{S_1}$ | $+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                           | $-9g_2^2\lambda_{S_1}$ | $+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                           | $-9g_2^2\lambda_{S_1}$ | $+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                           | $-9g_2^2\lambda_{S_1}$ | $+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+g_Y^2\left(6g_2^2-12\lambda_{S_1}\right)$ | $-9g_2^2\lambda_{S_1}$ | $+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+g_Y^2\left(6g_2^2-12\lambda_{S_1}\right)$ | $-9g_2^2\lambda_{S_1}$ | $+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                           | $-9g_2^2\lambda_{S_1}$ | $+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                           | $-9g_2^2\lambda_{S_1}$ | $+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                           | $-9g_2^2\lambda_{S_1}$ | $+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+g_Y^2\left(6g_2^2-12\lambda_{S_1}\right)$ | $-9g_2^2\lambda_{S_1}$ | $+\frac{\lambda_{12}^{\text{Adj}^2}}{4}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                      | —                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                      | —                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                      | —                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                      | —                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                      | —                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                           | —                      | —                                        |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                           | —                      | —                                        |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                           | —                      | —                                        |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                           | —                      | —                                        |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                           | —                      | —                                        |

Terms 5 of 10

| Column | Representative term structure                                                                                                                                                              |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $+ \frac{\lambda_{H1}^{Adj^2}}{4}$                                                                                                                                                         |
| T2     | $+ \lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}} (2 \operatorname{Tr} \left( y_1 \left( y_2 \right)^{\dagger} \right) + 24 \left( \lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}} \right)^{*} )$ |
| T3     | $+ 2 \left( \lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}} \right)^{*} \operatorname{Tr} \left( y_2 \left( y_1 \right)^{\dagger} \right)$                                                     |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                 | T2                                                                                                                                                                                                                                                                                                                              | T3 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                  | —                                                                                                                                                                                                                                                                                                                               | —  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                  | —                                                                                                                                                                                                                                                                                                                               | —  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                  | —                                                                                                                                                                                                                                                                                                                               | —  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                  | —                                                                                                                                                                                                                                                                                                                               | —  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                  | —                                                                                                                                                                                                                                                                                                                               | —  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+ \frac{\lambda_{H1}^{Adj^2}}{4}$ | —                                                                                                                                                                                                                                                                                                                               | —  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+ \frac{\lambda_{H1}^{Adj^2}}{4}$ | $+ \lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}} (2 \operatorname{Tr} \left( y_1 \left( y_2 \right)^{\dagger} \right) + 24 \left( \lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}} \right)^{*} ) + 2 \left( \lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}} \right)^{*} \operatorname{Tr} \left( y_2 \left( y_1 \right)^{\dagger} \right)$ |    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $+ \frac{\lambda_{H1}^{Adj^2}}{4}$ | —                                                                                                                                                                                                                                                                                                                               | —  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $+ \frac{\lambda_{H1}^{Adj^2}}{4}$ | —                                                                                                                                                                                                                                                                                                                               | —  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+ \frac{\lambda_{H1}^{Adj^2}}{4}$ | —                                                                                                                                                                                                                                                                                                                               | —  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+ \frac{\lambda_{H1}^{Adj^2}}{4}$ | —                                                                                                                                                                                                                                                                                                                               | —  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+ \frac{\lambda_{H1}^{Adj^2}}{4}$ | $+ \lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}} (3 \operatorname{Tr} \left( y_1 \left( y_2 \right)^{\dagger} \right) + 24 \left( \lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}} \right)^{*} ) + 3 \left( \lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}} \right)^{*} \operatorname{Tr} \left( y_2 \left( y_1 \right)^{\dagger} \right)$ |    |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $+ \frac{\lambda_{H1}^{Adj^2}}{4}$ | —                                                                                                                                                                                                                                                                                                                               | —  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $+ \frac{\lambda_{H1}^{Adj^2}}{4}$ | —                                                                                                                                                                                                                                                                                                                               | —  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+ \frac{\lambda_{H1}^{Adj^2}}{4}$ | —                                                                                                                                                                                                                                                                                                                               | —  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                  | —                                                                                                                                                                                                                                                                                                                               | —  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                  | —                                                                                                                                                                                                                                                                                                                               | —  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                  | —                                                                                                                                                                                                                                                                                                                               | —  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                  | —                                                                                                                                                                                                                                                                                                                               | —  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                  | —                                                                                                                                                                                                                                                                                                                               | —  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                  | —                                                                                                                                                                                                                                                                                                                               | —  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                  | —                                                                                                                                                                                                                                                                                                                               | —  |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                  | —                                                                                                                                                                                                                                                                                                                               | —  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                  | —                                                                                                                                                                                                                                                                                                                               | —  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$         | T1 | T2 | T3 |
|-------|----------|-----------|-----------|-----------|-----------|-------|---------------|----|----|----|
| T3-E  | 2        | 3         | 1         | 3         | 2         | 2     | $\frac{3}{2}$ | —  | —  | —  |

Terms 6 of 10

| Column | Representative term structure                                                                                                    |
|--------|----------------------------------------------------------------------------------------------------------------------------------|
| T1     | $+8\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left( \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$ |
| T2     | $+8\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \left( \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \right)^*$                         |
| T3     | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_{S_1} \right)$                                                                        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                               | T2                                                                                                       | T3                                                        |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                | —                                                                                                        | —                                                         |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                | —                                                                                                        | —                                                         |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                | —                                                                                                        | —                                                         |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                | —                                                                                                        | —                                                         |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                | —                                                                                                        | —                                                         |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                | —                                                                                                        | —                                                         |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+8\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left( \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$ | $+8\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \left( \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \right)^*$ | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_{S_1} \right)$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                | —                                                                                                        | —                                                         |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                | —                                                                                                        | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_{S_1} \right)$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                | —                                                                                                        | —                                                         |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                | —                                                                                                        | —                                                         |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+8\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left( \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$ | $+8\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \left( \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \right)^*$ | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_{S_1} \right)$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                | —                                                                                                        | —                                                         |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                | —                                                                                                        | $+g_Y^2 \left( \frac{3g_2^2}{2} - 3\lambda_{S_1} \right)$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                | —                                                                                                        | —                                                         |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                | —                                                                                                        | —                                                         |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                | —                                                                                                        | —                                                         |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                | —                                                                                                        | —                                                         |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                | —                                                                                                        | —                                                         |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                | —                                                                                                        | —                                                         |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                | —                                                                                                        | —                                                         |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                | —                                                                                                        | —                                                         |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                | —                                                                                                        | —                                                         |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                | —                                                                                                        | —                                                         |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                | —                                                                                                        | —                                                         |

Terms 7 of 10

| Column | Representative term structure                                                                                       |  |
|--------|---------------------------------------------------------------------------------------------------------------------|--|
| T1     | $\lambda_{S_1} (4 \operatorname{Tr} \left( y_1 \left( y_1 \right)^\dagger \right) + 14 \lambda_{S_1}^{\text{Adj}})$ |  |
| T2     | $+2 \lambda_{S_1}^{\text{Adj}} \operatorname{Tr} \left( y_1 \left( y_1 \right)^\dagger \right)$                     |  |
| T3     | $+21 g_2^4$                                                                                                         |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                  | T2                                                                                              | T3          |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                   | —                                                                                               | —           |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                   | —                                                                                               | —           |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                   | —                                                                                               | —           |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                   | —                                                                                               | —           |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                   | —                                                                                               | —           |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                   | —                                                                                               | —           |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                                                                   | —                                                                                               | —           |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                   | —                                                                                               | —           |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                   | —                                                                                               | —           |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                   | —                                                                                               | —           |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                   | —                                                                                               | —           |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                                                                   | —                                                                                               | —           |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                   | —                                                                                               | —           |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                   | —                                                                                               | —           |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                   | —                                                                                               | —           |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $\lambda_{S_1} (4 \operatorname{Tr} \left( y_1 \left( y_1 \right)^\dagger \right) + 14 \lambda_{S_1}^{\text{Adj}})$ | $+2 \lambda_{S_1}^{\text{Adj}} \operatorname{Tr} \left( y_1 \left( y_1 \right)^\dagger \right)$ | $+21 g_2^4$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $\lambda_{S_1} (4 \operatorname{Tr} \left( y_1 \left( y_1 \right)^\dagger \right) + 14 \lambda_{S_1}^{\text{Adj}})$ | $+2 \lambda_{S_1}^{\text{Adj}} \operatorname{Tr} \left( y_1 \left( y_1 \right)^\dagger \right)$ | $+21 g_2^4$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $\lambda_{S_1} (4 \operatorname{Tr} \left( y_1 \left( y_1 \right)^\dagger \right) + 14 \lambda_{S_1}^{\text{Adj}})$ | $+2 \lambda_{S_1}^{\text{Adj}} \operatorname{Tr} \left( y_1 \left( y_1 \right)^\dagger \right)$ | $+21 g_2^4$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $\lambda_{S_1} (4 \operatorname{Tr} \left( y_1 \left( y_1 \right)^\dagger \right) + 14 \lambda_{S_1}^{\text{Adj}})$ | $+2 \lambda_{S_1}^{\text{Adj}} \operatorname{Tr} \left( y_1 \left( y_1 \right)^\dagger \right)$ | $+21 g_2^4$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $\lambda_{S_1} (4 \operatorname{Tr} \left( y_1 \left( y_1 \right)^\dagger \right) + 14 \lambda_{S_1}^{\text{Adj}})$ | $+2 \lambda_{S_1}^{\text{Adj}} \operatorname{Tr} \left( y_1 \left( y_1 \right)^\dagger \right)$ | $+21 g_2^4$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\lambda_{S_1} (4 \operatorname{Tr} \left( y_1 \left( y_1 \right)^\dagger \right) + 14 \lambda_{S_1}^{\text{Adj}})$ | $+2 \lambda_{S_1}^{\text{Adj}} \operatorname{Tr} \left( y_1 \left( y_1 \right)^\dagger \right)$ | $+21 g_2^4$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\lambda_{S_1} (4 \operatorname{Tr} \left( y_1 \left( y_1 \right)^\dagger \right) + 14 \lambda_{S_1}^{\text{Adj}})$ | $+2 \lambda_{S_1}^{\text{Adj}} \operatorname{Tr} \left( y_1 \left( y_1 \right)^\dagger \right)$ | $+21 g_2^4$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\lambda_{S_1} (4 \operatorname{Tr} \left( y_1 \left( y_1 \right)^\dagger \right) + 14 \lambda_{S_1}^{\text{Adj}})$ | $+2 \lambda_{S_1}^{\text{Adj}} \operatorname{Tr} \left( y_1 \left( y_1 \right)^\dagger \right)$ | $+21 g_2^4$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\lambda_{S_1} (4 \operatorname{Tr} \left( y_1 \left( y_1 \right)^\dagger \right) + 14 \lambda_{S_1}^{\text{Adj}})$ | $+2 \lambda_{S_1}^{\text{Adj}} \operatorname{Tr} \left( y_1 \left( y_1 \right)^\dagger \right)$ | $+21 g_2^4$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\lambda_{S_1} (4 \operatorname{Tr} \left( y_1 \left( y_1 \right)^\dagger \right) + 14 \lambda_{S_1}^{\text{Adj}})$ | $+2 \lambda_{S_1}^{\text{Adj}} \operatorname{Tr} \left( y_1 \left( y_1 \right)^\dagger \right)$ | $+21 g_2^4$ |

Terms 8 of 10

| Column | Representative term structure                                            |
|--------|--------------------------------------------------------------------------|
| T1     | $+g_Y^2\left(12g_2^2-12\lambda_{S_1}-6\lambda_{S_1}^{\text{Adj}}\right)$ |
| T2     | $+g_2^2(-24\lambda_{S_1}-12\lambda_{S_1}^{\text{Adj}})$                  |
| T3     | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{2}$                                |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                       | T2                                                      | T3                                        |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------------|---------------------------------------------------------|-------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                        | —                                                       | —                                         |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                        | —                                                       | —                                         |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                        | —                                                       | —                                         |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                        | —                                                       | —                                         |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                        | —                                                       | —                                         |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                        | —                                                       | —                                         |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                        | —                                                       | —                                         |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                        | —                                                       | —                                         |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                        | —                                                       | —                                         |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                        | —                                                       | —                                         |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                        | —                                                       | —                                         |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                        | —                                                       | —                                         |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                        | —                                                       | —                                         |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                        | —                                                       | —                                         |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                        | —                                                       | —                                         |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $+g_Y^2\left(12g_2^2-12\lambda_{S_1}-6\lambda_{S_1}^{\text{Adj}}\right)$ | $+g_2^2(-24\lambda_{S_1}-12\lambda_{S_1}^{\text{Adj}})$ | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{2}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                        | $+g_2^2(-24\lambda_{S_1}-12\lambda_{S_1}^{\text{Adj}})$ | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{2}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                        | $+g_2^2(-24\lambda_{S_1}-12\lambda_{S_1}^{\text{Adj}})$ | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{2}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                        | $+g_2^2(-24\lambda_{S_1}-12\lambda_{S_1}^{\text{Adj}})$ | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{2}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $+g_Y^2\left(12g_2^2-12\lambda_{S_1}-6\lambda_{S_1}^{\text{Adj}}\right)$ | $+g_2^2(-24\lambda_{S_1}-12\lambda_{S_1}^{\text{Adj}})$ | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{2}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+g_Y^2\left(12g_2^2-12\lambda_{S_1}-6\lambda_{S_1}^{\text{Adj}}\right)$ | $+g_2^2(-24\lambda_{S_1}-12\lambda_{S_1}^{\text{Adj}})$ | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{2}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                        | $+g_2^2(-24\lambda_{S_1}-12\lambda_{S_1}^{\text{Adj}})$ | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{2}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                        | $+g_2^2(-24\lambda_{S_1}-12\lambda_{S_1}^{\text{Adj}})$ | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{2}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                        | $+g_2^2(-24\lambda_{S_1}-12\lambda_{S_1}^{\text{Adj}})$ | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{2}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+g_Y^2\left(12g_2^2-12\lambda_{S_1}-6\lambda_{S_1}^{\text{Adj}}\right)$ | $+g_2^2(-24\lambda_{S_1}-12\lambda_{S_1}^{\text{Adj}})$ | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{2}$ |

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| Column | Representative term structure                                                    |
|--------|----------------------------------------------------------------------------------|
| T1     | $+26\lambda_{S_1}^{\text{Adj}^2}$                                                |
| T2     | $+g_Y^2\left(3g_2^2-3\lambda_{S_1}-\frac{3\lambda_{S_1}^{\text{Adj}}}{2}\right)$ |
| T3     | $+8\lambda_{12}\lambda_{12}^{\text{Cross}}$                                      |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                | T2                                                                               | T3                                          |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------|----------------------------------------------------------------------------------|---------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                 | —                                                                                | —                                           |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                 | —                                                                                | —                                           |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                 | —                                                                                | —                                           |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                 | —                                                                                | —                                           |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                 | —                                                                                | —                                           |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                 | —                                                                                | —                                           |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                 | —                                                                                | —                                           |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                 | —                                                                                | —                                           |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                 | —                                                                                | —                                           |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                 | —                                                                                | —                                           |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                 | —                                                                                | —                                           |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                 | —                                                                                | —                                           |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                 | —                                                                                | —                                           |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                 | —                                                                                | —                                           |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                 | —                                                                                | —                                           |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $+26\lambda_{S_1}^{\text{Adj}^2}$ | —                                                                                | —                                           |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $+26\lambda_{S_1}^{\text{Adj}^2}$ | $+g_Y^2\left(3g_2^2-3\lambda_{S_1}-\frac{3\lambda_{S_1}^{\text{Adj}}}{2}\right)$ | —                                           |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $+26\lambda_{S_1}^{\text{Adj}^2}$ | —                                                                                | —                                           |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $+26\lambda_{S_1}^{\text{Adj}^2}$ | $+g_Y^2\left(3g_2^2-3\lambda_{S_1}-\frac{3\lambda_{S_1}^{\text{Adj}}}{2}\right)$ | —                                           |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $+26\lambda_{S_1}^{\text{Adj}^2}$ | —                                                                                | —                                           |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+26\lambda_{S_1}^{\text{Adj}^2}$ | —                                                                                | $+8\lambda_{12}\lambda_{12}^{\text{Cross}}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+26\lambda_{S_1}^{\text{Adj}^2}$ | $+g_Y^2\left(3g_2^2-3\lambda_{S_1}-\frac{3\lambda_{S_1}^{\text{Adj}}}{2}\right)$ | $+8\lambda_{12}\lambda_{12}^{\text{Cross}}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+26\lambda_{S_1}^{\text{Adj}^2}$ | —                                                                                | $+8\lambda_{12}\lambda_{12}^{\text{Cross}}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+26\lambda_{S_1}^{\text{Adj}^2}$ | $+g_Y^2\left(3g_2^2-3\lambda_{S_1}-\frac{3\lambda_{S_1}^{\text{Adj}}}{2}\right)$ | $+8\lambda_{12}\lambda_{12}^{\text{Cross}}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+26\lambda_{S_1}^{\text{Adj}^2}$ | —                                                                                | $+8\lambda_{12}\lambda_{12}^{\text{Cross}}$ |

Terms 10 of 10

| Column | Representative term structure     |
|--------|-----------------------------------|
| T1     | $+6\lambda_{12}^{\text{Adj}^2}$   |
| T2     | $+6\lambda_{12}^{\text{Cross}^2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                              | T2                                |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------|-----------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                               | —                                 |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                               | —                                 |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                               | —                                 |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                               | —                                 |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                               | —                                 |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                               | —                                 |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                               | —                                 |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                               | —                                 |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                               | —                                 |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                               | —                                 |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                               | —                                 |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                               | —                                 |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                               | —                                 |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                               | —                                 |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                               | —                                 |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                               | —                                 |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                               | —                                 |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                               | —                                 |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                               | —                                 |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                               | —                                 |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+6\lambda_{12}^{\text{Adj}^2}$ | $+6\lambda_{12}^{\text{Cross}^2}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+6\lambda_{12}^{\text{Adj}^2}$ | $+6\lambda_{12}^{\text{Cross}^2}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+6\lambda_{12}^{\text{Adj}^2}$ | $+6\lambda_{12}^{\text{Cross}^2}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+6\lambda_{12}^{\text{Adj}^2}$ | $+6\lambda_{12}^{\text{Cross}^2}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+6\lambda_{12}^{\text{Adj}^2}$ | $+6\lambda_{12}^{\text{Cross}^2}$ |

$\beta_{\lambda_{S_2}}^{(1)}$

Search name: lambda52

$16\pi^2\mu\frac{d\lambda_{S_2}}{d\mu}=\beta_{\lambda_{S_2}}^{(1)}$

Terms 1 of 10

| Column | Representative term structure                                                                    |
|--------|--------------------------------------------------------------------------------------------------|
| T1     | $\lambda_{S_2}(4\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+14\lambda_{S_2}^{\text{Adj}})$ |
| T2     | $+2\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$                  |
| T3     | $-3\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$                |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                               | T2                                                                              | T3                                                                                |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\lambda_{S_2}(4\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+14\lambda_{S_2}^{\text{Adj}})$ | $+2\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$ | $-3\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\lambda_{S_2}(4\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+14\lambda_{S_2}^{\text{Adj}})$ | $+2\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$ | $-3\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\lambda_{S_2}(4\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+14\lambda_{S_2}^{\text{Adj}})$ | $+2\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$ | $-3\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\lambda_{S_2}(4\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+14\lambda_{S_2}^{\text{Adj}})$ | $+2\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$ | $-3\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\lambda_{S_2}(4\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+14\lambda_{S_2}^{\text{Adj}})$ | $+2\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$ | $-3\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                | —                                                                               | $-4\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                                                | —                                                                               | $-4\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                | —                                                                               | $-4\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                | —                                                                               | $-4\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                | —                                                                               | $-4\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                | —                                                                               | $-5\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                                                | —                                                                               | $-5\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                | —                                                                               | $-5\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                | —                                                                               | $-5\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                | —                                                                               | $-5\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                | —                                                                               | $-8\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                | —                                                                               | $-8\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                | —                                                                               | $-8\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                | —                                                                               | $-8\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                | —                                                                               | $-8\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\lambda_{S_2}(4\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+14\lambda_{S_2}^{\text{Adj}})$ | $+2\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$ | $-3\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\lambda_{S_2}(4\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+14\lambda_{S_2}^{\text{Adj}})$ | $+2\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$ | $-3\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1                                                                             | T2                                                          | T3                                                  |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|--------------------------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------|
| T3-E  | 0        | 3         | 0             | 3         | 1             | 2     | $\frac{1}{2}$ | $\lambda_{S_2}(4 \text{Tr}(y_2 (y_2)^\dagger) + 14\lambda_{S_2}^{\text{Adj}})$ | $+2\lambda_{S_2}^{\text{Adj}} \text{Tr}(y_2 (y_2)^\dagger)$ | $-3 \text{Tr}(y_2 (y_2)^\dagger y_2 (y_2)^\dagger)$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | $\lambda_{S_2}(4 \text{Tr}(y_2 (y_2)^\dagger) + 14\lambda_{S_2}^{\text{Adj}})$ | $+2\lambda_{S_2}^{\text{Adj}} \text{Tr}(y_2 (y_2)^\dagger)$ | $-3 \text{Tr}(y_2 (y_2)^\dagger y_2 (y_2)^\dagger)$ |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | $\lambda_{S_2}(4 \text{Tr}(y_2 (y_2)^\dagger) + 14\lambda_{S_2}^{\text{Adj}})$ | $+2\lambda_{S_2}^{\text{Adj}} \text{Tr}(y_2 (y_2)^\dagger)$ | $-3 \text{Tr}(y_2 (y_2)^\dagger y_2 (y_2)^\dagger)$ |

Terms 2 of 10

| Column | Representative term structure                             |
|--------|-----------------------------------------------------------|
| T1     | $+21g_2^4$                                                |
| T2     | $+g_2^2(-24\lambda_{S_2} - 12\lambda_{S_2}^{\text{Adj}})$ |
| T3     | $+2\lambda_{12}^2$                                        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1         | T2                                                        | T3                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------|-----------------------------------------------------------|--------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+21g_2^4$ | $+g_2^2(-24\lambda_{S_2} - 12\lambda_{S_2}^{\text{Adj}})$ | $+2\lambda_{12}^2$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+21g_2^4$ | $+g_2^2(-24\lambda_{S_2} - 12\lambda_{S_2}^{\text{Adj}})$ | $+2\lambda_{12}^2$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+21g_2^4$ | $+g_2^2(-24\lambda_{S_2} - 12\lambda_{S_2}^{\text{Adj}})$ | $+2\lambda_{12}^2$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+21g_2^4$ | $+g_2^2(-24\lambda_{S_2} - 12\lambda_{S_2}^{\text{Adj}})$ | $+2\lambda_{12}^2$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+21g_2^4$ | $+g_2^2(-24\lambda_{S_2} - 12\lambda_{S_2}^{\text{Adj}})$ | $+2\lambda_{12}^2$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —          | —                                                         | $+4\lambda_{12}^2$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —          | —                                                         | $+4\lambda_{12}^2$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —          | —                                                         | $+4\lambda_{12}^2$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —          | —                                                         | $+4\lambda_{12}^2$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —          | —                                                         | $+4\lambda_{12}^2$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —          | —                                                         | $+4\lambda_{12}^2$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —          | —                                                         | $+4\lambda_{12}^2$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —          | —                                                         | $+4\lambda_{12}^2$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —          | —                                                         | $+4\lambda_{12}^2$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —          | —                                                         | $+4\lambda_{12}^2$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —          | —                                                         | $+6\lambda_{12}^2$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —          | —                                                         | $+6\lambda_{12}^2$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —          | —                                                         | $+6\lambda_{12}^2$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —          | —                                                         | $+6\lambda_{12}^2$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —          | —                                                         | $+6\lambda_{12}^2$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+21g_2^4$ | $+g_2^2(-24\lambda_{S_2} - 12\lambda_{S_2}^{\text{Adj}})$ | $+6\lambda_{12}^2$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+21g_2^4$ | $+g_2^2(-24\lambda_{S_2} - 12\lambda_{S_2}^{\text{Adj}})$ | $+6\lambda_{12}^2$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+21g_2^4$ | $+g_2^2(-24\lambda_{S_2} - 12\lambda_{S_2}^{\text{Adj}})$ | $+6\lambda_{12}^2$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+21g_2^4$ | $+g_2^2(-24\lambda_{S_2} - 12\lambda_{S_2}^{\text{Adj}})$ | $+6\lambda_{12}^2$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+21g_2^4$ | $+g_2^2(-24\lambda_{S_2} - 12\lambda_{S_2}^{\text{Adj}})$ | $+6\lambda_{12}^2$ |

Terms 3 of 10

| Column | Representative term structure      |
|--------|------------------------------------|
| T1     | $+4\lambda_{H2}^2$                 |
| T2     | $+\frac{3\lambda_{H2}^{Adj^2}}{2}$ |
| T3     | $+14\lambda_{S2}^2$                |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$       | $d_{S2}$ | $Y_{S2}$      | $d_F$ | $Y_F$          | T1                 | T2                                 | T3                  |
|-------|----------|----------|----------------|----------|---------------|-------|----------------|--------------------|------------------------------------|---------------------|
| T3-A  | -2       | 1        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+4\lambda_{H2}^2$ | $+\frac{3\lambda_{H2}^{Adj^2}}{2}$ | $+14\lambda_{S2}^2$ |
| T3-A  | -1       | 1        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+4\lambda_{H2}^2$ | $+\frac{3\lambda_{H2}^{Adj^2}}{2}$ | $+14\lambda_{S2}^2$ |
| T3-A  | 0        | 1        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+4\lambda_{H2}^2$ | $+\frac{3\lambda_{H2}^{Adj^2}}{2}$ | $+14\lambda_{S2}^2$ |
| T3-A  | 1        | 1        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+4\lambda_{H2}^2$ | $+\frac{3\lambda_{H2}^{Adj^2}}{2}$ | $+14\lambda_{S2}^2$ |
| T3-A  | 2        | 1        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+4\lambda_{H2}^2$ | $+\frac{3\lambda_{H2}^{Adj^2}}{2}$ | $+14\lambda_{S2}^2$ |
| T3-B  | -2       | 2        | -1             | 2        | 0             | 1     | $-\frac{1}{2}$ | $+4\lambda_{H2}^2$ | —                                  | $+12\lambda_{S2}^2$ |
| T3-B  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 1     | 0              | $+4\lambda_{H2}^2$ | —                                  | $+12\lambda_{S2}^2$ |
| T3-B  | 0        | 2        | 0              | 2        | 1             | 1     | $\frac{1}{2}$  | $+4\lambda_{H2}^2$ | —                                  | $+12\lambda_{S2}^2$ |
| T3-B  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 1     | 1              | $+4\lambda_{H2}^2$ | —                                  | $+12\lambda_{S2}^2$ |
| T3-B  | 2        | 2        | 1              | 2        | 2             | 1     | $\frac{3}{2}$  | $+4\lambda_{H2}^2$ | —                                  | $+12\lambda_{S2}^2$ |
| T3-C  | -2       | 2        | -1             | 2        | 0             | 3     | $-\frac{1}{2}$ | $+4\lambda_{H2}^2$ | —                                  | $+12\lambda_{S2}^2$ |
| T3-C  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 3     | 0              | $+4\lambda_{H2}^2$ | —                                  | $+12\lambda_{S2}^2$ |
| T3-C  | 0        | 2        | 0              | 2        | 1             | 3     | $\frac{1}{2}$  | $+4\lambda_{H2}^2$ | —                                  | $+12\lambda_{S2}^2$ |
| T3-C  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 3     | 1              | $+4\lambda_{H2}^2$ | —                                  | $+12\lambda_{S2}^2$ |
| T3-C  | 2        | 2        | 1              | 2        | 2             | 3     | $\frac{3}{2}$  | $+4\lambda_{H2}^2$ | —                                  | $+12\lambda_{S2}^2$ |
| T3-D  | -2       | 3        | -1             | 1        | 0             | 2     | $-\frac{1}{2}$ | $+4\lambda_{H2}^2$ | —                                  | $+10\lambda_{S2}^2$ |
| T3-D  | -1       | 3        | $-\frac{1}{2}$ | 1        | $\frac{1}{2}$ | 2     | 0              | $+4\lambda_{H2}^2$ | —                                  | $+10\lambda_{S2}^2$ |
| T3-D  | 0        | 3        | 0              | 1        | 1             | 2     | $\frac{1}{2}$  | $+4\lambda_{H2}^2$ | —                                  | $+10\lambda_{S2}^2$ |
| T3-D  | 1        | 3        | $\frac{1}{2}$  | 1        | $\frac{3}{2}$ | 2     | 1              | $+4\lambda_{H2}^2$ | —                                  | $+10\lambda_{S2}^2$ |
| T3-D  | 2        | 3        | 1              | 1        | 2             | 2     | $\frac{3}{2}$  | $+4\lambda_{H2}^2$ | —                                  | $+10\lambda_{S2}^2$ |
| T3-E  | -2       | 3        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+4\lambda_{H2}^2$ | $+\frac{3\lambda_{H2}^{Adj^2}}{2}$ | $+14\lambda_{S2}^2$ |
| T3-E  | -1       | 3        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+4\lambda_{H2}^2$ | $+\frac{3\lambda_{H2}^{Adj^2}}{2}$ | $+14\lambda_{S2}^2$ |
| T3-E  | 0        | 3        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+4\lambda_{H2}^2$ | $+\frac{3\lambda_{H2}^{Adj^2}}{2}$ | $+14\lambda_{S2}^2$ |
| T3-E  | 1        | 3        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+4\lambda_{H2}^2$ | $+\frac{3\lambda_{H2}^{Adj^2}}{2}$ | $+14\lambda_{S2}^2$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$         | T1                 | T2                                         | T3                   |
|-------|----------|-----------|-----------|-----------|-----------|-------|---------------|--------------------|--------------------------------------------|----------------------|
| T3-E  | 2        | 3         | 1         | 3         | 2         | 2     | $\frac{3}{2}$ | $+4\lambda_{H2}^2$ | $+ \frac{3\lambda_{H2}^{\text{Adj}^2}}{2}$ | $+14\lambda_{S_2}^2$ |

Terms 4 of 10

| Column | Representative term structure                                                    |
|--------|----------------------------------------------------------------------------------|
| T1     | $+26\lambda_{S_2}^{\text{Adj}^2}$                                                |
| T2     | $+g_Y^2\left(3g_2^2-3\lambda_{S_2}-\frac{3\lambda_{S_2}^{\text{Adj}}}{2}\right)$ |
| T3     | $+\frac{3g_Y^4}{4}$                                                              |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                | T2                                                                                  | T3                    |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------|-------------------------------------------------------------------------------------|-----------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+26\lambda_{S_2}^{\text{Adj}^2}$ | —                                                                                   | —                     |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+26\lambda_{S_2}^{\text{Adj}^2}$ | $+g_Y^2\left(3g_2^2-3\lambda_{S_2}-\frac{3\lambda_{S_2}^{\text{Adj}}}{2}\right)$    | $+\frac{3g_Y^4}{4}$   |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+26\lambda_{S_2}^{\text{Adj}^2}$ | —                                                                                   | —                     |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+26\lambda_{S_2}^{\text{Adj}^2}$ | $+g_Y^2\left(27g_2^2-27\lambda_{S_2}-\frac{27\lambda_{S_2}^{\text{Adj}}}{2}\right)$ | $+\frac{243g_Y^4}{4}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+26\lambda_{S_2}^{\text{Adj}^2}$ | —                                                                                   | —                     |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                 | —                                                                                   | —                     |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                 | —                                                                                   | $+\frac{3g_Y^4}{4}$   |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                 | —                                                                                   | —                     |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                 | —                                                                                   | $+\frac{243g_Y^4}{4}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                 | —                                                                                   | —                     |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                 | —                                                                                   | —                     |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                 | —                                                                                   | $+\frac{3g_Y^4}{4}$   |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                 | —                                                                                   | —                     |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                 | —                                                                                   | $+\frac{243g_Y^4}{4}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                 | —                                                                                   | —                     |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                 | —                                                                                   | —                     |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                 | —                                                                                   | $+\frac{3g_Y^4}{4}$   |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                 | —                                                                                   | —                     |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                 | —                                                                                   | $+\frac{243g_Y^4}{4}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                 | —                                                                                   | —                     |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+26\lambda_{S_2}^{\text{Adj}^2}$ | —                                                                                   | —                     |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+26\lambda_{S_2}^{\text{Adj}^2}$ | $+g_Y^2\left(3g_2^2-3\lambda_{S_2}-\frac{3\lambda_{S_2}^{\text{Adj}}}{2}\right)$    | $+\frac{3g_Y^4}{4}$   |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+26\lambda_{S_2}^{\text{Adj}^2}$ | —                                                                                   | —                     |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1                                | T2                                                                                         | T3                    |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|-----------------------------------|--------------------------------------------------------------------------------------------|-----------------------|
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | $+26\lambda_{S_2}^{\text{Adj}^2}$ | $+g_Y^2 \left( 27g_2^2 - 27\lambda_{S_2} - \frac{27\lambda_{S_2}^{\text{Adj}}}{2} \right)$ | $+\frac{243g_Y^4}{4}$ |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | $+26\lambda_{S_2}^{\text{Adj}^2}$ | —                                                                                          | —                     |

Terms 5 of 10

| Column | Representative term structure                                            |
|--------|--------------------------------------------------------------------------|
| T1     | $+g_Y^2\left(12g_2^2-12\lambda_{S_2}-6\lambda_{S_2}^{\text{Adj}}\right)$ |
| T2     | $+12g_Y^4$                                                               |
| T3     | $4\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                        | T2          | T3                                                                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------------------|-------------|--------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                         | —           | —                                                                  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                         | —           | —                                                                  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+g_Y^2\left(12g_2^2-12\lambda_{S_2}-6\lambda_{S_2}^{\text{Adj}}\right)$  | $+12g_Y^4$  | —                                                                  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                         | —           | —                                                                  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+g_Y^2\left(48g_2^2-48\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}}\right)$ | $+192g_Y^4$ | —                                                                  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                         | —           | $4\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                         | —           | $+4\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                         | $+12g_Y^4$  | $4\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                         | —           | $4\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                         | $+192g_Y^4$ | $4\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                         | —           | $6\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                         | —           | $+6\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                         | $+12g_Y^4$  | $6\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                         | —           | $6\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                         | $+192g_Y^4$ | $6\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                         | —           | $8\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                         | —           | $8\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                         | $+12g_Y^4$  | $8\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                         | —           | $8\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                         | $+192g_Y^4$ | $8\lambda_{S_2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)$  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                         | —           | —                                                                  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                         | —           | —                                                                  |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+g_Y^2\left(12g_2^2-12\lambda_{S_2}-6\lambda_{S_2}^{\text{Adj}}\right)$  | $+12g_Y^4$  | —                                                                  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                         | —           | —                                                                  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+g_Y^2\left(48g_2^2-48\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}}\right)$ | $+192g_Y^4$ | —                                                                  |

Terms 6 of 10

| Column | Representative term structure      |
|--------|------------------------------------|
| T1     | $+ \frac{9g_2^4}{4}$               |
| T2     | $-9g_2^2\lambda_{S_2}$             |
| T3     | $+ \frac{\lambda_{12}^{Adj^2}}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                   | T2                     | T3                                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------|------------------------|------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                    | —                      | —                                  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                    | —                      | —                                  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                    | —                      | —                                  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                    | —                      | —                                  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                    | —                      | —                                  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+ \frac{9g_2^4}{4}$ | $-9g_2^2\lambda_{S_2}$ | $+ \frac{\lambda_{12}^{Adj^2}}{4}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+ \frac{9g_2^4}{4}$ | $-9g_2^2\lambda_{S_2}$ | $+ \frac{\lambda_{12}^{Adj^2}}{4}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $+ \frac{9g_2^4}{4}$ | $-9g_2^2\lambda_{S_2}$ | $+ \frac{\lambda_{12}^{Adj^2}}{4}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $+ \frac{9g_2^4}{4}$ | $-9g_2^2\lambda_{S_2}$ | $+ \frac{\lambda_{12}^{Adj^2}}{4}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+ \frac{9g_2^4}{4}$ | $-9g_2^2\lambda_{S_2}$ | $+ \frac{\lambda_{12}^{Adj^2}}{4}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+ \frac{9g_2^4}{4}$ | $-9g_2^2\lambda_{S_2}$ | $+ \frac{\lambda_{12}^{Adj^2}}{4}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+ \frac{9g_2^4}{4}$ | $-9g_2^2\lambda_{S_2}$ | $+ \frac{\lambda_{12}^{Adj^2}}{4}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $+ \frac{9g_2^4}{4}$ | $-9g_2^2\lambda_{S_2}$ | $+ \frac{\lambda_{12}^{Adj^2}}{4}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $+ \frac{9g_2^4}{4}$ | $-9g_2^2\lambda_{S_2}$ | $+ \frac{\lambda_{12}^{Adj^2}}{4}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+ \frac{9g_2^4}{4}$ | $-9g_2^2\lambda_{S_2}$ | $+ \frac{\lambda_{12}^{Adj^2}}{4}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                    | —                      | —                                  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                    | —                      | —                                  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                    | —                      | —                                  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                    | —                      | —                                  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                    | —                      | —                                  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                    | —                      | —                                  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                    | —                      | —                                  |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                    | —                      | —                                  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                    | —                      | —                                  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                    | —                      | —                                  |

Terms 7 of 10

| Column | Representative term structure                                                                                                                                        |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $+ \frac{\lambda_{H2}^{Adj^2}}{4}$                                                                                                                                   |
| T2     | $\lambda_{S_1^\dagger S_2 S_2^\dagger 2} (2 \text{Tr} \left( y_1 \left( y_2 \right)^\dagger \right) + 24 \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* )$ |
| T3     | $+ 2 \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* \text{Tr} \left( y_2 \left( y_1 \right)^\dagger \right)$                                               |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                 | T2                                                                                                                                                                                                                                                                                        | T3 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                  | —                                                                                                                                                                                                                                                                                         | —  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                  | —                                                                                                                                                                                                                                                                                         | —  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                  | —                                                                                                                                                                                                                                                                                         | —  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                  | —                                                                                                                                                                                                                                                                                         | —  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                  | —                                                                                                                                                                                                                                                                                         | —  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+ \frac{\lambda_{H2}^{Adj^2}}{4}$ | —                                                                                                                                                                                                                                                                                         | —  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+ \frac{\lambda_{H2}^{Adj^2}}{4}$ | $\lambda_{S_1^\dagger S_2 S_2^\dagger 2} (2 \text{Tr} \left( y_1 \left( y_2 \right)^\dagger \right) + 24 \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* ) + 2 \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* \text{Tr} \left( y_2 \left( y_1 \right)^\dagger \right)$ |    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $+ \frac{\lambda_{H2}^{Adj^2}}{4}$ | —                                                                                                                                                                                                                                                                                         | —  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $+ \frac{\lambda_{H2}^{Adj^2}}{4}$ | —                                                                                                                                                                                                                                                                                         | —  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+ \frac{\lambda_{H2}^{Adj^2}}{4}$ | —                                                                                                                                                                                                                                                                                         | —  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+ \frac{\lambda_{H2}^{Adj^2}}{4}$ | —                                                                                                                                                                                                                                                                                         | —  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+ \frac{\lambda_{H2}^{Adj^2}}{4}$ | $\lambda_{S_1^\dagger S_2 S_2^\dagger 2} (3 \text{Tr} \left( y_1 \left( y_2 \right)^\dagger \right) + 24 \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* ) + 3 \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* \text{Tr} \left( y_2 \left( y_1 \right)^\dagger \right)$ |    |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $+ \frac{\lambda_{H2}^{Adj^2}}{4}$ | —                                                                                                                                                                                                                                                                                         | —  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $+ \frac{\lambda_{H2}^{Adj^2}}{4}$ | —                                                                                                                                                                                                                                                                                         | —  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+ \frac{\lambda_{H2}^{Adj^2}}{4}$ | —                                                                                                                                                                                                                                                                                         | —  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                  | —                                                                                                                                                                                                                                                                                         | —  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                  | —                                                                                                                                                                                                                                                                                         | —  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                  | —                                                                                                                                                                                                                                                                                         | —  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                  | —                                                                                                                                                                                                                                                                                         | —  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                  | —                                                                                                                                                                                                                                                                                         | —  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                  | —                                                                                                                                                                                                                                                                                         | —  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                  | —                                                                                                                                                                                                                                                                                         | —  |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                  | —                                                                                                                                                                                                                                                                                         | —  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                  | —                                                                                                                                                                                                                                                                                         | —  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$         | T1 | T2 | T3 |
|-------|----------|-----------|-----------|-----------|-----------|-------|---------------|----|----|----|
| T3-E  | 2        | 3         | 1         | 3         | 2         | 2     | $\frac{3}{2}$ | —  | —  | —  |

Terms 8 of 10

| Column | Representative term structure                                                                          |
|--------|--------------------------------------------------------------------------------------------------------|
| T1     | $+8\lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*$         |
| T2     | $+8\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\right)^*$ |
| T3     | $+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_2}\right)$                                                |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                             | T2                                                                                                     | T3                                                        |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                              | —                                                                                                      | —                                                         |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                              | —                                                                                                      | —                                                         |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                              | —                                                                                                      | —                                                         |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                              | —                                                                                                      | —                                                         |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                              | —                                                                                                      | —                                                         |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                              | —                                                                                                      | —                                                         |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+8\lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*$ | $+8\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\right)^*$ | $+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_2}\right)$   |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                              | —                                                                                                      | —                                                         |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                              | —                                                                                                      | $+g_Y^2 \left(\frac{27g_2^2}{2} - 27\lambda_{S_2}\right)$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                              | —                                                                                                      | —                                                         |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                              | —                                                                                                      | —                                                         |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+8\lambda_{H^\dagger H^\dagger S_2 S_2} \left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*$ | $+8\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\right)^*$ | $+g_Y^2 \left(\frac{3g_2^2}{2} - 3\lambda_{S_2}\right)$   |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                              | —                                                                                                      | —                                                         |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                              | —                                                                                                      | $+g_Y^2 \left(\frac{27g_2^2}{2} - 27\lambda_{S_2}\right)$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                              | —                                                                                                      | —                                                         |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                              | —                                                                                                      | —                                                         |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                              | —                                                                                                      | —                                                         |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                              | —                                                                                                      | —                                                         |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                              | —                                                                                                      | —                                                         |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                              | —                                                                                                      | —                                                         |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                              | —                                                                                                      | —                                                         |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                              | —                                                                                                      | —                                                         |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                              | —                                                                                                      | —                                                         |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                              | —                                                                                                      | —                                                         |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                              | —                                                                                                      | —                                                         |

Terms 9 of 10

| Column | Representative term structure               |
|--------|---------------------------------------------|
| T1     | $+g_Y^2\left(6g_2^2-12\lambda_{S_2}\right)$ |
| T2     | $-3g_Y^2\lambda_{S_2}$                      |
| T3     | $+8\lambda_{12}\lambda_{12}^{\text{Cross}}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                           | T2                      | T3                                          |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------|-------------------------|---------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                       | —                                           |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                       | —                                           |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                       | —                                           |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                       | —                                           |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                       | —                                           |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                            | —                       | —                                           |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                            | —                       | —                                           |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $+g_Y^2\left(6g_2^2-12\lambda_{S_2}\right)$  | —                       | —                                           |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                            | —                       | —                                           |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+g_Y^2\left(24g_2^2-48\lambda_{S_2}\right)$ | —                       | —                                           |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                            | —                       | —                                           |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                            | —                       | —                                           |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $+g_Y^2\left(6g_2^2-12\lambda_{S_2}\right)$  | —                       | —                                           |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                            | —                       | —                                           |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+g_Y^2\left(24g_2^2-48\lambda_{S_2}\right)$ | —                       | —                                           |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                       | —                                           |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                            | $-3g_Y^2\lambda_{S_2}$  | —                                           |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                            | $-12g_Y^2\lambda_{S_2}$ | —                                           |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                            | $-27g_Y^2\lambda_{S_2}$ | —                                           |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                            | $-48g_Y^2\lambda_{S_2}$ | —                                           |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                            | —                       | $+8\lambda_{12}\lambda_{12}^{\text{Cross}}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                            | —                       | $+8\lambda_{12}\lambda_{12}^{\text{Cross}}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                            | —                       | $+8\lambda_{12}\lambda_{12}^{\text{Cross}}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                            | —                       | $+8\lambda_{12}\lambda_{12}^{\text{Cross}}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                            | —                       | $+8\lambda_{12}\lambda_{12}^{\text{Cross}}$ |

Terms 10 of 10

| Column | Representative term structure     |
|--------|-----------------------------------|
| T1     | $+6\lambda_{12}^{\text{Adj}^2}$   |
| T2     | $+6\lambda_{12}^{\text{Cross}^2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                              | T2                                |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------|-----------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                               | —                                 |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                               | —                                 |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                               | —                                 |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                               | —                                 |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                               | —                                 |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                               | —                                 |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                               | —                                 |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                               | —                                 |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                               | —                                 |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                               | —                                 |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                               | —                                 |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                               | —                                 |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                               | —                                 |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                               | —                                 |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                               | —                                 |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                               | —                                 |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                               | —                                 |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                               | —                                 |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                               | —                                 |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                               | —                                 |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+6\lambda_{12}^{\text{Adj}^2}$ | $+6\lambda_{12}^{\text{Cross}^2}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+6\lambda_{12}^{\text{Adj}^2}$ | $+6\lambda_{12}^{\text{Cross}^2}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+6\lambda_{12}^{\text{Adj}^2}$ | $+6\lambda_{12}^{\text{Cross}^2}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+6\lambda_{12}^{\text{Adj}^2}$ | $+6\lambda_{12}^{\text{Cross}^2}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+6\lambda_{12}^{\text{Adj}^2}$ | $+6\lambda_{12}^{\text{Cross}^2}$ |



| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1 | T2                                                                                                  | T3                                              |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|----|-----------------------------------------------------------------------------------------------------|-------------------------------------------------|
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | —  | $-2 \operatorname{Tr} \left( y_1 \left( y_1 \right)^\dagger Y_e \left( Y_e \right)^\dagger \right)$ | $+4 \lambda_{T3} \left( \lambda_{T3} \right)^*$ |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | —  | $-2 \operatorname{Tr} \left( y_1 \left( y_1 \right)^\dagger Y_e \left( Y_e \right)^\dagger \right)$ | $+4 \lambda_{T3} \left( \lambda_{T3} \right)^*$ |

Terms 2 of 8

| Column | Representative term structure    |
|--------|----------------------------------|
| T1     | $-\frac{9g_2^2\lambda_{H1}}{2}$  |
| T2     | $+3g_Y^4$                        |
| T3     | $-\frac{15g_Y^2\lambda_{H1}}{2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                               | T2        | T3                               |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------|-----------|----------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{9g_2^2\lambda_{H1}}{2}$  | $+3g_Y^4$ | $-\frac{15g_Y^2\lambda_{H1}}{2}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{9g_2^2\lambda_{H1}}{2}$  | —         | —                                |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{9g_2^2\lambda_{H1}}{2}$  | —         | $-\frac{3g_Y^2\lambda_{H1}}{2}$  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{9g_2^2\lambda_{H1}}{2}$  | —         | —                                |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{9g_2^2\lambda_{H1}}{2}$  | $+3g_Y^4$ | $-\frac{15g_Y^2\lambda_{H1}}{2}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                | $+3g_Y^4$ | $-\frac{15g_Y^2\lambda_{H1}}{2}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                | —         | —                                |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                | —         | $-\frac{3g_Y^2\lambda_{H1}}{2}$  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                | —         | —                                |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                | $+3g_Y^4$ | $-\frac{15g_Y^2\lambda_{H1}}{2}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                | $+3g_Y^4$ | $-\frac{15g_Y^2\lambda_{H1}}{2}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                | —         | —                                |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                | —         | $-\frac{3g_Y^2\lambda_{H1}}{2}$  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                | —         | —                                |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                | $+3g_Y^4$ | $-\frac{15g_Y^2\lambda_{H1}}{2}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{33g_2^2\lambda_{H1}}{2}$ | $+3g_Y^4$ | $-\frac{15g_Y^2\lambda_{H1}}{2}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-\frac{33g_2^2\lambda_{H1}}{2}$ | —         | —                                |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-\frac{33g_2^2\lambda_{H1}}{2}$ | —         | $-\frac{3g_Y^2\lambda_{H1}}{2}$  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-\frac{33g_2^2\lambda_{H1}}{2}$ | —         | —                                |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-\frac{33g_2^2\lambda_{H1}}{2}$ | $+3g_Y^4$ | $-\frac{15g_Y^2\lambda_{H1}}{2}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{33g_2^2\lambda_{H1}}{2}$ | $+3g_Y^4$ | $-\frac{15g_Y^2\lambda_{H1}}{2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1                               | T2        | T3                               |
|-------|----------|-----------|----------------|-----------|---------------|-------|---------------|----------------------------------|-----------|----------------------------------|
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0             | $-\frac{33g_2^2\lambda_{H1}}{2}$ | —         | —                                |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$ | $-\frac{33g_2^2\lambda_{H1}}{2}$ | —         | $-\frac{3g_Y^2\lambda_{H1}}{2}$  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1             | $-\frac{33g_2^2\lambda_{H1}}{2}$ | —         | —                                |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$ | $-\frac{33g_2^2\lambda_{H1}}{2}$ | $+3g_Y^4$ | $-\frac{15g_Y^2\lambda_{H1}}{2}$ |

Terms 3 of 8

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $+6\lambda_{12}\lambda_{H2}$  |
| T2     | $+6\lambda_H\lambda_{H1}$     |
| T3     | $+4\lambda_{H1}^2$            |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$       | $d_{S2}$ | $Y_{S2}$      | $d_F$ | $Y_F$          | T1                           | T2                        | T3                 |
|-------|----------|----------|----------------|----------|---------------|-------|----------------|------------------------------|---------------------------|--------------------|
| T3-A  | -2       | 1        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+6\lambda_{12}\lambda_{H2}$ | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ |
| T3-A  | -1       | 1        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+6\lambda_{12}\lambda_{H2}$ | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ |
| T3-A  | 0        | 1        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+6\lambda_{12}\lambda_{H2}$ | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ |
| T3-A  | 1        | 1        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+6\lambda_{12}\lambda_{H2}$ | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ |
| T3-A  | 2        | 1        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+6\lambda_{12}\lambda_{H2}$ | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ |
| T3-B  | -2       | 2        | -1             | 2        | 0             | 1     | $-\frac{1}{2}$ | $+4\lambda_{12}\lambda_{H2}$ | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ |
| T3-B  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 1     | 0              | $+4\lambda_{12}\lambda_{H2}$ | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ |
| T3-B  | 0        | 2        | 0              | 2        | 1             | 1     | $\frac{1}{2}$  | $+4\lambda_{12}\lambda_{H2}$ | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ |
| T3-B  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 1     | 1              | $+4\lambda_{12}\lambda_{H2}$ | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ |
| T3-B  | 2        | 2        | 1              | 2        | 2             | 1     | $\frac{3}{2}$  | $+4\lambda_{12}\lambda_{H2}$ | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ |
| T3-C  | -2       | 2        | -1             | 2        | 0             | 3     | $-\frac{1}{2}$ | $+4\lambda_{12}\lambda_{H2}$ | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ |
| T3-C  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 3     | 0              | $+4\lambda_{12}\lambda_{H2}$ | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ |
| T3-C  | 0        | 2        | 0              | 2        | 1             | 3     | $\frac{1}{2}$  | $+4\lambda_{12}\lambda_{H2}$ | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ |
| T3-C  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 3     | 1              | $+4\lambda_{12}\lambda_{H2}$ | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ |
| T3-C  | 2        | 2        | 1              | 2        | 2             | 3     | $\frac{3}{2}$  | $+4\lambda_{12}\lambda_{H2}$ | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ |
| T3-D  | -2       | 3        | -1             | 1        | 0             | 2     | $-\frac{1}{2}$ | $+2\lambda_{12}\lambda_{H2}$ | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ |
| T3-D  | -1       | 3        | $-\frac{1}{2}$ | 1        | $\frac{1}{2}$ | 2     | 0              | $+2\lambda_{12}\lambda_{H2}$ | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ |
| T3-D  | 0        | 3        | 0              | 1        | 1             | 2     | $\frac{1}{2}$  | $+2\lambda_{12}\lambda_{H2}$ | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ |
| T3-D  | 1        | 3        | $\frac{1}{2}$  | 1        | $\frac{3}{2}$ | 2     | 1              | $+2\lambda_{12}\lambda_{H2}$ | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ |
| T3-D  | 2        | 3        | 1              | 1        | 2             | 2     | $\frac{3}{2}$  | $+2\lambda_{12}\lambda_{H2}$ | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ |
| T3-E  | -2       | 3        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | —                            | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ |
| T3-E  | -1       | 3        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | —                            | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ |
| T3-E  | 0        | 3        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | —                            | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ |
| T3-E  | 1        | 3        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | —                            | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ |
| T3-E  | 2        | 3        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | —                            | $+6\lambda_H\lambda_{H1}$ | $+4\lambda_{H1}^2$ |

Terms 4 of 8

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $+4\lambda_{H1}\lambda_{S1}$  |
| T2     | $+\frac{3g_Y^4}{4}$           |
| T3     | $-3g_Y^2\lambda_{H1}$         |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$       | $d_{S2}$ | $Y_{S2}$      | $d_F$ | $Y_F$          | T1                           | T2                  | T3                    |
|-------|----------|----------|----------------|----------|---------------|-------|----------------|------------------------------|---------------------|-----------------------|
| T3-A  | -2       | 1        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+4\lambda_{H1}\lambda_{S1}$ | —                   | —                     |
| T3-A  | -1       | 1        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+4\lambda_{H1}\lambda_{S1}$ | $+\frac{3g_Y^4}{4}$ | $-3g_Y^2\lambda_{H1}$ |
| T3-A  | 0        | 1        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+4\lambda_{H1}\lambda_{S1}$ | —                   | —                     |
| T3-A  | 1        | 1        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+4\lambda_{H1}\lambda_{S1}$ | $+\frac{3g_Y^4}{4}$ | $-3g_Y^2\lambda_{H1}$ |
| T3-A  | 2        | 1        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+4\lambda_{H1}\lambda_{S1}$ | —                   | —                     |
| T3-B  | -2       | 2        | -1             | 2        | 0             | 1     | $-\frac{1}{2}$ | $+6\lambda_{H1}\lambda_{S1}$ | —                   | —                     |
| T3-B  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 1     | 0              | $+6\lambda_{H1}\lambda_{S1}$ | $+\frac{3g_Y^4}{4}$ | $-3g_Y^2\lambda_{H1}$ |
| T3-B  | 0        | 2        | 0              | 2        | 1             | 1     | $\frac{1}{2}$  | $+6\lambda_{H1}\lambda_{S1}$ | —                   | —                     |
| T3-B  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 1     | 1              | $+6\lambda_{H1}\lambda_{S1}$ | $+\frac{3g_Y^4}{4}$ | $-3g_Y^2\lambda_{H1}$ |
| T3-B  | 2        | 2        | 1              | 2        | 2             | 1     | $\frac{3}{2}$  | $+6\lambda_{H1}\lambda_{S1}$ | —                   | —                     |
| T3-C  | -2       | 2        | -1             | 2        | 0             | 3     | $-\frac{1}{2}$ | $+6\lambda_{H1}\lambda_{S1}$ | —                   | —                     |
| T3-C  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 3     | 0              | $+6\lambda_{H1}\lambda_{S1}$ | $+\frac{3g_Y^4}{4}$ | $-3g_Y^2\lambda_{H1}$ |
| T3-C  | 0        | 2        | 0              | 2        | 1             | 3     | $\frac{1}{2}$  | $+6\lambda_{H1}\lambda_{S1}$ | —                   | —                     |
| T3-C  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 3     | 1              | $+6\lambda_{H1}\lambda_{S1}$ | $+\frac{3g_Y^4}{4}$ | $-3g_Y^2\lambda_{H1}$ |
| T3-C  | 2        | 2        | 1              | 2        | 2             | 3     | $\frac{3}{2}$  | $+6\lambda_{H1}\lambda_{S1}$ | —                   | —                     |
| T3-D  | -2       | 3        | -1             | 1        | 0             | 2     | $-\frac{1}{2}$ | $+8\lambda_{H1}\lambda_{S1}$ | —                   | —                     |
| T3-D  | -1       | 3        | $-\frac{1}{2}$ | 1        | $\frac{1}{2}$ | 2     | 0              | $+8\lambda_{H1}\lambda_{S1}$ | $+\frac{3g_Y^4}{4}$ | $-3g_Y^2\lambda_{H1}$ |
| T3-D  | 0        | 3        | 0              | 1        | 1             | 2     | $\frac{1}{2}$  | $+8\lambda_{H1}\lambda_{S1}$ | —                   | —                     |
| T3-D  | 1        | 3        | $\frac{1}{2}$  | 1        | $\frac{3}{2}$ | 2     | 1              | $+8\lambda_{H1}\lambda_{S1}$ | $+\frac{3g_Y^4}{4}$ | $-3g_Y^2\lambda_{H1}$ |
| T3-D  | 2        | 3        | 1              | 1        | 2             | 2     | $\frac{3}{2}$  | $+8\lambda_{H1}\lambda_{S1}$ | —                   | —                     |
| T3-E  | -2       | 3        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+8\lambda_{H1}\lambda_{S1}$ | —                   | —                     |
| T3-E  | -1       | 3        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+8\lambda_{H1}\lambda_{S1}$ | $+\frac{3g_Y^4}{4}$ | $-3g_Y^2\lambda_{H1}$ |
| T3-E  | 0        | 3        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+8\lambda_{H1}\lambda_{S1}$ | —                   | —                     |
| T3-E  | 1        | 3        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+8\lambda_{H1}\lambda_{S1}$ | $+\frac{3g_Y^4}{4}$ | $-3g_Y^2\lambda_{H1}$ |
| T3-E  | 2        | 3        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+8\lambda_{H1}\lambda_{S1}$ | —                   | —                     |

Terms 5 of 8

| Column |                                           | Representative term structure |
|--------|-------------------------------------------|-------------------------------|
| T1     | $+\frac{9g_2^4}{4}$                       |                               |
| T2     | $-9g_2^2\lambda_{H1}$                     |                               |
| T3     | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$ |                               |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                  | T2                    | T3                                        |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------|-----------------------|-------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                   | —                     | —                                         |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                   | —                     | —                                         |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                   | —                     | —                                         |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                   | —                     | —                                         |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                   | —                     | —                                         |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+\frac{9g_2^4}{4}$ | $-9g_2^2\lambda_{H1}$ | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+\frac{9g_2^4}{4}$ | $-9g_2^2\lambda_{H1}$ | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $+\frac{9g_2^4}{4}$ | $-9g_2^2\lambda_{H1}$ | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $+\frac{9g_2^4}{4}$ | $-9g_2^2\lambda_{H1}$ | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+\frac{9g_2^4}{4}$ | $-9g_2^2\lambda_{H1}$ | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+\frac{9g_2^4}{4}$ | $-9g_2^2\lambda_{H1}$ | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+\frac{9g_2^4}{4}$ | $-9g_2^2\lambda_{H1}$ | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $+\frac{9g_2^4}{4}$ | $-9g_2^2\lambda_{H1}$ | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $+\frac{9g_2^4}{4}$ | $-9g_2^2\lambda_{H1}$ | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+\frac{9g_2^4}{4}$ | $-9g_2^2\lambda_{H1}$ | $+\frac{3\lambda_{H1}^{\text{Adj}^2}}{4}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                   | —                     | —                                         |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                   | —                     | —                                         |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                   | —                     | —                                         |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                   | —                     | —                                         |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                   | —                     | —                                         |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                   | —                     | —                                         |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                   | —                     | —                                         |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                   | —                     | —                                         |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                   | —                     | —                                         |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                   | —                     | —                                         |

Terms 6 of 8

| Column | Representative term structure                                                                                                                                                                                                                                                 |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)+4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)^*\right)$ |
| T2     | $+\frac{1}{2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$                                                                                                                                     |
| T3     | $+24\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$                                                                                                                                                |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                                                                                                                                                                                                                                                                                   | T2                                                                                                                             | T3 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|----|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                    | —                                                                                                                              | —  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                                                    | —                                                                                                                              | —  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                    | —                                                                                                                              | —  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                    | —                                                                                                                              | —  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                    | —                                                                                                                              | —  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                    | —                                                                                                                              | —  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)+4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)^*\right)+\frac{1}{2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$ | $+24\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$ |    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                    | —                                                                                                                              | —  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                    | —                                                                                                                              | —  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                    | —                                                                                                                              | —  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                    | —                                                                                                                              | —  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{3}{4}\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)+4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)^*\right)+\frac{3}{4}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$ | $+24\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$ |    |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                    | —                                                                                                                              | —  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                    | —                                                                                                                              | —  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                    | —                                                                                                                              | —  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                    | —                                                                                                                              | —  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                                                    | —                                                                                                                              | —  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                    | —                                                                                                                              | —  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                    | —                                                                                                                              | —  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                    | —                                                                                                                              | —  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                    | —                                                                                                                              | —  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                                                    | —                                                                                                                              | —  |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                    | —                                                                                                                              | —  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                    | —                                                                                                                              | —  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                    | —                                                                                                                              | —  |

## Terms 7 of 8

| Column | Representative term structure                                                                                                                                                        |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $+3\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$                                                             |
| T2     | $\lambda_{H1} (2 \text{Tr} (y_1 (y_1)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 4\lambda_{S_1}^{\text{Adj}})$ |
| T3     | $+6g_2^4$                                                                                                                                                                            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                       | T2                                                                                                                                                                                            | T3 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                        | —                                                                                                                                                                                             | —  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                        | —                                                                                                                                                                                             | —  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                        | —                                                                                                                                                                                             | —  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                        | —                                                                                                                                                                                             | —  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                        | —                                                                                                                                                                                             | —  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                        | —                                                                                                                                                                                             | —  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+3\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ | —                                                                                                                                                                                             | —  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                        | —                                                                                                                                                                                             | —  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                        | —                                                                                                                                                                                             | —  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                        | —                                                                                                                                                                                             | —  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                        | —                                                                                                                                                                                             | —  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+3\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ | —                                                                                                                                                                                             | —  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                        | —                                                                                                                                                                                             | —  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                        | —                                                                                                                                                                                             | —  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                        | —                                                                                                                                                                                             | —  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                        | $\lambda_{H1} (2 \text{Tr} (y_1 (y_1)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 4\lambda_{S_1}^{\text{Adj}}) + 6g_2^4$ |    |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                        | $\lambda_{H1} (2 \text{Tr} (y_1 (y_1)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 4\lambda_{S_1}^{\text{Adj}}) + 6g_2^4$ |    |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                        | $\lambda_{H1} (2 \text{Tr} (y_1 (y_1)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 4\lambda_{S_1}^{\text{Adj}}) + 6g_2^4$ |    |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                        | $\lambda_{H1} (2 \text{Tr} (y_1 (y_1)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 4\lambda_{S_1}^{\text{Adj}}) + 6g_2^4$ |    |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                        | $\lambda_{H1} (2 \text{Tr} (y_1 (y_1)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 4\lambda_{S_1}^{\text{Adj}}) + 6g_2^4$ |    |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                        | $\lambda_{H1} (2 \text{Tr} (y_1 (y_1)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 4\lambda_{S_1}^{\text{Adj}}) + 6g_2^4$ |    |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                        | $\lambda_{H1} (2 \text{Tr} (y_1 (y_1)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 4\lambda_{S_1}^{\text{Adj}}) + 6g_2^4$ |    |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                        | $\lambda_{H1} (2 \text{Tr} (y_1 (y_1)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 4\lambda_{S_1}^{\text{Adj}}) + 6g_2^4$ |    |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                        | $\lambda_{H1} (2 \text{Tr} (y_1 (y_1)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 4\lambda_{S_1}^{\text{Adj}}) + 6g_2^4$ |    |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                        | $\lambda_{H1} (2 \text{Tr} (y_1 (y_1)^\dagger) + 6 \text{Tr} (Y_d (Y_d)^\dagger) + 2 \text{Tr} (Y_e (Y_e)^\dagger) + 6 \text{Tr} (Y_u (Y_u)^\dagger) + 4\lambda_{S_1}^{\text{Adj}}) + 6g_2^4$ |    |

Terms 8 of 8

| Column | Representative term structure                               |
|--------|-------------------------------------------------------------|
| T1     | $+6\lambda_{H1}^{\text{Adj}^2}$                             |
| T2     | $+\lambda_{H2}(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$ |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$       | $d_{S2}$ | $Y_{S2}$      | $d_F$ | $Y_F$          | T1                              | T2                                                          |
|-------|----------|----------|----------------|----------|---------------|-------|----------------|---------------------------------|-------------------------------------------------------------|
| T3-A  | -2       | 1        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | —                               | —                                                           |
| T3-A  | -1       | 1        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | —                               | —                                                           |
| T3-A  | 0        | 1        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | —                               | —                                                           |
| T3-A  | 1        | 1        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | —                               | —                                                           |
| T3-A  | 2        | 1        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | —                               | —                                                           |
| T3-B  | -2       | 2        | -1             | 2        | 0             | 1     | $-\frac{1}{2}$ | —                               | —                                                           |
| T3-B  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 1     | 0              | —                               | —                                                           |
| T3-B  | 0        | 2        | 0              | 2        | 1             | 1     | $\frac{1}{2}$  | —                               | —                                                           |
| T3-B  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 1     | 1              | —                               | —                                                           |
| T3-B  | 2        | 2        | 1              | 2        | 2             | 1     | $\frac{3}{2}$  | —                               | —                                                           |
| T3-C  | -2       | 2        | -1             | 2        | 0             | 3     | $-\frac{1}{2}$ | —                               | —                                                           |
| T3-C  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 3     | 0              | —                               | —                                                           |
| T3-C  | 0        | 2        | 0              | 2        | 1             | 3     | $\frac{1}{2}$  | —                               | —                                                           |
| T3-C  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 3     | 1              | —                               | —                                                           |
| T3-C  | 2        | 2        | 1              | 2        | 2             | 3     | $\frac{3}{2}$  | —                               | —                                                           |
| T3-D  | -2       | 3        | -1             | 1        | 0             | 2     | $-\frac{1}{2}$ | $+6\lambda_{H1}^{\text{Adj}^2}$ | —                                                           |
| T3-D  | -1       | 3        | $-\frac{1}{2}$ | 1        | $\frac{1}{2}$ | 2     | 0              | $+6\lambda_{H1}^{\text{Adj}^2}$ | —                                                           |
| T3-D  | 0        | 3        | 0              | 1        | 1             | 2     | $\frac{1}{2}$  | $+6\lambda_{H1}^{\text{Adj}^2}$ | —                                                           |
| T3-D  | 1        | 3        | $\frac{1}{2}$  | 1        | $\frac{3}{2}$ | 2     | 1              | $+6\lambda_{H1}^{\text{Adj}^2}$ | —                                                           |
| T3-D  | 2        | 3        | 1              | 1        | 2             | 2     | $\frac{3}{2}$  | $+6\lambda_{H1}^{\text{Adj}^2}$ | —                                                           |
| T3-E  | -2       | 3        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+6\lambda_{H1}^{\text{Adj}^2}$ | $+\lambda_{H2}(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$ |
| T3-E  | -1       | 3        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+6\lambda_{H1}^{\text{Adj}^2}$ | $+\lambda_{H2}(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 0        | 3        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+6\lambda_{H1}^{\text{Adj}^2}$ | $+\lambda_{H2}(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 1        | 3        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+6\lambda_{H1}^{\text{Adj}^2}$ | $+\lambda_{H2}(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 2        | 3        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+6\lambda_{H1}^{\text{Adj}^2}$ | $+\lambda_{H2}(6\lambda_{12}+4\lambda_{12}^{\text{Cross}})$ |

$$\beta_{\lambda_{H2}}^{(1)}$$

Search name: lambda#2

$$16\pi^2\,\mu\frac{d\lambda_{H2}}{d\mu}=\beta_{\lambda_{H2}}^{(1)}$$

## Terms 1 of 8

| Column | Representative term structure                                                                                                                                                                                   |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_{H2}(2\,\text{Tr}\big(y_2\,(y_2)^\dagger\big)+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big)+4\lambda_{S_2}^{\text{Adj}})$ |
| T2     | $-2\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$                                                                                                                                                 |
| T3     | $+4\lambda_{T3}\,(\lambda_{T3})^*$                                                                                                                                                                              |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                                                                                      | T2                                                              | T3                                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\lambda_{H2}(2\,\text{Tr}\big(y_2\,(y_2)^{\dagger}\big)+6\,\text{Tr}\big(Y_d\,(Y_d)^{\dagger}\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^{\dagger}\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^{\dagger}\big)+4\lambda_{S_2}^{\text{Adj}})$ | $-2\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$ | $+4\lambda_{T3}\,(\lambda_{T3})^*$  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\lambda_{H2}(2\,\text{Tr}\big(y_2\,(y_2)^{\dagger}\big)+6\,\text{Tr}\big(Y_d\,(Y_d)^{\dagger}\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^{\dagger}\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^{\dagger}\big)+4\lambda_{S_2}^{\text{Adj}})$ | $-2\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$ | $+4\lambda_{T3}\,(\lambda_{T3})^*$  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\lambda_{H2}(2\,\text{Tr}\big(y_2\,(y_2)^{\dagger}\big)+6\,\text{Tr}\big(Y_d\,(Y_d)^{\dagger}\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^{\dagger}\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^{\dagger}\big)+4\lambda_{S_2}^{\text{Adj}})$ | $-2\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$ | $+4\lambda_{T3}\,(\lambda_{T3})^*$  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\lambda_{H2}(2\,\text{Tr}\big(y_2\,(y_2)^{\dagger}\big)+6\,\text{Tr}\big(Y_d\,(Y_d)^{\dagger}\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^{\dagger}\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^{\dagger}\big)+4\lambda_{S_2}^{\text{Adj}})$ | $-2\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$ | $+4\lambda_{T3}\,(\lambda_{T3})^*$  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\lambda_{H2}(2\,\text{Tr}\big(y_2\,(y_2)^{\dagger}\big)+6\,\text{Tr}\big(Y_d\,(Y_d)^{\dagger}\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^{\dagger}\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^{\dagger}\big)+4\lambda_{S_2}^{\text{Adj}})$ | $-2\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$ | $+4\lambda_{T3}\,(\lambda_{T3})^*$  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                       | $-2\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$ | $+6\lambda_{T3}\,(\lambda_{T3})^*$  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                                                                                                                                                                       | $-2\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$ | $+6\lambda_{T3}\,(\lambda_{T3})^*$  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                                                                                       | $-2\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$ | $+6\lambda_{T3}\,(\lambda_{T3})^*$  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                                                                                       | $-2\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$ | $+6\lambda_{T3}\,(\lambda_{T3})^*$  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                                                                                       | $-2\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$ | $+6\lambda_{T3}\,(\lambda_{T3})^*$  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                       | $-3\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$ | $+6\lambda_{T3}\,(\lambda_{T3})^*$  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                                                                                                                                                                       | $-3\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$ | $+6\lambda_{T3}\,(\lambda_{T3})^*$  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                                                                                       | $-3\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$ | $+6\lambda_{T3}\,(\lambda_{T3})^*$  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                                                                                       | $-3\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$ | $+6\lambda_{T3}\,(\lambda_{T3})^*$  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                                                                                       | $-3\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$ | $+6\lambda_{T3}\,(\lambda_{T3})^*$  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                       | $-4\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$ | $+12\lambda_{T3}\,(\lambda_{T3})^*$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                       | $-4\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$ | $+12\lambda_{T3}\,(\lambda_{T3})^*$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                       | $-4\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$ | $+12\lambda_{T3}\,(\lambda_{T3})^*$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                       | $-4\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$ | $+12\lambda_{T3}\,(\lambda_{T3})^*$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                       | $-4\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$ | $+12\lambda_{T3}\,(\lambda_{T3})^*$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\lambda_{H2}(2\,\text{Tr}\big(y_2\,(y_2)^{\dagger}\big)+6\,\text{Tr}\big(Y_d\,(Y_d)^{\dagger}\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^{\dagger}\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^{\dagger}\big)+4\lambda_{S_2}^{\text{Adj}})$ | $-2\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$ | $+4\lambda_{T3}\,(\lambda_{T3})^*$  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\lambda_{H2}(2\,\text{Tr}\big(y_2\,(y_2)^{\dagger}\big)+6\,\text{Tr}\big(Y_d\,(Y_d)^{\dagger}\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^{\dagger}\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^{\dagger}\big)+4\lambda_{S_2}^{\text{Adj}})$ | $-2\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$ | $+4\lambda_{T3}\,(\lambda_{T3})^*$  |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\lambda_{H2}(2\,\text{Tr}\big(y_2\,(y_2)^{\dagger}\big)+6\,\text{Tr}\big(Y_d\,(Y_d)^{\dagger}\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^{\dagger}\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^{\dagger}\big)+4\lambda_{S_2}^{\text{Adj}})$ | $-2\,\text{Tr}\big(y_2\,(y_2)^\dagger\,Y_e\,(Y_e)^\dagger\big)$ | $+4\lambda_{T3}\,(\lambda_{T3})^*$  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1                                                                                                                                                              | T2                                               | T3                               |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|----------------------------------|
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | $\lambda_{H2}(2\text{Tr}(y_2(y_2)^\dagger) + 6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger) + 4\lambda_{S_2}^{Ad})$ | $-2\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$ | $+4\lambda_{T3}(\lambda_{T3})^*$ |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | $\lambda_{H2}(2\text{Tr}(y_2(y_2)^\dagger) + 6\text{Tr}(Y_d(Y_d)^\dagger) + 2\text{Tr}(Y_e(Y_e)^\dagger) + 6\text{Tr}(Y_u(Y_u)^\dagger) + 4\lambda_{S_2}^{Ad})$ | $-2\text{Tr}(y_2(y_2)^\dagger Y_e(Y_e)^\dagger)$ | $+4\lambda_{T3}(\lambda_{T3})^*$ |

Terms 2 of 8

| Column | Representative term structure    |
|--------|----------------------------------|
| T1     | $+6g_2^4$                        |
| T2     | $-\frac{33g_2^2\lambda_{H2}}{2}$ |
| T3     | $-\frac{3g_Y^2\lambda_{H2}}{2}$  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1        | T2                               | T3                               |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------|----------------------------------|----------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+6g_2^4$ | $-\frac{33g_2^2\lambda_{H2}}{2}$ | $-\frac{3g_Y^2\lambda_{H2}}{2}$  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+6g_2^4$ | $-\frac{33g_2^2\lambda_{H2}}{2}$ | —                                |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+6g_2^4$ | $-\frac{33g_2^2\lambda_{H2}}{2}$ | $-\frac{15g_Y^2\lambda_{H2}}{2}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+6g_2^4$ | $-\frac{33g_2^2\lambda_{H2}}{2}$ | —                                |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+6g_2^4$ | $-\frac{33g_2^2\lambda_{H2}}{2}$ | $-\frac{51g_Y^2\lambda_{H2}}{2}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —         | —                                | $-\frac{3g_Y^2\lambda_{H2}}{2}$  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —         | —                                | —                                |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —         | —                                | $-\frac{15g_Y^2\lambda_{H2}}{2}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —         | —                                | —                                |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —         | —                                | $-\frac{51g_Y^2\lambda_{H2}}{2}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —         | —                                | $-\frac{3g_Y^2\lambda_{H2}}{2}$  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —         | —                                | —                                |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —         | —                                | $-\frac{15g_Y^2\lambda_{H2}}{2}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —         | —                                | —                                |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —         | —                                | $-\frac{51g_Y^2\lambda_{H2}}{2}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —         | $-\frac{9g_2^2\lambda_{H2}}{2}$  | $-\frac{3g_Y^2\lambda_{H2}}{2}$  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —         | $-\frac{9g_2^2\lambda_{H2}}{2}$  | —                                |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —         | $-\frac{9g_2^2\lambda_{H2}}{2}$  | $-\frac{15g_Y^2\lambda_{H2}}{2}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —         | $-\frac{9g_2^2\lambda_{H2}}{2}$  | —                                |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —         | $-\frac{9g_2^2\lambda_{H2}}{2}$  | $-\frac{51g_Y^2\lambda_{H2}}{2}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+6g_2^4$ | $-\frac{33g_2^2\lambda_{H2}}{2}$ | $-\frac{3g_Y^2\lambda_{H2}}{2}$  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1        | T2                               | T3                               |
|-------|----------|-----------|----------------|-----------|---------------|-------|---------------|-----------|----------------------------------|----------------------------------|
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0             | $+6g_2^4$ | $-\frac{33g_2^2\lambda_{H2}}{2}$ | —                                |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$ | $+6g_2^4$ | $-\frac{33g_2^2\lambda_{H2}}{2}$ | $-\frac{15g_Y^2\lambda_{H2}}{2}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1             | $+6g_2^4$ | $-\frac{33g_2^2\lambda_{H2}}{2}$ | —                                |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$ | $+6g_2^4$ | $-\frac{33g_2^2\lambda_{H2}}{2}$ | $-\frac{51g_Y^2\lambda_{H2}}{2}$ |

Terms 3 of 8

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $+2\lambda_{12}\lambda_{H1}$  |
| T2     | $+6\lambda_H\lambda_{H2}$     |
| T3     | $+4\lambda_{H2}^2$            |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$       | $d_{S2}$ | $Y_{S2}$      | $d_F$ | $Y_F$          | T1                           | T2                        | T3                 |
|-------|----------|----------|----------------|----------|---------------|-------|----------------|------------------------------|---------------------------|--------------------|
| T3-A  | -2       | 1        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+2\lambda_{12}\lambda_{H1}$ | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ |
| T3-A  | -1       | 1        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+2\lambda_{12}\lambda_{H1}$ | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ |
| T3-A  | 0        | 1        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+2\lambda_{12}\lambda_{H1}$ | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ |
| T3-A  | 1        | 1        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+2\lambda_{12}\lambda_{H1}$ | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ |
| T3-A  | 2        | 1        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+2\lambda_{12}\lambda_{H1}$ | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ |
| T3-B  | -2       | 2        | -1             | 2        | 0             | 1     | $-\frac{1}{2}$ | $+4\lambda_{12}\lambda_{H1}$ | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ |
| T3-B  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 1     | 0              | $+4\lambda_{12}\lambda_{H1}$ | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ |
| T3-B  | 0        | 2        | 0              | 2        | 1             | 1     | $\frac{1}{2}$  | $+4\lambda_{12}\lambda_{H1}$ | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ |
| T3-B  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 1     | 1              | $+4\lambda_{12}\lambda_{H1}$ | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ |
| T3-B  | 2        | 2        | 1              | 2        | 2             | 1     | $\frac{3}{2}$  | $+4\lambda_{12}\lambda_{H1}$ | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ |
| T3-C  | -2       | 2        | -1             | 2        | 0             | 3     | $-\frac{1}{2}$ | $+4\lambda_{12}\lambda_{H1}$ | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ |
| T3-C  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 3     | 0              | $+4\lambda_{12}\lambda_{H1}$ | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ |
| T3-C  | 0        | 2        | 0              | 2        | 1             | 3     | $\frac{1}{2}$  | $+4\lambda_{12}\lambda_{H1}$ | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ |
| T3-C  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 3     | 1              | $+4\lambda_{12}\lambda_{H1}$ | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ |
| T3-C  | 2        | 2        | 1              | 2        | 2             | 3     | $\frac{3}{2}$  | $+4\lambda_{12}\lambda_{H1}$ | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ |
| T3-D  | -2       | 3        | -1             | 1        | 0             | 2     | $-\frac{1}{2}$ | $+6\lambda_{12}\lambda_{H1}$ | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ |
| T3-D  | -1       | 3        | $-\frac{1}{2}$ | 1        | $\frac{1}{2}$ | 2     | 0              | $+6\lambda_{12}\lambda_{H1}$ | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ |
| T3-D  | 0        | 3        | 0              | 1        | 1             | 2     | $\frac{1}{2}$  | $+6\lambda_{12}\lambda_{H1}$ | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ |
| T3-D  | 1        | 3        | $\frac{1}{2}$  | 1        | $\frac{3}{2}$ | 2     | 1              | $+6\lambda_{12}\lambda_{H1}$ | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ |
| T3-D  | 2        | 3        | 1              | 1        | 2             | 2     | $\frac{3}{2}$  | $+6\lambda_{12}\lambda_{H1}$ | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ |
| T3-E  | -2       | 3        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | —                            | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ |
| T3-E  | -1       | 3        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | —                            | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ |
| T3-E  | 0        | 3        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | —                            | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ |
| T3-E  | 1        | 3        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | —                            | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ |
| T3-E  | 2        | 3        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | —                            | $+6\lambda_H\lambda_{H2}$ | $+4\lambda_{H2}^2$ |

Terms 4 of 8

| Column | Representative term structure   |
|--------|---------------------------------|
| T1     | $+8\lambda_{H2}\lambda_{S2}$    |
| T2     | $+6\lambda_{H2}^{\text{Adj}^2}$ |
| T3     | $+\frac{3g_Y^4}{4}$             |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$       | $d_{S2}$ | $Y_{S2}$      | $d_F$ | $Y_F$          | T1                           | T2                              | T3                   |
|-------|----------|----------|----------------|----------|---------------|-------|----------------|------------------------------|---------------------------------|----------------------|
| T3-A  | -2       | 1        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+8\lambda_{H2}\lambda_{S2}$ | $+6\lambda_{H2}^{\text{Adj}^2}$ | —                    |
| T3-A  | -1       | 1        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+8\lambda_{H2}\lambda_{S2}$ | $+6\lambda_{H2}^{\text{Adj}^2}$ | $+\frac{3g_Y^4}{4}$  |
| T3-A  | 0        | 1        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+8\lambda_{H2}\lambda_{S2}$ | $+6\lambda_{H2}^{\text{Adj}^2}$ | —                    |
| T3-A  | 1        | 1        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+8\lambda_{H2}\lambda_{S2}$ | $+6\lambda_{H2}^{\text{Adj}^2}$ | $+\frac{27g_Y^4}{4}$ |
| T3-A  | 2        | 1        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+8\lambda_{H2}\lambda_{S2}$ | $+6\lambda_{H2}^{\text{Adj}^2}$ | —                    |
| T3-B  | -2       | 2        | -1             | 2        | 0             | 1     | $-\frac{1}{2}$ | $+6\lambda_{H2}\lambda_{S2}$ | —                               | —                    |
| T3-B  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 1     | 0              | $+6\lambda_{H2}\lambda_{S2}$ | —                               | $+\frac{3g_Y^4}{4}$  |
| T3-B  | 0        | 2        | 0              | 2        | 1             | 1     | $\frac{1}{2}$  | $+6\lambda_{H2}\lambda_{S2}$ | —                               | —                    |
| T3-B  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 1     | 1              | $+6\lambda_{H2}\lambda_{S2}$ | —                               | $+\frac{27g_Y^4}{4}$ |
| T3-B  | 2        | 2        | 1              | 2        | 2             | 1     | $\frac{3}{2}$  | $+6\lambda_{H2}\lambda_{S2}$ | —                               | —                    |
| T3-C  | -2       | 2        | -1             | 2        | 0             | 3     | $-\frac{1}{2}$ | $+6\lambda_{H2}\lambda_{S2}$ | —                               | —                    |
| T3-C  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 3     | 0              | $+6\lambda_{H2}\lambda_{S2}$ | —                               | $+\frac{3g_Y^4}{4}$  |
| T3-C  | 0        | 2        | 0              | 2        | 1             | 3     | $\frac{1}{2}$  | $+6\lambda_{H2}\lambda_{S2}$ | —                               | —                    |
| T3-C  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 3     | 1              | $+6\lambda_{H2}\lambda_{S2}$ | —                               | $+\frac{27g_Y^4}{4}$ |
| T3-C  | 2        | 2        | 1              | 2        | 2             | 3     | $\frac{3}{2}$  | $+6\lambda_{H2}\lambda_{S2}$ | —                               | —                    |
| T3-D  | -2       | 3        | -1             | 1        | 0             | 2     | $-\frac{1}{2}$ | $+4\lambda_{H2}\lambda_{S2}$ | —                               | —                    |
| T3-D  | -1       | 3        | $-\frac{1}{2}$ | 1        | $\frac{1}{2}$ | 2     | 0              | $+4\lambda_{H2}\lambda_{S2}$ | —                               | $+\frac{3g_Y^4}{4}$  |
| T3-D  | 0        | 3        | 0              | 1        | 1             | 2     | $\frac{1}{2}$  | $+4\lambda_{H2}\lambda_{S2}$ | —                               | —                    |
| T3-D  | 1        | 3        | $\frac{1}{2}$  | 1        | $\frac{3}{2}$ | 2     | 1              | $+4\lambda_{H2}\lambda_{S2}$ | —                               | $+\frac{27g_Y^4}{4}$ |
| T3-D  | 2        | 3        | 1              | 1        | 2             | 2     | $\frac{3}{2}$  | $+4\lambda_{H2}\lambda_{S2}$ | —                               | —                    |
| T3-E  | -2       | 3        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+8\lambda_{H2}\lambda_{S2}$ | $+6\lambda_{H2}^{\text{Adj}^2}$ | —                    |
| T3-E  | -1       | 3        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+8\lambda_{H2}\lambda_{S2}$ | $+6\lambda_{H2}^{\text{Adj}^2}$ | $+\frac{3g_Y^4}{4}$  |
| T3-E  | 0        | 3        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+8\lambda_{H2}\lambda_{S2}$ | $+6\lambda_{H2}^{\text{Adj}^2}$ | —                    |
| T3-E  | 1        | 3        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+8\lambda_{H2}\lambda_{S2}$ | $+6\lambda_{H2}^{\text{Adj}^2}$ | $+\frac{27g_Y^4}{4}$ |
| T3-E  | 2        | 3        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+8\lambda_{H2}\lambda_{S2}$ | $+6\lambda_{H2}^{\text{Adj}^2}$ | —                    |

## Terms 5 of 8

| Column |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  | Representative term structure |  |  |  |  |  |  |  |  |  |
|--------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|--|--|-------------------------------|--|--|--|--|--|--|--|--|--|
| T1     |  | $-3g_Y^2\lambda_{H2}$                                                                                                                                                                                                       |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
| T2     |  | $+3g_Y^4$                                                                                                                                                                                                                   |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
| T3     |  | $\lambda_{H2}(2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$ |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
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|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |
|        |  |                                                                                                                                                                                                                             |  |  |  |  |  |  |  |                               |  |  |  |  |  |  |  |  |  |

Terms 6 of 8

| Column | Representative term structure             |
|--------|-------------------------------------------|
| T1     | $+\frac{9g_2^4}{4}$                       |
| T2     | $-9g_2^2\lambda_{H2}$                     |
| T3     | $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                  | T2                    | T3                                        |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------|-----------------------|-------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                   | —                     | —                                         |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                   | —                     | —                                         |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                   | —                     | —                                         |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                   | —                     | —                                         |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                   | —                     | —                                         |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+\frac{9g_2^4}{4}$ | $-9g_2^2\lambda_{H2}$ | $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+\frac{9g_2^4}{4}$ | $-9g_2^2\lambda_{H2}$ | $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $+\frac{9g_2^4}{4}$ | $-9g_2^2\lambda_{H2}$ | $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $+\frac{9g_2^4}{4}$ | $-9g_2^2\lambda_{H2}$ | $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+\frac{9g_2^4}{4}$ | $-9g_2^2\lambda_{H2}$ | $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+\frac{9g_2^4}{4}$ | $-9g_2^2\lambda_{H2}$ | $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+\frac{9g_2^4}{4}$ | $-9g_2^2\lambda_{H2}$ | $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $+\frac{9g_2^4}{4}$ | $-9g_2^2\lambda_{H2}$ | $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $+\frac{9g_2^4}{4}$ | $-9g_2^2\lambda_{H2}$ | $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+\frac{9g_2^4}{4}$ | $-9g_2^2\lambda_{H2}$ | $+\frac{3\lambda_{H2}^{\text{Adj}^2}}{4}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                   | —                     | —                                         |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                   | —                     | —                                         |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                   | —                     | —                                         |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                   | —                     | —                                         |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                   | —                     | —                                         |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                   | —                     | —                                         |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                   | —                     | —                                         |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                   | —                     | —                                         |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                   | —                     | —                                         |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                   | —                     | —                                         |

Terms 7 of 8

| Column | Representative term structure                                                                                                                                                                                                                                               |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)+4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*\right)$ |
| T2     | $+\frac{1}{2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$                                                                                                                                   |
| T3     | $+3\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$                                                                                                                                                       |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                                                                                                                                          | T2                                                                                                                                        | T3                                                                                                                    |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                           | —                                                                                                                                         | —                                                                                                                     |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                           | —                                                                                                                                         | —                                                                                                                     |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                           | —                                                                                                                                         | —                                                                                                                     |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                           | —                                                                                                                                         | —                                                                                                                     |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                           | —                                                                                                                                         | —                                                                                                                     |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                           | —                                                                                                                                         | —                                                                                                                     |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)+4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*\right)$ | $+\frac{1}{2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$ | $+3\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                           | —                                                                                                                                         | —                                                                                                                     |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                                                                                                                                           | —                                                                                                                                         | —                                                                                                                     |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                           | —                                                                                                                                         | —                                                                                                                     |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                           | —                                                                                                                                         | —                                                                                                                     |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{3}{4}\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)+4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*\right)$ | $+\frac{3}{4}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$ | $+3\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                           | —                                                                                                                                         | —                                                                                                                     |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                                                                                                                                           | —                                                                                                                                         | —                                                                                                                     |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                           | —                                                                                                                                         | —                                                                                                                     |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                           | —                                                                                                                                         | —                                                                                                                     |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                           | —                                                                                                                                         | —                                                                                                                     |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                           | —                                                                                                                                         | —                                                                                                                     |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                           | —                                                                                                                                         | —                                                                                                                     |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                           | —                                                                                                                                         | —                                                                                                                     |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                           | —                                                                                                                                         | —                                                                                                                     |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                           | —                                                                                                                                         | —                                                                                                                     |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                           | —                                                                                                                                         | —                                                                                                                     |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                           | —                                                                                                                                         | —                                                                                                                     |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                           | —                                                                                                                                         | —                                                                                                                     |

Terms 8 of 8

| Column | Representative term structure                                                                     |
|--------|---------------------------------------------------------------------------------------------------|
| T1     | $+24\lambda_{H^\dagger H^\dagger S_2 S_2} \left( \lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$ |
| T2     | $+\lambda_{H1}(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$                                     |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                | T2                                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                 | —                                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                 | —                                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                 | —                                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                 | —                                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                 | —                                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                 | —                                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+24\lambda_{H^\dagger H^\dagger S_2 S_2} \left( \lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$ | —                                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                 | —                                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                 | —                                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                 | —                                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                 | —                                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+24\lambda_{H^\dagger H^\dagger S_2 S_2} \left( \lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$ | —                                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                 | —                                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                 | —                                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                 | —                                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                 | —                                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                 | —                                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                 | —                                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                 | —                                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                 | —                                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                 | $+\lambda_{H1}(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                 | $+\lambda_{H1}(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                 | $+\lambda_{H1}(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                 | $+\lambda_{H1}(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                 | $+\lambda_{H1}(6\lambda_{12} + 4\lambda_{12}^{\text{Cross}})$ |

$\beta_{\lambda_{12}}^{(1)}$

Search name: lambda12

$16\pi^2\,\mu\frac{d\lambda_{12}}{d\mu}=\beta_{\lambda_{12}}^{(1)}$

Terms 1 of 12

| Column | Representative term structure                                                                                                 |
|--------|-------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_{12}(4\,\text{Tr}\big(y_1\,(y_1)^\dagger\big)+2\,\text{Tr}\big(y_2\,(y_2)^\dagger\big)+4\lambda_{S_2}^{\text{Adj}})$ |
| T2     | $-4\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$                                                               |
| T3     | $+4\lambda_{T3}\,(\lambda_{T3})^*$                                                                                            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                            | T2                                                              | T3                                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\lambda_{12}(4\,\text{Tr}\big(y_1\,(y_1)^\dagger\big)+2\,\text{Tr}\big(y_2\,(y_2)^\dagger\big)+4\lambda_{S_2}^{\text{Adj}})$ | $-4\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$ | $+4\lambda_{T3}\,(\lambda_{T3})^*$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\lambda_{12}(4\,\text{Tr}\big(y_1\,(y_1)^\dagger\big)+2\,\text{Tr}\big(y_2\,(y_2)^\dagger\big)+4\lambda_{S_2}^{\text{Adj}})$ | $-4\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$ | $+4\lambda_{T3}\,(\lambda_{T3})^*$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\lambda_{12}(4\,\text{Tr}\big(y_1\,(y_1)^\dagger\big)+2\,\text{Tr}\big(y_2\,(y_2)^\dagger\big)+4\lambda_{S_2}^{\text{Adj}})$ | $-4\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$ | $+4\lambda_{T3}\,(\lambda_{T3})^*$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\lambda_{12}(4\,\text{Tr}\big(y_1\,(y_1)^\dagger\big)+2\,\text{Tr}\big(y_2\,(y_2)^\dagger\big)+4\lambda_{S_2}^{\text{Adj}})$ | $-4\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$ | $+4\lambda_{T3}\,(\lambda_{T3})^*$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\lambda_{12}(4\,\text{Tr}\big(y_1\,(y_1)^\dagger\big)+2\,\text{Tr}\big(y_2\,(y_2)^\dagger\big)+4\lambda_{S_2}^{\text{Adj}})$ | $-4\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$ | $+4\lambda_{T3}\,(\lambda_{T3})^*$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                             | $-2\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$ | $+3\lambda_{T3}\,(\lambda_{T3})^*$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                                                                             | $-2\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$ | $+3\lambda_{T3}\,(\lambda_{T3})^*$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                             | $-2\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$ | $+3\lambda_{T3}\,(\lambda_{T3})^*$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                             | $-2\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$ | $+3\lambda_{T3}\,(\lambda_{T3})^*$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                             | $-2\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$ | $+3\lambda_{T3}\,(\lambda_{T3})^*$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                             | —                                                               | $+3\lambda_{T3}\,(\lambda_{T3})^*$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                                                                             | —                                                               | $+3\lambda_{T3}\,(\lambda_{T3})^*$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                             | —                                                               | $+3\lambda_{T3}\,(\lambda_{T3})^*$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                             | —                                                               | $+3\lambda_{T3}\,(\lambda_{T3})^*$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                             | —                                                               | $+3\lambda_{T3}\,(\lambda_{T3})^*$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                             | $-4\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$ | $+4\lambda_{T3}\,(\lambda_{T3})^*$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                             | $-4\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$ | $+4\lambda_{T3}\,(\lambda_{T3})^*$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                             | $-4\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$ | $+4\lambda_{T3}\,(\lambda_{T3})^*$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                             | $-4\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$ | $+4\lambda_{T3}\,(\lambda_{T3})^*$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                             | $-4\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$ | $+4\lambda_{T3}\,(\lambda_{T3})^*$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                             | $-2\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$ | —                                  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                             | $-2\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$ | —                                  |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                             | $-2\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$ | —                                  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                             | $-2\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$ | —                                  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$         | T1 | T2                                                                                   | T3 |
|-------|----------|-----------|-----------|-----------|-----------|-------|---------------|----|--------------------------------------------------------------------------------------|----|
| T3-E  | 2        | 3         | 1         | 3         | 2         | 2     | $\frac{3}{2}$ | —  | $-2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$ | —  |

Terms 2 of 12

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $-12g_2^2\lambda_{12}$        |
| T2     | $-6g_Y^2\lambda_{12}$         |
| T3     | $+4\lambda_{12}^2$            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                     | T2                     | T3                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------|------------------------|--------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-12g_2^2\lambda_{12}$ | $-6g_Y^2\lambda_{12}$  | $+4\lambda_{12}^2$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-12g_2^2\lambda_{12}$ | $-3g_Y^2\lambda_{12}$  | $+4\lambda_{12}^2$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-12g_2^2\lambda_{12}$ | $-6g_Y^2\lambda_{12}$  | $+4\lambda_{12}^2$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-12g_2^2\lambda_{12}$ | $-15g_Y^2\lambda_{12}$ | $+4\lambda_{12}^2$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-12g_2^2\lambda_{12}$ | $-30g_Y^2\lambda_{12}$ | $+4\lambda_{12}^2$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-9g_2^2\lambda_{12}$  | $-6g_Y^2\lambda_{12}$  | $+4\lambda_{12}^2$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-9g_2^2\lambda_{12}$  | $-3g_Y^2\lambda_{12}$  | $+4\lambda_{12}^2$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $-9g_2^2\lambda_{12}$  | $-6g_Y^2\lambda_{12}$  | $+4\lambda_{12}^2$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-9g_2^2\lambda_{12}$  | $-15g_Y^2\lambda_{12}$ | $+4\lambda_{12}^2$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-9g_2^2\lambda_{12}$  | $-30g_Y^2\lambda_{12}$ | $+4\lambda_{12}^2$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-9g_2^2\lambda_{12}$  | $-6g_Y^2\lambda_{12}$  | $+4\lambda_{12}^2$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-9g_2^2\lambda_{12}$  | $-3g_Y^2\lambda_{12}$  | $+4\lambda_{12}^2$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-9g_2^2\lambda_{12}$  | $-6g_Y^2\lambda_{12}$  | $+4\lambda_{12}^2$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-9g_2^2\lambda_{12}$  | $-15g_Y^2\lambda_{12}$ | $+4\lambda_{12}^2$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-9g_2^2\lambda_{12}$  | $-30g_Y^2\lambda_{12}$ | $+4\lambda_{12}^2$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $-12g_2^2\lambda_{12}$ | $-6g_Y^2\lambda_{12}$  | $+4\lambda_{12}^2$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-12g_2^2\lambda_{12}$ | $-3g_Y^2\lambda_{12}$  | $+4\lambda_{12}^2$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $-12g_2^2\lambda_{12}$ | $-6g_Y^2\lambda_{12}$  | $+4\lambda_{12}^2$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $-12g_2^2\lambda_{12}$ | $-15g_Y^2\lambda_{12}$ | $+4\lambda_{12}^2$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $-12g_2^2\lambda_{12}$ | $-30g_Y^2\lambda_{12}$ | $+4\lambda_{12}^2$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                      | —                      | $+4\lambda_{12}^2$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                      | —                      | $+4\lambda_{12}^2$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                      | —                      | $+4\lambda_{12}^2$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                      | —                      | $+4\lambda_{12}^2$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                      | —                      | $+4\lambda_{12}^2$ |

Terms 3 of 12

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $+4\lambda_{12}\lambda_{S_1}$ |
| T2     | $+8\lambda_{12}\lambda_{S_2}$ |
| T3     | $+4\lambda_{H1}\lambda_{H2}$  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                            | T2                            | T3                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------|-------------------------------|------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+4\lambda_{12}\lambda_{S_1}$ | $+8\lambda_{12}\lambda_{S_2}$ | $+4\lambda_{H1}\lambda_{H2}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+4\lambda_{12}\lambda_{S_1}$ | $+8\lambda_{12}\lambda_{S_2}$ | $+4\lambda_{H1}\lambda_{H2}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+4\lambda_{12}\lambda_{S_1}$ | $+8\lambda_{12}\lambda_{S_2}$ | $+4\lambda_{H1}\lambda_{H2}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+4\lambda_{12}\lambda_{S_1}$ | $+8\lambda_{12}\lambda_{S_2}$ | $+4\lambda_{H1}\lambda_{H2}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+4\lambda_{12}\lambda_{S_1}$ | $+8\lambda_{12}\lambda_{S_2}$ | $+4\lambda_{H1}\lambda_{H2}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+6\lambda_{12}\lambda_{S_1}$ | $+6\lambda_{12}\lambda_{S_2}$ | $+4\lambda_{H1}\lambda_{H2}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+6\lambda_{12}\lambda_{S_1}$ | $+6\lambda_{12}\lambda_{S_2}$ | $+4\lambda_{H1}\lambda_{H2}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $+6\lambda_{12}\lambda_{S_1}$ | $+6\lambda_{12}\lambda_{S_2}$ | $+4\lambda_{H1}\lambda_{H2}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $+6\lambda_{12}\lambda_{S_1}$ | $+6\lambda_{12}\lambda_{S_2}$ | $+4\lambda_{H1}\lambda_{H2}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+6\lambda_{12}\lambda_{S_1}$ | $+6\lambda_{12}\lambda_{S_2}$ | $+4\lambda_{H1}\lambda_{H2}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+6\lambda_{12}\lambda_{S_1}$ | $+6\lambda_{12}\lambda_{S_2}$ | $+4\lambda_{H1}\lambda_{H2}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+6\lambda_{12}\lambda_{S_1}$ | $+6\lambda_{12}\lambda_{S_2}$ | $+4\lambda_{H1}\lambda_{H2}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $+6\lambda_{12}\lambda_{S_1}$ | $+6\lambda_{12}\lambda_{S_2}$ | $+4\lambda_{H1}\lambda_{H2}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $+6\lambda_{12}\lambda_{S_1}$ | $+6\lambda_{12}\lambda_{S_2}$ | $+4\lambda_{H1}\lambda_{H2}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+6\lambda_{12}\lambda_{S_1}$ | $+6\lambda_{12}\lambda_{S_2}$ | $+4\lambda_{H1}\lambda_{H2}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $+8\lambda_{12}\lambda_{S_1}$ | $+4\lambda_{12}\lambda_{S_2}$ | $+4\lambda_{H1}\lambda_{H2}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $+8\lambda_{12}\lambda_{S_1}$ | $+4\lambda_{12}\lambda_{S_2}$ | $+4\lambda_{H1}\lambda_{H2}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $+8\lambda_{12}\lambda_{S_1}$ | $+4\lambda_{12}\lambda_{S_2}$ | $+4\lambda_{H1}\lambda_{H2}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $+8\lambda_{12}\lambda_{S_1}$ | $+4\lambda_{12}\lambda_{S_2}$ | $+4\lambda_{H1}\lambda_{H2}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $+8\lambda_{12}\lambda_{S_1}$ | $+4\lambda_{12}\lambda_{S_2}$ | $+4\lambda_{H1}\lambda_{H2}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                             | —                             | $+4\lambda_{H1}\lambda_{H2}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                             | —                             | $+4\lambda_{H1}\lambda_{H2}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                             | —                             | $+4\lambda_{H1}\lambda_{H2}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                             | —                             | $+4\lambda_{H1}\lambda_{H2}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                             | —                             | $+4\lambda_{H1}\lambda_{H2}$ |

Terms 4 of 12

| Column | Representative term structure                                                        |
|--------|--------------------------------------------------------------------------------------|
| T1     | $-4\text{Tr}\left(y_1\left(y_2\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$ |
| T2     | $+\frac{3g_Y^4}{4}$                                                                  |
| T3     | $+48g_Y^4$                                                                           |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                   | T2                   | T3         |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------------------------|----------------------|------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                    | —                    | —          |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-4\text{Tr}\left(y_1\left(y_2\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$ | $+\frac{3g_Y^4}{4}$  | —          |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                    | —                    | —          |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                    | $+\frac{27g_Y^4}{4}$ | —          |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                    | —                    | $+48g_Y^4$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                    | —                    | —          |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-4\text{Tr}\left(y_1\left(y_2\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$ | $+\frac{3g_Y^4}{4}$  | —          |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                    | —                    | —          |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                    | $+\frac{27g_Y^4}{4}$ | —          |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                    | —                    | $+48g_Y^4$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                    | —                    | —          |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-3\text{Tr}\left(y_1\left(y_2\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$ | $+\frac{3g_Y^4}{4}$  | —          |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                    | —                    | —          |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                    | $+\frac{27g_Y^4}{4}$ | —          |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                    | —                    | $+48g_Y^4$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                    | —                    | —          |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $-4\text{Tr}\left(y_1\left(y_2\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$ | $+\frac{3g_Y^4}{4}$  | —          |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                    | —                    | —          |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                    | $+\frac{27g_Y^4}{4}$ | —          |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                    | —                    | $+48g_Y^4$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                    | —                    | —          |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-2\text{Tr}\left(y_1\left(y_2\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$ | $+\frac{3g_Y^4}{4}$  | —          |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                    | —                    | —          |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                    | $+\frac{27g_Y^4}{4}$ | —          |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                    | —                    | $+48g_Y^4$ |

Terms 5 of 12

| Column | Representative term structure                                                                                             |
|--------|---------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_{12}(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right))$ |
| T2     | $+\frac{9g_2^4}{4}$                                                                                                       |
| T3     | $+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$                                                                                 |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                        | T2                  | T3                                        |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                         | —                   | —                                         |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                         | —                   | —                                         |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                         | —                   | —                                         |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                         | —                   | —                                         |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                         | —                   | —                                         |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\lambda_{12}(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right))$ | $+\frac{9g_2^4}{4}$ | $+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\lambda_{12}(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right))$ | $+\frac{9g_2^4}{4}$ | $+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\lambda_{12}(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right))$ | $+\frac{9g_2^4}{4}$ | $+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\lambda_{12}(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right))$ | $+\frac{9g_2^4}{4}$ | $+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\lambda_{12}(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right))$ | $+\frac{9g_2^4}{4}$ | $+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $\lambda_{12}(3\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+3\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right))$ | $+\frac{9g_2^4}{4}$ | $+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $\lambda_{12}(3\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+3\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right))$ | $+\frac{9g_2^4}{4}$ | $+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $\lambda_{12}(3\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+3\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right))$ | $+\frac{9g_2^4}{4}$ | $+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $\lambda_{12}(3\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+3\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right))$ | $+\frac{9g_2^4}{4}$ | $+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $\lambda_{12}(3\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+3\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right))$ | $+\frac{9g_2^4}{4}$ | $+\frac{3\lambda_{12}^{\text{Adj}^2}}{4}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                         | —                   | —                                         |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                         | —                   | —                                         |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                         | —                   | —                                         |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                         | —                   | —                                         |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                         | —                   | —                                         |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                         | —                   | —                                         |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                         | —                   | —                                         |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                         | —                   | —                                         |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                         | —                   | —                                         |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                         | —                   | —                                         |

Terms 6 of 12

| Column | Representative term structure                                                                                                                                                                                                                             |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $+\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \left( \frac{3}{2} \text{Tr} \left( y_1 \ (y_2)^\dagger \right) + 9 \left( \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \right)$ |
| T2     | $+\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left( \frac{3}{2} \text{Tr} \left( y_1 \ (y_2)^\dagger \right) + 9 \left( \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \right)$ |
| T3     | $+\frac{3}{2} \left( \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \text{Tr} \left( y_2 \ (y_1)^\dagger \right)$                                                            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | T2                                                                                                                                                                                                                                                                                                                                                                                                                                                      | T3                                                                                                                                                                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | —                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —                                                                                                                                                                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | —                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —                                                                                                                                                                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | —                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —                                                                                                                                                                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | —                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —                                                                                                                                                                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | —                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —                                                                                                                                                                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | —                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —                                                                                                                                                                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \left( \frac{3}{2} \text{Tr} \left( y_1 \ (y_2)^\dagger \right) + 9 \left( \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \right) + \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left( \frac{3}{2} \text{Tr} \left( y_1 \ (y_2)^\dagger \right) + 9 \left( \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \right) + \frac{3}{2} \left( \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \text{Tr} \left( y_2 \ (y_1)^\dagger \right)$ | $+\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left( \frac{3}{2} \text{Tr} \left( y_1 \ (y_2)^\dagger \right) + 9 \left( \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \right) + \frac{3}{2} \left( \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \text{Tr} \left( y_2 \ (y_1)^\dagger \right)$ | $+\frac{3}{2} \left( \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \text{Tr} \left( y_2 \ (y_1)^\dagger \right)$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | —                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —                                                                                                                                                                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | —                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —                                                                                                                                                                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | —                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —                                                                                                                                                                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | —                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —                                                                                                                                                                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+\lambda_{S_1^\dagger S_2 S_2^\dagger 2} \left( \frac{9}{4} \text{Tr} \left( y_1 \ (y_2)^\dagger \right) + 9 \left( \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \right) + \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left( \frac{9}{4} \text{Tr} \left( y_1 \ (y_2)^\dagger \right) + 9 \left( \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \right) + \frac{9}{4} \left( \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \text{Tr} \left( y_2 \ (y_1)^\dagger \right)$ | $+\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left( \frac{9}{4} \text{Tr} \left( y_1 \ (y_2)^\dagger \right) + 9 \left( \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \right) + \frac{9}{4} \left( \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \text{Tr} \left( y_2 \ (y_1)^\dagger \right)$ | $+\frac{9}{4} \left( \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \text{Tr} \left( y_2 \ (y_1)^\dagger \right)$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | —                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —                                                                                                                                                                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | —                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —                                                                                                                                                                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | —                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —                                                                                                                                                                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | —                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —                                                                                                                                                                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | —                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —                                                                                                                                                                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | —                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —                                                                                                                                                                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | —                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —                                                                                                                                                                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | —                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —                                                                                                                                                                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | —                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —                                                                                                                                                                                              |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | —                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —                                                                                                                                                                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | —                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —                                                                                                                                                                                              |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | —                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —                                                                                                                                                                                              |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | —                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —                                                                                                                                                                                              |

## Terms 7 of 12

| Column | Representative term structure                                                                                                                |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $+2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ |
| T2     | $+24\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \left( \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \right)^*$                                    |
| T3     | $-\frac{9}{2} \text{Tr} \left( y_1 (y_1)^\dagger y_2 (y_2)^\dagger \right)$                                                                  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                           | T2                                                                                                        | T3                                                                          |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                            | —                                                                                                         | —                                                                           |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                            | —                                                                                                         | —                                                                           |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                            | —                                                                                                         | —                                                                           |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                            | —                                                                                                         | —                                                                           |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                            | —                                                                                                         | —                                                                           |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                            | —                                                                                                         | —                                                                           |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ | $+24\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \left( \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \right)^*$ | —                                                                           |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                            | —                                                                                                         | —                                                                           |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                            | —                                                                                                         | —                                                                           |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                            | —                                                                                                         | —                                                                           |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                            | —                                                                                                         | $-\frac{9}{2} \text{Tr} \left( y_1 (y_1)^\dagger y_2 (y_2)^\dagger \right)$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ | $+24\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \left( \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}} \right)^*$ | $-\frac{9}{2} \text{Tr} \left( y_1 (y_1)^\dagger y_2 (y_2)^\dagger \right)$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                            | —                                                                                                         | $-\frac{9}{2} \text{Tr} \left( y_1 (y_1)^\dagger y_2 (y_2)^\dagger \right)$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                            | —                                                                                                         | $-\frac{9}{2} \text{Tr} \left( y_1 (y_1)^\dagger y_2 (y_2)^\dagger \right)$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                            | —                                                                                                         | $-\frac{9}{2} \text{Tr} \left( y_1 (y_1)^\dagger y_2 (y_2)^\dagger \right)$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                            | —                                                                                                         | —                                                                           |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                            | —                                                                                                         | —                                                                           |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                            | —                                                                                                         | —                                                                           |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                            | —                                                                                                         | —                                                                           |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                            | —                                                                                                         | —                                                                           |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                            | —                                                                                                         | —                                                                           |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                            | —                                                                                                         | —                                                                           |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                            | —                                                                                                         | —                                                                           |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                            | —                                                                                                         | —                                                                           |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                            | —                                                                                                         | —                                                                           |

## Terms 8 of 12

| Column | Representative term structure                                                                                                                                                                                       |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_{12}(2 \operatorname{Tr}(y_1 (y_1)^\dagger) + 4 \operatorname{Tr}(y_2 (y_2)^\dagger) + 4\lambda_{S_1}^{\text{Adj}})$                                                                                       |
| T2     | $\lambda_{12} \left( 2 \operatorname{Tr}(y_1 (y_1)^\dagger) + 2 \operatorname{Tr}(y_2 (y_2)^\dagger) + \frac{16\lambda_{12}^{\text{Cross}}}{3} + 4\lambda_{S_1}^{\text{Adj}} + 4\lambda_{S_2}^{\text{Adj}} \right)$ |
| T3     | $+\lambda_{12}^{\text{Cross}} \left( \frac{4}{3} \operatorname{Tr}(y_1 (y_1)^\dagger) + \frac{4}{3} \operatorname{Tr}(y_2 (y_2)^\dagger) \right)$                                                                   |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                            | T2                                                                                                                                                                                                                                                                                                                                                                   | T3 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                             | —                                                                                                                                                                                                                                                                                                                                                                    | —  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                             | —                                                                                                                                                                                                                                                                                                                                                                    | —  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                             | —                                                                                                                                                                                                                                                                                                                                                                    | —  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                             | —                                                                                                                                                                                                                                                                                                                                                                    | —  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                             | —                                                                                                                                                                                                                                                                                                                                                                    | —  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                             | —                                                                                                                                                                                                                                                                                                                                                                    | —  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                                                                             | —                                                                                                                                                                                                                                                                                                                                                                    | —  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                             | —                                                                                                                                                                                                                                                                                                                                                                    | —  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                             | —                                                                                                                                                                                                                                                                                                                                                                    | —  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                             | —                                                                                                                                                                                                                                                                                                                                                                    | —  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                             | —                                                                                                                                                                                                                                                                                                                                                                    | —  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                                                                             | —                                                                                                                                                                                                                                                                                                                                                                    | —  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                             | —                                                                                                                                                                                                                                                                                                                                                                    | —  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                             | —                                                                                                                                                                                                                                                                                                                                                                    | —  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                             | —                                                                                                                                                                                                                                                                                                                                                                    | —  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $\lambda_{12}(2 \operatorname{Tr}(y_1 (y_1)^\dagger) + 4 \operatorname{Tr}(y_2 (y_2)^\dagger) + 4\lambda_{S_1}^{\text{Adj}})$ | —                                                                                                                                                                                                                                                                                                                                                                    | —  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $\lambda_{12}(2 \operatorname{Tr}(y_1 (y_1)^\dagger) + 4 \operatorname{Tr}(y_2 (y_2)^\dagger) + 4\lambda_{S_1}^{\text{Adj}})$ | —                                                                                                                                                                                                                                                                                                                                                                    | —  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $\lambda_{12}(2 \operatorname{Tr}(y_1 (y_1)^\dagger) + 4 \operatorname{Tr}(y_2 (y_2)^\dagger) + 4\lambda_{S_1}^{\text{Adj}})$ | —                                                                                                                                                                                                                                                                                                                                                                    | —  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $\lambda_{12}(2 \operatorname{Tr}(y_1 (y_1)^\dagger) + 4 \operatorname{Tr}(y_2 (y_2)^\dagger) + 4\lambda_{S_1}^{\text{Adj}})$ | —                                                                                                                                                                                                                                                                                                                                                                    | —  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $\lambda_{12}(2 \operatorname{Tr}(y_1 (y_1)^\dagger) + 4 \operatorname{Tr}(y_2 (y_2)^\dagger) + 4\lambda_{S_1}^{\text{Adj}})$ | —                                                                                                                                                                                                                                                                                                                                                                    | —  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                             | $\lambda_{12} \left( 2 \operatorname{Tr}(y_1 (y_1)^\dagger) + 2 \operatorname{Tr}(y_2 (y_2)^\dagger) + \frac{16\lambda_{12}^{\text{Cross}}}{3} + 4\lambda_{S_1}^{\text{Adj}} + 4\lambda_{S_2}^{\text{Adj}} \right) + \lambda_{12}^{\text{Cross}} \left( \frac{4}{3} \operatorname{Tr}(y_1 (y_1)^\dagger) + \frac{4}{3} \operatorname{Tr}(y_2 (y_2)^\dagger) \right)$ |    |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                             | $\lambda_{12} \left( 2 \operatorname{Tr}(y_1 (y_1)^\dagger) + 2 \operatorname{Tr}(y_2 (y_2)^\dagger) + \frac{16\lambda_{12}^{\text{Cross}}}{3} + 4\lambda_{S_1}^{\text{Adj}} + 4\lambda_{S_2}^{\text{Adj}} \right) + \lambda_{12}^{\text{Cross}} \left( \frac{4}{3} \operatorname{Tr}(y_1 (y_1)^\dagger) + \frac{4}{3} \operatorname{Tr}(y_2 (y_2)^\dagger) \right)$ |    |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                             | $\lambda_{12} \left( 2 \operatorname{Tr}(y_1 (y_1)^\dagger) + 2 \operatorname{Tr}(y_2 (y_2)^\dagger) + \frac{16\lambda_{12}^{\text{Cross}}}{3} + 4\lambda_{S_1}^{\text{Adj}} + 4\lambda_{S_2}^{\text{Adj}} \right) + \lambda_{12}^{\text{Cross}} \left( \frac{4}{3} \operatorname{Tr}(y_1 (y_1)^\dagger) + \frac{4}{3} \operatorname{Tr}(y_2 (y_2)^\dagger) \right)$ |    |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                             | $\lambda_{12} \left( 2 \operatorname{Tr}(y_1 (y_1)^\dagger) + 2 \operatorname{Tr}(y_2 (y_2)^\dagger) + \frac{16\lambda_{12}^{\text{Cross}}}{3} + 4\lambda_{S_1}^{\text{Adj}} + 4\lambda_{S_2}^{\text{Adj}} \right) + \lambda_{12}^{\text{Cross}} \left( \frac{4}{3} \operatorname{Tr}(y_1 (y_1)^\dagger) + \frac{4}{3} \operatorname{Tr}(y_2 (y_2)^\dagger) \right)$ |    |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                             | $\lambda_{12} \left( 2 \operatorname{Tr}(y_1 (y_1)^\dagger) + 2 \operatorname{Tr}(y_2 (y_2)^\dagger) + \frac{16\lambda_{12}^{\text{Cross}}}{3} + 4\lambda_{S_1}^{\text{Adj}} + 4\lambda_{S_2}^{\text{Adj}} \right) + \lambda_{12}^{\text{Cross}} \left( \frac{4}{3} \operatorname{Tr}(y_1 (y_1)^\dagger) + \frac{4}{3} \operatorname{Tr}(y_2 (y_2)^\dagger) \right)$ |    |

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| Column | Representative term structure                                 |
|--------|---------------------------------------------------------------|
| T1     | $+ \frac{4}{3} \lambda_{T3} (\lambda_{T3})^*$                 |
| T2     | $+ 16 g_2^4$                                                  |
| T3     | $+ g_2^2 (-24 \lambda_{12} - 16 \lambda_{12}^{\text{Cross}})$ |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$       | $d_{S2}$ | $Y_{S2}$      | $d_F$ | $Y_F$          | T1                                            | T2           | T3                                                            |
|-------|----------|----------|----------------|----------|---------------|-------|----------------|-----------------------------------------------|--------------|---------------------------------------------------------------|
| T3-A  | -2       | 1        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | —                                             | —            | —                                                             |
| T3-A  | -1       | 1        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | —                                             | —            | —                                                             |
| T3-A  | 0        | 1        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | —                                             | —            | —                                                             |
| T3-A  | 1        | 1        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | —                                             | —            | —                                                             |
| T3-A  | 2        | 1        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | —                                             | —            | —                                                             |
| T3-B  | -2       | 2        | -1             | 2        | 0             | 1     | $-\frac{1}{2}$ | —                                             | —            | —                                                             |
| T3-B  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 1     | 0              | —                                             | —            | —                                                             |
| T3-B  | 0        | 2        | 0              | 2        | 1             | 1     | $\frac{1}{2}$  | —                                             | —            | —                                                             |
| T3-B  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 1     | 1              | —                                             | —            | —                                                             |
| T3-B  | 2        | 2        | 1              | 2        | 2             | 1     | $\frac{3}{2}$  | —                                             | —            | —                                                             |
| T3-C  | -2       | 2        | -1             | 2        | 0             | 3     | $-\frac{1}{2}$ | —                                             | —            | —                                                             |
| T3-C  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 3     | 0              | —                                             | —            | —                                                             |
| T3-C  | 0        | 2        | 0              | 2        | 1             | 3     | $\frac{1}{2}$  | —                                             | —            | —                                                             |
| T3-C  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 3     | 1              | —                                             | —            | —                                                             |
| T3-C  | 2        | 2        | 1              | 2        | 2             | 3     | $\frac{3}{2}$  | —                                             | —            | —                                                             |
| T3-D  | -2       | 3        | -1             | 1        | 0             | 2     | $-\frac{1}{2}$ | —                                             | —            | —                                                             |
| T3-D  | -1       | 3        | $-\frac{1}{2}$ | 1        | $\frac{1}{2}$ | 2     | 0              | —                                             | —            | —                                                             |
| T3-D  | 0        | 3        | 0              | 1        | 1             | 2     | $\frac{1}{2}$  | —                                             | —            | —                                                             |
| T3-D  | 1        | 3        | $\frac{1}{2}$  | 1        | $\frac{3}{2}$ | 2     | 1              | —                                             | —            | —                                                             |
| T3-D  | 2        | 3        | 1              | 1        | 2             | 2     | $\frac{3}{2}$  | —                                             | —            | —                                                             |
| T3-E  | -2       | 3        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+ \frac{4}{3} \lambda_{T3} (\lambda_{T3})^*$ | $+ 16 g_2^4$ | $+ g_2^2 (-24 \lambda_{12} - 16 \lambda_{12}^{\text{Cross}})$ |
| T3-E  | -1       | 3        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+ \frac{4}{3} \lambda_{T3} (\lambda_{T3})^*$ | $+ 16 g_2^4$ | $+ g_2^2 (-24 \lambda_{12} - 16 \lambda_{12}^{\text{Cross}})$ |
| T3-E  | 0        | 3        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+ \frac{4}{3} \lambda_{T3} (\lambda_{T3})^*$ | $+ 16 g_2^4$ | $+ g_2^2 (-24 \lambda_{12} - 16 \lambda_{12}^{\text{Cross}})$ |
| T3-E  | 1        | 3        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+ \frac{4}{3} \lambda_{T3} (\lambda_{T3})^*$ | $+ 16 g_2^4$ | $+ g_2^2 (-24 \lambda_{12} - 16 \lambda_{12}^{\text{Cross}})$ |
| T3-E  | 2        | 3        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+ \frac{4}{3} \lambda_{T3} (\lambda_{T3})^*$ | $+ 16 g_2^4$ | $+ g_2^2 (-24 \lambda_{12} - 16 \lambda_{12}^{\text{Cross}})$ |

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| Column | Representative term structure                                                      |
|--------|------------------------------------------------------------------------------------|
| T1     | $+g_Y^2(-6\lambda_{12}-4\lambda_{12}^{\text{Cross}})$                              |
| T2     | $+\lambda_{S_1}\left(8\lambda_{12}+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$ |
| T3     | $+\lambda_{S_2}\left(8\lambda_{12}+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                      | T2                                                                                 | T3                                                                                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                       | —                                                                                  | —                                                                                  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                       | —                                                                                  | —                                                                                  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                       | —                                                                                  | —                                                                                  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                       | —                                                                                  | —                                                                                  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                       | —                                                                                  | —                                                                                  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                       | —                                                                                  | —                                                                                  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                       | —                                                                                  | —                                                                                  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                       | —                                                                                  | —                                                                                  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                       | —                                                                                  | —                                                                                  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                       | —                                                                                  | —                                                                                  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                       | —                                                                                  | —                                                                                  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                       | —                                                                                  | —                                                                                  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                       | —                                                                                  | —                                                                                  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                       | —                                                                                  | —                                                                                  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                       | —                                                                                  | —                                                                                  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                       | —                                                                                  | —                                                                                  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                       | —                                                                                  | —                                                                                  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                       | —                                                                                  | —                                                                                  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                       | —                                                                                  | —                                                                                  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                       | —                                                                                  | —                                                                                  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+g_Y^2(-6\lambda_{12}-4\lambda_{12}^{\text{Cross}})$   | $+\lambda_{S_1}\left(8\lambda_{12}+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$ | $+\lambda_{S_2}\left(8\lambda_{12}+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+g_Y^2(-3\lambda_{12}-2\lambda_{12}^{\text{Cross}})$   | $+\lambda_{S_1}\left(8\lambda_{12}+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$ | $+\lambda_{S_2}\left(8\lambda_{12}+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+g_Y^2(-6\lambda_{12}-4\lambda_{12}^{\text{Cross}})$   | $+\lambda_{S_1}\left(8\lambda_{12}+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$ | $+\lambda_{S_2}\left(8\lambda_{12}+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+g_Y^2(-15\lambda_{12}-10\lambda_{12}^{\text{Cross}})$ | $+\lambda_{S_1}\left(8\lambda_{12}+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$ | $+\lambda_{S_2}\left(8\lambda_{12}+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+g_Y^2(-30\lambda_{12}-20\lambda_{12}^{\text{Cross}})$ | $+\lambda_{S_1}\left(8\lambda_{12}+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$ | $+\lambda_{S_2}\left(8\lambda_{12}+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$ |

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| Column | Representative term structure                                       |
|--------|---------------------------------------------------------------------|
| T1     | $+16\lambda_{12}^{\text{Adj}^2}$                                    |
| T2     | $+\frac{32\lambda_{12}^{\text{Cross}^2}}{3}$                        |
| T3     | $+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}}{3}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                               | T2                                           | T3                                                                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------|----------------------------------------------|---------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                | —                                            | —                                                                   |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                | —                                            | —                                                                   |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                | —                                            | —                                                                   |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                | —                                            | —                                                                   |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                | —                                            | —                                                                   |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                | —                                            | —                                                                   |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                | —                                            | —                                                                   |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                | —                                            | —                                                                   |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                | —                                            | —                                                                   |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                | —                                            | —                                                                   |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                | —                                            | —                                                                   |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                | —                                            | —                                                                   |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                | —                                            | —                                                                   |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                | —                                            | —                                                                   |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                | —                                            | —                                                                   |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                | —                                            | —                                                                   |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                | —                                            | —                                                                   |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                | —                                            | —                                                                   |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                | —                                            | —                                                                   |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                | —                                            | —                                                                   |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+16\lambda_{12}^{\text{Adj}^2}$ | $+\frac{32\lambda_{12}^{\text{Cross}^2}}{3}$ | $+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}}{3}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+16\lambda_{12}^{\text{Adj}^2}$ | $+\frac{32\lambda_{12}^{\text{Cross}^2}}{3}$ | $+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}}{3}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+16\lambda_{12}^{\text{Adj}^2}$ | $+\frac{32\lambda_{12}^{\text{Cross}^2}}{3}$ | $+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}}{3}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+16\lambda_{12}^{\text{Adj}^2}$ | $+\frac{32\lambda_{12}^{\text{Cross}^2}}{3}$ | $+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}}{3}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+16\lambda_{12}^{\text{Adj}^2}$ | $+\frac{32\lambda_{12}^{\text{Cross}^2}}{3}$ | $+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}}{3}$ |

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| Column | Representative term structure                                       |
|--------|---------------------------------------------------------------------|
| T1     | $+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}}{3}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                   |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                   |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                   |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                   |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                   |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                   |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                   |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                   |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                   |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                   |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                   |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                   |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                   |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                   |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                   |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                   |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                   |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                   |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                   |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                   |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}}{3}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}}{3}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}}{3}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}}{3}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+\frac{8\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}}{3}$ |

$\beta_{\lambda_{T3}}^{(1)}$

Search name: lambdaT3

$16\pi^2\,\mu\frac{d\lambda_{T3}}{d\mu}=\beta_{\lambda_{T3}}^{(1)}$

Terms 1 of 6

| Column | Representative term structure                                                                                                                                                                                                                                                                                              |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_{T3}(2\,\text{Tr}\left(y_1\,\left(y_1\right)^\dagger\right)+\text{Tr}\left(y_2\,\left(y_2\right)^\dagger\right)+6\,\text{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right)+2\,\text{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right)+6\,\text{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right)+4\lambda_{H2}^{\text{Adj}})$ |
| T2     | $-\frac{21g_2^2\lambda_{T3}}{2}$                                                                                                                                                                                                                                                                                           |
| T3     | $-\frac{9g_Y^2\lambda_{T3}}{2}$                                                                                                                                                                                                                                                                                            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                                                                                                                                                                                         | T2                               | T3                               |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|----------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\lambda_{T3}(2\,\text{Tr}\left(y_1\,\left(y_1\right)^\dagger\right)+\text{Tr}\left(y_2\,\left(y_2\right)^\dagger\right)+6\,\text{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right)+2\,\text{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right)+6\,\text{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right)+4\lambda_{H2}^{\text{Adj}})$ | $-\frac{21g_2^2\lambda_{T3}}{2}$ | $-\frac{9g_Y^2\lambda_{T3}}{2}$  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\lambda_{T3}(2\,\text{Tr}\left(y_1\,\left(y_1\right)^\dagger\right)+\text{Tr}\left(y_2\,\left(y_2\right)^\dagger\right)+6\,\text{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right)+2\,\text{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right)+6\,\text{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right)+4\lambda_{H2}^{\text{Adj}})$ | $-\frac{21g_2^2\lambda_{T3}}{2}$ | —                                |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\lambda_{T3}(2\,\text{Tr}\left(y_1\,\left(y_1\right)^\dagger\right)+\text{Tr}\left(y_2\,\left(y_2\right)^\dagger\right)+6\,\text{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right)+2\,\text{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right)+6\,\text{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right)+4\lambda_{H2}^{\text{Adj}})$ | $-\frac{21g_2^2\lambda_{T3}}{2}$ | $-\frac{9g_Y^2\lambda_{T3}}{2}$  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\lambda_{T3}(2\,\text{Tr}\left(y_1\,\left(y_1\right)^\dagger\right)+\text{Tr}\left(y_2\,\left(y_2\right)^\dagger\right)+6\,\text{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right)+2\,\text{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right)+6\,\text{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right)+4\lambda_{H2}^{\text{Adj}})$ | $-\frac{21g_2^2\lambda_{T3}}{2}$ | —                                |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\lambda_{T3}(2\,\text{Tr}\left(y_1\,\left(y_1\right)^\dagger\right)+\text{Tr}\left(y_2\,\left(y_2\right)^\dagger\right)+6\,\text{Tr}\left(Y_d\,\left(Y_d\right)^\dagger\right)+2\,\text{Tr}\left(Y_e\,\left(Y_e\right)^\dagger\right)+6\,\text{Tr}\left(Y_u\,\left(Y_u\right)^\dagger\right)+4\lambda_{H2}^{\text{Adj}})$ | $-\frac{21g_2^2\lambda_{T3}}{2}$ | $-\frac{33g_Y^2\lambda_{T3}}{2}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                          | —                                | $-\frac{9g_Y^2\lambda_{T3}}{2}$  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                                                                                                                                                                                                                                                                          | —                                | —                                |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                          | —                                | $-\frac{9g_Y^2\lambda_{T3}}{2}$  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                                                                                                                                                                                          | —                                | —                                |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                          | —                                | $-\frac{33g_Y^2\lambda_{T3}}{2}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                          | —                                | $-\frac{9g_Y^2\lambda_{T3}}{2}$  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                                                                                                                                                                                                                                                                          | —                                | —                                |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                          | —                                | $-\frac{9g_Y^2\lambda_{T3}}{2}$  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                                                                                                                                                                                          | —                                | —                                |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                          | —                                | $-\frac{33g_Y^2\lambda_{T3}}{2}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                          | $-\frac{21g_2^2\lambda_{T3}}{2}$ | $-\frac{9g_Y^2\lambda_{T3}}{2}$  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                                                          | $-\frac{21g_2^2\lambda_{T3}}{2}$ | —                                |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                          | $-\frac{21g_2^2\lambda_{T3}}{2}$ | $-\frac{9g_Y^2\lambda_{T3}}{2}$  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                                                          | $-\frac{21g_2^2\lambda_{T3}}{2}$ | —                                |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1 | T2                               | T3                               |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----|----------------------------------|----------------------------------|
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —  | $-\frac{21g_2^2\lambda_{T3}}{2}$ | $-\frac{33g_Y^2\lambda_{T3}}{2}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —  | $-\frac{33g_2^2\lambda_{T3}}{2}$ | $-\frac{9g_Y^2\lambda_{T3}}{2}$  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —  | $-\frac{33g_2^2\lambda_{T3}}{2}$ | —                                |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —  | $-\frac{33g_2^2\lambda_{T3}}{2}$ | $-\frac{9g_Y^2\lambda_{T3}}{2}$  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —  | $-\frac{33g_2^2\lambda_{T3}}{2}$ | —                                |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —  | $-\frac{33g_2^2\lambda_{T3}}{2}$ | $-\frac{33g_Y^2\lambda_{T3}}{2}$ |

Terms 2 of 6

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $+2\lambda_{12}\lambda_{T3}$  |
| T2     | $+2\lambda_H\lambda_{T3}$     |
| T3     | $+4\lambda_{H1}\lambda_{T3}$  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                           | T2                        | T3                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------|---------------------------|------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+2\lambda_{12}\lambda_{T3}$ | $+2\lambda_H\lambda_{T3}$ | $+4\lambda_{H1}\lambda_{T3}$ |

Terms 3 of 6

| Column | Representative term structure                                                                                                                                                                                                                                                                                                                                                               |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $+4\lambda_{H2}\lambda_{T3}$                                                                                                                                                                                                                                                                                                                                                                |
| T2     | $-3g_Y^2\lambda_{T3}$                                                                                                                                                                                                                                                                                                                                                                       |
| T3     | $\lambda_{T3}\left(\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{\text{Adj}}}{6}+\frac{2\lambda_{H1}^{\text{Adj}}}{3}+2\lambda_{H2}^{\text{Adj}}\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                           | T2                    | T3                                                                                                                                                                                                                                                                                                                                                                                          |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+4\lambda_{H2}\lambda_{T3}$ | —                     | —                                                                                                                                                                                                                                                                                                                                                                                           |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+4\lambda_{H2}\lambda_{T3}$ | $-3g_Y^2\lambda_{T3}$ | —                                                                                                                                                                                                                                                                                                                                                                                           |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+4\lambda_{H2}\lambda_{T3}$ | —                     | —                                                                                                                                                                                                                                                                                                                                                                                           |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+4\lambda_{H2}\lambda_{T3}$ | $-9g_Y^2\lambda_{T3}$ | —                                                                                                                                                                                                                                                                                                                                                                                           |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+4\lambda_{H2}\lambda_{T3}$ | —                     | —                                                                                                                                                                                                                                                                                                                                                                                           |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+4\lambda_{H2}\lambda_{T3}$ | —                     | $\lambda_{T3}\left(\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{\text{Adj}}}{6}+\frac{2\lambda_{H1}^{\text{Adj}}}{3}+2\lambda_{H2}^{\text{Adj}}\right)$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+4\lambda_{H2}\lambda_{T3}$ | $-3g_Y^2\lambda_{T3}$ | $\lambda_{T3}\left(\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{\text{Adj}}}{6}+\frac{2\lambda_{H1}^{\text{Adj}}}{3}+2\lambda_{H2}^{\text{Adj}}\right)$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $+4\lambda_{H2}\lambda_{T3}$ | —                     | $\lambda_{T3}\left(\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{\text{Adj}}}{6}+\frac{2\lambda_{H1}^{\text{Adj}}}{3}+2\lambda_{H2}^{\text{Adj}}\right)$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $+4\lambda_{H2}\lambda_{T3}$ | $-9g_Y^2\lambda_{T3}$ | $\lambda_{T3}\left(\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{\text{Adj}}}{6}+\frac{2\lambda_{H1}^{\text{Adj}}}{3}+2\lambda_{H2}^{\text{Adj}}\right)$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+4\lambda_{H2}\lambda_{T3}$ | —                     | $\lambda_{T3}\left(\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{\text{Adj}}}{6}+\frac{2\lambda_{H1}^{\text{Adj}}}{3}+2\lambda_{H2}^{\text{Adj}}\right)$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+4\lambda_{H2}\lambda_{T3}$ | —                     | —                                                                                                                                                                                                                                                                                                                                                                                           |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+4\lambda_{H2}\lambda_{T3}$ | $-3g_Y^2\lambda_{T3}$ | —                                                                                                                                                                                                                                                                                                                                                                                           |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $+4\lambda_{H2}\lambda_{T3}$ | —                     | —                                                                                                                                                                                                                                                                                                                                                                                           |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $+4\lambda_{H2}\lambda_{T3}$ | $-9g_Y^2\lambda_{T3}$ | —                                                                                                                                                                                                                                                                                                                                                                                           |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+4\lambda_{H2}\lambda_{T3}$ | —                     | —                                                                                                                                                                                                                                                                                                                                                                                           |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $+4\lambda_{H2}\lambda_{T3}$ | —                     | —                                                                                                                                                                                                                                                                                                                                                                                           |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $+4\lambda_{H2}\lambda_{T3}$ | $-3g_Y^2\lambda_{T3}$ | —                                                                                                                                                                                                                                                                                                                                                                                           |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $+4\lambda_{H2}\lambda_{T3}$ | —                     | —                                                                                                                                                                                                                                                                                                                                                                                           |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $+4\lambda_{H2}\lambda_{T3}$ | $-9g_Y^2\lambda_{T3}$ | —                                                                                                                                                                                                                                                                                                                                                                                           |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $+4\lambda_{H2}\lambda_{T3}$ | —                     | —                                                                                                                                                                                                                                                                                                                                                                                           |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+4\lambda_{H2}\lambda_{T3}$ | —                     | —                                                                                                                                                                                                                                                                                                                                                                                           |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+4\lambda_{H2}\lambda_{T3}$ | $-3g_Y^2\lambda_{T3}$ | —                                                                                                                                                                                                                                                                                                                                                                                           |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+4\lambda_{H2}\lambda_{T3}$ | —                     | —                                                                                                                                                                                                                                                                                                                                                                                           |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+4\lambda_{H2}\lambda_{T3}$ | $-9g_Y^2\lambda_{T3}$ | —                                                                                                                                                                                                                                                                                                                                                                                           |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+4\lambda_{H2}\lambda_{T3}$ | —                     | —                                                                                                                                                                                                                                                                                                                                                                                           |

Terms 4 of 6

| Column | Representative term structure                                                                                             |
|--------|---------------------------------------------------------------------------------------------------------------------------|
| T1     | $-9g_2^2\lambda_{T3}$                                                                                                     |
| T2     | $+2\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$ |
| T3     | $+2\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)$                 |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                    | T2                                                                                                                        | T3                                                                                                        |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                     | —                                                                                                                         | —                                                                                                         |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                     | —                                                                                                                         | —                                                                                                         |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                     | —                                                                                                                         | —                                                                                                         |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                     | —                                                                                                                         | —                                                                                                         |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                     | —                                                                                                                         | —                                                                                                         |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-9g_2^2\lambda_{T3}$ | —                                                                                                                         | —                                                                                                         |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-9g_2^2\lambda_{T3}$ | $+2\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$ | $+2\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $-9g_2^2\lambda_{T3}$ | —                                                                                                                         | —                                                                                                         |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-9g_2^2\lambda_{T3}$ | —                                                                                                                         | —                                                                                                         |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-9g_2^2\lambda_{T3}$ | —                                                                                                                         | —                                                                                                         |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-9g_2^2\lambda_{T3}$ | —                                                                                                                         | —                                                                                                         |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-9g_2^2\lambda_{T3}$ | $+3\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$ | $+3\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-9g_2^2\lambda_{T3}$ | —                                                                                                                         | —                                                                                                         |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-9g_2^2\lambda_{T3}$ | —                                                                                                                         | —                                                                                                         |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-9g_2^2\lambda_{T3}$ | —                                                                                                                         | —                                                                                                         |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                     | —                                                                                                                         | —                                                                                                         |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                     | —                                                                                                                         | —                                                                                                         |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                     | —                                                                                                                         | —                                                                                                         |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                     | —                                                                                                                         | —                                                                                                         |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                     | —                                                                                                                         | —                                                                                                         |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                     | —                                                                                                                         | —                                                                                                         |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                     | —                                                                                                                         | —                                                                                                         |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                     | —                                                                                                                         | —                                                                                                         |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                     | —                                                                                                                         | —                                                                                                         |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                     | —                                                                                                                         | —                                                                                                         |

Terms 5 of 6

| Column | Representative term structure                                                                                                                                                                         |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left( \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$                                                                |
| T2     | $+4\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger} \left( \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$                                                                                  |
| T3     | $+4 \left( \lambda_{H^\dagger H^\dagger S_2 S_2} \right)^* \left( \lambda_{S_1^\dagger S_2 S_2^{\dagger 2}} \right)^* - \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                     | T2                                                                                                                   | T3                                                                                                                                                                                                    |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                      | —                                                                                                                    | —                                                                                                                                                                                                     |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                      | —                                                                                                                    | —                                                                                                                                                                                                     |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                      | —                                                                                                                    | —                                                                                                                                                                                                     |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                      | —                                                                                                                    | —                                                                                                                                                                                                     |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                      | —                                                                                                                    | —                                                                                                                                                                                                     |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                      | —                                                                                                                    | —                                                                                                                                                                                                     |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left( \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$ | $+4\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger} \left( \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$ | $+4 \left( \lambda_{H^\dagger H^\dagger S_2 S_2} \right)^* \left( \lambda_{S_1^\dagger S_2 S_2^{\dagger 2}} \right)^* - \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                      | —                                                                                                                    | —                                                                                                                                                                                                     |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                      | —                                                                                                                    | —                                                                                                                                                                                                     |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                      | —                                                                                                                    | —                                                                                                                                                                                                     |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                      | —                                                                                                                    | —                                                                                                                                                                                                     |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left( \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$ | $+4\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger} \left( \lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \right)^*$ | $+4 \left( \lambda_{H^\dagger H^\dagger S_2 S_2} \right)^* \left( \lambda_{S_1^\dagger S_2 S_2^{\dagger 2}} \right)^* - \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                      | —                                                                                                                    | —                                                                                                                                                                                                     |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                      | —                                                                                                                    | —                                                                                                                                                                                                     |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                      | —                                                                                                                    | —                                                                                                                                                                                                     |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                      | —                                                                                                                    | —                                                                                                                                                                                                     |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                      | —                                                                                                                    | —                                                                                                                                                                                                     |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                      | —                                                                                                                    | —                                                                                                                                                                                                     |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                      | —                                                                                                                    | —                                                                                                                                                                                                     |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                      | —                                                                                                                    | —                                                                                                                                                                                                     |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                      | —                                                                                                                    | —                                                                                                                                                                                                     |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                      | —                                                                                                                    | —                                                                                                                                                                                                     |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                      | —                                                                                                                    | —                                                                                                                                                                                                     |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                      | —                                                                                                                    | —                                                                                                                                                                                                     |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                      | —                                                                                                                    | —                                                                                                                                                                                                     |



$\beta_{\lambda_{12}^{\text{Adj}}}^{(1)}$

Search name: lambda12Adj

$$16\pi^2\mu\frac{d\lambda_{12}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{12}^{\text{Adj}}}^{(1)}$$

Terms 1 of 8

| Column | Representative term structure                                                                              |  |  |  |  |  |  |  |
|--------|------------------------------------------------------------------------------------------------------------|--|--|--|--|--|--|--|
| T1     | $\lambda_{12}^{\text{Adj}}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+2\text{Tr}\big(y_2\ (y_2)^\dagger\big))$ |  |  |  |  |  |  |  |
| T2     | $-\frac{8}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$                                    |  |  |  |  |  |  |  |
| T3     | $+\frac{4}{3}\lambda_{T3}\ (\lambda_{T3})^*$                                                               |  |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                         | T2                                                                      | T3                                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                          | —                                                                       | —                                            |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                          | —                                                                       | —                                            |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                          | —                                                                       | —                                            |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                          | —                                                                       | —                                            |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                          | —                                                                       | —                                            |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\lambda_{12}^{\text{Adj}}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+2\text{Tr}\big(y_2\ (y_2)^\dagger\big))$ | $-\frac{8}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$ | $+\frac{4}{3}\lambda_{T3}\ (\lambda_{T3})^*$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\lambda_{12}^{\text{Adj}}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+2\text{Tr}\big(y_2\ (y_2)^\dagger\big))$ | $-\frac{8}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$ | $+\frac{4}{3}\lambda_{T3}\ (\lambda_{T3})^*$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\lambda_{12}^{\text{Adj}}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+2\text{Tr}\big(y_2\ (y_2)^\dagger\big))$ | $-\frac{8}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$ | $+\frac{4}{3}\lambda_{T3}\ (\lambda_{T3})^*$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\lambda_{12}^{\text{Adj}}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+2\text{Tr}\big(y_2\ (y_2)^\dagger\big))$ | $-\frac{8}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$ | $+\frac{4}{3}\lambda_{T3}\ (\lambda_{T3})^*$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\lambda_{12}^{\text{Adj}}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+2\text{Tr}\big(y_2\ (y_2)^\dagger\big))$ | $-\frac{8}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$ | $+\frac{4}{3}\lambda_{T3}\ (\lambda_{T3})^*$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $\lambda_{12}^{\text{Adj}}(3\text{Tr}\big(y_1\ (y_1)^\dagger\big)+3\text{Tr}\big(y_2\ (y_2)^\dagger\big))$ | $-\frac{2}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$ | $+\frac{4}{3}\lambda_{T3}\ (\lambda_{T3})^*$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $\lambda_{12}^{\text{Adj}}(3\text{Tr}\big(y_1\ (y_1)^\dagger\big)+3\text{Tr}\big(y_2\ (y_2)^\dagger\big))$ | $-\frac{2}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$ | $+\frac{4}{3}\lambda_{T3}\ (\lambda_{T3})^*$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $\lambda_{12}^{\text{Adj}}(3\text{Tr}\big(y_1\ (y_1)^\dagger\big)+3\text{Tr}\big(y_2\ (y_2)^\dagger\big))$ | $-\frac{2}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$ | $+\frac{4}{3}\lambda_{T3}\ (\lambda_{T3})^*$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $\lambda_{12}^{\text{Adj}}(3\text{Tr}\big(y_1\ (y_1)^\dagger\big)+3\text{Tr}\big(y_2\ (y_2)^\dagger\big))$ | $-\frac{2}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$ | $+\frac{4}{3}\lambda_{T3}\ (\lambda_{T3})^*$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $\lambda_{12}^{\text{Adj}}(3\text{Tr}\big(y_1\ (y_1)^\dagger\big)+3\text{Tr}\big(y_2\ (y_2)^\dagger\big))$ | $-\frac{2}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$ | $+\frac{4}{3}\lambda_{T3}\ (\lambda_{T3})^*$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                          | —                                                                       | —                                            |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                          | —                                                                       | —                                            |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                          | —                                                                       | —                                            |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                          | —                                                                       | —                                            |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                          | —                                                                       | —                                            |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                          | $+\frac{2}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$ | $-\frac{2}{9}\lambda_{T3}\ (\lambda_{T3})^*$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                          | $+\frac{2}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$ | $-\frac{2}{9}\lambda_{T3}\ (\lambda_{T3})^*$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                          | $+\frac{2}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\big)$ | $-\frac{2}{9}\lambda_{T3}\ (\lambda_{T3})^*$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1 | T2                                                                         | T3                                           |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|----|----------------------------------------------------------------------------|----------------------------------------------|
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | —  | $+\frac{2}{3}\text{Tr}\left(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\right)$ | $-\frac{2}{9}\lambda_{T3}\ (\lambda_{T3})^*$ |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | —  | $+\frac{2}{3}\text{Tr}\left(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\right)$ | $-\frac{2}{9}\lambda_{T3}\ (\lambda_{T3})^*$ |

Terms 2 of 8

| Column | Representative term structure             |  |
|--------|-------------------------------------------|--|
| T1     | $-9g_2^2\lambda_{12}^{\text{Adj}}$        |  |
| T2     | $-6g_Y^2\lambda_{12}^{\text{Adj}}$        |  |
| T3     | $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                  | T2                                 | T3                                        |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------|------------------------------------|-------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                   | —                                  | —                                         |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                   | —                                  | —                                         |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                   | —                                  | —                                         |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                   | —                                  | —                                         |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                   | —                                  | —                                         |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $-9g_2^2\lambda_{12}^{\text{Adj}}$  | $-6g_Y^2\lambda_{12}^{\text{Adj}}$ | $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-9g_2^2\lambda_{12}^{\text{Adj}}$  | —                                  | $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $-9g_2^2\lambda_{12}^{\text{Adj}}$  | $-6g_Y^2\lambda_{12}^{\text{Adj}}$ | $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $-9g_2^2\lambda_{12}^{\text{Adj}}$  | —                                  | $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $-9g_2^2\lambda_{12}^{\text{Adj}}$  | —                                  | $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $-9g_2^2\lambda_{12}^{\text{Adj}}$  | $-6g_Y^2\lambda_{12}^{\text{Adj}}$ | $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-9g_2^2\lambda_{12}^{\text{Adj}}$  | —                                  | $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $-9g_2^2\lambda_{12}^{\text{Adj}}$  | $-6g_Y^2\lambda_{12}^{\text{Adj}}$ | $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $-9g_2^2\lambda_{12}^{\text{Adj}}$  | —                                  | $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $-9g_2^2\lambda_{12}^{\text{Adj}}$  | —                                  | $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                   | —                                  | —                                         |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                   | —                                  | —                                         |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                   | —                                  | —                                         |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                   | —                                  | —                                         |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                   | —                                  | —                                         |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-24g_2^2\lambda_{12}^{\text{Adj}}$ | $-6g_Y^2\lambda_{12}^{\text{Adj}}$ | $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-24g_2^2\lambda_{12}^{\text{Adj}}$ | —                                  | $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-24g_2^2\lambda_{12}^{\text{Adj}}$ | $-6g_Y^2\lambda_{12}^{\text{Adj}}$ | $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-24g_2^2\lambda_{12}^{\text{Adj}}$ | —                                  | $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-24g_2^2\lambda_{12}^{\text{Adj}}$ | —                                  | $+8\lambda_{12}\lambda_{12}^{\text{Adj}}$ |

Terms 3 of 8

| Column | Representative term structure                         |  |
|--------|-------------------------------------------------------|--|
| T1     | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$            |  |
| T2     | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$            |  |
| T3     | $+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                         | T2                                         | T3                                                    |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------|--------------------------------------------|-------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                          | —                                          | —                                                     |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                          | —                                          | —                                                     |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                          | —                                          | —                                                     |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                          | —                                          | —                                                     |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                          | —                                          | —                                                     |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$ | $+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$ | $+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$ | $+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$ | $+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$ | $+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$ | $+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$ | $+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$ | $+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$ | $+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$ | $+\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                          | —                                          | —                                                     |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                          | —                                          | —                                                     |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                          | —                                          | —                                                     |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                          | —                                          | —                                                     |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                          | —                                          | —                                                     |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$ | —                                                     |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$ | —                                                     |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$ | —                                                     |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$ | —                                                     |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_1}$ | $+2\lambda_{12}^{\text{Adj}}\lambda_{S_2}$ | —                                                     |

Terms 4 of 8

| Column | Representative term structure                                                                                                                                                                                                                                          |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $+ \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \left( \frac{2}{3} \text{Tr} \left( y_1 \ (y_2)^\dagger \right) + \frac{4}{3} (5 \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* ) \right)$             |
| T2     | $+ \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left( \frac{2}{3} \text{Tr} \left( y_1 \ (y_2)^\dagger \right) + \frac{4}{3} \left( \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + 5 \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \right)$ |
| T3     | $+ \frac{2}{3} \left( \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \text{Tr} \left( y_2 \ (y_1)^\dagger \right)$                                                                        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | T2 | T3 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+ \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \left( \frac{2}{3} \text{Tr} \left( y_1 \ (y_2)^\dagger \right) + \frac{4}{3} (5 \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* ) \right) + \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left( \frac{2}{3} \text{Tr} \left( y_1 \ (y_2)^\dagger \right) + \frac{4}{3} \left( \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + 5 \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \right) + \frac{2}{3} \left( \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* \right) \text{Tr} \left( y_2 \ (y_1)^\dagger \right)$ |    |    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |

Terms 5 of 8

| Column | Representative term structure                                                                                  |  |
|--------|----------------------------------------------------------------------------------------------------------------|--|
| T1     | $+\frac{64}{3}\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\left(\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\right)^*$ |  |
| T2     | $+g_Y^2\left(-6g_2^2-3\lambda_{12}^{\text{Adj}}\right)$                                                        |  |
| T3     | $+g_Y^2\left(18g_2^2-15\lambda_{12}^{\text{Adj}}\right)$                                                       |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                             | T2                                                      | T3                                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                              | —                                                       | —                                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                              | —                                                       | —                                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                              | —                                                       | —                                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                              | —                                                       | —                                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                              | —                                                       | —                                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                              | —                                                       | —                                                        |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+\frac{64}{3}\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\left(\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\right)^*$ | $+g_Y^2\left(-6g_2^2-3\lambda_{12}^{\text{Adj}}\right)$ | —                                                        |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                              | —                                                       | —                                                        |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                              | —                                                       | $+g_Y^2\left(18g_2^2-15\lambda_{12}^{\text{Adj}}\right)$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                              | —                                                       | $+g_Y^2\left(48g_2^2-30\lambda_{12}^{\text{Adj}}\right)$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                              | —                                                       | —                                                        |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+\frac{64}{3}\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\left(\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\right)^*$ | $+g_Y^2\left(-6g_2^2-3\lambda_{12}^{\text{Adj}}\right)$ | —                                                        |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                              | —                                                       | —                                                        |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                              | —                                                       | $+g_Y^2\left(18g_2^2-15\lambda_{12}^{\text{Adj}}\right)$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                              | —                                                       | $+g_Y^2\left(48g_2^2-30\lambda_{12}^{\text{Adj}}\right)$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                              | —                                                       | —                                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                              | —                                                       | —                                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                              | —                                                       | —                                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                              | —                                                       | —                                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                              | —                                                       | —                                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                              | —                                                       | —                                                        |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                              | $+g_Y^2\left(-2g_2^2-3\lambda_{12}^{\text{Adj}}\right)$ | —                                                        |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                              | —                                                       | —                                                        |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                              | —                                                       | $+g_Y^2\left(6g_2^2-15\lambda_{12}^{\text{Adj}}\right)$  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                              | —                                                       | $+g_Y^2\left(16g_2^2-30\lambda_{12}^{\text{Adj}}\right)$ |

Terms 6 of 8

| Column | Representative term structure                                                                                                                                                                                                                              |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $+ \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \left( \text{Tr} \left( y_1 \left( y_2 \right)^\dagger \right) + \frac{4}{3} (5 \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* ) \right)$  |
| T2     | $+ \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left( \text{Tr} \left( y_1 \left( y_2 \right)^\dagger \right) + \frac{4}{3} ( \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + 5 \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* ) \right)$ |
| T3     | $+ ( \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* ) \text{Tr} \left( y_2 \left( y_1 \right)^\dagger \right)$                                                                        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | T2 | T3 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+ \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \left( \text{Tr} \left( y_1 \left( y_2 \right)^\dagger \right) + \frac{4}{3} (5 \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* ) \right) + \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left( \text{Tr} \left( y_1 \left( y_2 \right)^\dagger \right) + \frac{4}{3} ( \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + 5 \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* ) \right) + ( \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* ) \text{Tr} \left( y_2 \left( y_1 \right)^\dagger \right)$ |    |    |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |

Terms 7 of 8

| Column | Representative term structure                                                                                                                                                             |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $-\frac{8}{3}\text{Tr}\left(y_1\left(y_2\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$                                                                                            |
| T2     | $\lambda_{12}^{\text{Adj}}\left(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)$ |
| T3     | $+16\lambda_{12}^{\text{Adj}}\lambda_{S_1}^{\text{Adj}}$                                                                                                                                  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                             | T2                                                                                                                                                                                                                                              | T3 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                              | —                                                                                                                                                                                                                                               | —  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                              | —                                                                                                                                                                                                                                               | —  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                              | —                                                                                                                                                                                                                                               | —  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                              | —                                                                                                                                                                                                                                               | —  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                              | —                                                                                                                                                                                                                                               | —  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                              | —                                                                                                                                                                                                                                               | —  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                                              | —                                                                                                                                                                                                                                               | —  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                              | —                                                                                                                                                                                                                                               | —  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                              | —                                                                                                                                                                                                                                               | —  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                              | —                                                                                                                                                                                                                                               | —  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                              | —                                                                                                                                                                                                                                               | —  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-\frac{8}{3}\text{Tr}\left(y_1\left(y_2\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$ | —                                                                                                                                                                                                                                               | —  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                              | —                                                                                                                                                                                                                                               | —  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                              | —                                                                                                                                                                                                                                               | —  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                              | —                                                                                                                                                                                                                                               | —  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                              | —                                                                                                                                                                                                                                               | —  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                              | —                                                                                                                                                                                                                                               | —  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                              | —                                                                                                                                                                                                                                               | —  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                              | —                                                                                                                                                                                                                                               | —  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                              | —                                                                                                                                                                                                                                               | —  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                              | $\lambda_{12}^{\text{Adj}}\left(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)+16\lambda_{12}^{\text{Adj}}\lambda_{S_1}^{\text{Adj}}$ |    |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+\frac{2}{3}\text{Tr}\left(y_1\left(y_2\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$ | $\lambda_{12}^{\text{Adj}}\left(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)+16\lambda_{12}^{\text{Adj}}\lambda_{S_1}^{\text{Adj}}$ |    |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                              | $\lambda_{12}^{\text{Adj}}\left(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)+16\lambda_{12}^{\text{Adj}}\lambda_{S_1}^{\text{Adj}}$ |    |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                              | $\lambda_{12}^{\text{Adj}}\left(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)+16\lambda_{12}^{\text{Adj}}\lambda_{S_1}^{\text{Adj}}$ |    |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                              | $\lambda_{12}^{\text{Adj}}\left(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+\frac{16\lambda_{12}^{\text{Cross}}}{3}\right)+16\lambda_{12}^{\text{Adj}}\lambda_{S_1}^{\text{Adj}}$ |    |

Terms 8 of 8

| Column | Representative term structure                            |
|--------|----------------------------------------------------------|
| T1     | $+16\lambda_{12}^{\text{Adj}}\lambda_{S_2}^{\text{Adj}}$ |
| T2     | $+3\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$   |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                       | T2                                                     |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------|--------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                        | —                                                      |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                        | —                                                      |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                        | —                                                      |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                        | —                                                      |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                        | —                                                      |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                        | —                                                      |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                        | —                                                      |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                        | —                                                      |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                        | —                                                      |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                        | —                                                      |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                        | —                                                      |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                        | —                                                      |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                        | —                                                      |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                        | —                                                      |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                        | —                                                      |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                        | —                                                      |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                        | —                                                      |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                        | —                                                      |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                        | —                                                      |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                        | —                                                      |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+16\lambda_{12}^{\text{Adj}}\lambda_{S_2}^{\text{Adj}}$ | $+3\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+16\lambda_{12}^{\text{Adj}}\lambda_{S_2}^{\text{Adj}}$ | $+3\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+16\lambda_{12}^{\text{Adj}}\lambda_{S_2}^{\text{Adj}}$ | $+3\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+16\lambda_{12}^{\text{Adj}}\lambda_{S_2}^{\text{Adj}}$ | $+3\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+16\lambda_{12}^{\text{Adj}}\lambda_{S_2}^{\text{Adj}}$ | $+3\lambda_{H1}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |

$\beta_{\lambda_{12}^{\text{Cross}}}^{(1)}$

Search name: lambda12Cross

$$16\pi^2\mu\frac{d\lambda_{12}^{\text{Cross}}}{d\mu}=\beta_{\lambda_{12}^{\text{Cross}}}^{(1)}$$

Terms 1 of 6

| Column | Representative term structure                                                                                                                                                                                                               |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_{12}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+8\lambda_{12}^{\text{Cross}}+\lambda_{S_1}^{\text{Adj}}+\lambda_{S_2}^{\text{Adj}}\right)$ |
| T2     | $+\lambda_{12}^{\text{Cross}}(2\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+2\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right))$                                                                                                   |
| T3     | $-\frac{7}{6}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$                                                                                                                                              |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                                                                                                                                                                                                                                                    | T2                                                                                          | T3 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|----|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                     | —                                                                                           | —  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                     | —                                                                                           | —  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                     | —                                                                                           | —  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                     | —                                                                                           | —  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                     | —                                                                                           | —  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                     | —                                                                                           | —  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                     | —                                                                                           | —  |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                     | —                                                                                           | —  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                     | —                                                                                           | —  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                     | —                                                                                           | —  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                     | —                                                                                           | —  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                     | —                                                                                           | —  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                     | —                                                                                           | —  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                     | —                                                                                           | —  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                     | —                                                                                           | —  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                     | —                                                                                           | —  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                     | —                                                                                           | —  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                     | —                                                                                           | —  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                     | —                                                                                           | —  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                     | —                                                                                           | —  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\lambda_{12}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+8\lambda_{12}^{\text{Cross}}+\lambda_{S_1}^{\text{Adj}}+\lambda_{S_2}^{\text{Adj}}\right)+\lambda_{12}^{\text{Cross}}\left(2\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)\right)$ | $-\frac{7}{6}\text{Tr}\left(y_1\left(y_1\right)^\dagger y_2\left(y_2\right)^\dagger\right)$ |    |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\lambda_{12}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+8\lambda_{12}^{\text{Cross}}+\lambda_{S_1}^{\text{Adj}}+\lambda_{S_2}^{\text{Adj}}\right)+\lambda_{12}^{\text{Cross}}\left(2\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)\right)$ | $-\frac{7}{6}\text{Tr}\left(y_1\left(y_1\right)^\dagger y_2\left(y_2\right)^\dagger\right)$ |    |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\lambda_{12}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+8\lambda_{12}^{\text{Cross}}+\lambda_{S_1}^{\text{Adj}}+\lambda_{S_2}^{\text{Adj}}\right)+\lambda_{12}^{\text{Cross}}\left(2\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)\right)$ | $-\frac{7}{6}\text{Tr}\left(y_1\left(y_1\right)^\dagger y_2\left(y_2\right)^\dagger\right)$ |    |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\lambda_{12}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+8\lambda_{12}^{\text{Cross}}+\lambda_{S_1}^{\text{Adj}}+\lambda_{S_2}^{\text{Adj}}\right)+\lambda_{12}^{\text{Cross}}\left(2\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)\right)$ | $-\frac{7}{6}\text{Tr}\left(y_1\left(y_1\right)^\dagger y_2\left(y_2\right)^\dagger\right)$ |    |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\lambda_{12}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+8\lambda_{12}^{\text{Cross}}+\lambda_{S_1}^{\text{Adj}}+\lambda_{S_2}^{\text{Adj}}\right)+\lambda_{12}^{\text{Cross}}\left(2\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)\right)$ | $-\frac{7}{6}\text{Tr}\left(y_1\left(y_1\right)^\dagger y_2\left(y_2\right)^\dagger\right)$ |    |

Terms 2 of 6

| Column | Representative term structure                          |
|--------|--------------------------------------------------------|
| T1     | $-\frac{1}{2}\lambda_{T3}\left(\lambda_{T3}\right)^*$  |
| T2     | $+\frac{33g_2^4}{4}$                                   |
| T3     | $+g_2^2(-6\lambda_{12}-24\lambda_{12}^{\text{Cross}})$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                    | T2                   | T3                                                     |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------|----------------------|--------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                     | —                    | —                                                      |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                     | —                    | —                                                      |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                     | —                    | —                                                      |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                     | —                    | —                                                      |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                     | —                    | —                                                      |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                     | —                    | —                                                      |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                     | —                    | —                                                      |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                     | —                    | —                                                      |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                     | —                    | —                                                      |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                     | —                    | —                                                      |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                     | —                    | —                                                      |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                     | —                    | —                                                      |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                     | —                    | —                                                      |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                     | —                    | —                                                      |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                     | —                    | —                                                      |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                     | —                    | —                                                      |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                     | —                    | —                                                      |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                     | —                    | —                                                      |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                     | —                    | —                                                      |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                     | —                    | —                                                      |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{1}{2}\lambda_{T3}\left(\lambda_{T3}\right)^*$ | $+\frac{33g_2^4}{4}$ | $+g_2^2(-6\lambda_{12}-24\lambda_{12}^{\text{Cross}})$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{1}{2}\lambda_{T3}\left(\lambda_{T3}\right)^*$ | $+\frac{33g_2^4}{4}$ | $+g_2^2(-6\lambda_{12}-24\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{1}{2}\lambda_{T3}\left(\lambda_{T3}\right)^*$ | $+\frac{33g_2^4}{4}$ | $+g_2^2(-6\lambda_{12}-24\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{1}{2}\lambda_{T3}\left(\lambda_{T3}\right)^*$ | $+\frac{33g_2^4}{4}$ | $+g_2^2(-6\lambda_{12}-24\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{1}{2}\lambda_{T3}\left(\lambda_{T3}\right)^*$ | $+\frac{33g_2^4}{4}$ | $+g_2^2(-6\lambda_{12}-24\lambda_{12}^{\text{Cross}})$ |

Terms 3 of 6

| Column | Representative term structure                                              |
|--------|----------------------------------------------------------------------------|
| T1     | $+g_Y^2\left(-\frac{3\lambda_{12}}{2}-6\lambda_{12}^{\text{Cross}}\right)$ |
| T2     | $+\lambda_{12}^2$                                                          |
| T3     | $+\lambda_{S_1}(2\lambda_{12}+3\lambda_{12}^{\text{Cross}})$               |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                           | T2                | T3                                                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------------------------|-------------------|--------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                            | —                 | —                                                            |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                            | —                 | —                                                            |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                            | —                 | —                                                            |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                            | —                 | —                                                            |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                            | —                 | —                                                            |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                            | —                 | —                                                            |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                            | —                 | —                                                            |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                            | —                 | —                                                            |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                            | —                 | —                                                            |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                            | —                 | —                                                            |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                            | —                 | —                                                            |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                            | —                 | —                                                            |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                            | —                 | —                                                            |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                            | —                 | —                                                            |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                            | —                 | —                                                            |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                            | —                 | —                                                            |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                            | —                 | —                                                            |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                            | —                 | —                                                            |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                            | —                 | —                                                            |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                            | —                 | —                                                            |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+g_Y^2\left(-\frac{3\lambda_{12}}{2}-6\lambda_{12}^{\text{Cross}}\right)$   | $+\lambda_{12}^2$ | $+\lambda_{S_1}(2\lambda_{12}+3\lambda_{12}^{\text{Cross}})$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+g_Y^2\left(-\frac{3\lambda_{12}}{4}-3\lambda_{12}^{\text{Cross}}\right)$   | $+\lambda_{12}^2$ | $+\lambda_{S_1}(2\lambda_{12}+3\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+g_Y^2\left(-\frac{3\lambda_{12}}{2}-6\lambda_{12}^{\text{Cross}}\right)$   | $+\lambda_{12}^2$ | $+\lambda_{S_1}(2\lambda_{12}+3\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+g_Y^2\left(-\frac{15\lambda_{12}}{4}-15\lambda_{12}^{\text{Cross}}\right)$ | $+\lambda_{12}^2$ | $+\lambda_{S_1}(2\lambda_{12}+3\lambda_{12}^{\text{Cross}})$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+g_Y^2\left(-\frac{15\lambda_{12}}{2}-30\lambda_{12}^{\text{Cross}}\right)$ | $+\lambda_{12}^2$ | $+\lambda_{S_1}(2\lambda_{12}+3\lambda_{12}^{\text{Cross}})$ |

Terms 4 of 6

| Column | Representative term structure                                    |  |
|--------|------------------------------------------------------------------|--|
| T1     | $+ \lambda_{S_2} (2\lambda_{12} + 3\lambda_{12}^{\text{Cross}})$ |  |
| T2     | $+ 4\lambda_{12}^{\text{Adj}^2}$                                 |  |
| T3     | $+ 11\lambda_{12}^{\text{Cross}^2}$                              |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$                                                                         | T1                               | T2                               | T3                                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|-------------------------------------------------------------------------------|----------------------------------|----------------------------------|-------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$                                                                | —                                | —                                | —                                   |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0                                                                             | —                                | —                                | —                                   |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$                                                                 | —                                | —                                | —                                   |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1                                                                             | —                                | —                                | —                                   |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$                                                                 | —                                | —                                | —                                   |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$                                                                | —                                | —                                | —                                   |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0                                                                             | —                                | —                                | —                                   |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$                                                                 | —                                | —                                | —                                   |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1                                                                             | —                                | —                                | —                                   |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$                                                                 | —                                | —                                | —                                   |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$                                                                | —                                | —                                | —                                   |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0                                                                             | —                                | —                                | —                                   |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$                                                                 | —                                | —                                | —                                   |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1                                                                             | —                                | —                                | —                                   |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$                                                                 | —                                | —                                | —                                   |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$                                                                | —                                | —                                | —                                   |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0                                                                             | —                                | —                                | —                                   |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$                                                                 | —                                | —                                | —                                   |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1                                                                             | —                                | —                                | —                                   |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$                                                                 | —                                | —                                | —                                   |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2} + \lambda_{S_2} (2\lambda_{12} + 3\lambda_{12}^{\text{Cross}})$ | $+ 4\lambda_{12}^{\text{Adj}^2}$ | $+ 4\lambda_{12}^{\text{Adj}^2}$ | $+ 11\lambda_{12}^{\text{Cross}^2}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | $0 + \lambda_{S_2} (2\lambda_{12} + 3\lambda_{12}^{\text{Cross}})$            | $+ 4\lambda_{12}^{\text{Adj}^2}$ | $+ 4\lambda_{12}^{\text{Adj}^2}$ | $+ 11\lambda_{12}^{\text{Cross}^2}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2} + \lambda_{S_2} (2\lambda_{12} + 3\lambda_{12}^{\text{Cross}})$  | $+ 4\lambda_{12}^{\text{Adj}^2}$ | $+ 4\lambda_{12}^{\text{Adj}^2}$ | $+ 11\lambda_{12}^{\text{Cross}^2}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | $1 + \lambda_{S_2} (2\lambda_{12} + 3\lambda_{12}^{\text{Cross}})$            | $+ 4\lambda_{12}^{\text{Adj}^2}$ | $+ 4\lambda_{12}^{\text{Adj}^2}$ | $+ 11\lambda_{12}^{\text{Cross}^2}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2} + \lambda_{S_2} (2\lambda_{12} + 3\lambda_{12}^{\text{Cross}})$  | $+ 4\lambda_{12}^{\text{Adj}^2}$ | $+ 4\lambda_{12}^{\text{Adj}^2}$ | $+ 11\lambda_{12}^{\text{Cross}^2}$ |

Terms 5 of 6

| Column | Representative term structure                            |
|--------|----------------------------------------------------------|
| T1     | $-\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}$ |
| T2     | $-\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}$ |
| T3     | $+\lambda_{H1}\lambda_{H2}$                              |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                       | T2                                                       | T3                          |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------|----------------------------------------------------------|-----------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                        | —                                                        | —                           |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                        | —                                                        | —                           |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                        | —                                                        | —                           |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                        | —                                                        | —                           |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                        | —                                                        | —                           |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                        | —                                                        | —                           |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                        | —                                                        | —                           |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                        | —                                                        | —                           |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                        | —                                                        | —                           |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                        | —                                                        | —                           |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                        | —                                                        | —                           |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                        | —                                                        | —                           |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                        | —                                                        | —                           |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                        | —                                                        | —                           |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                        | —                                                        | —                           |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                        | —                                                        | —                           |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                        | —                                                        | —                           |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                        | —                                                        | —                           |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                        | —                                                        | —                           |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                        | —                                                        | —                           |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}$ | $-\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}$ | $+\lambda_{H1}\lambda_{H2}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}$ | $-\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}$ | $+\lambda_{H1}\lambda_{H2}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}$ | $-\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}$ | $+\lambda_{H1}\lambda_{H2}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}$ | $-\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}$ | $+\lambda_{H1}\lambda_{H2}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\lambda_{12}^{\text{Cross}}\lambda_{S_1}^{\text{Adj}}$ | $-\lambda_{12}^{\text{Cross}}\lambda_{S_2}^{\text{Adj}}$ | $+\lambda_{H1}\lambda_{H2}$ |

Terms 6 of 6

| Column | Representative term structure                                                                  |
|--------|------------------------------------------------------------------------------------------------|
| T1     | $-\frac{7}{6}\text{Tr}\left(y_1\left(y_2\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$ |
| T2     | $+\frac{3g_Y^4}{16}$                                                                           |
| T3     | $+12g_Y^4$                                                                                     |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                             | T2                    | T3         |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------------------------------------------|-----------------------|------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                              | —                     | —          |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                              | —                     | —          |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                              | —                     | —          |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                              | —                     | —          |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                              | —                     | —          |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                              | —                     | —          |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                                              | —                     | —          |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                              | —                     | —          |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                              | —                     | —          |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                              | —                     | —          |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                              | —                     | —          |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                                              | —                     | —          |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                              | —                     | —          |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                              | —                     | —          |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                              | —                     | —          |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                              | —                     | —          |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                              | —                     | —          |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                              | —                     | —          |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                              | —                     | —          |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                              | —                     | —          |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                              | —                     | —          |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{7}{6}\text{Tr}\left(y_1\left(y_2\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$ | $+\frac{3g_Y^4}{16}$  | —          |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                              | —                     | —          |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                              | $+\frac{27g_Y^4}{16}$ | —          |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                              | —                     | $+12g_Y^4$ |

$$\beta_{\lambda_{H1}^{\text{Adj}}}^{(1)}$$

Search name: lambdaAff1Adj

$$16\pi^2\,\mu\frac{d\lambda_{H1}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{H1}^{\text{Adj}}}^{(1)}$$

# Terms 1 of 8

| Column | Representative term structure                                                                                                                                                            |  |  |  |  |  |  |  |  |  |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|--|--|--|--|--|
| T1     | $\lambda_{H1}^{\text{Adj}}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$ |  |  |  |  |  |  |  |  |  |
| T2     | $+\frac{8}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$                                                                                                                  |  |  |  |  |  |  |  |  |  |
| T3     | $+\frac{16}{3}\lambda_{T3}\ (\lambda_{T3})^*$                                                                                                                                            |  |  |  |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                                                       | T2                                                                      | T3                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                        | —                                                                       | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                        | —                                                                       | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                        | —                                                                       | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                        | —                                                                       | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                        | —                                                                       | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $\lambda_{H1}^{\text{Adj}}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$ | $+\frac{8}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$ | $+\frac{16}{3}\lambda_{T3}\ (\lambda_{T3})^*$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\lambda_{H1}^{\text{Adj}}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$ | $+\frac{8}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$ | $+\frac{16}{3}\lambda_{T3}\ (\lambda_{T3})^*$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $\lambda_{H1}^{\text{Adj}}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$ | $+\frac{8}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$ | $+\frac{16}{3}\lambda_{T3}\ (\lambda_{T3})^*$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $\lambda_{H1}^{\text{Adj}}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$ | $+\frac{8}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$ | $+\frac{16}{3}\lambda_{T3}\ (\lambda_{T3})^*$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $\lambda_{H1}^{\text{Adj}}(2\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$ | $+\frac{8}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$ | $+\frac{16}{3}\lambda_{T3}\ (\lambda_{T3})^*$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $\lambda_{H1}^{\text{Adj}}(3\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$ | $-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$ | $+\frac{16}{3}\lambda_{T3}\ (\lambda_{T3})^*$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $\lambda_{H1}^{\text{Adj}}(3\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$ | $-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$ | $+\frac{16}{3}\lambda_{T3}\ (\lambda_{T3})^*$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $\lambda_{H1}^{\text{Adj}}(3\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$ | $-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$ | $+\frac{16}{3}\lambda_{T3}\ (\lambda_{T3})^*$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $\lambda_{H1}^{\text{Adj}}(3\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$ | $-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$ | $+\frac{16}{3}\lambda_{T3}\ (\lambda_{T3})^*$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $\lambda_{H1}^{\text{Adj}}(3\text{Tr}\big(y_1\ (y_1)^\dagger\big)+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$ | $-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$ | $+\frac{16}{3}\lambda_{T3}\ (\lambda_{T3})^*$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                        | $-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$ | $-\frac{8}{3}\lambda_{T3}\ (\lambda_{T3})^*$  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                        | $-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$ | $-\frac{8}{3}\lambda_{T3}\ (\lambda_{T3})^*$  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                        | $-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$ | $-\frac{8}{3}\lambda_{T3}\ (\lambda_{T3})^*$  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                        | $-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$ | $-\frac{8}{3}\lambda_{T3}\ (\lambda_{T3})^*$  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                        | $-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$ | $-\frac{8}{3}\lambda_{T3}\ (\lambda_{T3})^*$  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                        | $-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$ | $-\frac{4}{3}\lambda_{T3}\ (\lambda_{T3})^*$  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1 | T2                                                                          | T3                                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|---------------|----|-----------------------------------------------------------------------------|----------------------------------------------|
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0             | —  | $-\frac{4}{3} \text{Tr} \left( y_1 (y_1)^\dagger Y_e (Y_e)^\dagger \right)$ | $-\frac{4}{3} \lambda_{T3} (\lambda_{T3})^*$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$ | —  | $-\frac{4}{3} \text{Tr} \left( y_1 (y_1)^\dagger Y_e (Y_e)^\dagger \right)$ | $-\frac{4}{3} \lambda_{T3} (\lambda_{T3})^*$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1             | —  | $-\frac{4}{3} \text{Tr} \left( y_1 (y_1)^\dagger Y_e (Y_e)^\dagger \right)$ | $-\frac{4}{3} \lambda_{T3} (\lambda_{T3})^*$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$ | —  | $-\frac{4}{3} \text{Tr} \left( y_1 (y_1)^\dagger Y_e (Y_e)^\dagger \right)$ | $-\frac{4}{3} \lambda_{T3} (\lambda_{T3})^*$ |

Terms 2 of 8

| Column | Representative term structure                                       |
|--------|---------------------------------------------------------------------|
| T1     | $+g_Y^2\left(-12g_2^2-\frac{15\lambda_{H1}^{\text{Adj}}}{2}\right)$ |
| T2     | $-9g_2^2\lambda_{H1}^{\text{Adj}}$                                  |
| T3     | $+\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$               |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                  | T2                                 | T3                                                    |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------------|------------------------------------|-------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                   | —                                  | —                                                     |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                   | —                                  | —                                                     |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                   | —                                  | —                                                     |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                   | —                                  | —                                                     |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                   | —                                  | —                                                     |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+g_Y^2\left(-12g_2^2-\frac{15\lambda_{H1}^{\text{Adj}}}{2}\right)$ | $-9g_2^2\lambda_{H1}^{\text{Adj}}$ | $+\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                   | $-9g_2^2\lambda_{H1}^{\text{Adj}}$ | $+\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                   | $-9g_2^2\lambda_{H1}^{\text{Adj}}$ | $+\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                   | $-9g_2^2\lambda_{H1}^{\text{Adj}}$ | $+\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                   | $-9g_2^2\lambda_{H1}^{\text{Adj}}$ | $+\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+g_Y^2\left(-12g_2^2-\frac{15\lambda_{H1}^{\text{Adj}}}{2}\right)$ | $-9g_2^2\lambda_{H1}^{\text{Adj}}$ | $+\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                   | $-9g_2^2\lambda_{H1}^{\text{Adj}}$ | $+\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                   | $-9g_2^2\lambda_{H1}^{\text{Adj}}$ | $+\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                   | $-9g_2^2\lambda_{H1}^{\text{Adj}}$ | $+\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                   | $-9g_2^2\lambda_{H1}^{\text{Adj}}$ | $+\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $+g_Y^2\left(-4g_2^2-\frac{15\lambda_{H1}^{\text{Adj}}}{2}\right)$  | —                                  | —                                                     |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                   | —                                  | —                                                     |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                   | —                                  | —                                                     |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                   | —                                  | —                                                     |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                   | —                                  | —                                                     |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+g_Y^2\left(-4g_2^2-\frac{15\lambda_{H1}^{\text{Adj}}}{2}\right)$  | —                                  | —                                                     |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                   | —                                  | —                                                     |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                   | —                                  | —                                                     |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                   | —                                  | —                                                     |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                   | —                                  | —                                                     |

Terms 3 of 8

| Column | Representative term structure             |  |
|--------|-------------------------------------------|--|
| T1     | $+2\lambda_H\lambda_{H1}^{\text{Adj}}$    |  |
| T2     | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ |  |
| T3     | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ |  |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$       | $d_{S2}$ | $Y_{S2}$      | $d_F$ | $Y_F$          | T1                                     | T2                                        | T3                                        |
|-------|----------|----------|----------------|----------|---------------|-------|----------------|----------------------------------------|-------------------------------------------|-------------------------------------------|
| T3-A  | -2       | 1        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | —                                      | —                                         | —                                         |
| T3-A  | -1       | 1        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | —                                      | —                                         | —                                         |
| T3-A  | 0        | 1        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | —                                      | —                                         | —                                         |
| T3-A  | 1        | 1        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | —                                      | —                                         | —                                         |
| T3-A  | 2        | 1        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | —                                      | —                                         | —                                         |
| T3-B  | -2       | 2        | -1             | 2        | 0             | 1     | $-\frac{1}{2}$ | $+2\lambda_H\lambda_{H1}^{\text{Adj}}$ | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ |
| T3-B  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 1     | 0              | $+2\lambda_H\lambda_{H1}^{\text{Adj}}$ | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ |
| T3-B  | 0        | 2        | 0              | 2        | 1             | 1     | $\frac{1}{2}$  | $+2\lambda_H\lambda_{H1}^{\text{Adj}}$ | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ |
| T3-B  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 1     | 1              | $+2\lambda_H\lambda_{H1}^{\text{Adj}}$ | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ |
| T3-B  | 2        | 2        | 1              | 2        | 2             | 1     | $\frac{3}{2}$  | $+2\lambda_H\lambda_{H1}^{\text{Adj}}$ | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ |
| T3-C  | -2       | 2        | -1             | 2        | 0             | 3     | $-\frac{1}{2}$ | $+2\lambda_H\lambda_{H1}^{\text{Adj}}$ | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ |
| T3-C  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 3     | 0              | $+2\lambda_H\lambda_{H1}^{\text{Adj}}$ | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ |
| T3-C  | 0        | 2        | 0              | 2        | 1             | 3     | $\frac{1}{2}$  | $+2\lambda_H\lambda_{H1}^{\text{Adj}}$ | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ |
| T3-C  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 3     | 1              | $+2\lambda_H\lambda_{H1}^{\text{Adj}}$ | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ |
| T3-C  | 2        | 2        | 1              | 2        | 2             | 3     | $\frac{3}{2}$  | $+2\lambda_H\lambda_{H1}^{\text{Adj}}$ | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ |
| T3-D  | -2       | 3        | -1             | 1        | 0             | 2     | $-\frac{1}{2}$ | $+2\lambda_H\lambda_{H1}^{\text{Adj}}$ | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ |
| T3-D  | -1       | 3        | $-\frac{1}{2}$ | 1        | $\frac{1}{2}$ | 2     | 0              | $+2\lambda_H\lambda_{H1}^{\text{Adj}}$ | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ |
| T3-D  | 0        | 3        | 0              | 1        | 1             | 2     | $\frac{1}{2}$  | $+2\lambda_H\lambda_{H1}^{\text{Adj}}$ | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ |
| T3-D  | 1        | 3        | $\frac{1}{2}$  | 1        | $\frac{3}{2}$ | 2     | 1              | $+2\lambda_H\lambda_{H1}^{\text{Adj}}$ | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ |
| T3-D  | 2        | 3        | 1              | 1        | 2             | 2     | $\frac{3}{2}$  | $+2\lambda_H\lambda_{H1}^{\text{Adj}}$ | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ |
| T3-E  | -2       | 3        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+2\lambda_H\lambda_{H1}^{\text{Adj}}$ | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ |
| T3-E  | -1       | 3        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+2\lambda_H\lambda_{H1}^{\text{Adj}}$ | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ |
| T3-E  | 0        | 3        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+2\lambda_H\lambda_{H1}^{\text{Adj}}$ | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ |
| T3-E  | 1        | 3        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+2\lambda_H\lambda_{H1}^{\text{Adj}}$ | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ |
| T3-E  | 2        | 3        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+2\lambda_H\lambda_{H1}^{\text{Adj}}$ | $+8\lambda_{H1}\lambda_{H1}^{\text{Adj}}$ | $+2\lambda_{H1}^{\text{Adj}}\lambda_{S1}$ |

Terms 4 of 8

| Column | Representative term structure                                                                                                                                                                                                                                                                         |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(-\frac{2}{3}\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)-\frac{4}{3}\left(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+\left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)^*\right)\right)$ |
| T2     | $-\frac{2}{3}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$                                                                                                                                                             |
| T3     | $+\frac{64}{3}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$                                                                                                                                                              |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                                                                                                                                                                                                                                                                                                           | T2                                                                                                                                       | T3                                                                                                                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                            | —                                                                                                                                        | —                                                                                                                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                                                                            | —                                                                                                                                        | —                                                                                                                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                            | —                                                                                                                                        | —                                                                                                                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                            | —                                                                                                                                        | —                                                                                                                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                            | —                                                                                                                                        | —                                                                                                                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                            | —                                                                                                                                        | —                                                                                                                                        |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(-\frac{2}{3}\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)-\frac{4}{3}\left(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+\left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)^*\right)\right)-\frac{2}{3}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$ | $+\frac{64}{3}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$ |                                                                                                                                          |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                            | —                                                                                                                                        | —                                                                                                                                        |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                            | —                                                                                                                                        | —                                                                                                                                        |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                            | —                                                                                                                                        | —                                                                                                                                        |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                            | —                                                                                                                                        | —                                                                                                                                        |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                                                                            | —                                                                                                                                        | $+\frac{64}{3}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\left(\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}\right)^*$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                            | —                                                                                                                                        | —                                                                                                                                        |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                            | —                                                                                                                                        | —                                                                                                                                        |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                            | —                                                                                                                                        | —                                                                                                                                        |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                            | —                                                                                                                                        | —                                                                                                                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                                                                            | —                                                                                                                                        | —                                                                                                                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                            | —                                                                                                                                        | —                                                                                                                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                            | —                                                                                                                                        | —                                                                                                                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                            | —                                                                                                                                        | —                                                                                                                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                            | —                                                                                                                                        | —                                                                                                                                        |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                                                                            | —                                                                                                                                        | —                                                                                                                                        |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                            | —                                                                                                                                        | —                                                                                                                                        |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                            | —                                                                                                                                        | —                                                                                                                                        |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                            | —                                                                                                                                        | —                                                                                                                                        |

## Terms 5 of 8

| Column | Representative term structure                                                                                                      |  |
|--------|------------------------------------------------------------------------------------------------------------------------------------|--|
| T1     | $-\frac{4}{3}\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ |  |
| T2     | $+g_Y^2 \left( -6g_2^2 - 3\lambda_{H1}^{\text{Adj}} \right)$                                                                       |  |
| T3     | $-\frac{3g_Y^2 \lambda_{H1}^{\text{Adj}}}{2}$                                                                                      |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                 | T2                                                           | T3                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                  | —                                                            | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                  | —                                                            | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                  | —                                                            | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                  | —                                                            | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                  | —                                                            | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                  | —                                                            | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{4}{3}\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ | $+g_Y^2 \left( -6g_2^2 - 3\lambda_{H1}^{\text{Adj}} \right)$ | —                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                  | —                                                            | $-\frac{3g_Y^2 \lambda_{H1}^{\text{Adj}}}{2}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                  | —                                                            | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                  | —                                                            | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                  | —                                                            | —                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-\frac{4}{3}\lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ | $+g_Y^2 \left( -6g_2^2 - 3\lambda_{H1}^{\text{Adj}} \right)$ | —                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                  | —                                                            | $-\frac{3g_Y^2 \lambda_{H1}^{\text{Adj}}}{2}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                  | —                                                            | —                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                  | —                                                            | —                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                  | —                                                            | —                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                  | $+g_Y^2 \left( -2g_2^2 - 3\lambda_{H1}^{\text{Adj}} \right)$ | —                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                  | —                                                            | $-\frac{3g_Y^2 \lambda_{H1}^{\text{Adj}}}{2}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                  | —                                                            | —                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                  | —                                                            | —                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                  | —                                                            | —                                             |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                  | $+g_Y^2 \left( -2g_2^2 - 3\lambda_{H1}^{\text{Adj}} \right)$ | —                                             |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                  | —                                                            | $-\frac{3g_Y^2 \lambda_{H1}^{\text{Adj}}}{2}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                  | —                                                            | —                                             |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                  | —                                                            | —                                             |

Terms 6 of 8

| Column | Representative term structure                                                                                                                                                                                                                                                              |  |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| T1     | $+g_Y^2\left(6g_2^2-3\lambda_{H1}^{\text{Adj}}\right)$                                                                                                                                                                                                                                     |  |
| T2     | $+g_Y^2\left(12g_2^2-\frac{15\lambda_{H1}^{\text{Adj}}}{2}\right)$                                                                                                                                                                                                                         |  |
| T3     | $+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(-\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)-\frac{4}{3}\left(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+\left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)^*\right)\right)$ |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                 | T2 | T3                                                                                                                                                                                                                                                                                         |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                  | —  | —                                                                                                                                                                                                                                                                                          |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                  | —  | —                                                                                                                                                                                                                                                                                          |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                  | —  | —                                                                                                                                                                                                                                                                                          |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                  | —  | —                                                                                                                                                                                                                                                                                          |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                  | —  | —                                                                                                                                                                                                                                                                                          |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                  | —  | —                                                                                                                                                                                                                                                                                          |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                  | —  | —                                                                                                                                                                                                                                                                                          |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                  | —  | —                                                                                                                                                                                                                                                                                          |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $+g_Y^2\left(6g_2^2-3\lambda_{H1}^{\text{Adj}}\right)$             | —  | —                                                                                                                                                                                                                                                                                          |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+g_Y^2\left(12g_2^2-\frac{15\lambda_{H1}^{\text{Adj}}}{2}\right)$ | —  | —                                                                                                                                                                                                                                                                                          |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                  | —  | —                                                                                                                                                                                                                                                                                          |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                  | —  | $+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(-\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)-\frac{4}{3}\left(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+\left(\lambda_{S_1 S_1^{\dagger 2} S_2^\dagger}\right)^*\right)\right)$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                  | —  | —                                                                                                                                                                                                                                                                                          |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $+g_Y^2\left(6g_2^2-3\lambda_{H1}^{\text{Adj}}\right)$             | —  | —                                                                                                                                                                                                                                                                                          |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+g_Y^2\left(12g_2^2-\frac{15\lambda_{H1}^{\text{Adj}}}{2}\right)$ | —  | —                                                                                                                                                                                                                                                                                          |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                  | —  | —                                                                                                                                                                                                                                                                                          |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                  | —  | —                                                                                                                                                                                                                                                                                          |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                  | —  | —                                                                                                                                                                                                                                                                                          |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $+g_Y^2\left(2g_2^2-3\lambda_{H1}^{\text{Adj}}\right)$             | —  | —                                                                                                                                                                                                                                                                                          |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $+g_Y^2\left(4g_2^2-\frac{15\lambda_{H1}^{\text{Adj}}}{2}\right)$  | —  | —                                                                                                                                                                                                                                                                                          |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                  | —  | —                                                                                                                                                                                                                                                                                          |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                  | —  | —                                                                                                                                                                                                                                                                                          |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                  | —  | —                                                                                                                                                                                                                                                                                          |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+g_Y^2\left(2g_2^2-3\lambda_{H1}^{\text{Adj}}\right)$             | —  | —                                                                                                                                                                                                                                                                                          |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+g_Y^2\left(4g_2^2-\frac{15\lambda_{H1}^{\text{Adj}}}{2}\right)$  | —  | —                                                                                                                                                                                                                                                                                          |

Terms 7 of 8

| Column | Representative term structure                                                                                                                                                                                                                                         |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$                                                                                                                                        |
| T2     | $\lambda_{H1}^{\text{Adj}}(2\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S_1}^{\text{Adj}})$ |
| T3     | $-\frac{33g_2^2\lambda_{H1}^{\text{Adj}}}{2}$                                                                                                                                                                                                                         |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                             | T2                                                                                                                                                                                                                                                                    | T3                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                              | —                                                                                                                                                                                                                                                                     | —                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                              | —                                                                                                                                                                                                                                                                     | —                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                              | —                                                                                                                                                                                                                                                                     | —                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                              | —                                                                                                                                                                                                                                                                     | —                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                              | —                                                                                                                                                                                                                                                                     | —                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                              | —                                                                                                                                                                                                                                                                     | —                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                                                                              | —                                                                                                                                                                                                                                                                     | —                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                              | —                                                                                                                                                                                                                                                                     | —                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                              | —                                                                                                                                                                                                                                                                     | —                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                              | —                                                                                                                                                                                                                                                                     | —                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                              | —                                                                                                                                                                                                                                                                     | —                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$ | —                                                                                                                                                                                                                                                                     | —                                             |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                              | —                                                                                                                                                                                                                                                                     | —                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                              | —                                                                                                                                                                                                                                                                     | —                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                              | —                                                                                                                                                                                                                                                                     | —                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                              | $\lambda_{H1}^{\text{Adj}}(2\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S_1}^{\text{Adj}})$ | $-\frac{33g_2^2\lambda_{H1}^{\text{Adj}}}{2}$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                              | $\lambda_{H1}^{\text{Adj}}(2\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S_1}^{\text{Adj}})$ | $-\frac{33g_2^2\lambda_{H1}^{\text{Adj}}}{2}$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                              | $\lambda_{H1}^{\text{Adj}}(2\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S_1}^{\text{Adj}})$ | $-\frac{33g_2^2\lambda_{H1}^{\text{Adj}}}{2}$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                              | $\lambda_{H1}^{\text{Adj}}(2\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S_1}^{\text{Adj}})$ | $-\frac{33g_2^2\lambda_{H1}^{\text{Adj}}}{2}$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                              | $\lambda_{H1}^{\text{Adj}}(2\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S_1}^{\text{Adj}})$ | $-\frac{33g_2^2\lambda_{H1}^{\text{Adj}}}{2}$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                              | $\lambda_{H1}^{\text{Adj}}(2\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S_1}^{\text{Adj}})$ | $-\frac{33g_2^2\lambda_{H1}^{\text{Adj}}}{2}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                              | $\lambda_{H1}^{\text{Adj}}(2\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S_1}^{\text{Adj}})$ | $-\frac{33g_2^2\lambda_{H1}^{\text{Adj}}}{2}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                              | $\lambda_{H1}^{\text{Adj}}(2\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S_1}^{\text{Adj}})$ | $-\frac{33g_2^2\lambda_{H1}^{\text{Adj}}}{2}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                              | $\lambda_{H1}^{\text{Adj}}(2\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S_1}^{\text{Adj}})$ | $-\frac{33g_2^2\lambda_{H1}^{\text{Adj}}}{2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$         | T1 | T2                                                                                                                                                                                                                                                                           | T3                                            |
|-------|----------|-----------|-----------|-----------|-----------|-------|---------------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| T3-E  | 2        | 3         | 1         | 3         | 2         | 2     | $\frac{3}{2}$ | —  | $\lambda_{H1}^{\text{Adj}}(2\text{Tr}\left(y_l\left(y_l\right)^{\dagger}\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)+16\lambda_{S1}^{\text{Adj}})$ | $-\frac{33g_2^2\lambda_{H1}^{\text{Adj}}}{2}$ |

Terms 8 of 8

| Column | Representative term structure                           |
|--------|---------------------------------------------------------|
| T1     | $+12\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                      |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                       |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                       |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                       |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                       |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                       |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                       |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                       |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                       |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                       |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                       |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                       |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                       |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                       |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                       |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                       |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                       |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                       |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                       |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                       |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                       |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+12\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+12\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+12\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+12\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+12\lambda_{12}^{\text{Adj}}\lambda_{H2}^{\text{Adj}}$ |

$\beta_{\lambda_{H2}^{\text{Adj}}}^{(1)}$

Search name: lambdaH2Adj

$$16\pi^2\mu\frac{d\lambda_{H2}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{H2}^{\text{Adj}}}^{(1)}$$

Terms 1 of 7

| Column | Representative term structure                                                                                                                                                                                                 |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_{H2}^{\text{Adj}}(2\text{Tr}\big(y_2\ (y_2)^{\dagger}\big)+6\text{Tr}\big(Y_d\ (Y_d)^{\dagger}\big)+2\text{Tr}\big(Y_e\ (Y_e)^{\dagger}\big)+6\text{Tr}\big(Y_u\ (Y_u)^{\dagger}\big)+16\lambda_{S_2}^{\text{Adj}})$ |
| T2     | $+\frac{4}{3}\text{Tr}\big(y_2\ (y_2)^{\dagger}\ Y_e\ (Y_e)^{\dagger}\big)$                                                                                                                                                   |
| T3     | $+\frac{8}{3}\lambda_{T3}\ (\lambda_{T3})^*$                                                                                                                                                                                  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                                                                                            | T2                                                                          | T3                                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|----------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\lambda_{H2}^{\text{Adj}}(2\text{Tr}\big(y_2\ (y_2)^{\dagger}\big)+6\text{Tr}\big(Y_d\ (Y_d)^{\dagger}\big)+2\text{Tr}\big(Y_e\ (Y_e)^{\dagger}\big)+6\text{Tr}\big(Y_u\ (Y_u)^{\dagger}\big)+16\lambda_{S_2}^{\text{Adj}})$ | $+\frac{4}{3}\text{Tr}\big(y_2\ (y_2)^{\dagger}\ Y_e\ (Y_e)^{\dagger}\big)$ | $+\frac{8}{3}\lambda_{T3}\ (\lambda_{T3})^*$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\lambda_{H2}^{\text{Adj}}(2\text{Tr}\big(y_2\ (y_2)^{\dagger}\big)+6\text{Tr}\big(Y_d\ (Y_d)^{\dagger}\big)+2\text{Tr}\big(Y_e\ (Y_e)^{\dagger}\big)+6\text{Tr}\big(Y_u\ (Y_u)^{\dagger}\big)+16\lambda_{S_2}^{\text{Adj}})$ | $+\frac{4}{3}\text{Tr}\big(y_2\ (y_2)^{\dagger}\ Y_e\ (Y_e)^{\dagger}\big)$ | $+\frac{8}{3}\lambda_{T3}\ (\lambda_{T3})^*$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\lambda_{H2}^{\text{Adj}}(2\text{Tr}\big(y_2\ (y_2)^{\dagger}\big)+6\text{Tr}\big(Y_d\ (Y_d)^{\dagger}\big)+2\text{Tr}\big(Y_e\ (Y_e)^{\dagger}\big)+6\text{Tr}\big(Y_u\ (Y_u)^{\dagger}\big)+16\lambda_{S_2}^{\text{Adj}})$ | $+\frac{4}{3}\text{Tr}\big(y_2\ (y_2)^{\dagger}\ Y_e\ (Y_e)^{\dagger}\big)$ | $+\frac{8}{3}\lambda_{T3}\ (\lambda_{T3})^*$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\lambda_{H2}^{\text{Adj}}(2\text{Tr}\big(y_2\ (y_2)^{\dagger}\big)+6\text{Tr}\big(Y_d\ (Y_d)^{\dagger}\big)+2\text{Tr}\big(Y_e\ (Y_e)^{\dagger}\big)+6\text{Tr}\big(Y_u\ (Y_u)^{\dagger}\big)+16\lambda_{S_2}^{\text{Adj}})$ | $+\frac{4}{3}\text{Tr}\big(y_2\ (y_2)^{\dagger}\ Y_e\ (Y_e)^{\dagger}\big)$ | $+\frac{8}{3}\lambda_{T3}\ (\lambda_{T3})^*$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\lambda_{H2}^{\text{Adj}}(2\text{Tr}\big(y_2\ (y_2)^{\dagger}\big)+6\text{Tr}\big(Y_d\ (Y_d)^{\dagger}\big)+2\text{Tr}\big(Y_e\ (Y_e)^{\dagger}\big)+6\text{Tr}\big(Y_u\ (Y_u)^{\dagger}\big)+16\lambda_{S_2}^{\text{Adj}})$ | $+\frac{4}{3}\text{Tr}\big(y_2\ (y_2)^{\dagger}\ Y_e\ (Y_e)^{\dagger}\big)$ | $+\frac{8}{3}\lambda_{T3}\ (\lambda_{T3})^*$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                             | —                                                                           | —                                            |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                                                                                                                                                                             | —                                                                           | —                                            |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                                                                                             | —                                                                           | —                                            |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                                                                                             | —                                                                           | —                                            |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                                                                                             | —                                                                           | —                                            |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                             | —                                                                           | —                                            |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                                                                                                                                                                             | —                                                                           | —                                            |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                                                                                             | —                                                                           | —                                            |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                                                                                             | —                                                                           | —                                            |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                                                                                             | —                                                                           | —                                            |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                             | —                                                                           | —                                            |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                             | —                                                                           | —                                            |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                             | —                                                                           | —                                            |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                             | —                                                                           | —                                            |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                             | —                                                                           | —                                            |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\lambda_{H2}^{\text{Adj}}(2\text{Tr}\big(y_2\ (y_2)^{\dagger}\big)+6\text{Tr}\big(Y_d\ (Y_d)^{\dagger}\big)+2\text{Tr}\big(Y_e\ (Y_e)^{\dagger}\big)+6\text{Tr}\big(Y_u\ (Y_u)^{\dagger}\big)+16\lambda_{S_2}^{\text{Adj}})$ | $+\frac{4}{3}\text{Tr}\big(y_2\ (y_2)^{\dagger}\ Y_e\ (Y_e)^{\dagger}\big)$ | $-\frac{8}{9}\lambda_{T3}\ (\lambda_{T3})^*$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\lambda_{H2}^{\text{Adj}}(2\text{Tr}\big(y_2\ (y_2)^{\dagger}\big)+6\text{Tr}\big(Y_d\ (Y_d)^{\dagger}\big)+2\text{Tr}\big(Y_e\ (Y_e)^{\dagger}\big)+6\text{Tr}\big(Y_u\ (Y_u)^{\dagger}\big)+16\lambda_{S_2}^{\text{Adj}})$ | $+\frac{4}{3}\text{Tr}\big(y_2\ (y_2)^{\dagger}\ Y_e\ (Y_e)^{\dagger}\big)$ | $-\frac{8}{9}\lambda_{T3}\ (\lambda_{T3})^*$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\lambda_{H2}^{\text{Adj}}(2\text{Tr}\big(y_2\ (y_2)^{\dagger}\big)+6\text{Tr}\big(Y_d\ (Y_d)^{\dagger}\big)+2\text{Tr}\big(Y_e\ (Y_e)^{\dagger}\big)+6\text{Tr}\big(Y_u\ (Y_u)^{\dagger}\big)+16\lambda_{S_2}^{\text{Adj}})$ | $+\frac{4}{3}\text{Tr}\big(y_2\ (y_2)^{\dagger}\ Y_e\ (Y_e)^{\dagger}\big)$ | $-\frac{8}{9}\lambda_{T3}\ (\lambda_{T3})^*$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\lambda_{H2}^{\text{Adj}}(2\text{Tr}\big(y_2\ (y_2)^{\dagger}\big)+6\text{Tr}\big(Y_d\ (Y_d)^{\dagger}\big)+2\text{Tr}\big(Y_e\ (Y_e)^{\dagger}\big)+6\text{Tr}\big(Y_u\ (Y_u)^{\dagger}\big)+16\lambda_{S_2}^{\text{Adj}})$ | $+\frac{4}{3}\text{Tr}\big(y_2\ (y_2)^{\dagger}\ Y_e\ (Y_e)^{\dagger}\big)$ | $-\frac{8}{9}\lambda_{T3}\ (\lambda_{T3})^*$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$         | T1                                                                                                                                                                                           | T2                                                                 | T3                                           |
|-------|----------|-----------|-----------|-----------|-----------|-------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|----------------------------------------------|
| T3-E  | 2        | 3         | 1         | 3         | 2         | 2     | $\frac{3}{2}$ | $\lambda_{H_2}^{Adj} (2 \text{Tr} (y_2)^{\dagger}) + 6 \text{Tr} (Y_d (Y_d)^{\dagger}) + 2 \text{Tr} (Y_e (Y_e)^{\dagger}) + 6 \text{Tr} (Y_\nu (Y_\nu)^{\dagger}) + 16 \lambda_{S_2}^{Adj}$ | $+\frac{4}{3} \text{Tr} (y_2 (y_2)^{\dagger} Y_e (Y_e)^{\dagger})$ | $-\frac{8}{9} \lambda_{T3} (\lambda_{T3})^*$ |

Terms 2 of 7

| Column | Representative term structure                 |
|--------|-----------------------------------------------|
| T1     | $-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$ |
| T2     | $-\frac{3g_Y^2\lambda_{H2}^{\text{Adj}}}{2}$  |
| T3     | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                            | T2                                           | T3                                     |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------|----------------------------------------------|----------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$ | $-\frac{3g_Y^2\lambda_{H2}^{\text{Adj}}}{2}$ | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$ | —                                            | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$ | —                                            | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$ | —                                            | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$ | —                                            | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                             | $-\frac{3g_Y^2\lambda_{H2}^{\text{Adj}}}{2}$ | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                             | —                                            | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                             | —                                            | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                             | —                                            | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                             | —                                            | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                             | $-\frac{3g_Y^2\lambda_{H2}^{\text{Adj}}}{2}$ | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                             | —                                            | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                             | —                                            | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                             | —                                            | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                             | —                                            | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                             | —                                            | —                                      |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                             | —                                            | —                                      |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                             | —                                            | —                                      |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                             | —                                            | —                                      |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                             | —                                            | —                                      |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$ | $-\frac{3g_Y^2\lambda_{H2}^{\text{Adj}}}{2}$ | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$ | —                                            | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$ | —                                            | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$ | —                                            | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$         | T1                                            | T2 | T3                                     |
|-------|----------|-----------|-----------|-----------|-----------|-------|---------------|-----------------------------------------------|----|----------------------------------------|
| T3-E  | 2        | 3         | 1         | 3         | 2         | 2     | $\frac{3}{2}$ | $-\frac{33g_2^2\lambda_{H2}^{\text{Adj}}}{2}$ | —  | $+2\lambda_H\lambda_{H2}^{\text{Adj}}$ |

Terms 3 of 7

| Column | Representative term structure                          |
|--------|--------------------------------------------------------|
| T1     | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$              |
| T2     | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$              |
| T3     | $+g_Y^2\left(2g_2^2-3\lambda_{H2}^{\text{Adj}}\right)$ |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$       | $d_{S2}$ | $Y_{S2}$      | $d_F$ | $Y_F$          | T1                                        | T2                                        | T3                                                       |
|-------|----------|----------|----------------|----------|---------------|-------|----------------|-------------------------------------------|-------------------------------------------|----------------------------------------------------------|
| T3-A  | -2       | 1        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | —                                                        |
| T3-A  | -1       | 1        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | $+g_Y^2\left(2g_2^2-3\lambda_{H2}^{\text{Adj}}\right)$   |
| T3-A  | 0        | 1        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | —                                                        |
| T3-A  | 1        | 1        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | $+g_Y^2\left(6g_2^2-15\lambda_{H2}^{\text{Adj}}\right)$  |
| T3-A  | 2        | 1        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | —                                                        |
| T3-B  | -2       | 2        | -1             | 2        | 0             | 1     | $-\frac{1}{2}$ | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | —                                                        |
| T3-B  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 1     | 0              | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | $+g_Y^2\left(6g_2^2-3\lambda_{H2}^{\text{Adj}}\right)$   |
| T3-B  | 0        | 2        | 0              | 2        | 1             | 1     | $\frac{1}{2}$  | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | —                                                        |
| T3-B  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 1     | 1              | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | $+g_Y^2\left(18g_2^2-15\lambda_{H2}^{\text{Adj}}\right)$ |
| T3-B  | 2        | 2        | 1              | 2        | 2             | 1     | $\frac{3}{2}$  | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | —                                                        |
| T3-C  | -2       | 2        | -1             | 2        | 0             | 3     | $-\frac{1}{2}$ | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | —                                                        |
| T3-C  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 3     | 0              | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | $+g_Y^2\left(6g_2^2-3\lambda_{H2}^{\text{Adj}}\right)$   |
| T3-C  | 0        | 2        | 0              | 2        | 1             | 3     | $\frac{1}{2}$  | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | —                                                        |
| T3-C  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 3     | 1              | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | $+g_Y^2\left(18g_2^2-15\lambda_{H2}^{\text{Adj}}\right)$ |
| T3-C  | 2        | 2        | 1              | 2        | 2             | 3     | $\frac{3}{2}$  | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | —                                                        |
| T3-D  | -2       | 3        | -1             | 1        | 0             | 2     | $-\frac{1}{2}$ | —                                         | —                                         | —                                                        |
| T3-D  | -1       | 3        | $-\frac{1}{2}$ | 1        | $\frac{1}{2}$ | 2     | 0              | —                                         | —                                         | —                                                        |
| T3-D  | 0        | 3        | 0              | 1        | 1             | 2     | $\frac{1}{2}$  | —                                         | —                                         | —                                                        |
| T3-D  | 1        | 3        | $\frac{1}{2}$  | 1        | $\frac{3}{2}$ | 2     | 1              | —                                         | —                                         | —                                                        |
| T3-D  | 2        | 3        | 1              | 1        | 2             | 2     | $\frac{3}{2}$  | —                                         | —                                         | —                                                        |
| T3-E  | -2       | 3        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | —                                                        |
| T3-E  | -1       | 3        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | $+g_Y^2\left(2g_2^2-3\lambda_{H2}^{\text{Adj}}\right)$   |
| T3-E  | 0        | 3        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | —                                                        |
| T3-E  | 1        | 3        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | $+g_Y^2\left(6g_2^2-15\lambda_{H2}^{\text{Adj}}\right)$  |
| T3-E  | 2        | 3        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+8\lambda_{H2}\lambda_{H2}^{\text{Adj}}$ | $+2\lambda_{H2}^{\text{Adj}}\lambda_{S2}$ | —                                                        |

Terms 4 of 7

| Column | Representative term structure                                                                                                                                                                                                            |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $+g_Y{}^2\left(4g_2{}^2-\frac{15\lambda_{H2}^{\text{Adj}}}{2}\right)$                                                                                                                                                                    |
| T2     | $\lambda_{H2}^{\text{Adj}}(2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$ |
| T3     | $+8\text{Tr}\left(y_2\left(y_2\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$                                                                                                                                                        |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                     | T2                                                                                                                                                                                                                                       | T3                                                                                |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                      | —                                                                                                                                                                                                                                        | —                                                                                 |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                      | —                                                                                                                                                                                                                                        | —                                                                                 |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+g_Y{}^2\left(4g_2{}^2-\frac{15\lambda_{H2}^{\text{Adj}}}{2}\right)$  | —                                                                                                                                                                                                                                        | —                                                                                 |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                      | —                                                                                                                                                                                                                                        | —                                                                                 |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+g_Y{}^2\left(8g_2{}^2-\frac{51\lambda_{H2}^{\text{Adj}}}{2}\right)$  | —                                                                                                                                                                                                                                        | —                                                                                 |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                      | $\lambda_{H2}^{\text{Adj}}(2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$ | $+8\text{Tr}\left(y_2\left(y_2\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                      | $\lambda_{H2}^{\text{Adj}}(2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$ | $+8\text{Tr}\left(y_2\left(y_2\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $+g_Y{}^2\left(12g_2{}^2-\frac{15\lambda_{H2}^{\text{Adj}}}{2}\right)$ | $\lambda_{H2}^{\text{Adj}}(2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$ | $+8\text{Tr}\left(y_2\left(y_2\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                      | $\lambda_{H2}^{\text{Adj}}(2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$ | $+8\text{Tr}\left(y_2\left(y_2\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+g_Y{}^2\left(24g_2{}^2-\frac{51\lambda_{H2}^{\text{Adj}}}{2}\right)$ | $\lambda_{H2}^{\text{Adj}}(2\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$ | $+8\text{Tr}\left(y_2\left(y_2\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                      | $\lambda_{H2}^{\text{Adj}}(3\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$ | $-4\text{Tr}\left(y_2\left(y_2\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                      | $\lambda_{H2}^{\text{Adj}}(3\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$ | $-4\text{Tr}\left(y_2\left(y_2\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $+g_Y{}^2\left(12g_2{}^2-\frac{15\lambda_{H2}^{\text{Adj}}}{2}\right)$ | $\lambda_{H2}^{\text{Adj}}(3\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$ | $-4\text{Tr}\left(y_2\left(y_2\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                      | $\lambda_{H2}^{\text{Adj}}(3\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$ | $-4\text{Tr}\left(y_2\left(y_2\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+g_Y{}^2\left(24g_2{}^2-\frac{51\lambda_{H2}^{\text{Adj}}}{2}\right)$ | $\lambda_{H2}^{\text{Adj}}(3\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$ | $-4\text{Tr}\left(y_2\left(y_2\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                      | —                                                                                                                                                                                                                                        | —                                                                                 |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                      | —                                                                                                                                                                                                                                        | —                                                                                 |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                      | —                                                                                                                                                                                                                                        | —                                                                                 |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                      | —                                                                                                                                                                                                                                        | —                                                                                 |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                      | —                                                                                                                                                                                                                                        | —                                                                                 |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                      | —                                                                                                                                                                                                                                        | —                                                                                 |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                      | —                                                                                                                                                                                                                                        | —                                                                                 |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+g_Y{}^2\left(4g_2{}^2-\frac{15\lambda_{H2}^{\text{Adj}}}{2}\right)$  | —                                                                                                                                                                                                                                        | —                                                                                 |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                      | —                                                                                                                                                                                                                                        | —                                                                                 |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$         | T1                                                                     | T2 | T3 |
|-------|----------|-----------|-----------|-----------|-----------|-------|---------------|------------------------------------------------------------------------|----|----|
| T3-E  | 2        | 3         | 1         | 3         | 2         | 2     | $\frac{3}{2}$ | $+g_Y{}^2\left(8g_2{}^2-\frac{51\lambda_{H^2}^{\text{Adj}}}{2}\right)$ | —  | —  |

Terms 5 of 7

| Column | Representative term structure                         |
|--------|-------------------------------------------------------|
| T1     | $+16\lambda_{T3}(\lambda_{T3})^*$                     |
| T2     | $-9g_2^2\lambda_{H2}^{\text{Adj}}$                    |
| T3     | $+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                | T2                                 | T3                                                    |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------|------------------------------------|-------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                 | —                                  | —                                                     |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                 | —                                  | —                                                     |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                 | —                                  | —                                                     |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                 | —                                  | —                                                     |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                 | —                                  | —                                                     |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | $+16\lambda_{T3}(\lambda_{T3})^*$ | $-9g_2^2\lambda_{H2}^{\text{Adj}}$ | $+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$ |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+16\lambda_{T3}(\lambda_{T3})^*$ | $-9g_2^2\lambda_{H2}^{\text{Adj}}$ | $+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | $+16\lambda_{T3}(\lambda_{T3})^*$ | $-9g_2^2\lambda_{H2}^{\text{Adj}}$ | $+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$ |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | $+16\lambda_{T3}(\lambda_{T3})^*$ | $-9g_2^2\lambda_{H2}^{\text{Adj}}$ | $+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$ |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | $+16\lambda_{T3}(\lambda_{T3})^*$ | $-9g_2^2\lambda_{H2}^{\text{Adj}}$ | $+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$ |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | $+16\lambda_{T3}(\lambda_{T3})^*$ | $-9g_2^2\lambda_{H2}^{\text{Adj}}$ | $+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$ |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+16\lambda_{T3}(\lambda_{T3})^*$ | $-9g_2^2\lambda_{H2}^{\text{Adj}}$ | $+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | $+16\lambda_{T3}(\lambda_{T3})^*$ | $-9g_2^2\lambda_{H2}^{\text{Adj}}$ | $+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$ |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | $+16\lambda_{T3}(\lambda_{T3})^*$ | $-9g_2^2\lambda_{H2}^{\text{Adj}}$ | $+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$ |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | $+16\lambda_{T3}(\lambda_{T3})^*$ | $-9g_2^2\lambda_{H2}^{\text{Adj}}$ | $+\lambda_{12}^{\text{Adj}}\lambda_{H1}^{\text{Adj}}$ |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                 | —                                  | —                                                     |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                 | —                                  | —                                                     |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                 | —                                  | —                                                     |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                 | —                                  | —                                                     |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                 | —                                  | —                                                     |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                 | —                                  | —                                                     |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                 | —                                  | —                                                     |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                 | —                                  | —                                                     |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                 | —                                  | —                                                     |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                 | —                                  | —                                                     |

## Terms 6 of 7

| Column | Representative term structure                                                                                                                                                                                                              |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(-2\text{Tr}(y_1 (y_2)^\dagger) - 4(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^* + \left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)^*))$ |
| T2     | $-2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^* \text{Tr}(y_2 (y_1)^\dagger)$                                                                                                                                |
| T3     | $-4\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$                                                                                                                    |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                                                                                                         | T2                                                                                                          | T3                                                                                                                      |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                          | —                                                                                                           | —                                                                                                                       |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                          | —                                                                                                           | —                                                                                                                       |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                          | —                                                                                                           | —                                                                                                                       |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                          | —                                                                                                           | —                                                                                                                       |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                          | —                                                                                                           | —                                                                                                                       |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                          | —                                                                                                           | —                                                                                                                       |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(-2\text{Tr}(y_1 (y_2)^\dagger) - 4(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^* + \left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)^*))$ | $-2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^* \text{Tr}(y_2 (y_1)^\dagger)$ | $-4\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                          | —                                                                                                           | —                                                                                                                       |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                                                                                                          | —                                                                                                           | —                                                                                                                       |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                          | —                                                                                                           | —                                                                                                                       |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                          | —                                                                                                           | —                                                                                                                       |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(-3\text{Tr}(y_1 (y_2)^\dagger) - 4(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^* + \left(\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\right)^*))$ | $-3\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^* \text{Tr}(y_2 (y_1)^\dagger)$ | $-4\lambda_{S_1^\dagger S_2 S_2^{\dagger 2}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                          | —                                                                                                           | —                                                                                                                       |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                                                                                                          | —                                                                                                           | —                                                                                                                       |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                          | —                                                                                                           | —                                                                                                                       |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                          | —                                                                                                           | —                                                                                                                       |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                          | —                                                                                                           | —                                                                                                                       |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                          | —                                                                                                           | —                                                                                                                       |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                          | —                                                                                                           | —                                                                                                                       |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                          | —                                                                                                           | —                                                                                                                       |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                          | —                                                                                                           | —                                                                                                                       |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                          | —                                                                                                           | —                                                                                                                       |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                          | —                                                                                                           | —                                                                                                                       |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                          | —                                                                                                           | —                                                                                                                       |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                          | —                                                                                                           | —                                                                                                                       |

Terms 7 of 7

| Column | Representative term structure                                                                     |
|--------|---------------------------------------------------------------------------------------------------|
| T1     | $+64\lambda_{H^\dagger H^\dagger S_2 S_2} \left( \lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$ |
| T2     | $+12\lambda_{12}^{\text{Adj}} \lambda_{H1}^{\text{Adj}}$                                          |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                | T2                                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                 | —                                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                 | —                                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                 | —                                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                 | —                                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                 | —                                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                 | —                                                        |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+64\lambda_{H^\dagger H^\dagger S_2 S_2} \left( \lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$ | —                                                        |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                 | —                                                        |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                 | —                                                        |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                 | —                                                        |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                 | —                                                        |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+64\lambda_{H^\dagger H^\dagger S_2 S_2} \left( \lambda_{H^\dagger H^\dagger S_2 S_2} \right)^*$ | —                                                        |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                 | —                                                        |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                 | —                                                        |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                 | —                                                        |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                 | —                                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                 | —                                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                 | —                                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                 | —                                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                 | —                                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                 | $+12\lambda_{12}^{\text{Adj}} \lambda_{H1}^{\text{Adj}}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                 | $+12\lambda_{12}^{\text{Adj}} \lambda_{H1}^{\text{Adj}}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                 | $+12\lambda_{12}^{\text{Adj}} \lambda_{H1}^{\text{Adj}}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                 | $+12\lambda_{12}^{\text{Adj}} \lambda_{H1}^{\text{Adj}}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                 | $+12\lambda_{12}^{\text{Adj}} \lambda_{H1}^{\text{Adj}}$ |

$\beta_{\lambda}^{(1)}_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$

Search name: lambdaHdagSibarSibar

$$16\pi^2\,\mu\frac{d\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}{d\mu}=\beta_{\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}^{(1)}$$

Terms 1 of 4

| Column | Representative term structure                                                                                                                                                                                               |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} (2\text{Tr}\big(y_1\ (y_1)^\dagger\big) + 6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big) + 2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big) + 6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$ |
| T2     | $+ (\lambda_{T3})^* \text{Tr}\big(y_2\ (y_1)^\dagger\big)$                                                                                                                                                                  |
| T3     | $-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (\lambda_{T3})^*$                                                                                                                                            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                                                                                                                                                   | T2 | T3                                                                               |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                    | —  | —                                                                                |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                    | —  | —                                                                                |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                    | —  | —                                                                                |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                    | —  | —                                                                                |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                    | —  | —                                                                                |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                    | —  | —                                                                                |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} (2\text{Tr}\big(y_1\ (y_1)^\dagger\big) + 6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big) + 2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big) + 6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)) + (\lambda_{T3})^* \text{Tr}\big(y_2\ (y_1)^\dagger\big)$ |    | $-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (\lambda_{T3})^*$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                    | —  | —                                                                                |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                                                                                                                                                    | —  | —                                                                                |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                    | —  | —                                                                                |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                    | —  | —                                                                                |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} (3\text{Tr}\big(y_1\ (y_1)^\dagger\big) + 6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big) + 2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big) + 6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$                                                          | —  | $-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (\lambda_{T3})^*$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                    | —  | —                                                                                |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                                                                                                                                                    | —  | —                                                                                |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                    | —  | —                                                                                |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                    | —  | —                                                                                |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                    | —  | —                                                                                |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                    | —  | —                                                                                |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                    | —  | —                                                                                |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                    | —  | —                                                                                |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                    | —  | —                                                                                |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                    | —  | —                                                                                |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                    | —  | —                                                                                |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                    | —  | —                                                                                |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                    | —  | —                                                                                |

Terms 2 of 4

| Column | Representative term structure                                    |
|--------|------------------------------------------------------------------|
| T1     | $+2\lambda_{S_1S_1^\dagger{}^2S_2^\dagger}(\lambda_{T3})^*$      |
| T2     | $-9g_2{}^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$ |
| T3     | $-3g_Y{}^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                          | T2                                                               | T3                                                               |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------------|------------------------------------------------------------------|------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                           | —                                                                | —                                                                |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                           | —                                                                | —                                                                |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                           | —                                                                | —                                                                |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                           | —                                                                | —                                                                |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                           | —                                                                | —                                                                |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                           | —                                                                | —                                                                |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+2\lambda_{S_1S_1^\dagger{}^2S_2^\dagger}(\lambda_{T3})^*$ | $-9g_2{}^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$ | $-3g_Y{}^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                           | —                                                                | —                                                                |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                           | —                                                                | —                                                                |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                           | —                                                                | —                                                                |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                           | —                                                                | —                                                                |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+2\lambda_{S_1S_1^\dagger{}^2S_2^\dagger}(\lambda_{T3})^*$ | $-9g_2{}^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$ | $-3g_Y{}^2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                           | —                                                                | —                                                                |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                           | —                                                                | —                                                                |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                           | —                                                                | —                                                                |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                           | —                                                                | —                                                                |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                           | —                                                                | —                                                                |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                           | —                                                                | —                                                                |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                           | —                                                                | —                                                                |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                           | —                                                                | —                                                                |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                           | —                                                                | —                                                                |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                           | —                                                                | —                                                                |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                           | —                                                                | —                                                                |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                           | —                                                                | —                                                                |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                           | —                                                                | —                                                                |

Terms 3 of 4

| Column | Representative term structure                                                                |
|--------|----------------------------------------------------------------------------------------------|
| T1     | $+2\lambda_H\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$                           |
| T2     | $+8\lambda_{H1}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$                        |
| T3     | $+\frac{4\lambda_{H1}^{\text{Adj}}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}{3}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                 | T2                                                                    | T3                                                                                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                  | —                                                                     | —                                                                                            |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                  | —                                                                     | —                                                                                            |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                  | —                                                                     | —                                                                                            |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                  | —                                                                     | —                                                                                            |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                  | —                                                                     | —                                                                                            |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                  | —                                                                     | —                                                                                            |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+2\lambda_H\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$ | $+8\lambda_{H1}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$ | $+\frac{4\lambda_{H1}^{\text{Adj}}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}{3}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                  | —                                                                     | —                                                                                            |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                  | —                                                                     | —                                                                                            |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                  | —                                                                     | —                                                                                            |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                  | —                                                                     | —                                                                                            |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+2\lambda_H\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$ | $+8\lambda_{H1}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}$ | $+\frac{4\lambda_{H1}^{\text{Adj}}\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}}{3}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                  | —                                                                     | —                                                                                            |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                  | —                                                                     | —                                                                                            |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                  | —                                                                     | —                                                                                            |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                  | —                                                                     | —                                                                                            |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                  | —                                                                     | —                                                                                            |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                  | —                                                                     | —                                                                                            |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                  | —                                                                     | —                                                                                            |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                  | —                                                                     | —                                                                                            |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                  | —                                                                     | —                                                                                            |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                  | —                                                                     | —                                                                                            |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                  | —                                                                     | —                                                                                            |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                  | —                                                                     | —                                                                                            |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                  | —                                                                     | —                                                                                            |

Terms 4 of 4

| Column | Representative term structure                                                       |
|--------|-------------------------------------------------------------------------------------|
| T1     | $+2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \lambda_{S_1}$             |
| T2     | $+4\lambda_{H^\dagger H^\dagger S_2 S_2} \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$ |
| T3     | $+\frac{3}{2}(\lambda_{T3})^* \text{Tr}\Big(y_2 \ (y_1)^\dagger\Big)$               |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                      | T2                                                                                  | T3                                                                    |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                       | —                                                                                   | —                                                                     |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                       | —                                                                                   | —                                                                     |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                       | —                                                                                   | —                                                                     |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                       | —                                                                                   | —                                                                     |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                       | —                                                                                   | —                                                                     |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                       | —                                                                                   | —                                                                     |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \lambda_{S_1}$ | $+4\lambda_{H^\dagger H^\dagger S_2 S_2} \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$ | —                                                                     |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                       | —                                                                                   | —                                                                     |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                       | —                                                                                   | —                                                                     |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                       | —                                                                                   | —                                                                     |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                       | —                                                                                   | —                                                                     |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+2\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \lambda_{S_1}$ | $+4\lambda_{H^\dagger H^\dagger S_2 S_2} \lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}$ | $+\frac{3}{2}(\lambda_{T3})^* \text{Tr}\Big(y_2 \ (y_1)^\dagger\Big)$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                       | —                                                                                   | —                                                                     |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                       | —                                                                                   | —                                                                     |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                       | —                                                                                   | —                                                                     |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                       | —                                                                                   | —                                                                     |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                       | —                                                                                   | —                                                                     |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                       | —                                                                                   | —                                                                     |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                       | —                                                                                   | —                                                                     |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                       | —                                                                                   | —                                                                     |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                       | —                                                                                   | —                                                                     |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                       | —                                                                                   | —                                                                     |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                       | —                                                                                   | —                                                                     |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                       | —                                                                                   | —                                                                     |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                       | —                                                                                   | —                                                                     |

$$\beta_{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}^{(1)}$$

Search name:  $\lambda_{\text{bdaHd}gS_1\text{bar}S_2\text{barCross}}$

$$16\pi^2\mu\frac{d\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{d\mu}=\beta_{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}^{(1)}$$

# Terms 1 of 5

| Column | Representative term structure                                                                                                                                                                                                  |  |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| T1     | $\lambda_{H1}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+3\lambda_{S_1S_1^{\dagger 2}S_2^\dagger}\right)$                                  |  |
| T2     | $+\lambda_{H1}^{\text{Adj}}\left(-\frac{1}{8}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)-\frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}-\frac{\lambda_{S_1S_1^{\dagger 2}S_2^\dagger}}{4}\right)$ |  |
| T3     | $+\lambda_{H2}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+3\lambda_{S_1^\dagger S_2S_2^{\dagger 2}}\right)$                                |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | T2 | T3 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —  | —  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —  | —  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —  | —  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —  | —  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —  | —  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —  | —  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\lambda_{H1}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+3\lambda_{S_1S_1^{\dagger 2}S_2^\dagger}\right)+\lambda_{H1}^{\text{Adj}}\left(-\frac{1}{8}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)-\frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}-\frac{\lambda_{S_1S_1^{\dagger 2}S_2^\dagger}}{4}\right)+\lambda_{H2}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+3\lambda_{S_1^\dagger S_2S_2^{\dagger 2}}\right)$  |    |    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —  | —  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —  | —  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —  | —  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —  | —  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $\lambda_{H1}\left(\frac{3}{4}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+3\lambda_{S_1S_1^{\dagger 2}S_2^\dagger}\right)+\lambda_{H1}^{\text{Adj}}\left(-\frac{3}{16}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)-\frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}-\frac{\lambda_{S_1S_1^{\dagger 2}S_2^\dagger}}{4}\right)+\lambda_{H2}\left(\frac{3}{4}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+3\lambda_{S_1^\dagger S_2S_2^{\dagger 2}}\right)$ |    |    |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —  | —  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —  | —  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —  | —  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —  | —  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —  | —  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —  | —  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —  | —  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —  | —  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —  | —  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —  | —  |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —  | —  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | —  | —  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$ | $d_{S_2}$ | $Y_{S_2}$ | $d_F$ | $Y_F$         | T1 | T2 | T3 |
|-------|----------|-----------|-----------|-----------|-----------|-------|---------------|----|----|----|
| T3-E  | 2        | 3         | 1         | 3         | 2         | 2     | $\frac{3}{2}$ | —  | —  | —  |

Terms 2 of 5

| Column | Representative term structure                                                                                                                                                                                                                                                                                                                         |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $+\lambda_{H2}^{\text{Adj}}\left(-\frac{3}{8}\text{Tr}\left(y_2\left(y_1\right)^{\dagger}\right)-\frac{3\lambda^{\text{Cross}}_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}}{2}-\frac{3\lambda_{S_1^{\dagger}S_2S_2^{\dagger}{}^2}}{4}\right)$                                                                                                            |
| T2     | $+\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}\left(\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)\right)$ |
| T3     | $-4\text{Tr}\left(y_2\left(y_1\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$                                                                                                                                                                                                                                                                  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | T2                                                                                   | T3 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|----|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | —                                                                                    | —  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | —                                                                                    | —  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | —                                                                                    | —  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | —                                                                                    | —  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | —                                                                                    | —  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | —                                                                                    | —  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Adj}}\left(-\frac{3}{8}\text{Tr}\left(y_2\left(y_1\right)^{\dagger}\right)-\frac{3\lambda^{\text{Cross}}_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}}{2}-\frac{3\lambda_{S_1^{\dagger}S_2S_2^{\dagger}{}^2}}{4}\right)+\lambda_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}^{\text{Cross}}\left(\text{Tr}\left(y_1\left(y_1\right)^{\dagger}\right)+\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)\right)$ | $-4\text{Tr}\left(y_2\left(y_1\right)^{\dagger}Y_e\left(Y_e\right)^{\dagger}\right)$ |    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | —                                                                                    | —  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | —                                                                                    | —  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | —                                                                                    | —  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | —                                                                                    | —  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+\lambda_{H2}^{\text{Adj}}\left(-\frac{9}{16}\text{Tr}\left(y_2\left(y_1\right)^{\dagger}\right)-\frac{3\lambda^{\text{Cross}}_{HH^{\dagger}S_1^{\dagger}S_2^{\dagger}}}{2}-\frac{3\lambda_{S_1^{\dagger}S_2S_2^{\dagger}{}^2}}{4}\right)$                                                                                                                                                                                                                                                                                                                                                                                       | —                                                                                    | —  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | —                                                                                    | —  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | —                                                                                    | —  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | —                                                                                    | —  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | —                                                                                    | —  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | —                                                                                    | —  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | —                                                                                    | —  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | —                                                                                    | —  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | —                                                                                    | —  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | —                                                                                    | —  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | —                                                                                    | —  |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | —                                                                                    | —  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | —                                                                                    | —  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | —                                                                                    | —  |

Terms 3 of 5

| Column | Representative term structure                                                                                          |
|--------|------------------------------------------------------------------------------------------------------------------------|
| T1     | $+6\lambda_{S_1^\dagger 2 S_2^\dagger 2} \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ |
| T2     | $-6 \left( \lambda_{H^\dagger H^\dagger S_2 S_2} \right)^* (\lambda_{T3})^*$                                           |
| T3     | $-9g_2^2 \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$                                                  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                     | T2                                                                           | T3                                                                    |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                      | —                                                                            | —                                                                     |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                      | —                                                                            | —                                                                     |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                      | —                                                                            | —                                                                     |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                      | —                                                                            | —                                                                     |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                      | —                                                                            | —                                                                     |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                      | —                                                                            | —                                                                     |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+6\lambda_{S_1^\dagger 2 S_2^\dagger 2} \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ | $-6 \left( \lambda_{H^\dagger H^\dagger S_2 S_2} \right)^* (\lambda_{T3})^*$ | $-9g_2^2 \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                      | —                                                                            | —                                                                     |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                      | —                                                                            | —                                                                     |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                      | —                                                                            | —                                                                     |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                      | —                                                                            | —                                                                     |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+6\lambda_{S_1^\dagger 2 S_2^\dagger 2} \left( \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$ | $-6 \left( \lambda_{H^\dagger H^\dagger S_2 S_2} \right)^* (\lambda_{T3})^*$ | $-9g_2^2 \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                      | —                                                                            | —                                                                     |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                      | —                                                                            | —                                                                     |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                      | —                                                                            | —                                                                     |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                      | —                                                                            | —                                                                     |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                      | —                                                                            | —                                                                     |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                      | —                                                                            | —                                                                     |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                      | —                                                                            | —                                                                     |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                      | —                                                                            | —                                                                     |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                      | —                                                                            | —                                                                     |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                      | —                                                                            | —                                                                     |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                      | —                                                                            | —                                                                     |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                      | —                                                                            | —                                                                     |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                      | —                                                                            | —                                                                     |

Terms 4 of 5

| Column | Representative term structure                                               |
|--------|-----------------------------------------------------------------------------|
| T1     | $-3g_Y^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$        |
| T2     | $+2\lambda_{12}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$ |
| T3     | $+3\lambda_H\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$    |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                   | T2                                                                          | T3                                                                       |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------|--------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                    | —                                                                           | —                                                                        |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                    | —                                                                           | —                                                                        |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                    | —                                                                           | —                                                                        |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                    | —                                                                           | —                                                                        |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                    | —                                                                           | —                                                                        |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                    | —                                                                           | —                                                                        |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-3g_Y^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$ | $+2\lambda_{12}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$ | $+3\lambda_H\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                    | —                                                                           | —                                                                        |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                    | —                                                                           | —                                                                        |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                    | —                                                                           | —                                                                        |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                    | —                                                                           | —                                                                        |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-3g_Y^2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$ | $+2\lambda_{12}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$ | $+3\lambda_H\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                    | —                                                                           | —                                                                        |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                    | —                                                                           | —                                                                        |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                    | —                                                                           | —                                                                        |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                    | —                                                                           | —                                                                        |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                    | —                                                                           | —                                                                        |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                    | —                                                                           | —                                                                        |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                    | —                                                                           | —                                                                        |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                    | —                                                                           | —                                                                        |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                    | —                                                                           | —                                                                        |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                    | —                                                                           | —                                                                        |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                    | —                                                                           | —                                                                        |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                    | —                                                                           | —                                                                        |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                    | —                                                                           | —                                                                        |

Terms 5 of 5

| Column | Representative term structure                                                                                                                                                                                                                                                                                                                                 |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $-6\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}^{\lambda_{T3}}$                                                                                                                                                                                                                                                                                      |
| T2     | $+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{3}{2}\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+\frac{3}{2}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                       | T2                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                        | —                                                                                                                                                                                                                                                                                                                                                             |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                        | —                                                                                                                                                                                                                                                                                                                                                             |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                        | —                                                                                                                                                                                                                                                                                                                                                             |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                        | —                                                                                                                                                                                                                                                                                                                                                             |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                        | —                                                                                                                                                                                                                                                                                                                                                             |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                        | —                                                                                                                                                                                                                                                                                                                                                             |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-6\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}^{\lambda_{T3}}$ | —                                                                                                                                                                                                                                                                                                                                                             |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                        | —                                                                                                                                                                                                                                                                                                                                                             |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                        | —                                                                                                                                                                                                                                                                                                                                                             |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                        | —                                                                                                                                                                                                                                                                                                                                                             |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                        | —                                                                                                                                                                                                                                                                                                                                                             |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-6\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}^{\lambda_{T3}}$ | $+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{2}{3}\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+\frac{2}{3}\text{Tr}\left(y_2\left(y_2\right)^\dagger\right)+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                        | —                                                                                                                                                                                                                                                                                                                                                             |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                        | —                                                                                                                                                                                                                                                                                                                                                             |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                        | —                                                                                                                                                                                                                                                                                                                                                             |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                        | —                                                                                                                                                                                                                                                                                                                                                             |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                        | —                                                                                                                                                                                                                                                                                                                                                             |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                        | —                                                                                                                                                                                                                                                                                                                                                             |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                        | —                                                                                                                                                                                                                                                                                                                                                             |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                        | —                                                                                                                                                                                                                                                                                                                                                             |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                        | —                                                                                                                                                                                                                                                                                                                                                             |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                        | —                                                                                                                                                                                                                                                                                                                                                             |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                        | —                                                                                                                                                                                                                                                                                                                                                             |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                        | —                                                                                                                                                                                                                                                                                                                                                             |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                        | —                                                                                                                                                                                                                                                                                                                                                             |

$\beta_{\lambda_{H^\dagger H^\dagger S_2 S_2}}^{(1)}$

Search name: lambdaHHdagS2S2

$$16\pi^2\,\mu\frac{d\lambda_{H^\dagger H^\dagger S_2 S_2}}{d\mu}=\beta_{\lambda_{H^\dagger H^\dagger S_2 S_2}}^{(1)}$$

Terms 1 of 4

| Column | Representative term structure                                                                                                                                                                                               |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_{H^\dagger H^\dagger S_2 S_2} (2\, \text{Tr}\big(y_2\, (y_2)^\dagger\big) + 6\, \text{Tr}\big(Y_d\, (Y_d)^\dagger\big) + 2\, \text{Tr}\big(Y_e\, (Y_e)^\dagger\big) + 6\, \text{Tr}\big(Y_u\, (Y_u)^\dagger\big))$ |
| T2     | $+ (\lambda_{T3})^* \, \text{Tr}\big(y_1\, (y_2)^\dagger\big)$                                                                                                                                                              |
| T3     | $+ 4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\right)^*$                                                                                                         |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                                                                                                                                                       | T2 | T3                                                                                                                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|---------------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                        | —  | —                                                                                                                   |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                        | —  | —                                                                                                                   |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                        | —  | —                                                                                                                   |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                        | —  | —                                                                                                                   |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                        | —  | —                                                                                                                   |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                        | —  | —                                                                                                                   |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\lambda_{H^\dagger H^\dagger S_2 S_2} (2\, \text{Tr}\big(y_2\, (y_2)^\dagger\big) + 6\, \text{Tr}\big(Y_d\, (Y_d)^\dagger\big) + 2\, \text{Tr}\big(Y_e\, (Y_e)^\dagger\big) + 6\, \text{Tr}\big(Y_u\, (Y_u)^\dagger\big)) + (\lambda_{T3})^* \, \text{Tr}\big(y_1\, (y_2)^\dagger\big)$ |    | $+ 4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\right)^*$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                        | —  | —                                                                                                                   |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                                                                                                                                                        | —  | —                                                                                                                   |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                        | —  | —                                                                                                                   |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                        | —  | —                                                                                                                   |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $\lambda_{H^\dagger H^\dagger S_2 S_2} (3\, \text{Tr}\big(y_2\, (y_2)^\dagger\big) + 6\, \text{Tr}\big(Y_d\, (Y_d)^\dagger\big) + 2\, \text{Tr}\big(Y_e\, (Y_e)^\dagger\big) + 6\, \text{Tr}\big(Y_u\, (Y_u)^\dagger\big))$                                                              | —  | $+ 4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger} \left(\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}\right)^*$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                        | —  | —                                                                                                                   |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                                                                                                                                                        | —  | —                                                                                                                   |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                        | —  | —                                                                                                                   |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                        | —  | —                                                                                                                   |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                        | —  | —                                                                                                                   |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                        | —  | —                                                                                                                   |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                        | —  | —                                                                                                                   |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                        | —  | —                                                                                                                   |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                        | —  | —                                                                                                                   |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                        | —  | —                                                                                                                   |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                        | —  | —                                                                                                                   |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                        | —  | —                                                                                                                   |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                        | —  | —                                                                                                                   |

Terms 2 of 4

| Column | Representative term structure                                                                                                                                                 |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $+2\left(\lambda_{T3}\right)^*\left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\right)$ |
| T2     | $-9g_2^2\lambda_{H^\dagger H^\dagger S_2 S_2}$                                                                                                                                |
| T3     | $-3g_Y^2\lambda_{H^\dagger H^\dagger S_2 S_2}$                                                                                                                                |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                                            | T2                                             | T3                                             |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                             | —                                              | —                                              |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                             | —                                              | —                                              |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                             | —                                              | —                                              |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                             | —                                              | —                                              |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                             | —                                              | —                                              |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                                             | —                                              | —                                              |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+2\left(\lambda_{T3}\right)^*\left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\right)$ | $-9g_2^2\lambda_{H^\dagger H^\dagger S_2 S_2}$ | $-3g_Y^2\lambda_{H^\dagger H^\dagger S_2 S_2}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                                             | —                                              | —                                              |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                                             | —                                              | —                                              |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                                             | —                                              | —                                              |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                                             | —                                              | —                                              |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+2\left(\lambda_{T3}\right)^*\left(\left(\lambda_{S_1^\dagger S_2 S_2^\dagger 2}\right)^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\right)$ | $-9g_2^2\lambda_{H^\dagger H^\dagger S_2 S_2}$ | $-3g_Y^2\lambda_{H^\dagger H^\dagger S_2 S_2}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                                             | —                                              | —                                              |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                                             | —                                              | —                                              |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                                             | —                                              | —                                              |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                             | —                                              | —                                              |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                             | —                                              | —                                              |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                             | —                                              | —                                              |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                             | —                                              | —                                              |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                             | —                                              | —                                              |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                             | —                                              | —                                              |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                             | —                                              | —                                              |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                             | —                                              | —                                              |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                             | —                                              | —                                              |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                             | —                                              | —                                              |

Terms 3 of 4

| Column | Representative term structure                                      |
|--------|--------------------------------------------------------------------|
| T1     | $+2\lambda_H\lambda_{H^\dagger H^\dagger S_2 S_2}$                 |
| T2     | $+8\lambda_{H2}\lambda_{H^\dagger H^\dagger S_2 S_2}$              |
| T3     | $+4\lambda_{H2}^{\text{Adj}}\lambda_{H^\dagger H^\dagger S_2 S_2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                 | T2                                                    | T3                                                                 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------------------------|-------------------------------------------------------|--------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                  | —                                                     | —                                                                  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                  | —                                                     | —                                                                  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                  | —                                                     | —                                                                  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                  | —                                                     | —                                                                  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                  | —                                                     | —                                                                  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                  | —                                                     | —                                                                  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+2\lambda_H\lambda_{H^\dagger H^\dagger S_2 S_2}$ | $+8\lambda_{H2}\lambda_{H^\dagger H^\dagger S_2 S_2}$ | $+4\lambda_{H2}^{\text{Adj}}\lambda_{H^\dagger H^\dagger S_2 S_2}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                  | —                                                     | —                                                                  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                  | —                                                     | —                                                                  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                  | —                                                     | —                                                                  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                  | —                                                     | —                                                                  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+2\lambda_H\lambda_{H^\dagger H^\dagger S_2 S_2}$ | $+8\lambda_{H2}\lambda_{H^\dagger H^\dagger S_2 S_2}$ | $+4\lambda_{H2}^{\text{Adj}}\lambda_{H^\dagger H^\dagger S_2 S_2}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                  | —                                                     | —                                                                  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                  | —                                                     | —                                                                  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                  | —                                                     | —                                                                  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                  | —                                                     | —                                                                  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                  | —                                                     | —                                                                  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                  | —                                                     | —                                                                  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                  | —                                                     | —                                                                  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                  | —                                                     | —                                                                  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                  | —                                                     | —                                                                  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                  | —                                                     | —                                                                  |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                  | —                                                     | —                                                                  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                  | —                                                     | —                                                                  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                  | —                                                     | —                                                                  |

Terms 4 of 4

| Column | Representative term structure                                        |
|--------|----------------------------------------------------------------------|
| T1     | $+2\lambda_{H^\dagger H^\dagger S_2 S_2} \lambda_{S_2}$              |
| T2     | $+\frac{3}{2}(\lambda_{T3})^* \text{Tr}\big(y_1\ (y_2)^\dagger\big)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                      | T2                                                                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------|----------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                       | —                                                                    |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                       | —                                                                    |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                       | —                                                                    |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                       | —                                                                    |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                       | —                                                                    |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                       | —                                                                    |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+2\lambda_{H^\dagger H^\dagger S_2 S_2} \lambda_{S_2}$ | —                                                                    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                       | —                                                                    |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                       | —                                                                    |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                       | —                                                                    |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                       | —                                                                    |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+2\lambda_{H^\dagger H^\dagger S_2 S_2} \lambda_{S_2}$ | $+\frac{3}{2}(\lambda_{T3})^* \text{Tr}\big(y_1\ (y_2)^\dagger\big)$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                       | —                                                                    |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                       | —                                                                    |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                       | —                                                                    |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                       | —                                                                    |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                       | —                                                                    |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                       | —                                                                    |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                       | —                                                                    |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                       | —                                                                    |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                       | —                                                                    |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                       | —                                                                    |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                       | —                                                                    |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                       | —                                                                    |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                       | —                                                                    |

$\beta_{\lambda_{S_1}^{\text{Adj}}}^{(1)}$

Search name: lambdaS1Adj

$$16\pi^2\mu\frac{d\lambda_{S_1}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{S_1}^{\text{Adj}}}^{(1)}$$

Terms 1 of 6

| Column | Representative term structure                                                                                         |
|--------|-----------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_{S_1}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+13\lambda_{S_1}^{\text{Adj}}\right)$ |
| T2     | $+4\lambda_{S_1}^{\text{Adj}}\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)$                                       |
| T3     | $-\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$                                      |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                    | T2                                                                              | T3                                                                               |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                     | —                                                                               | —                                                                                |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                     | —                                                                               | —                                                                                |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                     | —                                                                               | —                                                                                |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                     | —                                                                               | —                                                                                |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                     | —                                                                               | —                                                                                |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                     | —                                                                               | —                                                                                |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                                                                     | —                                                                               | —                                                                                |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                     | —                                                                               | —                                                                                |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                     | —                                                                               | —                                                                                |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                     | —                                                                               | —                                                                                |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                     | —                                                                               | —                                                                                |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                                                                     | —                                                                               | —                                                                                |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                     | —                                                                               | —                                                                                |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                     | —                                                                               | —                                                                                |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                     | —                                                                               | —                                                                                |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $\lambda_{S_1}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+13\lambda_{S_1}^{\text{Adj}}\right)$ | $+4\lambda_{S_1}^{\text{Adj}}\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)$ | $-\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $\lambda_{S_1}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+13\lambda_{S_1}^{\text{Adj}}\right)$ | $+4\lambda_{S_1}^{\text{Adj}}\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)$ | $-\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $\lambda_{S_1}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+13\lambda_{S_1}^{\text{Adj}}\right)$ | $+4\lambda_{S_1}^{\text{Adj}}\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)$ | $-\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $\lambda_{S_1}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+13\lambda_{S_1}^{\text{Adj}}\right)$ | $+4\lambda_{S_1}^{\text{Adj}}\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)$ | $-\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $\lambda_{S_1}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+13\lambda_{S_1}^{\text{Adj}}\right)$ | $+4\lambda_{S_1}^{\text{Adj}}\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)$ | $-\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\lambda_{S_1}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+13\lambda_{S_1}^{\text{Adj}}\right)$ | $+4\lambda_{S_1}^{\text{Adj}}\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)$ | $-\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\lambda_{S_1}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+13\lambda_{S_1}^{\text{Adj}}\right)$ | $+4\lambda_{S_1}^{\text{Adj}}\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)$ | $-\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\lambda_{S_1}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)+13\lambda_{S_1}^{\text{Adj}}\right)$ | $+4\lambda_{S_1}^{\text{Adj}}\text{Tr}\left(y_1\left(y_1\right)^\dagger\right)$ | $-\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1                                                                                                                    | T2                                                                         | T3                                                              |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------|
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | $\lambda_{S_1} \left( \frac{1}{2} \text{Tr} \left( y_1 (y_1)^\dagger \right) + 13 \lambda_{S_1}^{\text{Adj}} \right)$ | $+4 \lambda_{S_1}^{\text{Adj}} \text{Tr} \left( y_1 (y_1)^\dagger \right)$ | $-\text{Tr} \left( y_1 (y_1)^\dagger y_1 (y_1)^\dagger \right)$ |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | $\lambda_{S_1} \left( \frac{1}{2} \text{Tr} \left( y_1 (y_1)^\dagger \right) + 13 \lambda_{S_1}^{\text{Adj}} \right)$ | $+4 \lambda_{S_1}^{\text{Adj}} \text{Tr} \left( y_1 (y_1)^\dagger \right)$ | $-\text{Tr} \left( y_1 (y_1)^\dagger y_1 (y_1)^\dagger \right)$ |

Terms 2 of 6

| Column | Representative term structure                                                            |
|--------|------------------------------------------------------------------------------------------|
| T1     | $+ \frac{3g_2^4}{4}$                                                                     |
| T2     | $+g_Y^2 \left( 9g_2^2 - \frac{3\lambda_{S_1}}{2} - 12\lambda_{S_1}^{\text{Adj}} \right)$ |
| T3     | $+g_2^2 (-3\lambda_{S_1} - 24\lambda_{S_1}^{\text{Adj}})$                                |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                   | T2                                                                                       | T3                                                        |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                    | —                                                                                        | —                                                         |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                    | —                                                                                        | —                                                         |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                    | —                                                                                        | —                                                         |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                    | —                                                                                        | —                                                         |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                    | —                                                                                        | —                                                         |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                    | —                                                                                        | —                                                         |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                    | —                                                                                        | —                                                         |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                    | —                                                                                        | —                                                         |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                    | —                                                                                        | —                                                         |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                    | —                                                                                        | —                                                         |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                    | —                                                                                        | —                                                         |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                    | —                                                                                        | —                                                         |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                    | —                                                                                        | —                                                         |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                    | —                                                                                        | —                                                         |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                    | —                                                                                        | —                                                         |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | $+ \frac{3g_2^4}{4}$ | $+g_Y^2 \left( 9g_2^2 - \frac{3\lambda_{S_1}}{2} - 12\lambda_{S_1}^{\text{Adj}} \right)$ | $+g_2^2 (-3\lambda_{S_1} - 24\lambda_{S_1}^{\text{Adj}})$ |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $+ \frac{3g_2^4}{4}$ | —                                                                                        | $+g_2^2 (-3\lambda_{S_1} - 24\lambda_{S_1}^{\text{Adj}})$ |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | $+ \frac{3g_2^4}{4}$ | —                                                                                        | $+g_2^2 (-3\lambda_{S_1} - 24\lambda_{S_1}^{\text{Adj}})$ |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $+ \frac{3g_2^4}{4}$ | —                                                                                        | $+g_2^2 (-3\lambda_{S_1} - 24\lambda_{S_1}^{\text{Adj}})$ |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | $+ \frac{3g_2^4}{4}$ | $+g_Y^2 \left( 9g_2^2 - \frac{3\lambda_{S_1}}{2} - 12\lambda_{S_1}^{\text{Adj}} \right)$ | $+g_2^2 (-3\lambda_{S_1} - 24\lambda_{S_1}^{\text{Adj}})$ |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+ \frac{3g_2^4}{4}$ | $+g_Y^2 \left( 9g_2^2 - \frac{3\lambda_{S_1}}{2} - 12\lambda_{S_1}^{\text{Adj}} \right)$ | $+g_2^2 (-3\lambda_{S_1} - 24\lambda_{S_1}^{\text{Adj}})$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+ \frac{3g_2^4}{4}$ | —                                                                                        | $+g_2^2 (-3\lambda_{S_1} - 24\lambda_{S_1}^{\text{Adj}})$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+ \frac{3g_2^4}{4}$ | —                                                                                        | $+g_2^2 (-3\lambda_{S_1} - 24\lambda_{S_1}^{\text{Adj}})$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+ \frac{3g_2^4}{4}$ | —                                                                                        | $+g_2^2 (-3\lambda_{S_1} - 24\lambda_{S_1}^{\text{Adj}})$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+ \frac{3g_2^4}{4}$ | $+g_Y^2 \left( 9g_2^2 - \frac{3\lambda_{S_1}}{2} - 12\lambda_{S_1}^{\text{Adj}} \right)$ | $+g_2^2 (-3\lambda_{S_1} - 24\lambda_{S_1}^{\text{Adj}})$ |

Terms 3 of 6

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $+\frac{3g_Y^4}{2}$           |
| T2     | $+\frac{\lambda_{12}^2}{4}$   |
| T3     | $+\frac{\lambda_{H1}^2}{2}$   |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$       | $d_{S2}$ | $Y_{S2}$      | $d_F$ | $Y_F$          | T1                   | T2                          | T3                          |
|-------|----------|----------|----------------|----------|---------------|-------|----------------|----------------------|-----------------------------|-----------------------------|
| T3-A  | -2       | 1        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | —                    | —                           | —                           |
| T3-A  | -1       | 1        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | —                    | —                           | —                           |
| T3-A  | 0        | 1        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | —                    | —                           | —                           |
| T3-A  | 1        | 1        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | —                    | —                           | —                           |
| T3-A  | 2        | 1        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | —                    | —                           | —                           |
| T3-B  | -2       | 2        | -1             | 2        | 0             | 1     | $-\frac{1}{2}$ | —                    | —                           | —                           |
| T3-B  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 1     | 0              | —                    | —                           | —                           |
| T3-B  | 0        | 2        | 0              | 2        | 1             | 1     | $\frac{1}{2}$  | —                    | —                           | —                           |
| T3-B  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 1     | 1              | —                    | —                           | —                           |
| T3-B  | 2        | 2        | 1              | 2        | 2             | 1     | $\frac{3}{2}$  | —                    | —                           | —                           |
| T3-C  | -2       | 2        | -1             | 2        | 0             | 3     | $-\frac{1}{2}$ | —                    | —                           | —                           |
| T3-C  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 3     | 0              | —                    | —                           | —                           |
| T3-C  | 0        | 2        | 0              | 2        | 1             | 3     | $\frac{1}{2}$  | —                    | —                           | —                           |
| T3-C  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 3     | 1              | —                    | —                           | —                           |
| T3-C  | 2        | 2        | 1              | 2        | 2             | 3     | $\frac{3}{2}$  | —                    | —                           | —                           |
| T3-D  | -2       | 3        | -1             | 1        | 0             | 2     | $-\frac{1}{2}$ | $+\frac{3g_Y^4}{2}$  | $+\frac{\lambda_{12}^2}{4}$ | $+\frac{\lambda_{H1}^2}{2}$ |
| T3-D  | -1       | 3        | $-\frac{1}{2}$ | 1        | $\frac{1}{2}$ | 2     | 0              | $+\frac{3g_Y^4}{32}$ | $+\frac{\lambda_{12}^2}{4}$ | $+\frac{\lambda_{H1}^2}{2}$ |
| T3-D  | 0        | 3        | 0              | 1        | 1             | 2     | $\frac{1}{2}$  | —                    | $+\frac{\lambda_{12}^2}{4}$ | $+\frac{\lambda_{H1}^2}{2}$ |
| T3-D  | 1        | 3        | $\frac{1}{2}$  | 1        | $\frac{3}{2}$ | 2     | 1              | $+\frac{3g_Y^4}{32}$ | $+\frac{\lambda_{12}^2}{4}$ | $+\frac{\lambda_{H1}^2}{2}$ |
| T3-D  | 2        | 3        | 1              | 1        | 2             | 2     | $\frac{3}{2}$  | $+\frac{3g_Y^4}{2}$  | $+\frac{\lambda_{12}^2}{4}$ | $+\frac{\lambda_{H1}^2}{2}$ |
| T3-E  | -2       | 3        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+\frac{3g_Y^4}{2}$  | —                           | $+\frac{\lambda_{H1}^2}{2}$ |
| T3-E  | -1       | 3        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+\frac{3g_Y^4}{32}$ | —                           | $+\frac{\lambda_{H1}^2}{2}$ |
| T3-E  | 0        | 3        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | —                    | —                           | $+\frac{\lambda_{H1}^2}{2}$ |
| T3-E  | 1        | 3        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+\frac{3g_Y^4}{32}$ | —                           | $+\frac{\lambda_{H1}^2}{2}$ |
| T3-E  | 2        | 3        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+\frac{3g_Y^4}{2}$  | —                           | $+\frac{\lambda_{H1}^2}{2}$ |

Terms 4 of 6

| Column | Representative term structure    |  |
|--------|----------------------------------|--|
| T1     | $+3\lambda_{H1}^{\text{Adj}^2}$  |  |
| T2     | $+\frac{7\lambda_{S1}^2}{4}$     |  |
| T3     | $+22\lambda_{S1}^{\text{Adj}^2}$ |  |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$       | $d_{S2}$ | $Y_{S2}$      | $d_F$ | $Y_F$          | T1                              | T2                           | T3                               |
|-------|----------|----------|----------------|----------|---------------|-------|----------------|---------------------------------|------------------------------|----------------------------------|
| T3-A  | -2       | 1        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | —                               | —                            | —                                |
| T3-A  | -1       | 1        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | —                               | —                            | —                                |
| T3-A  | 0        | 1        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | —                               | —                            | —                                |
| T3-A  | 1        | 1        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | —                               | —                            | —                                |
| T3-A  | 2        | 1        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | —                               | —                            | —                                |
| T3-B  | -2       | 2        | -1             | 2        | 0             | 1     | $-\frac{1}{2}$ | —                               | —                            | —                                |
| T3-B  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 1     | 0              | —                               | —                            | —                                |
| T3-B  | 0        | 2        | 0              | 2        | 1             | 1     | $\frac{1}{2}$  | —                               | —                            | —                                |
| T3-B  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 1     | 1              | —                               | —                            | —                                |
| T3-B  | 2        | 2        | 1              | 2        | 2             | 1     | $\frac{3}{2}$  | —                               | —                            | —                                |
| T3-C  | -2       | 2        | -1             | 2        | 0             | 3     | $-\frac{1}{2}$ | —                               | —                            | —                                |
| T3-C  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 3     | 0              | —                               | —                            | —                                |
| T3-C  | 0        | 2        | 0              | 2        | 1             | 3     | $\frac{1}{2}$  | —                               | —                            | —                                |
| T3-C  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 3     | 1              | —                               | —                            | —                                |
| T3-C  | 2        | 2        | 1              | 2        | 2             | 3     | $\frac{3}{2}$  | —                               | —                            | —                                |
| T3-D  | -2       | 3        | -1             | 1        | 0             | 2     | $-\frac{1}{2}$ | $+3\lambda_{H1}^{\text{Adj}^2}$ | $+\frac{7\lambda_{S1}^2}{4}$ | $+22\lambda_{S1}^{\text{Adj}^2}$ |
| T3-D  | -1       | 3        | $-\frac{1}{2}$ | 1        | $\frac{1}{2}$ | 2     | 0              | $+3\lambda_{H1}^{\text{Adj}^2}$ | $+\frac{7\lambda_{S1}^2}{4}$ | $+22\lambda_{S1}^{\text{Adj}^2}$ |
| T3-D  | 0        | 3        | 0              | 1        | 1             | 2     | $\frac{1}{2}$  | $+3\lambda_{H1}^{\text{Adj}^2}$ | $+\frac{7\lambda_{S1}^2}{4}$ | $+22\lambda_{S1}^{\text{Adj}^2}$ |
| T3-D  | 1        | 3        | $\frac{1}{2}$  | 1        | $\frac{3}{2}$ | 2     | 1              | $+3\lambda_{H1}^{\text{Adj}^2}$ | $+\frac{7\lambda_{S1}^2}{4}$ | $+22\lambda_{S1}^{\text{Adj}^2}$ |
| T3-D  | 2        | 3        | 1              | 1        | 2             | 2     | $\frac{3}{2}$  | $+3\lambda_{H1}^{\text{Adj}^2}$ | $+\frac{7\lambda_{S1}^2}{4}$ | $+22\lambda_{S1}^{\text{Adj}^2}$ |
| T3-E  | -2       | 3        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+3\lambda_{H1}^{\text{Adj}^2}$ | $+\frac{7\lambda_{S1}^2}{4}$ | $+22\lambda_{S1}^{\text{Adj}^2}$ |
| T3-E  | -1       | 3        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+3\lambda_{H1}^{\text{Adj}^2}$ | $+\frac{7\lambda_{S1}^2}{4}$ | $+22\lambda_{S1}^{\text{Adj}^2}$ |
| T3-E  | 0        | 3        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+3\lambda_{H1}^{\text{Adj}^2}$ | $+\frac{7\lambda_{S1}^2}{4}$ | $+22\lambda_{S1}^{\text{Adj}^2}$ |
| T3-E  | 1        | 3        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+3\lambda_{H1}^{\text{Adj}^2}$ | $+\frac{7\lambda_{S1}^2}{4}$ | $+22\lambda_{S1}^{\text{Adj}^2}$ |
| T3-E  | 2        | 3        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+3\lambda_{H1}^{\text{Adj}^2}$ | $+\frac{7\lambda_{S1}^2}{4}$ | $+22\lambda_{S1}^{\text{Adj}^2}$ |

Terms 5 of 6

| Column | Representative term structure                                                              |
|--------|--------------------------------------------------------------------------------------------|
| T1     | $+g_Y^2\left(\frac{9g_2^2}{4}-\frac{3\lambda_{S_1}}{8}-3\lambda_{S_1}^{\text{Adj}}\right)$ |
| T2     | $+\frac{3\lambda_{12}^2}{4}$                                                               |
| T3     | $+\lambda_{12}\lambda_{12}^{\text{Cross}}$                                                 |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                         | T2                           | T3                                         |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------------------------------|------------------------------|--------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                          | —                            | —                                          |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                          | —                            | —                                          |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                          | —                            | —                                          |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                          | —                            | —                                          |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                          | —                            | —                                          |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                          | —                            | —                                          |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                                          | —                            | —                                          |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                          | —                            | —                                          |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                          | —                            | —                                          |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                          | —                            | —                                          |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                          | —                            | —                                          |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                                          | —                            | —                                          |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                          | —                            | —                                          |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                          | —                            | —                                          |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                          | —                            | —                                          |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                          | —                            | —                                          |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | $+g_Y^2\left(\frac{9g_2^2}{4}-\frac{3\lambda_{S_1}}{8}-3\lambda_{S_1}^{\text{Adj}}\right)$ | —                            | —                                          |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                          | —                            | —                                          |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | $+g_Y^2\left(\frac{9g_2^2}{4}-\frac{3\lambda_{S_1}}{8}-3\lambda_{S_1}^{\text{Adj}}\right)$ | —                            | —                                          |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                          | —                            | —                                          |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                          | $+\frac{3\lambda_{12}^2}{4}$ | $+\lambda_{12}\lambda_{12}^{\text{Cross}}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+g_Y^2\left(\frac{9g_2^2}{4}-\frac{3\lambda_{S_1}}{8}-3\lambda_{S_1}^{\text{Adj}}\right)$ | $+\frac{3\lambda_{12}^2}{4}$ | $+\lambda_{12}\lambda_{12}^{\text{Cross}}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                          | $+\frac{3\lambda_{12}^2}{4}$ | $+\lambda_{12}\lambda_{12}^{\text{Cross}}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+g_Y^2\left(\frac{9g_2^2}{4}-\frac{3\lambda_{S_1}}{8}-3\lambda_{S_1}^{\text{Adj}}\right)$ | $+\frac{3\lambda_{12}^2}{4}$ | $+\lambda_{12}\lambda_{12}^{\text{Cross}}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                          | $+\frac{3\lambda_{12}^2}{4}$ | $+\lambda_{12}\lambda_{12}^{\text{Cross}}$ |

Terms 6 of 6

| Column | Representative term structure              |
|--------|--------------------------------------------|
| T1     | $+12\lambda_{12}^{\text{Adj}^2}$           |
| T2     | $-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                               | T2                                         |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|----------------------------------|--------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                | —                                          |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                | —                                          |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                | —                                          |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                | —                                          |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                | —                                          |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                | —                                          |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                | —                                          |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                | —                                          |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                | —                                          |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                | —                                          |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                | —                                          |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                | —                                          |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                | —                                          |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                | —                                          |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                | —                                          |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                | —                                          |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                | —                                          |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                | —                                          |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                | —                                          |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                | —                                          |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+12\lambda_{12}^{\text{Adj}^2}$ | $-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+12\lambda_{12}^{\text{Adj}^2}$ | $-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+12\lambda_{12}^{\text{Adj}^2}$ | $-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+12\lambda_{12}^{\text{Adj}^2}$ | $-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+12\lambda_{12}^{\text{Adj}^2}$ | $-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$ |

$$16\pi^2\,\mu\frac{d\lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}}}{d\mu}=\beta_{\lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}}}^{(1)}$$

Terms 1 of 5

| Column | Representative term structure                                                                                                                                                                              |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_{12}(\text{Tr}\big(y_2\ (y_1)^{\dagger}\big)+6\lambda_{S_1^{\dagger} S_2 S_2^{\dagger 2}}+6\lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}})$                                                          |
| T2     | $+ \lambda_{12}^{\text{Adj}}\left(\frac{1}{12}\text{Tr}\big(y_2\ (y_1)^{\dagger}\big)+\frac{\lambda_{S_1^{\dagger} S_2 S_2^{\dagger 2}}}{6}+\frac{5\lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}}}{6}\right)$ |
| T3     | $+ \lambda_{S_1}(\text{Tr}\big(y_2\ (y_1)^{\dagger}\big)+12\lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}})$                                                                                                   |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                                                                                                                                                                                                                                                                                                                            | T2 | T3 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\lambda_{12}(\text{Tr}\big(y_2\ (y_1)^{\dagger}\big)+6\lambda_{S_1^{\dagger} S_2 S_2^{\dagger 2}}+6\lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}})+\lambda_{12}^{\text{Adj}}\left(\frac{1}{12}\text{Tr}\big(y_2\ (y_1)^{\dagger}\big)+\frac{\lambda_{S_1^{\dagger} S_2 S_2^{\dagger 2}}}{6}+\frac{5\lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}}}{6}\right)+\lambda_{S_1}(\text{Tr}\big(y_2\ (y_1)^{\dagger}\big)+12\lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}})$ |    |    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+\lambda_{12}^{\text{Adj}}\left(\frac{1}{8}\text{Tr}\big(y_2\ (y_1)^{\dagger}\big)+\frac{\lambda_{S_1^{\dagger} S_2 S_2^{\dagger 2}}}{6}+\frac{5\lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}}}{6}\right)$                                                                                                                                                                                                                                                      | —  | —  |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —  | —  |

Terms 2 of 5

| Column | Representative term structure                                                                                                                                                                                                                |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $+ \lambda_{S_1^\dagger 2 S_2^\dagger 2} (2 \operatorname{Tr} \left( y_1 \left( y_2 \right)^\dagger \right) + 4 ( \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + 5 \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* ) )$ |
| T2     | $+ \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} (3 \operatorname{Tr} \left( y_1 \left( y_1 \right)^\dagger \right) + \operatorname{Tr} \left( y_2 \left( y_2 \right)^\dagger \right) )$                                                           |
| T3     | $- 4 \operatorname{Tr} \left( y_1 \left( y_1 \right)^\dagger y_2 \left( y_1 \right)^\dagger \right)$                                                                                                                                         |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                                                                                                                                                                                                                                                                                            | T2                                                                                                   | T3                                                                                                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                             | —                                                                                                    | —                                                                                                    |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                                                             | —                                                                                                    | —                                                                                                    |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                             | —                                                                                                    | —                                                                                                    |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                             | —                                                                                                    | —                                                                                                    |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                             | —                                                                                                    | —                                                                                                    |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                             | —                                                                                                    | —                                                                                                    |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+ \lambda_{S_1^\dagger 2 S_2^\dagger 2} (2 \operatorname{Tr} \left( y_1 \left( y_2 \right)^\dagger \right) + 4 ( \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + 5 \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* ) ) + \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} (3 \operatorname{Tr} \left( y_1 \left( y_1 \right)^\dagger \right) + \operatorname{Tr} \left( y_2 \left( y_2 \right)^\dagger \right) )$ | $- 4 \operatorname{Tr} \left( y_1 \left( y_1 \right)^\dagger y_2 \left( y_1 \right)^\dagger \right)$ |                                                                                                      |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                             | —                                                                                                    | —                                                                                                    |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                             | —                                                                                                    | —                                                                                                    |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                             | —                                                                                                    | —                                                                                                    |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                             | —                                                                                                    | —                                                                                                    |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+ \lambda_{S_1^\dagger 2 S_2^\dagger 2} (3 \operatorname{Tr} \left( y_1 \left( y_2 \right)^\dagger \right) + 4 ( \left( \lambda_{S_1^\dagger S_2 S_2^\dagger 2} \right)^* + 5 \left( \lambda_{S_1 S_1^\dagger 2 S_2^\dagger} \right)^* ) )$                                                                                                                                                                                  | —                                                                                                    | $- 5 \operatorname{Tr} \left( y_1 \left( y_1 \right)^\dagger y_2 \left( y_1 \right)^\dagger \right)$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                             | —                                                                                                    | —                                                                                                    |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                             | —                                                                                                    | —                                                                                                    |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                             | —                                                                                                    | —                                                                                                    |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                             | —                                                                                                    | —                                                                                                    |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                                                             | —                                                                                                    | —                                                                                                    |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                             | —                                                                                                    | —                                                                                                    |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                             | —                                                                                                    | —                                                                                                    |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                             | —                                                                                                    | —                                                                                                    |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                             | —                                                                                                    | —                                                                                                    |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                                                             | —                                                                                                    | —                                                                                                    |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                             | —                                                                                                    | —                                                                                                    |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                             | —                                                                                                    | —                                                                                                    |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                             | —                                                                                                    | —                                                                                                    |

Terms 3 of 5

| Column | Representative term structure                                               |
|--------|-----------------------------------------------------------------------------|
| T1     | $-9g_2^2\lambda_{S_1S_1^\dagger S_2^\dagger}$                               |
| T2     | $-3g_Y^2\lambda_{S_1S_1^\dagger S_2^\dagger}$                               |
| T3     | $+2\lambda_{H1}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                            | T2                                            | T3                                                                          |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                             | —                                             | —                                                                           |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                             | —                                             | —                                                                           |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                             | —                                             | —                                                                           |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                             | —                                             | —                                                                           |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                             | —                                             | —                                                                           |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                             | —                                             | —                                                                           |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-9g_2^2\lambda_{S_1S_1^\dagger S_2^\dagger}$ | $-3g_Y^2\lambda_{S_1S_1^\dagger S_2^\dagger}$ | $+2\lambda_{H1}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                             | —                                             | —                                                                           |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                             | —                                             | —                                                                           |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                             | —                                             | —                                                                           |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                             | —                                             | —                                                                           |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-9g_2^2\lambda_{S_1S_1^\dagger S_2^\dagger}$ | $-3g_Y^2\lambda_{S_1S_1^\dagger S_2^\dagger}$ | $+2\lambda_{H1}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                             | —                                             | —                                                                           |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                             | —                                             | —                                                                           |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                             | —                                             | —                                                                           |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                             | —                                             | —                                                                           |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                             | —                                             | —                                                                           |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                             | —                                             | —                                                                           |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                             | —                                             | —                                                                           |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                             | —                                             | —                                                                           |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                             | —                                             | —                                                                           |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                             | —                                             | —                                                                           |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                             | —                                             | —                                                                           |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                             | —                                             | —                                                                           |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                             | —                                             | —                                                                           |

Terms 4 of 5

| Column | Representative term structure                                                                                                                                             |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $-\frac{\lambda_{H1}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{6}$                                                                         |
| T2     | $+4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}^{\lambda T3}$                                                                                                    |
| T3     | $\lambda_{12}\left(\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+6\lambda_{S_1^\dagger S_2 S_2^\dagger 2}+6\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                | T2                                                                     | T3                                                                                                                                                                        |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                 | —                                                                      | —                                                                                                                                                                         |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                 | —                                                                      | —                                                                                                                                                                         |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                 | —                                                                      | —                                                                                                                                                                         |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                 | —                                                                      | —                                                                                                                                                                         |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                 | —                                                                      | —                                                                                                                                                                         |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                 | —                                                                      | —                                                                                                                                                                         |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-\frac{\lambda_{H1}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{6}$ | $+4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}^{\lambda T3}$ | —                                                                                                                                                                         |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                 | —                                                                      | —                                                                                                                                                                         |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                 | —                                                                      | —                                                                                                                                                                         |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                 | —                                                                      | —                                                                                                                                                                         |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                 | —                                                                      | —                                                                                                                                                                         |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-\frac{\lambda_{H1}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{6}$ | $+4\lambda_{H^\dagger H^\dagger S_1^\dagger S_1^\dagger}^{\lambda T3}$ | $\lambda_{12}\left(\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+6\lambda_{S_1^\dagger S_2 S_2^\dagger 2}+6\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                 | —                                                                      | —                                                                                                                                                                         |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                 | —                                                                      | —                                                                                                                                                                         |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                 | —                                                                      | —                                                                                                                                                                         |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                 | —                                                                      | —                                                                                                                                                                         |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                 | —                                                                      | —                                                                                                                                                                         |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                 | —                                                                      | —                                                                                                                                                                         |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                 | —                                                                      | —                                                                                                                                                                         |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                 | —                                                                      | —                                                                                                                                                                         |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                 | —                                                                      | —                                                                                                                                                                         |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                 | —                                                                      | —                                                                                                                                                                         |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                 | —                                                                      | —                                                                                                                                                                         |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                 | —                                                                      | —                                                                                                                                                                         |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                 | —                                                                      | —                                                                                                                                                                         |

Terms 5 of 5

| Column | Representative term structure                                                                                                                                                                        |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $+ \lambda_{S_1} \left( \frac{3}{2} \text{Tr} \left( y_2 \left( y_1 \right)^\dagger \right) + 12 \lambda_{S_1 S_1^\dagger^2 S_2^\dagger} \right)$                                                    |
| T2     | $+ \lambda_{S_1 S_1^\dagger^2 S_2^\dagger} \left( \frac{9}{2} \text{Tr} \left( y_1 \left( y_1 \right)^\dagger \right) + \frac{3}{2} \text{Tr} \left( y_2 \left( y_2 \right)^\dagger \right) \right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                | T2                                                                                                                                                                                                   |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                 | —                                                                                                                                                                                                    |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                 | —                                                                                                                                                                                                    |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                 | —                                                                                                                                                                                                    |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                 | —                                                                                                                                                                                                    |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                 | —                                                                                                                                                                                                    |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                 | —                                                                                                                                                                                                    |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                                                                                                 | —                                                                                                                                                                                                    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                 | —                                                                                                                                                                                                    |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                 | —                                                                                                                                                                                                    |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                 | —                                                                                                                                                                                                    |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                 | —                                                                                                                                                                                                    |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+ \lambda_{S_1} \left( \frac{3}{2} \text{Tr} \left( y_2 \left( y_1 \right)^\dagger \right) + 12 \lambda_{S_1 S_1^\dagger^2 S_2^\dagger} \right)$ | $+ \lambda_{S_1 S_1^\dagger^2 S_2^\dagger} \left( \frac{9}{2} \text{Tr} \left( y_1 \left( y_1 \right)^\dagger \right) + \frac{3}{2} \text{Tr} \left( y_2 \left( y_2 \right)^\dagger \right) \right)$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                 | —                                                                                                                                                                                                    |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                 | —                                                                                                                                                                                                    |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                 | —                                                                                                                                                                                                    |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                 | —                                                                                                                                                                                                    |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                 | —                                                                                                                                                                                                    |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                 | —                                                                                                                                                                                                    |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                 | —                                                                                                                                                                                                    |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                 | —                                                                                                                                                                                                    |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                 | —                                                                                                                                                                                                    |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                 | —                                                                                                                                                                                                    |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                 | —                                                                                                                                                                                                    |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                 | —                                                                                                                                                                                                    |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                 | —                                                                                                                                                                                                    |

$\beta_{\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}}^{(1)}$

Search name: lambdaS1bar2S2bar2

$$16\pi^2\,\mu\frac{d\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}}{d\mu}=\beta_{\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}}^{(1)}$$

Terms 1 of 6

| Column | Representative term structure                                                                                                          |
|--------|----------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}(2\,\text{Tr}\big(y_1\ (y_1)^{\dagger}\big)+2\,\text{Tr}\big(y_2\ (y_2)^{\dagger}\big))$     |
| T2     | $+ \lambda_{S_1^{\dagger} S_2 S_2^{\dagger 2}}(\text{Tr}\big(y_2\ (y_1)^{\dagger}\big))+2\lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}})$ |
| T3     | $+ \lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}}\text{Tr}\big(y_2\ (y_1)^{\dagger}\big)$                                                 |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                 | T2                                                                                                                                     | T3                                                                                     |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                  | —                                                                                                                                      | —                                                                                      |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                  | —                                                                                                                                      | —                                                                                      |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                  | —                                                                                                                                      | —                                                                                      |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                  | —                                                                                                                                      | —                                                                                      |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                  | —                                                                                                                                      | —                                                                                      |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                  | —                                                                                                                                      | —                                                                                      |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}(2\,\text{Tr}\big(y_1\ (y_1)^{\dagger}\big)+2\,\text{Tr}\big(y_2\ (y_2)^{\dagger}\big))$ | $+ \lambda_{S_1^{\dagger} S_2 S_2^{\dagger 2}}(\text{Tr}\big(y_2\ (y_1)^{\dagger}\big))+2\lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}})$ | $+ \lambda_{S_1 S_1^{\dagger 2} S_2^{\dagger}}\text{Tr}\big(y_2\ (y_1)^{\dagger}\big)$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                  | —                                                                                                                                      | —                                                                                      |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                  | —                                                                                                                                      | —                                                                                      |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                  | —                                                                                                                                      | —                                                                                      |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                  | —                                                                                                                                      | —                                                                                      |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $\lambda_{S_1^{\dagger 2} S_2^{\dagger 2}}(3\,\text{Tr}\big(y_1\ (y_1)^{\dagger}\big)+3\,\text{Tr}\big(y_2\ (y_2)^{\dagger}\big))$ | —                                                                                                                                      | —                                                                                      |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                  | —                                                                                                                                      | —                                                                                      |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                  | —                                                                                                                                      | —                                                                                      |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                  | —                                                                                                                                      | —                                                                                      |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                  | —                                                                                                                                      | —                                                                                      |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                  | —                                                                                                                                      | —                                                                                      |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                  | —                                                                                                                                      | —                                                                                      |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                  | —                                                                                                                                      | —                                                                                      |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                  | —                                                                                                                                      | —                                                                                      |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                  | —                                                                                                                                      | —                                                                                      |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                  | —                                                                                                                                      | —                                                                                      |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                  | —                                                                                                                                      | —                                                                                      |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                  | —                                                                                                                                      | —                                                                                      |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                  | —                                                                                                                                      | —                                                                                      |

Terms 2 of 6

| Column | Representative term structure                                                                                       |
|--------|---------------------------------------------------------------------------------------------------------------------|
| T1     | $-2\text{Tr}\left(y_2\left(y_1\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$                                |
| T2     | $+4\lambda_{H^{\dagger}H^{\dagger}S_1^{\dagger}S_1^{\dagger}}\left(\lambda_{H^{\dagger}H^{\dagger}S_2S_2}\right)^*$ |
| T3     | $-9g_2^2\lambda_{S_1^{\dagger}2S_2^{\dagger}2}$                                                                     |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                   | T2                                                                                                                  | T3                                              |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                    | —                                                                                                                   | —                                               |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                    | —                                                                                                                   | —                                               |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                    | —                                                                                                                   | —                                               |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                    | —                                                                                                                   | —                                               |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                    | —                                                                                                                   | —                                               |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                    | —                                                                                                                   | —                                               |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-2\text{Tr}\left(y_2\left(y_1\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$ | $+4\lambda_{H^{\dagger}H^{\dagger}S_1^{\dagger}S_1^{\dagger}}\left(\lambda_{H^{\dagger}H^{\dagger}S_2S_2}\right)^*$ | $-9g_2^2\lambda_{S_1^{\dagger}2S_2^{\dagger}2}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                    | —                                                                                                                   | —                                               |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                    | —                                                                                                                   | —                                               |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                    | —                                                                                                                   | —                                               |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                    | —                                                                                                                   | —                                               |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                                    | $+4\lambda_{H^{\dagger}H^{\dagger}S_1^{\dagger}S_1^{\dagger}}\left(\lambda_{H^{\dagger}H^{\dagger}S_2S_2}\right)^*$ | $-9g_2^2\lambda_{S_1^{\dagger}2S_2^{\dagger}2}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                    | —                                                                                                                   | —                                               |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                    | —                                                                                                                   | —                                               |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                    | —                                                                                                                   | —                                               |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                    | —                                                                                                                   | —                                               |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                    | —                                                                                                                   | —                                               |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                    | —                                                                                                                   | —                                               |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                    | —                                                                                                                   | —                                               |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                    | —                                                                                                                   | —                                               |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                    | —                                                                                                                   | —                                               |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                    | —                                                                                                                   | —                                               |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                    | —                                                                                                                   | —                                               |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                    | —                                                                                                                   | —                                               |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                    | —                                                                                                                   | —                                               |

Terms 3 of 6

| Column | Representative term structure                                                |
|--------|------------------------------------------------------------------------------|
| T1     | $-3g_Y^2\lambda_{S_1^\dagger 2 S_2^\dagger 2}$                               |
| T2     | $+8\lambda_{12}\lambda_{S_1^\dagger 2 S_2^\dagger 2}$                        |
| T3     | $+\frac{4\lambda_{12}^{\text{Adj}}\lambda_{S_1^\dagger 2 S_2^\dagger 2}}{3}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                             | T2                                                    | T3                                                                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------|-------------------------------------------------------|------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                                     | —                                                                            |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                                     | —                                                                            |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                                     | —                                                                            |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                                     | —                                                                            |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                                     | —                                                                            |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                              | —                                                     | —                                                                            |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $-3g_Y^2\lambda_{S_1^\dagger 2 S_2^\dagger 2}$ | $+8\lambda_{12}\lambda_{S_1^\dagger 2 S_2^\dagger 2}$ | $+\frac{4\lambda_{12}^{\text{Adj}}\lambda_{S_1^\dagger 2 S_2^\dagger 2}}{3}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                              | —                                                     | —                                                                            |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                              | —                                                     | —                                                                            |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                              | —                                                     | —                                                                            |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                              | —                                                     | —                                                                            |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $-3g_Y^2\lambda_{S_1^\dagger 2 S_2^\dagger 2}$ | $+8\lambda_{12}\lambda_{S_1^\dagger 2 S_2^\dagger 2}$ | $+\frac{4\lambda_{12}^{\text{Adj}}\lambda_{S_1^\dagger 2 S_2^\dagger 2}}{3}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                              | —                                                     | —                                                                            |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                              | —                                                     | —                                                                            |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                              | —                                                     | —                                                                            |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                                     | —                                                                            |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                                     | —                                                                            |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                                     | —                                                                            |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                                     | —                                                                            |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                                     | —                                                                            |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                              | —                                                     | —                                                                            |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                              | —                                                     | —                                                                            |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                              | —                                                     | —                                                                            |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                              | —                                                     | —                                                                            |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                              | —                                                     | —                                                                            |

Terms 4 of 6

| Column | Representative term structure                                      |
|--------|--------------------------------------------------------------------|
| T1     | $+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}{}^2$ |
| T2     | $+2\lambda_{S_1}\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}$          |
| T3     | $+2\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\lambda_{S_2}$          |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                 | T2                                                        | T3                                                        |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                  | —                                                         | —                                                         |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                  | —                                                         | —                                                         |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                  | —                                                         | —                                                         |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                  | —                                                         | —                                                         |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                  | —                                                         | —                                                         |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                  | —                                                         | —                                                         |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}{}^2$ | $+2\lambda_{S_1}\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}$ | $+2\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\lambda_{S_2}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                  | —                                                         | —                                                         |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                  | —                                                         | —                                                         |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                  | —                                                         | —                                                         |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                  | —                                                         | —                                                         |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}{}^2$ | $+2\lambda_{S_1}\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}$ | $+2\lambda_{S_1^{\dagger 2}S_2^{\dagger 2}}\lambda_{S_2}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                  | —                                                         | —                                                         |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                  | —                                                         | —                                                         |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                  | —                                                         | —                                                         |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                  | —                                                         | —                                                         |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                  | —                                                         | —                                                         |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                  | —                                                         | —                                                         |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                  | —                                                         | —                                                         |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                  | —                                                         | —                                                         |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                  | —                                                         | —                                                         |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                  | —                                                         | —                                                         |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                  | —                                                         | —                                                         |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                  | —                                                         | —                                                         |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                  | —                                                         | —                                                         |

Terms 5 of 6

| Column | Representative term structure                                                                                                                                    |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $+5\lambda_{S_1^\dagger S_2 S_2^\dagger}{}^2$                                                                                                                    |
| T2     | $+5\lambda_{S_1 S_1^\dagger S_2^\dagger}{}^2$                                                                                                                    |
| T3     | $+\lambda_{S_1^\dagger S_2 S_2^\dagger}{}^2\left(\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+2\lambda_{S_1 S_1^\dagger S_2^\dagger}{}^2\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                            | T2                                            | T3                                                                                                                                                               |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                             | —                                             | —                                                                                                                                                                |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                             | —                                             | —                                                                                                                                                                |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                             | —                                             | —                                                                                                                                                                |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                             | —                                             | —                                                                                                                                                                |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                             | —                                             | —                                                                                                                                                                |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                             | —                                             | —                                                                                                                                                                |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+5\lambda_{S_1^\dagger S_2 S_2^\dagger}{}^2$ | $+5\lambda_{S_1 S_1^\dagger S_2^\dagger}{}^2$ | —                                                                                                                                                                |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                             | —                                             | —                                                                                                                                                                |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                             | —                                             | —                                                                                                                                                                |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                             | —                                             | —                                                                                                                                                                |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                             | —                                             | —                                                                                                                                                                |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+5\lambda_{S_1^\dagger S_2 S_2^\dagger}{}^2$ | $+5\lambda_{S_1 S_1^\dagger S_2^\dagger}{}^2$ | $+\lambda_{S_1^\dagger S_2 S_2^\dagger}{}^2\left(\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+2\lambda_{S_1 S_1^\dagger S_2^\dagger}{}^2\right)$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                             | —                                             | —                                                                                                                                                                |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                             | —                                             | —                                                                                                                                                                |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                             | —                                             | —                                                                                                                                                                |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                             | —                                             | —                                                                                                                                                                |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                             | —                                             | —                                                                                                                                                                |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                             | —                                             | —                                                                                                                                                                |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                             | —                                             | —                                                                                                                                                                |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                             | —                                             | —                                                                                                                                                                |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                             | —                                             | —                                                                                                                                                                |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                             | —                                             | —                                                                                                                                                                |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                             | —                                             | —                                                                                                                                                                |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                             | —                                             | —                                                                                                                                                                |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                             | —                                             | —                                                                                                                                                                |

Terms 6 of 6

| Column | Representative term structure                                                            |
|--------|------------------------------------------------------------------------------------------|
| T1     | $+\frac{3}{2}\lambda_{S_1S_1^\dagger 2S_2^\dagger}\text{Tr}\big(y_2\ (y_1)^\dagger\big)$ |
| T2     | $-\frac{5}{2}\text{Tr}\big(y_2\ (y_1)^\dagger\ y_2\ (y_1)^\dagger\big)$                  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                       | T2                                                                      |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                        | —                                                                       |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                        | —                                                                       |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                        | —                                                                       |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                        | —                                                                       |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                        | —                                                                       |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                        | —                                                                       |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                                        | —                                                                       |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                        | —                                                                       |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                        | —                                                                       |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                        | —                                                                       |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                        | —                                                                       |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+\frac{3}{2}\lambda_{S_1S_1^\dagger 2S_2^\dagger}\text{Tr}\big(y_2\ (y_1)^\dagger\big)$ | $-\frac{5}{2}\text{Tr}\big(y_2\ (y_1)^\dagger\ y_2\ (y_1)^\dagger\big)$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                        | —                                                                       |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                        | —                                                                       |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                        | —                                                                       |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                        | —                                                                       |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                        | —                                                                       |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                        | —                                                                       |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                        | —                                                                       |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                        | —                                                                       |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                        | —                                                                       |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                        | —                                                                       |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                        | —                                                                       |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                        | —                                                                       |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                        | —                                                                       |

$$16\pi^2\,\mu\frac{d\lambda_{S_1^\dagger S_2 S_2^\dagger^2}}{d\mu}=\beta_{\lambda_{S_1^\dagger S_2 S_2^\dagger^2}}^{(1)}$$

Terms 1 of 5

| Column | Representative term structure                                                                                                                                                                       |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_{12}(\text{Tr}\big(y_2\ (y_1)^\dagger\big)+6\lambda_{S_1^\dagger S_2 S_2^\dagger^2}+6\lambda_{S_1 S_1^\dagger^2 S_2^\dagger})$                                                             |
| T2     | $+ \lambda_{12}^{\text{Adj}}\left(\frac{1}{12}\text{Tr}\big(y_2\ (y_1)^\dagger\big)+\frac{5\lambda_{S_1^\dagger S_2 S_2^\dagger^2}}{6}+\frac{\lambda_{S_1 S_1^\dagger^2 S_2^\dagger}}{6}\right)$    |
| T3     | $+ \lambda_{S_1^\dagger^2 S_2^\dagger^2}(2\text{Tr}\big(y_1\ (y_2)^\dagger\big)+4(5\left(\lambda_{S_1^\dagger S_2 S_2^\dagger^2}\right)^*+\left(\lambda_{S_1 S_1^\dagger^2 S_2^\dagger}\right)^*))$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | T2 | T3 |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $\lambda_{12}(\text{Tr}\big(y_2\ (y_1)^\dagger\big)+6\lambda_{S_1^\dagger S_2 S_2^\dagger^2}+6\lambda_{S_1 S_1^\dagger^2 S_2^\dagger})+\lambda_{12}^{\text{Adj}}\left(\frac{1}{12}\text{Tr}\big(y_2\ (y_1)^\dagger\big)+\frac{5\lambda_{S_1^\dagger S_2 S_2^\dagger^2}}{6}+\frac{\lambda_{S_1 S_1^\dagger^2 S_2^\dagger}}{6}\right)+\lambda_{S_1^\dagger^2 S_2^\dagger^2}(2\text{Tr}\big(y_1\ (y_2)^\dagger\big)+4(5\left(\lambda_{S_1^\dagger S_2 S_2^\dagger^2}\right)^*+\left(\lambda_{S_1 S_1^\dagger^2 S_2^\dagger}\right)^*))$ |    |    |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+\lambda_{12}^{\text{Adj}}\left(\frac{1}{8}\text{Tr}\big(y_2\ (y_1)^\dagger\big)+\frac{5\lambda_{S_1^\dagger S_2 S_2^\dagger^2}}{6}+\frac{\lambda_{S_1 S_1^\dagger^2 S_2^\dagger}}{6}\right)+\lambda_{S_1^\dagger^2 S_2^\dagger^2}(3\text{Tr}\big(y_1\ (y_2)^\dagger\big)+4(5\left(\lambda_{S_1^\dagger S_2 S_2^\dagger^2}\right)^*+\left(\lambda_{S_1 S_1^\dagger^2 S_2^\dagger}\right)^*))$                                                                                                                                       |    |    |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | —  | —  |

Terms 2 of 5

| Column | Representative term structure                                                                                                                                     |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $+ \lambda_{S_2} (\text{Tr} \left( y_2 \left( y_1 \right)^\dagger \right) + 12 \lambda_{S_1^\dagger S_2 S_2^\dagger 2})$                                          |
| T2     | $+ \lambda_{S_1^\dagger S_2 S_2^\dagger 2} (\text{Tr} \left( y_1 \left( y_1 \right)^\dagger \right) + 3 \text{Tr} \left( y_2 \left( y_2 \right)^\dagger \right))$ |
| T3     | $- 4 \text{Tr} \left( y_2 \left( y_1 \right)^\dagger y_2 \left( y_2 \right)^\dagger \right)$                                                                      |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                       | T2                                                                                                                                                                | T3                                                                                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                        | —                                                                                                                                                                 | —                                                                                            |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                        | —                                                                                                                                                                 | —                                                                                            |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                        | —                                                                                                                                                                 | —                                                                                            |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                        | —                                                                                                                                                                 | —                                                                                            |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                        | —                                                                                                                                                                 | —                                                                                            |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                        | —                                                                                                                                                                 | —                                                                                            |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+ \lambda_{S_2} (\text{Tr} \left( y_2 \left( y_1 \right)^\dagger \right) + 12 \lambda_{S_1^\dagger S_2 S_2^\dagger 2})$ | $+ \lambda_{S_1^\dagger S_2 S_2^\dagger 2} (\text{Tr} \left( y_1 \left( y_1 \right)^\dagger \right) + 3 \text{Tr} \left( y_2 \left( y_2 \right)^\dagger \right))$ | $- 4 \text{Tr} \left( y_2 \left( y_1 \right)^\dagger y_2 \left( y_2 \right)^\dagger \right)$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                        | —                                                                                                                                                                 | —                                                                                            |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                        | —                                                                                                                                                                 | —                                                                                            |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                        | —                                                                                                                                                                 | —                                                                                            |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                        | —                                                                                                                                                                 | —                                                                                            |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                                                                        | —                                                                                                                                                                 | $- 5 \text{Tr} \left( y_2 \left( y_1 \right)^\dagger y_2 \left( y_2 \right)^\dagger \right)$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                        | —                                                                                                                                                                 | —                                                                                            |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                        | —                                                                                                                                                                 | —                                                                                            |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                        | —                                                                                                                                                                 | —                                                                                            |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                        | —                                                                                                                                                                 | —                                                                                            |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                        | —                                                                                                                                                                 | —                                                                                            |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                        | —                                                                                                                                                                 | —                                                                                            |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                        | —                                                                                                                                                                 | —                                                                                            |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                        | —                                                                                                                                                                 | —                                                                                            |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                        | —                                                                                                                                                                 | —                                                                                            |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                        | —                                                                                                                                                                 | —                                                                                            |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                        | —                                                                                                                                                                 | —                                                                                            |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                        | —                                                                                                                                                                 | —                                                                                            |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                        | —                                                                                                                                                                 | —                                                                                            |

Terms 3 of 5

| Column | Representative term structure                                                       |
|--------|-------------------------------------------------------------------------------------|
| T1     | $+4\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*\left(\lambda_{T3}\right)^*$ |
| T2     | $-9g_2^2\lambda_{S_1^\dagger S_2 S_2^\dagger 2}$                                    |
| T3     | $-3g_Y^2\lambda_{S_1^\dagger S_2 S_2^\dagger 2}$                                    |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                  | T2                                               | T3                                               |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------------------------------------|--------------------------------------------------|--------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                   | —                                                | —                                                |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                   | —                                                | —                                                |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                   | —                                                | —                                                |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                   | —                                                | —                                                |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                   | —                                                | —                                                |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                   | —                                                | —                                                |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+4\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*\left(\lambda_{T3}\right)^*$ | $-9g_2^2\lambda_{S_1^\dagger S_2 S_2^\dagger 2}$ | $-3g_Y^2\lambda_{S_1^\dagger S_2 S_2^\dagger 2}$ |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                   | —                                                | —                                                |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                   | —                                                | —                                                |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                   | —                                                | —                                                |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                   | —                                                | —                                                |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+4\left(\lambda_{H^\dagger H^\dagger S_2 S_2}\right)^*\left(\lambda_{T3}\right)^*$ | $-9g_2^2\lambda_{S_1^\dagger S_2 S_2^\dagger 2}$ | $-3g_Y^2\lambda_{S_1^\dagger S_2 S_2^\dagger 2}$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                   | —                                                | —                                                |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                   | —                                                | —                                                |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                   | —                                                | —                                                |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                   | —                                                | —                                                |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                   | —                                                | —                                                |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                   | —                                                | —                                                |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                   | —                                                | —                                                |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                   | —                                                | —                                                |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                   | —                                                | —                                                |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                   | —                                                | —                                                |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                   | —                                                | —                                                |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                   | —                                                | —                                                |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                   | —                                                | —                                                |

## Terms 4 of 5

| Column | Representative term structure                                                                                                                                    |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $+2\lambda_{H2}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$                                                                                      |
| T2     | $-\frac{\lambda_{H2}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}$                                                                |
| T3     | $\lambda_{12}\left(\frac{3}{2}\text{Tr}\left(y_2\ (y_1)^\dagger\right)+6\lambda_{S_1^\dagger S_2 S_2^\dagger 2}+6\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                          | T2                                                                                                | T3                                                                                                                                                               |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                           | —                                                                                                 | —                                                                                                                                                                |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                           | —                                                                                                 | —                                                                                                                                                                |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                           | —                                                                                                 | —                                                                                                                                                                |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                           | —                                                                                                 | —                                                                                                                                                                |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                           | —                                                                                                 | —                                                                                                                                                                |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                           | —                                                                                                 | —                                                                                                                                                                |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | $+2\lambda_{H2}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$ | $-\frac{\lambda_{H2}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}$ | —                                                                                                                                                                |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                           | —                                                                                                 | —                                                                                                                                                                |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                           | —                                                                                                 | —                                                                                                                                                                |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                           | —                                                                                                 | —                                                                                                                                                                |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                           | —                                                                                                 | —                                                                                                                                                                |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+2\lambda_{H2}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$ | $-\frac{\lambda_{H2}^{\text{Adj}}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}$ | $\lambda_{12}\left(\frac{3}{2}\text{Tr}\left(y_2\ (y_1)^\dagger\right)+6\lambda_{S_1^\dagger S_2 S_2^\dagger 2}+6\lambda_{S_1 S_1^\dagger 2 S_2^\dagger}\right)$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                           | —                                                                                                 | —                                                                                                                                                                |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                           | —                                                                                                 | —                                                                                                                                                                |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                           | —                                                                                                 | —                                                                                                                                                                |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                           | —                                                                                                 | —                                                                                                                                                                |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                           | —                                                                                                 | —                                                                                                                                                                |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                           | —                                                                                                 | —                                                                                                                                                                |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                           | —                                                                                                 | —                                                                                                                                                                |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                           | —                                                                                                 | —                                                                                                                                                                |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                           | —                                                                                                 | —                                                                                                                                                                |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                           | —                                                                                                 | —                                                                                                                                                                |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                           | —                                                                                                 | —                                                                                                                                                                |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                           | —                                                                                                 | —                                                                                                                                                                |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                           | —                                                                                                 | —                                                                                                                                                                |

Terms 5 of 5

| Column | Representative term structure                                                                                                                                                |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1     | $+ \lambda_{S_2} \left( \frac{3}{2} \text{Tr} \left( y_2 \ (y_1)^\dagger \right) + 12 \lambda_{S_1^\dagger S_2 S_2^\dagger} \right)$                                         |
| T2     | $+ \lambda_{S_1^\dagger S_2 S_2^\dagger} \left( \frac{3}{2} \text{Tr} \left( y_1 \ (y_1)^\dagger \right) + \frac{9}{2} \text{Tr} \left( y_2 \ (y_2)^\dagger \right) \right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                                   | T2                                                                                                                                                                           |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                    | —                                                                                                                                                                            |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                    | —                                                                                                                                                                            |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                    | —                                                                                                                                                                            |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                    | —                                                                                                                                                                            |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                    | —                                                                                                                                                                            |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                                    | —                                                                                                                                                                            |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                                                                                    | —                                                                                                                                                                            |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                                    | —                                                                                                                                                                            |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                                    | —                                                                                                                                                                            |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                                    | —                                                                                                                                                                            |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                                    | —                                                                                                                                                                            |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | $+ \lambda_{S_2} \left( \frac{3}{2} \text{Tr} \left( y_2 \ (y_1)^\dagger \right) + 12 \lambda_{S_1^\dagger S_2 S_2^\dagger} \right)$ | $+ \lambda_{S_1^\dagger S_2 S_2^\dagger} \left( \frac{3}{2} \text{Tr} \left( y_1 \ (y_1)^\dagger \right) + \frac{9}{2} \text{Tr} \left( y_2 \ (y_2)^\dagger \right) \right)$ |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                                    | —                                                                                                                                                                            |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                                    | —                                                                                                                                                                            |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                                    | —                                                                                                                                                                            |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                    | —                                                                                                                                                                            |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                    | —                                                                                                                                                                            |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                    | —                                                                                                                                                                            |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                    | —                                                                                                                                                                            |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                    | —                                                                                                                                                                            |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                                    | —                                                                                                                                                                            |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                                    | —                                                                                                                                                                            |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                                    | —                                                                                                                                                                            |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                                    | —                                                                                                                                                                            |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                                    | —                                                                                                                                                                            |

$\beta_{\lambda_{S_2}^{\text{Adj}}}^{(1)}$

Search name: lambdaS2Adj

$$16\pi^2\mu\frac{d\lambda_{S_2}^{\text{Adj}}}{d\mu}=\beta_{\lambda_{S_2}^{\text{Adj}}}^{(1)}$$

Terms 1 of 7

| Column | Representative term structure                                                                                           |
|--------|-------------------------------------------------------------------------------------------------------------------------|
| T1     | $\lambda_{S_2}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+13\lambda_{S_2}^{\text{Adj}}\right)$ |
| T2     | $+4\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)$                                       |
| T3     | $-\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$                                     |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                                                      | T2                                                                                | T3                                                                                  |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\lambda_{S_2}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+13\lambda_{S_2}^{\text{Adj}}\right)$ | $+4\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)$ | $-\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\lambda_{S_2}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+13\lambda_{S_2}^{\text{Adj}}\right)$ | $+4\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)$ | $-\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\lambda_{S_2}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+13\lambda_{S_2}^{\text{Adj}}\right)$ | $+4\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)$ | $-\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $\lambda_{S_2}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+13\lambda_{S_2}^{\text{Adj}}\right)$ | $+4\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)$ | $-\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $\lambda_{S_2}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+13\lambda_{S_2}^{\text{Adj}}\right)$ | $+4\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)$ | $-\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                                                       | —                                                                                 | —                                                                                   |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                                                                       | —                                                                                 | —                                                                                   |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                                                       | —                                                                                 | —                                                                                   |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                                                       | —                                                                                 | —                                                                                   |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                                                       | —                                                                                 | —                                                                                   |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                                                       | —                                                                                 | —                                                                                   |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                                                                       | —                                                                                 | —                                                                                   |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                                                       | —                                                                                 | —                                                                                   |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                                                       | —                                                                                 | —                                                                                   |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                                                       | —                                                                                 | —                                                                                   |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                                                       | —                                                                                 | —                                                                                   |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                                                       | —                                                                                 | —                                                                                   |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                                                       | —                                                                                 | —                                                                                   |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                                                       | —                                                                                 | —                                                                                   |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                                                       | —                                                                                 | —                                                                                   |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $\lambda_{S_2}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+13\lambda_{S_2}^{\text{Adj}}\right)$ | $+4\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)$ | $-\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $\lambda_{S_2}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+13\lambda_{S_2}^{\text{Adj}}\right)$ | $+4\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)$ | $-\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $\lambda_{S_2}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)+13\lambda_{S_2}^{\text{Adj}}\right)$ | $+4\lambda_{S_2}^{\text{Adj}}\text{Tr}\left(y_2\left(y_2\right)^{\dagger}\right)$ | $-\text{Tr}\left(y_2\left(y_2\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$     | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$         | T1                                                                                                                    | T2                                                                         | T3                                                              |
|-------|----------|-----------|---------------|-----------|---------------|-------|---------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------|
| T3-E  | 1        | 3         | $\frac{1}{2}$ | 3         | $\frac{3}{2}$ | 2     | 1             | $\lambda_{S_2} \left( \frac{1}{2} \text{Tr} \left( y_2 (y_2)^\dagger \right) + 13 \lambda_{S_2}^{\text{Adj}} \right)$ | $+4 \lambda_{S_2}^{\text{Adj}} \text{Tr} \left( y_2 (y_2)^\dagger \right)$ | $-\text{Tr} \left( y_2 (y_2)^\dagger y_2 (y_2)^\dagger \right)$ |
| T3-E  | 2        | 3         | 1             | 3         | 2             | 2     | $\frac{3}{2}$ | $\lambda_{S_2} \left( \frac{1}{2} \text{Tr} \left( y_2 (y_2)^\dagger \right) + 13 \lambda_{S_2}^{\text{Adj}} \right)$ | $+4 \lambda_{S_2}^{\text{Adj}} \text{Tr} \left( y_2 (y_2)^\dagger \right)$ | $-\text{Tr} \left( y_2 (y_2)^\dagger y_2 (y_2)^\dagger \right)$ |

Terms 2 of 7

| Column | Representative term structure                          |
|--------|--------------------------------------------------------|
| T1     | $+\frac{3g_2^4}{4}$                                    |
| T2     | $+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$ |
| T3     | $+\frac{\lambda_{12}^2}{4}$                            |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                  | T2                                                     | T3                          |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|---------------------|--------------------------------------------------------|-----------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+\frac{3g_2^4}{4}$ | $+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$ | $+\frac{\lambda_{12}^2}{4}$ |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+\frac{3g_2^4}{4}$ | $+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$ | $+\frac{\lambda_{12}^2}{4}$ |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+\frac{3g_2^4}{4}$ | $+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$ | $+\frac{\lambda_{12}^2}{4}$ |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+\frac{3g_2^4}{4}$ | $+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$ | $+\frac{\lambda_{12}^2}{4}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+\frac{3g_2^4}{4}$ | $+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$ | $+\frac{\lambda_{12}^2}{4}$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                   | —                                                      | —                           |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                   | —                                                      | —                           |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                   | —                                                      | —                           |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                   | —                                                      | —                           |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                   | —                                                      | —                           |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                   | —                                                      | —                           |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                   | —                                                      | —                           |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                   | —                                                      | —                           |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                   | —                                                      | —                           |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                   | —                                                      | —                           |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                   | —                                                      | —                           |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                   | —                                                      | —                           |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                   | —                                                      | —                           |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                   | —                                                      | —                           |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                   | —                                                      | —                           |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+\frac{3g_2^4}{4}$ | $+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$ | —                           |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+\frac{3g_2^4}{4}$ | $+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$ | —                           |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+\frac{3g_2^4}{4}$ | $+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$ | —                           |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+\frac{3g_2^4}{4}$ | $+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$ | —                           |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+\frac{3g_2^4}{4}$ | $+g_2^2(-3\lambda_{S_2}-24\lambda_{S_2}^{\text{Adj}})$ | —                           |

Terms 3 of 7

| Column | Representative term structure |
|--------|-------------------------------|
| T1     | $+\frac{\lambda_{H2}^2}{2}$   |
| T2     | $+3\lambda_{H2}^{Adj^2}$      |
| T3     | $+\frac{7\lambda_{S2}^2}{4}$  |

| Model | $\alpha$ | $d_{S1}$ | $Y_{S1}$       | $d_{S2}$ | $Y_{S2}$      | $d_F$ | $Y_F$          | T1                          | T2                       | T3                           |
|-------|----------|----------|----------------|----------|---------------|-------|----------------|-----------------------------|--------------------------|------------------------------|
| T3-A  | -2       | 1        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+\frac{\lambda_{H2}^2}{2}$ | $+3\lambda_{H2}^{Adj^2}$ | $+\frac{7\lambda_{S2}^2}{4}$ |
| T3-A  | -1       | 1        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+\frac{\lambda_{H2}^2}{2}$ | $+3\lambda_{H2}^{Adj^2}$ | $+\frac{7\lambda_{S2}^2}{4}$ |
| T3-A  | 0        | 1        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+\frac{\lambda_{H2}^2}{2}$ | $+3\lambda_{H2}^{Adj^2}$ | $+\frac{7\lambda_{S2}^2}{4}$ |
| T3-A  | 1        | 1        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+\frac{\lambda_{H2}^2}{2}$ | $+3\lambda_{H2}^{Adj^2}$ | $+\frac{7\lambda_{S2}^2}{4}$ |
| T3-A  | 2        | 1        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+\frac{\lambda_{H2}^2}{2}$ | $+3\lambda_{H2}^{Adj^2}$ | $+\frac{7\lambda_{S2}^2}{4}$ |
| T3-B  | -2       | 2        | -1             | 2        | 0             | 1     | $-\frac{1}{2}$ | —                           | —                        | —                            |
| T3-B  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 1     | 0              | —                           | —                        | —                            |
| T3-B  | 0        | 2        | 0              | 2        | 1             | 1     | $\frac{1}{2}$  | —                           | —                        | —                            |
| T3-B  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 1     | 1              | —                           | —                        | —                            |
| T3-B  | 2        | 2        | 1              | 2        | 2             | 1     | $\frac{3}{2}$  | —                           | —                        | —                            |
| T3-C  | -2       | 2        | -1             | 2        | 0             | 3     | $-\frac{1}{2}$ | —                           | —                        | —                            |
| T3-C  | -1       | 2        | $-\frac{1}{2}$ | 2        | $\frac{1}{2}$ | 3     | 0              | —                           | —                        | —                            |
| T3-C  | 0        | 2        | 0              | 2        | 1             | 3     | $\frac{1}{2}$  | —                           | —                        | —                            |
| T3-C  | 1        | 2        | $\frac{1}{2}$  | 2        | $\frac{3}{2}$ | 3     | 1              | —                           | —                        | —                            |
| T3-C  | 2        | 2        | 1              | 2        | 2             | 3     | $\frac{3}{2}$  | —                           | —                        | —                            |
| T3-D  | -2       | 3        | -1             | 1        | 0             | 2     | $-\frac{1}{2}$ | —                           | —                        | —                            |
| T3-D  | -1       | 3        | $-\frac{1}{2}$ | 1        | $\frac{1}{2}$ | 2     | 0              | —                           | —                        | —                            |
| T3-D  | 0        | 3        | 0              | 1        | 1             | 2     | $\frac{1}{2}$  | —                           | —                        | —                            |
| T3-D  | 1        | 3        | $\frac{1}{2}$  | 1        | $\frac{3}{2}$ | 2     | 1              | —                           | —                        | —                            |
| T3-D  | 2        | 3        | 1              | 1        | 2             | 2     | $\frac{3}{2}$  | —                           | —                        | —                            |
| T3-E  | -2       | 3        | -1             | 3        | 0             | 2     | $-\frac{1}{2}$ | $+\frac{\lambda_{H2}^2}{2}$ | $+3\lambda_{H2}^{Adj^2}$ | $+\frac{7\lambda_{S2}^2}{4}$ |
| T3-E  | -1       | 3        | $-\frac{1}{2}$ | 3        | $\frac{1}{2}$ | 2     | 0              | $+\frac{\lambda_{H2}^2}{2}$ | $+3\lambda_{H2}^{Adj^2}$ | $+\frac{7\lambda_{S2}^2}{4}$ |
| T3-E  | 0        | 3        | 0              | 3        | 1             | 2     | $\frac{1}{2}$  | $+\frac{\lambda_{H2}^2}{2}$ | $+3\lambda_{H2}^{Adj^2}$ | $+\frac{7\lambda_{S2}^2}{4}$ |
| T3-E  | 1        | 3        | $\frac{1}{2}$  | 3        | $\frac{3}{2}$ | 2     | 1              | $+\frac{\lambda_{H2}^2}{2}$ | $+3\lambda_{H2}^{Adj^2}$ | $+\frac{7\lambda_{S2}^2}{4}$ |
| T3-E  | 2        | 3        | 1              | 3        | 2             | 2     | $\frac{3}{2}$  | $+\frac{\lambda_{H2}^2}{2}$ | $+3\lambda_{H2}^{Adj^2}$ | $+\frac{7\lambda_{S2}^2}{4}$ |

Terms 4 of 7

| Column | Representative term structure                                                              |
|--------|--------------------------------------------------------------------------------------------|
| T1     | $+22\lambda_{S_2}^{\text{Adj}^2}$                                                          |
| T2     | $+g_Y^2\left(\frac{9g_2^2}{4}-\frac{3\lambda_{S_2}}{8}-3\lambda_{S_2}^{\text{Adj}}\right)$ |
| T3     | $+\frac{3g_Y^4}{32}$                                                                       |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                | T2                                                                                            | T3                     |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------|-----------------------------------------------------------------------------------------------|------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+22\lambda_{S_2}^{\text{Adj}^2}$ | —                                                                                             | —                      |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+22\lambda_{S_2}^{\text{Adj}^2}$ | $+g_Y^2\left(\frac{9g_2^2}{4}-\frac{3\lambda_{S_2}}{8}-3\lambda_{S_2}^{\text{Adj}}\right)$    | $+\frac{3g_Y^4}{32}$   |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+22\lambda_{S_2}^{\text{Adj}^2}$ | —                                                                                             | $+\frac{3g_Y^4}{2}$    |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+22\lambda_{S_2}^{\text{Adj}^2}$ | $+g_Y^2\left(\frac{81g_2^2}{4}-\frac{27\lambda_{S_2}}{8}-27\lambda_{S_2}^{\text{Adj}}\right)$ | $+\frac{243g_Y^4}{32}$ |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+22\lambda_{S_2}^{\text{Adj}^2}$ | —                                                                                             | —                      |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                 | —                                                                                             | —                      |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                 | —                                                                                             | —                      |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                 | —                                                                                             | —                      |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                 | —                                                                                             | —                      |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                 | —                                                                                             | —                      |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                 | —                                                                                             | —                      |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                 | —                                                                                             | —                      |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                 | —                                                                                             | —                      |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                 | —                                                                                             | —                      |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                 | —                                                                                             | —                      |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                 | —                                                                                             | —                      |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                 | —                                                                                             | —                      |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                 | —                                                                                             | —                      |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                 | —                                                                                             | —                      |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                 | —                                                                                             | —                      |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+22\lambda_{S_2}^{\text{Adj}^2}$ | —                                                                                             | —                      |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+22\lambda_{S_2}^{\text{Adj}^2}$ | $+g_Y^2\left(\frac{9g_2^2}{4}-\frac{3\lambda_{S_2}}{8}-3\lambda_{S_2}^{\text{Adj}}\right)$    | $+\frac{3g_Y^4}{32}$   |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+22\lambda_{S_2}^{\text{Adj}^2}$ | —                                                                                             | $+\frac{3g_Y^4}{2}$    |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+22\lambda_{S_2}^{\text{Adj}^2}$ | $+g_Y^2\left(\frac{81g_2^2}{4}-\frac{27\lambda_{S_2}}{8}-27\lambda_{S_2}^{\text{Adj}}\right)$ | $+\frac{243g_Y^4}{32}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+22\lambda_{S_2}^{\text{Adj}^2}$ | —                                                                                             | —                      |

Terms 5 of 7

| Column | Representative term structure                                                     |  |  |  |  |  |  |  |
|--------|-----------------------------------------------------------------------------------|--|--|--|--|--|--|--|
| T1     | $+g_Y^2\left(9g_2^2-\frac{3\lambda_{S_2}}{2}-12\lambda_{S_2}^{\text{Adj}}\right)$ |  |  |  |  |  |  |  |
| T2     | $+g_Y^2\left(36g_2^2-6\lambda_{S_2}-48\lambda_{S_2}^{\text{Adj}}\right)$          |  |  |  |  |  |  |  |
| T3     | $+24g_Y^4$                                                                        |  |  |  |  |  |  |  |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                                                                | T2                                                                       | T3         |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------|------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                 | —                                                                        | —          |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                 | —                                                                        | —          |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+g_Y^2\left(9g_2^2-\frac{3\lambda_{S_2}}{2}-12\lambda_{S_2}^{\text{Adj}}\right)$ | —                                                                        | —          |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                 | —                                                                        | —          |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                 | $+g_Y^2\left(36g_2^2-6\lambda_{S_2}-48\lambda_{S_2}^{\text{Adj}}\right)$ | $+24g_Y^4$ |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                                                                 | —                                                                        | —          |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                                                                 | —                                                                        | —          |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                                                                 | —                                                                        | —          |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                                                                 | —                                                                        | —          |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                                                                 | —                                                                        | —          |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                                                                 | —                                                                        | —          |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                                                                 | —                                                                        | —          |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                                                                 | —                                                                        | —          |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                                                                 | —                                                                        | —          |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                                                                 | —                                                                        | —          |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                 | —                                                                        | —          |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                                                                 | —                                                                        | —          |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                                                                 | —                                                                        | —          |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                                                                 | —                                                                        | —          |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                                                                 | —                                                                        | —          |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                                                                 | —                                                                        | —          |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                                                                 | —                                                                        | —          |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+g_Y^2\left(9g_2^2-\frac{3\lambda_{S_2}}{2}-12\lambda_{S_2}^{\text{Adj}}\right)$ | —                                                                        | —          |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                                                                 | —                                                                        | —          |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                                                                 | $+g_Y^2\left(36g_2^2-6\lambda_{S_2}-48\lambda_{S_2}^{\text{Adj}}\right)$ | $+24g_Y^4$ |

Terms 6 of 7

| Column | Representative term structure              |
|--------|--------------------------------------------|
| T1     | $+\frac{3\lambda_{12}^2}{4}$               |
| T2     | $+\lambda_{12}\lambda_{12}^{\text{Cross}}$ |
| T3     | $+12\lambda_{12}^{\text{Adj}^2}$           |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                           | T2                                         | T3                               |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|------------------------------|--------------------------------------------|----------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                            | —                                          | —                                |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                            | —                                          | —                                |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                            | —                                          | —                                |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                            | —                                          | —                                |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                            | —                                          | —                                |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                            | —                                          | —                                |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                            | —                                          | —                                |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                            | —                                          | —                                |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                            | —                                          | —                                |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                            | —                                          | —                                |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                            | —                                          | —                                |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                            | —                                          | —                                |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                            | —                                          | —                                |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                            | —                                          | —                                |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                            | —                                          | —                                |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                            | —                                          | —                                |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                            | —                                          | —                                |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                            | —                                          | —                                |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                            | —                                          | —                                |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                            | —                                          | —                                |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $+\frac{3\lambda_{12}^2}{4}$ | $+\lambda_{12}\lambda_{12}^{\text{Cross}}$ | $+12\lambda_{12}^{\text{Adj}^2}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $+\frac{3\lambda_{12}^2}{4}$ | $+\lambda_{12}\lambda_{12}^{\text{Cross}}$ | $+12\lambda_{12}^{\text{Adj}^2}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $+\frac{3\lambda_{12}^2}{4}$ | $+\lambda_{12}\lambda_{12}^{\text{Cross}}$ | $+12\lambda_{12}^{\text{Adj}^2}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $+\frac{3\lambda_{12}^2}{4}$ | $+\lambda_{12}\lambda_{12}^{\text{Cross}}$ | $+12\lambda_{12}^{\text{Adj}^2}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $+\frac{3\lambda_{12}^2}{4}$ | $+\lambda_{12}\lambda_{12}^{\text{Cross}}$ | $+12\lambda_{12}^{\text{Adj}^2}$ |

Terms 7 of 7

| Column | Representative term structure              |
|--------|--------------------------------------------|
| T1     | $-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$ |

| Model | $\alpha$ | $d_{S_1}$ | $Y_{S_1}$      | $d_{S_2}$ | $Y_{S_2}$     | $d_F$ | $Y_F$          | T1                                         |
|-------|----------|-----------|----------------|-----------|---------------|-------|----------------|--------------------------------------------|
| T3-A  | -2       | 1         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | —                                          |
| T3-A  | -1       | 1         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | —                                          |
| T3-A  | 0        | 1         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | —                                          |
| T3-A  | 1        | 1         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | —                                          |
| T3-A  | 2        | 1         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | —                                          |
| T3-B  | -2       | 2         | -1             | 2         | 0             | 1     | $-\frac{1}{2}$ | —                                          |
| T3-B  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 1     | 0              | —                                          |
| T3-B  | 0        | 2         | 0              | 2         | 1             | 1     | $\frac{1}{2}$  | —                                          |
| T3-B  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 1     | 1              | —                                          |
| T3-B  | 2        | 2         | 1              | 2         | 2             | 1     | $\frac{3}{2}$  | —                                          |
| T3-C  | -2       | 2         | -1             | 2         | 0             | 3     | $-\frac{1}{2}$ | —                                          |
| T3-C  | -1       | 2         | $-\frac{1}{2}$ | 2         | $\frac{1}{2}$ | 3     | 0              | —                                          |
| T3-C  | 0        | 2         | 0              | 2         | 1             | 3     | $\frac{1}{2}$  | —                                          |
| T3-C  | 1        | 2         | $\frac{1}{2}$  | 2         | $\frac{3}{2}$ | 3     | 1              | —                                          |
| T3-C  | 2        | 2         | 1              | 2         | 2             | 3     | $\frac{3}{2}$  | —                                          |
| T3-D  | -2       | 3         | -1             | 1         | 0             | 2     | $-\frac{1}{2}$ | —                                          |
| T3-D  | -1       | 3         | $-\frac{1}{2}$ | 1         | $\frac{1}{2}$ | 2     | 0              | —                                          |
| T3-D  | 0        | 3         | 0              | 1         | 1             | 2     | $\frac{1}{2}$  | —                                          |
| T3-D  | 1        | 3         | $\frac{1}{2}$  | 1         | $\frac{3}{2}$ | 2     | 1              | —                                          |
| T3-D  | 2        | 3         | 1              | 1         | 2             | 2     | $\frac{3}{2}$  | —                                          |
| T3-E  | -2       | 3         | -1             | 3         | 0             | 2     | $-\frac{1}{2}$ | $-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$ |
| T3-E  | -1       | 3         | $-\frac{1}{2}$ | 3         | $\frac{1}{2}$ | 2     | 0              | $-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$ |
| T3-E  | 0        | 3         | 0              | 3         | 1             | 2     | $\frac{1}{2}$  | $-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$ |
| T3-E  | 1        | 3         | $\frac{1}{2}$  | 3         | $\frac{3}{2}$ | 2     | 1              | $-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$ |
| T3-E  | 2        | 3         | 1              | 3         | 2             | 2     | $\frac{3}{2}$  | $-\frac{\lambda_{12}^{\text{Cross}^2}}{2}$ |